

Advances in Quantum Interior-Point Methods for Sparse Problems: Conditioning Geometry, Query Lower Bounds, and Access-Model Frontiers

Abstract

Can quantum interior-point methods (QIPMs) outperform classical interior-point methods on sparse linear and semidefinite programs? We give a calibrated answer by separating geometry, data access, and output. First, a sublevel-chord law lower-bounds the reduced central-path Hessian condition number for every self-concordant barrier; canonical log and log-det barriers attain this floor up to instance constants, so barrier redesign cannot in general remove the geometric obstruction. Degenerate LP log-barrier tails nevertheless have clustered spectra that are benign for classical Krylov methods. On the same clusters, plain quantum block-encoding access in sufficiently large dimension has linear worst-case condition dependence, while norm-tight factor access unconditionally supports a square-root polynomial route in its model; only the corresponding end-to-end solve needs explicit overlap assumptions. Exact coupling and effective spectral-dimension laws then identify which beyond-condition-number solver is appropriate for a given Newton right-hand side.

For block log-det centers we prove a winner-take-all condensation transition, finite mass-conditioning-visibility laws, and matching raw-query bounds for sparse holonomy SDPs. Verified-sample decoders and polynomial symmetrization give accuracy-dependent state-conversion curves. Separate LP and SDP families show that loading or recovering optimization states can require linear queries even when the relevant reduced Newton condition number is one. Convex-mixture, electrical, Fourier, congruence, preconditioning, and state-interface frontiers delimit the gadget and access classes covered by these lower bounds. The surviving positive results are correspondingly conditional: residual-certified pathwise direction reuse, partition-free equilibration and face repair, implicit dual coordinates with compressed output, and block-angular border aggregation.

The results do not give an unconditional end-to-end quantum advantage or disadvantage for sparse optimization. They instead locate the remaining bottlenecks in loading, recovery, dynamic oracle maintenance, and output. Every result reported here was re-derived and re-verified; in that process we correct the simulator-norm and clock analysis in the QCPM preprint and several earlier claims from our own research program.

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1 Introduction

Sparse linear and semidefinite programs are an attractive test case for quantum optimization. Their data may be queried or applied without reading a dense matrix, and an interior-point method reduces optimization to a sequence of structured linear systems. A quantum linear-system solver (QLS) can, in favorable access models, prepare a state proportional to a Newton direction without first writing every coordinate. These observations motivate a simple question:

Do quantum interior-point methods beat classical interior-point methods on sparse problems?

The word “sparse” does not by itself determine the answer. An input matrix can be sparse while its normal equations, null-space basis, factorization, or inverse is dense. Conversely, a matrix that looks unfavorable through its condition number can have a clustered spectrum that a classical Krylov method uses efficiently. A quantum state proportional to a direction is not yet a classical iterate, and a preconditioned matrix oracle is not free merely because the matrix it represents is well conditioned. The relevant comparison must therefore name the Newton formulation, the input oracle, the block-encoding normalization, the right-hand side, the requested output, and the classical solver that receives the same structure.

This paper develops that comparison at three levels. At the *geometric* level, we ask which central-path conditioning is inherent in the optimization problem and which can be changed by selecting another barrier. At the *access-model* level, we ask what a classical or quantum linear solver can extract from the same spectral structure and what it costs to construct the operator and right-hand side it receives. At the *interface* level, we ask whether the output of one solve can be recovered, certified, and used to form the next solve. The resulting picture is neither a blanket quantum speedup nor a blanket no-go theorem. It is a set of sharp laws and scoped frontiers that identify where either conclusion would still have to be proved.

1.1 State of play

The first generation of QIPMs inserted a QLS into a classical path-following framework. Kerenidis and Prakash treated LPs and SDPs; Casares and Martin-Delgado developed a predictor–corrector LP method; Augustino et al. gave conic and semidefinite variants; and the Terlaky-group line developed inexact feasible, infeasible, orthogonal-subspace, and refinement-based methods [11–14, 38, 83, 106, 107, 137]. These methods made the potential role of QLS state preparation precise, but their ledgers also exposed condition, normalization, tomography, QRAM, and classical-update costs. Dalzell et al. emphasize the same end-to-end accounting issues in their QIPM review and resource analysis [51, 52].

Two newer lines change the architecture rather than merely improving the linear solver. The quantum central-path method of Augustino, Leng, Nannicini, Terlaky, and Wu uses Hamiltonian evolution and potential evaluation rather than repeated QLS tomography [15]. The self-concordant Schrödinger-operator program of Gribling et al. likewise seeks optimization without a Newton condition-number factor [70]. These approaches show why a conditioning theorem must be scoped to a named Newton matrix: a lower bound on a barrier Hessian is not a lower bound on every quantum optimization architecture.

Apers and Gribling take a different condition-number-free route. Their tall LP algorithms use quantum spectral approximation and local-norm estimation, return an explicit solution, and prove a row-query separation when the number of constraints is sufficiently larger than the variable dimension [10]. Within the QIPM evidence compared here, this is the only proved asymptotic quantum–classical separation. It is a row-query separation under a succinct row oracle, not automatically a wall-clock separation from the best nnz-aware sparse solver. Its requirements and high powers of the variable dimension also make it a specific positive regime rather than a generic consequence of sparsity.

The QLS primitive itself has evolved from HHL through the Childs–Kothari–Somma precision improvement and QSVT pseudoinversion [43, 65, 76]. Orsucci and Dunjko proved access-sensitive upper and lower bounds for positive-definite systems. Mori et al. proved an $\Omega(\kappa\sqrt{s})$ sparse-access lower bound at fixed constant error and restated an $\Omega(\kappa \log(1/\epsilon))$ constant-sparsity bound from earlier unpublished work of Harrow and Kothari. Dalzell, Li, and Su introduced truncation and filtering parameters that can be smaller than a worst-case condition number [53, 109, 118]. Somma and de Wolf’s related plain-versus-sum-of-squares separation concerns guided ground-state energy, not QLS, and we keep that distinction explicit in [section 4.3](#) [125]. These results make “the QLS cost” an incomplete phrase: cost depends on whether one has plain matrix access, factor access, sparse-entry access, or a stronger supplied oracle, and on how the right-hand side overlaps the hard spectral subspace.

Tomography is the second standard bottleneck. A path-following method usually needs a classical direction to update all primal and dual coordinates, while a QLS naturally returns only a normalized state. State-preparation-unitary tomography improves the dependence on readout accuracy, but still scales with the direction dimension [105, 131]. Avoiding that cost requires a different output contract and, crucially, a way to generate the next changing diagonal or Newton operator from the compressed representation. The interface cannot be removed by renaming a state as an iterate.

Classical sparsity exploitation sets a demanding baseline. Practical IPMs choose between normal equations and augmented systems, control fill, reuse symbolic factorizations and preconditioners, and adapt inner tolerances [67, 69, 136]. For bounded-treewidth structure, supplied elimination orders and specialized IPMs can be nearly linear in the problem size [56, 72]. A sparse-QIPM claim must therefore be compared with a structure-aware factorization or Krylov method, not with dense Gaussian elimination by default.

1.2 What a matched comparison counts

The algorithmic literature above cannot be compared by retaining only the dimension and the sparsity of the original constraint matrix. Its QIPM lines use different Newton formulations, state-preparation primitives, refinement schemes, and outputs. We use a raw-query/QRAM-write ledger throughout the paper, counting calls after the oracle is precisely defined and writes needed to build or refresh QRAM. We pair it with qualitative end-to-end audits of construction, reuse, recovery, and output. Gate and arithmetic work are populated only for selected results, so these audits are not a uniform wall-clock ledger. Neither accounting view substitutes for the other.

From sparse input to the solved operator. For an LP, a diagonal scaling of a sparse constraint matrix may produce a normal equation $AD^2A^\top$, an augmented KKT matrix, or an orthogonal-subspace system. Constant row sparsity of A need not imply constant sparsity of any of these objects: one dense column creates a clique in the normal-equation graph, and a null-space basis can be dense. The opposite can also occur, since a carefully chosen basis or augmented formulation can preserve structure that normal equations lose. Consequently our notation distinguishes $\text{nnz}(A)$, the sparsity s_H of the actual system H , and the access description of a factor G when $H = G^\dagger G$; the formal definitions are in [sections 2.1.1](#) and [2.2](#). A claim that starts from sparse A must prove the access properties of the operator it actually solves.

Condition and normalization. A block encoding presents H/α_H , so the inversion scale is governed by $\alpha_H\|H^{-1}\|$, not by the spectral condition number in isolation. Rescaling H changes neither this product nor the underlying problem. Multiplying separately encoded factors can multiply their normalizations, and forming a perfect preconditioned product can require the same input information that the original solve was meant to discover. The factor, congruence, and parity results of [section 12](#) make these exchanges exact on representative families. On the classical side, an effective preconditioner also has a construction and application cost, but it can often be reused and can exploit entries already stored explicitly. A matched comparison prices both constructions rather than granting either side an ideal inverse oracle.

Right-hand sides and solve accuracy. Worst-case QLS complexity is not always the right estimate for a Newton step. The direction depends on a particular RHS, and truncation can ignore a hard eigenspace only if the requested output error permits losing its solution mass. A residual guarantee, a normalized-state error, and a coordinatewise direction error are different contracts. They are connected by condition-dependent inequalities, so comparing a quantum state-error bound with a classical residual tolerance without conversion is not meaningful. The coupling and spectral-measure analysis in [section 5](#) evaluates these distinctions on exact IPM systems, while the output contracts of [section 2.3](#) state which error is being promised.

State preparation and readout. Under normalized operator and RHS access, a QLS call produces a quantum state proportional to a solution. A hybrid QIPM that updates a classical iterate must then estimate a full vector, recover its norm and scale, and ensure that the resulting step satisfies the outer neighborhood condition. Repeating the state preparation for tomography repeats every nonpersistent oracle and RHS cost. A method that requests only an observable or

compact certificate can avoid full readout, but then it must show how that object is sufficient for optimization and for constructing the next iteration. The distinction among classical-vector, compact, state, density, coordinate-oracle, and heralded outputs in [section 2.3](#) is designed to prevent a proof for one interface from being silently reused for another.

The classical comparator. For a sparse direct method, the per-step cost is controlled by realized fill, numerical factorization, and triangular solves, with a symbolic ordering normally reused. Multiple predictor or corrector RHSs can reuse the same factorization. For a Krylov method, the relevant costs are matrix–vector and preconditioner applications times a spectrum- and tolerance-dependent iteration count. Clustered eigenvalues can make that count far smaller than a worst-case $\sqrt{\kappa}$ estimate, as [proposition 4.2](#) demonstrates. Specialized graph, treewidth, block-angular, and low-rank structure can improve the baseline further. We therefore do not compare a quantum query count directly with dense cubic arithmetic or treat the act of writing a dense classical direction as the only classical cost.

Setup, refresh, and trajectory reuse. One-step costs do not automatically multiply by the IPM iteration count. A fixed sparse input can be read once, a symbolic factorization can persist, and a lazy oracle can combine stored data with a changing scalar path parameter. Conversely, a QRAM or block encoding of changing primal and slack diagonals may require per-step writes, and tomography may be repeated at every step. The distinction is formalized by the amortized accounting contract in [definition 2.27](#). Our online direct-sum lower bound applies only when new oracle information is genuinely fresh [theorem 12.14](#); the caching-ceiling argument in [section 12.5](#) shows why no unqualified T -times-one-step claim follows for a static LP.

Query separation versus time separation. Raw coefficient, sparse-entry, row, block, QRAM, and preconditioned-product queries reveal different amounts of information. A sublinear row-query algorithm can be a genuine oracle separation even when generating a queried row or performing local arithmetic prevents a time advantage. Conversely, an explicitly stored sparse matrix may have no interesting input-query cost, while its repeated numerical solves remain expensive. The Apers–Gribling result is positive in its declared row-query model; our parity and holonomy lower bounds are negative in their declared raw local-query models. Neither statement changes models for the sake of the comparison. This discipline is why the later theorems often conclude with a setup–solve–recovery inequality rather than with a single condition-number exponent.

1.3 Answer in outline

Our first answer is that central-path conditioning is geometry before it is barrier design. For a compact feasible slice, the diameter of an objective sublevel set and the length of its longest chord control the extreme eigenvalues of every self-concordant barrier Hessian. The sublevel chord law in [theorem 3.3](#) gives a condition-number floor at a fixed objective gap, and [theorem 3.5](#) shows that the canonical log and log-det barriers attain that floor up to explicit instance constants. The conclusion is minimax and gap matched: changing the barrier cannot in general remove the divergence. It does not say that every optimum is ill conditioned, nor does it cover a method that avoids barrier-Newton solves.

Our second answer is that a large condition number does not determine the relative classical and quantum costs. Degenerate LP log-barrier Hessians split into two intervals with bounded internal ratios [proposition 4.1](#). A classical residual polynomial can exploit the empty interval, and [proposition 4.2](#) gives polylogarithmic iteration dependence on the inter-cluster separation. Quantum polynomial transformations must also remain bounded on the domain imposed by their access model. Under plain block-encoding access the worst-case QLS cost is $\hat{\Theta}(\kappa)$, even

for two-point clusters when $N_{\text{QLS}} \gtrsim \kappa$. A norm-tight factor presentation unconditionally has a $\tilde{\Theta}(\sqrt{\kappa})$ polynomial route in its model; only the corresponding end-to-end solve needs the overlap hypotheses in [theorem 4.3](#). On-path right-hand sides can be much easier still: [proposition 4.5](#) and [corollary 4.6](#) show that the exact central-path right-hand side has $O(\mu^2)$ weak-cluster fraction, so a top-window solve meets the fixed relative-residual contract once the spectral gap is resolved.

This classical–quantum contrast is not paradoxical. CG is charged for matrix–vector products and can place roots in spectral gaps. QSVT is charged for a bounded transformation of a supplied normalized operator, and the normalization and symmetric-domain constraints remain part of the problem. The comparison reverses the naive view that a large κ is always bad for the classical method and automatically favorable to a quantum method. It also motivates the instance-selection laws of [section 5](#): one should measure right-hand-side coupling and spectral mass, not select a solver from κ alone.

Our third answer is that the decisive lower bounds often live outside the linear solve. The LP constructions of [section 9](#) can make the reduced Newton geometry condition one while hiding a parity-dependent affine translation or original-coordinate direction. The SDP constructions of [section 11](#) separate intrinsic value information, Schur solves, KKT access, and recovery. The preconditioning results of [section 12](#) show product laws: improving spectral condition can worsen factor normalization or physical-state sensitivity, and a state from one solve need not provide a coordinate oracle for the next. The right unit of analysis is therefore a complete setup–load–solve–recover–output ledger.

Condensation supplies a complementary positive and negative example. A shared trace budget drives winner-take-all mass onto the lowest-cost PSD block. That mass simultaneously controls reduced-Hessian conditioning, trace-distance visibility, and the query cost of state preparation. The transition has matching lower and upper query bounds on a bounded-incidence holonomy SDP, while small-success search gives a tight accuracy curve on its proved domain; the $\delta = o(k/G)$ sliver and the internal-state η_{int} -channel remain open [section 8.4](#). Thus output hardness is not inferred from condition alone: it is proved through a visible, decodable output contract.

Finally, our no-go results are frontier theorems, not universal impossibilities. Convex mixtures, cut resistance, bounded-arity Walsh rows, symmetry work identities, scalar congruences, and state-interface products rule out the natural gadget classes to which their hypotheses apply. The section records the five escape routes in [section 10.6](#). The positive modules of [section 13](#) inhabit some of those routes, but remain conditional and per-step: they require stable geometry, explicit access, compressed output, or a low-dimensional border. No section silently upgrades one such module to an end-to-end sparse-QIPM advantage.

1.4 Contributions

The technical contributions follow the paper’s order. Each item names the result that supports its headline and the boundary under which it should be read.

1. **Universal conditioning geometry ([section 3](#))**. We prove minimum-eigenvalue rigidity and the sublevel chord law for every self-concordant barrier [theorems 3.2](#) and [3.3](#), sharpen it to a width-by-width spectral sandwich [theorem 3.4](#), and show canonical LP and product-SDP barriers attain the resulting floor up to an explicit gap-matched factor [theorem 3.5](#). The conditioning dictionary includes positive-dimensional optimal faces, degenerate vertices, curved optima, singularity-degree exponents, and approximate centrality [corollaries 3.6, 3.8, 3.10](#) and [3.11](#).
2. **Clustered spectra and solver separation ([section 4](#))**. We prove the two-cluster LP tail [proposition 4.1](#), the cluster-sensitive CG bound [proposition 4.2](#), and the plain-versus-factor quantum polynomial frontier [theorem 4.3](#). The exact central-path right-hand side obeys a weak-mode cancellation law, yielding a condition-number-free top-window solve under the fixed

relative-residual contract once the spectral gap is resolved [proposition 4.5](#) and [corollary 4.6](#). The generic $\tilde{\Theta}(\sqrt{\kappa})$ factor-access QLS interpretation is explicitly rejected: the theorem gives a polynomial metric and an overlap-qualified solve, not such an unrestricted algorithm.

3. **Instance laws beyond κ** ([section 5](#)). Exact block inversion identifies when DLS filtering improves on worst-case conditioning [theorem 5.1](#); a truncation transition separates solution-state and residual contracts [proposition 5.3](#); and the effective-spectral-dimension theorem places truncation and filtering thresholds at dimensions four and eight [theorem 5.4](#). Stable active-range equilibration preserves the Newton state for $f \in \text{range } Q$ while removing the divergent tail when its projector is supplied [theorem 5.7](#).
4. **Block log-det condensation** ([section 6](#)). We derive the exact capacity transition and ground-multiplicity-dependent conditioning phases [theorems 6.2](#) and [6.4](#), prove finite mass-conditioning laws [theorem 6.5](#), and link them to trace-distance visibility and raw-query complexity [theorems 6.6](#) and [6.7](#). The transition extends to sublinear multiwinner sets and to a stated approximate-centrality model [theorems 6.9](#) and [6.10](#). These are statements about the equality-reduced Frobenius Hessian and trace-normalized output, not every preconditioned KKT representation.
5. **A sparse winner-take-all SDP** ([section 7](#)). A bounded-incidence qutrit-holonomy family realizes the condensation law with tight $\Theta(N\sqrt{G})$ quantum and $\Theta(NG)$ randomized value-query costs, a condition-one public root start, robust winner recovery [theorem 7.2](#), and a matching search-then-solve compact-output algorithm [theorem 7.3](#). The formulation tradeoff in [section 7.9](#) keeps the bounded-incidence claim separate from an isometric ambient conditioning claim.
6. **Accuracy curves for hidden-block state conversion** ([section 8](#)). A verified-sample decoder gives a mixed-output lower bound [theorem 8.1](#); detuned rejection preparation supplies the upper bound [theorem 8.2](#); and parity collapse yields the tight nontrivial conversion curve [theorem 8.7](#) and [corollary 8.9](#). The batch results separate nonamortizable vector materialization from scalar approximate counting [theorems 8.12](#) and [8.13](#). The theorem domains retain the loser-state discrepancy, the small-success sliver, and logarithmic fixed-point costs rather than extrapolating through them.
7. **LP loading and recovery hardness** ([section 9](#)). The gain-plateau family gives a linear robust primal-state lower bound [theorem 9.4](#). Prefix rigidification strengthens this to condition-one reduced Newton geometry with linear original-coordinate recovery cost [theorem 9.7](#). Complete KKT and Schur-preconditioned variants isolate public-right-hand-side, residual, bounded-data, and scale tradeoffs [theorems 9.8](#) to [9.11](#) and [9.13](#). None is presented as a condition-one QLS lower bound with free transformed oracles.
8. **Gadget-class frontiers** ([section 10](#)). A single-flip convex-mixture theorem gives the graph-free default obstruction [theorem 10.1](#). Cut-resistance and Walsh-row theorems quantify the volume or scale needed by signed-copy and bounded-arity constructions [theorems 10.4](#) and [10.5](#); gauge-flow, work, and Fourier identities locate the multiplier or input-incidence cost [theorems 10.6](#) to [10.8](#). The escape routes in [section 10.6](#) are part of the result.
9. **SDP-native loading and recovery hardness** ([section 11](#)). An intrinsic trace-normalized qutrit family has parity-hard value and a public, condition-one reduced Newton step whose primal and complete-KKT solution states are parity hard [theorems 11.1](#) and [11.2](#). Non-SOCP Schur families give an exact loading/accuracy frontier [theorems 11.3](#) and [11.4](#), and tree and group holonomy lift the mechanism to matrix-valued words [theorems 11.5](#) to [11.7](#). The proved query exponent for the S_3 family is still obtained on an abelian restriction; intrinsically nonabelian word hardness remains open.

10. **Preconditioning and state-interface products (section 12).** Whitening and reusable inverse access obey normalization tradeoffs [theorems 12.2 and 12.3](#); scalar congruence trades transformed condition against physical recovery [theorems 12.4 and 12.5](#); and the parity ledger charges setup, RHS preparation, solve, and recovery [theorems 12.7 and 12.8](#). State-to-next-solve and lazy-oracle theorems distinguish a one-use state from a reusable coordinate service [theorems 12.9, 12.13, 12.15 and 12.16](#).
11. **Conditional positive modules (section 13).** We prove a residual-certified pathwise refresh bound [theorem 13.1](#), a constructive stable active-range equilibration and exact face repair [theorems 13.6 and 13.7](#), an implicit slack oracle and conditional compressed dual QIPM [theorems 13.8 and 13.9](#), and a block-angular border solver whose $\sqrt{N}$ block-count exponent is optimal in the block-oracle model, as proved in [section 13.5](#). Every theorem keeps its tail, gap, forcing, normalization, and output assumptions visible.
12. **Reusable structural tools (section 14).** Involution and compact-group work identities [theorems 14.1 and 14.2](#), a noncommutative central-mixture sandwich [theorem 14.5](#), and rank and support barriers [theorems 14.8 and 14.9](#) explain several no-go mechanisms abstractly. Sparse OSS and treewidth-two KKT embeddings preserve known QLS restrictions [theorems 14.10 and 14.11](#), while exact-treewidth value hardness and a central-oracle light cone distinguish query, output, and physical-locality models [theorems 14.12 and 14.13](#).
13. **Corrections and dead ends (section 15).** We show that the global potential supremum used in the QCPM simulator ledger cannot scale as asserted [proposition 15.1](#), correct the independent clock/time-dilation error [equation \(693\)](#), derive the corrected norm scale and its speed tradeoff [theorem 15.3 and corollary 15.4](#), and give explicit densification and hidden-start counterexamples [theorems 15.5 and 15.6](#). The recorded dead ends prevent first-inverse norms, sparse metrics, exact Schur products, or one-step bounds from being reused as claims they do not prove.

1.5 Related work and positioning

Quantum interior-point methods. The QIPM literature spans QLS-based hybrid path following, iterative refinement, orthogonal-subspace formulations, condition-number-free sampling, and Hamiltonian central paths. Representative lines include Kerenidis–Prakash, Casares–Martin-Delgado, Augustino et al., Mohammadisiahroudi et al., Wu et al., Apers–Gribbling, and the reviews and resource ledgers of Dalzell et al. [[10, 14, 38, 52, 83, 107, 137](#)]. The QCPM and self-concordant-Schrödinger lines avoid the usual repeated-QLS architecture [[15, 70](#)]. We do not collapse these algorithms into a single complexity table: each uses a different access, condition, precision, and output contract. The per-section novelty paragraphs compare the exact theorem at hand with its nearest line of work.

Quantum linear systems. HHL introduced sparse QLS state preparation, CKS improved the precision dependence, and QSVT placed inversion in a general block-encoding calculus [[43, 65, 76](#)]. Orsucci–Dunjko’s positive-definite lower bounds ground our plain-access clustered comparison. Mori et al. prove $\Omega(\kappa\sqrt{s})$ at fixed constant error and restate an $\Omega(\kappa \log(1/\epsilon))$ constant-sparsity bound from earlier unpublished work of Harrow and Kothari. DLS provide the beyond-condition-number parameters evaluated in [section 5](#) [[53, 109, 118](#)]. Somma–de Wolf provide an analogous plain-versus-sum-of-squares resolution separation in a guided Hamiltonian problem, not a QLS theorem [[125](#)]. This access-model discipline is essential to [sections 4 and 12](#).

Quantum optimization lower bounds. Van Apeldoorn et al. establish general quantum SDP upper and lower bounds and Boolean-in-optimum constructions; Brandão–Svore initiated the quantum SDP solver line; and Apers–Gribbling’s Theorem 8.4 gives a sparse-LP value lower

bound already having a linear exponent in balanced dimensions [10, 30, 130]. Our parity and holonomy families do not claim parity, adversary composition, or the linear exponent as new. Their contribution is the conjunction with exact central paths, condition-one or bounded-condition formulations, robust output contracts, and complete loading/recovery ledgers described in sections 7, 9 and 11.

Classical interior-point structure. Wright’s work on logarithmic-barrier Hessians and numerical error, the standard primal–dual theory, and the Forsgren–Gill–Wright survey provide the conditioning and implementation background [63, 134–136]. Gondzio and collaborators document preconditioning, structure exploitation, and practical IPM evolution [67, 69]. Small-treewidth LP and SDP algorithms show why bounded graph width is a classical algorithmic asset rather than merely a sparse-oracle promise [56, 72]. The sublevel chord law differs from the classical log-barrier conditioning analyses by quantifying over all self-concordant barriers, while section 4 uses rather than ignores the classical cluster-sensitive Krylov advantage.

Positioning discipline. Many ingredients used below are established: self-concordant containment, weak-sharp and singularity-degree error bounds, matrix-Burg resolvents, Rayleigh–Jeans condensation, parity and search, group synchronization, Kantorovich inequalities, effective resistance, and sparse elimination. We credit them where each enters. Because the nearest prior work changes from one theorem package to the next, this introduction does not restate every collision or priority qualification. The detailed novelty and scope statements in sections 3.1, 6.11, 7.9, 11.5, 12.6, 13.7 and 14.6 are part of the claims.

1.6 Conventions, verification, and corrections

The distinctions formalized in section 2 are standing conventions, not optional caveats. In particular, input sparsity differs from Newton-system sparsity; κ_H differs from block-encoding normalization; reduced and ambient KKT conditioning are different quantities; and value, classical-vector, quantum-state, coordinate-oracle, and heralded outputs are different contracts. Query bounds count the setup and transformations named by their access model. An uncharged stronger oracle is treated as a stronger problem statement, not as an implementation of raw sparse access.

The verification process changed several statements inherited from preliminary research notes and identified two independent issues in a public preprint. The corrections relevant to reading the paper are:

1. section 15.1 corrects the QCPM v2 simulator-norm estimate and its independent clock/time-dilation error: a global supremum on a fixed truncation box is not controlled by concentration near the instantaneous center, and the time-varying kinetic coefficient requires the corrected clock in equation (693). Theorem 15.3 and corollary 15.4 give the resulting scale. This is a correction to that derivation, not a general impossibility result for Hamiltonian optimization.
2. section 4.3 corrects an overbroad factor-access inference in our earlier analysis. A square-root polynomial metric, or a two-stage solve with explicit overlap, does not imply a generic $\tilde{\Theta}(\sqrt{\kappa})$ factor-access QLS; the Orsucci–Dunjko instance rules out that unrestricted interpretation.
3. section 4.1 replaces a coarse three-scale picture of the degree-two SDP witness by the verified four-window spectrum. The correction changes the intermediate spectral description, not the condition-number exponent.
4. section 5.3 anchors effective spectral dimension at the smallest visible eigenvalue. This repairs the earlier shell formulation and is needed for the dimension-four and dimension-eight phase laws in theorem 5.4.

5. [sections 8.1, 8.2](#) and [8.4](#) replace the collision shortcut by a verified-sample decoder, restrict the classical linear-in-success law to its actual domain, keep the very-small-success sliver open, and attribute the k -marked small-success search bound to Dohotaru–Høyer rather than to Zalka’s one-marked theorem alone.
6. [section 13.2](#) corrects the earlier claim that two convergent leakage ledgers alone bound projective variation. Neither the ledgers nor an additional $O(\mu_t)$ tail bound suffices: an oscillating-face-displacement counterexample has convergent ledgers, the tail bound, and linearly growing realized variation, so finite realized variation stays a separate hypothesis.

The list is deliberately narrow. It records changes that affect the theorem statements or their interpretation, rather than presenting ordinary proof repairs as additional results. Further failed shortcuts and their valid replacements are collected in [section 15.4](#).

1.7 Organization and reading paths

The conventions are fixed in [section 2](#); thereafter we use the section pointers rather than redefining those contracts. Conditioning geometry, clustered spectra, and beyond-condition-number instance laws occupy [sections 3 to 5](#). Condensation, its sparse winner-take-all realization, and the associated state-conversion accuracy curve occupy [sections 6 to 8](#). The LP and SDP loading/recovery lower bounds, gadget frontiers, and preconditioning/interface obstructions are in [sections 9 to 12](#). The conditional positive modules are in [section 13](#), structural tools in [section 14](#), corrections and dead ends in [section 15](#), and the combined verdict and open frontier in [section 16](#).

Three shorter paths may be useful. A reader interested only in conditioning can read [sections 2 to 5](#). A reader interested only in lower bounds can read [sections 2 and 6 to 12](#). A reader interested only in algorithmic modules can read [sections 2, 5 and 13](#). In every path, the output and access definitions of [section 2](#) remain part of the theorem statements.

2 Preliminaries

This section fixes the optimization, access, and output models used throughout the paper. The distinctions are part of the problem specification: input sparsity is not Newton-system sparsity, the spectral condition number is not a block-encoding normalization, and a normalized quantum state is not an explicit solution vector.

2.1 Conic programs and central paths

For a linear program (LP) in standard form, the primal–dual pair is

$$\begin{aligned}
 (\text{P}_{\text{LP}}) \quad & \min_{x \in \mathbb{R}^n} \quad c^\top x \quad \text{subject to} \quad Ax = b, \quad x \geq 0, \\
 (\text{D}_{\text{LP}}) \quad & \max_{y \in \mathbb{R}^m, s \in \mathbb{R}^n} \quad b^\top y \quad \text{subject to} \quad A^\top y + s = c, \quad s \geq 0,
 \end{aligned} \tag{1}$$

Throughout the LP discussion, $A \in \mathbb{R}^{m \times n}$ has full row rank unless stated otherwise. We write $X = \text{diag}(x)$, $S = \text{diag}(s)$, and e for the all-ones vector. A feasible primal–dual point has gap

$$c^\top x - b^\top y = x^\top s. \tag{2}$$

The product-cone form of a semidefinite program (SDP) used below has q matrix blocks. Put $\mathcal{K} = \prod_{j=1}^q \mathbb{S}_+^{d_j}$ and equip it with $\langle X, S \rangle = \sum_j \text{tr}(X_j S_j)$. For symmetric data $C_j, A_{ij} \in \mathbb{S}^{d_j}$, define $\mathcal{A}(X)_i = \sum_j \text{tr}(A_{ij} X_j)$ and $(\mathcal{A}^* y)_j = \sum_i y_i A_{ij}$. The pair is

$$\begin{aligned}
 (\text{P}_{\text{SDP}}) \quad & \min_{X \in \mathcal{K}} \quad \langle C, X \rangle \quad \text{subject to} \quad \mathcal{A}(X) = b, \\
 (\text{D}_{\text{SDP}}) \quad & \max_{y \in \mathbb{R}^m, S \in \mathcal{K}} \quad b^\top y \quad \text{subject to} \quad \mathcal{A}^* y + S = C.
 \end{aligned} \tag{3}$$

Its primal–dual gap is $\langle C, X \rangle - b^\top y = \langle X, S \rangle$. The one-block formulation is the special case $q = 1$; scalar nonnegative variables can equivalently be included as 1×1 blocks. These are the standard conic dual pairs under the trace inner product [28].

Definition 2.1 (Strict feasibility). An LP is *strictly primal–dual feasible* if there are points satisfying (1) with $x > 0$ and $s > 0$. An SDP is strictly primal–dual feasible if there are feasible points with $X_j \succ 0$ and $S_j \succ 0$ for every block. We use *strict feasibility* for this two-sided condition unless one side is named. Because both sides are then feasible, their optimal values are finite; the two-sided Slater condition gives zero duality gap and attainment on both sides [28].

We next recall the barrier terminology in the normalization used by Nesterov and Nemirovskii [113]. For a C^3 convex function F on an open convex domain, define the local norms

$$\|h\|_x = \sqrt{D^2 F(x)[h, h]}. \quad (4)$$

On an affine feasible slice, derivatives and Hessians in (4) are restricted to its tangent space.

Definition 2.2 (ν -self-concordant barrier). A function F is a ν -self-concordant barrier for a convex set Q if $\text{dom } F = \text{relint } Q$, it tends to $+\infty$ at the relative boundary, its Hessian is positive definite on the tangent space, and, for all $x \in \text{relint } Q$ and tangent directions h ,

$$|D^3 F(x)[h, h, h]| \leq 2\|h\|_x^3, \quad |DF(x)[h]| \leq \sqrt{\nu} \|h\|_x. \quad (5)$$

The declared value ν is called a barrier parameter; it need not be the least admissible value.

The logarithmic barrier for the nonnegative orthant and the log-determinant barrier for the product PSD cone are, respectively,

$$F_{\log}(x) = -\sum_{i=1}^n \log x_i, \quad F_{\log \det}(X) = -\sum_{j=1}^q \log \det X_j. \quad (6)$$

They have parameters $\nu = n$ and $\nu = \sum_j d_j$. Both are logarithmically homogeneous: $F(tx) = F(x) - \nu \log t$ in the appropriate cone notation. The symbol ν always denotes this cone barrier's parameter, which for these canonical barriers equals the degree of logarithmic homogeneity. Restricting the cone barrier to an affine feasible slice preserves self-concordance with parameter at most ν , possibly with a smaller least parameter; we do not redefine ν after that restriction. The cone barrier parameter, rather than the ambient number of stored coefficients, controls the standard path-following iteration bound [113].

Definition 2.3 (Central path and its two parametrizations). Let F be a self-concordant barrier on the relative interior of the primal feasible region, suppose OPT is finite and attained, and suppose the barrier subproblem has a unique minimizer. The μ -parametrized primal central path is

$$x_F(\mu) = \arg \min \{c^\top x + \mu F(x) : Ax = b, x \in \text{dom } F\}, \quad \mu > 0. \quad (7)$$

Its primal objective gap is

$$g_F(\mu) = c^\top x_F(\mu) - \text{OPT}. \quad (8)$$

On any interval where g_F is one-to-one, the same curve indexed by $g > 0$ is the g -parametrized central path, denoted $x_F(g)$. For the canonical LP barrier, the primal–dual central path additionally satisfies

$$Ax = b, \quad A^\top y + s = c, \quad x, s > 0, \quad XSe = \mu e. \quad (9)$$

For the product-PSD log-det barrier it satisfies

$$\mathcal{A}(X) = b, \quad \mathcal{A}^* y + S = C, \quad X_j, S_j \succ 0, \quad S_j = \mu X_j^{-1} \quad (1 \leq j \leq q). \quad (10)$$

Thus the primal–dual gap on either canonical path is exactly $\nu\mu$, with ν the cone barrier parameter fixed above.

The first-order condition in (7) is the source of the dual slack relation $s = -\mu \nabla F(x)$ whenever that gradient is identified with a cone dual variable. For a general self-concordant barrier, $g_F(\mu) \leq \nu\mu$; equality need not hold [113].

Definition 2.4 (Trace-normalized log-det center). On the affine slice $\sum_j \text{tr } X_j = 1$, the *trace-normalized log-det center* at parameter μ is the minimizer of

$$\langle C, X \rangle - \mu \sum_j \log \det X_j, \quad X_j \succ 0, \quad \sum_j \text{tr } X_j = 1. \quad (11)$$

With the Lagrangian sign convention $\langle C, X \rangle - \mu \sum_j \log \det X_j - \lambda(\sum_j \text{tr } X_j - 1)$, stationarity reads $X_j = \mu(C_j - \lambda I)^{-1}$. This expression is admissible only for $\lambda < \min_j \lambda_{\min}(C_j)$, so every inverse is positive definite. The corresponding log-det Hessian is $D^2 F(X)[U, V] = \sum_j \text{tr}(X_j^{-1} U_j X_j^{-1} V_j)$ [28].

This is a specialization of the product-PSD central path, not a different optimization model. Trace normalization also makes $\bigoplus_j X_j$ a density operator, connecting the conic and quantum output conventions below.

Remark 2.5 (The convention for g). The conditioning sections use g only for the primal objective gap (8). Although an objective gap is sometimes called a “duality gap” informally, here *primal-dual gap* means $c^\top x - b^\top y = x^\top s$, which equals $\nu\mu$ on the canonical LP central path. In general $g_F(\mu) \leq \nu\mu$, and equality need not hold.

Definition 2.6 (Analytic center). If F attains a minimum over the relative interior of a feasible convex set, its minimizer is the *F-analytic center*. With the canonical barriers we simply say *analytic center*. For a bounded feasible region, the large- μ limit of (7), when it exists, is this center. For a face on the boundary of the original domain, “analytic center of the face” means the minimizer of a barrier defined on that face’s relative interior; it does not mean evaluating the original barrier at a boundary point [113].

For an LP point satisfying the two feasibility equations, put $\bar{\mu} = x^\top s/n$. The narrow neighborhood used by short-step analyses is the following standard one [136].

Definition 2.7 ($N_2(\theta)$ neighborhood). For $0 < \theta < 1$, the $N_2(\theta)$ *neighborhood* of the LP central path is

$$N_2(\theta) = \left\{ (x, y, s) : \begin{array}{l} Ax = b, \quad A^\top y + s = c, \quad x, s > 0, \quad \bar{\mu} = x^\top s/n, \\ \|XSe - \bar{\mu}e\|_2 \leq \theta \bar{\mu} \end{array} \right\}. \quad (12)$$

Unless an infeasible or matrix-valued analogue is explicitly named, $N_2(\theta)$ always means (12).

A short-step path-following method keeps a fixed small θ , changes the target complementarity by a $1 - \Theta(1/\sqrt{\nu})$ factor, and takes a full Newton step back into the neighborhood. This gives $O(\sqrt{\nu} \log(\nu/\epsilon_{\text{opt}}))$ Newton steps under standard initialization and scaling conventions; this statement is an iteration count, not a linear-solver, data-loading, or readout cost [113, 136].

2.1.1 Newton formulations and conditioning

At a positive, possibly infeasible LP iterate, define residuals

$$r_p = b - Ax, \quad r_d = c - A^\top y - s, \quad r_c = \hat{\mu}e - XSe, \quad (13)$$

where frequently $\hat{\mu} = \sigma \bar{\mu}$ for a centering parameter $\sigma \in [0, 1]$. Linearizing primal feasibility, dual feasibility, and complementarity gives the full primal-dual KKT system

$$\begin{bmatrix} A & 0 & 0 \\ 0 & A^\top & I \\ S & 0 & X \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta s \end{bmatrix} = \begin{bmatrix} r_p \\ r_d \\ r_c \end{bmatrix}. \quad (14)$$

This equation, with the residual signs in (13), is what we call the *standard infeasible-start Newton correction*; a feasible-start direction has $r_p = r_d = 0$ [136]. A full step eliminates the affine primal and dual residuals exactly, but complementarity is targeted only to first order. After a full step satisfying the third block row exactly, $(X + \Delta X)(S + \Delta S)e = \hat{\mu}e + \Delta X \Delta S e$. The quadratic term is uncontrolled by the linear system alone, so progress also requires a neighborhood invariant and a step-size rule.

Eliminating Δx and Δs gives the normal-equation coefficient matrix

$$H_{\text{NE}} = AXS^{-1}A^\top \quad (15)$$

and the equation

$$H_{\text{NE}}\Delta y = r_p - AS^{-1}r_c + AXS^{-1}r_d. \quad (16)$$

Under the standing rank assumption, it is positive definite for $x, s > 0$. Even when A is sparse, H_{NE} can be dense because two rows are coupled whenever they share a nonzero column. Accordingly, $\text{nnz}(A)$ and the maximum row/column sparsity s_H of the matrix actually sent to a solver are distinct parameters.

Two reformulations used later move an inexact-solve residual to a more useful part of the Newton equations. At a feasible iterate, if $V \in \mathbb{R}^{n \times (n-m)}$ has full column rank and range $V = \ker A$, a feasible direction can be represented as $\Delta x = V\lambda$ and $\Delta s = -A^\top \Delta y$. Substitution in the third row of (14) gives the *orthogonal-subspaces system (OSS)*

$$\underbrace{\begin{bmatrix} -XA^\top & SV \end{bmatrix}}_{H_{\text{OSS}}} \begin{bmatrix} \Delta y \\ \lambda \end{bmatrix} = r_c. \quad (17)$$

Every approximate coefficient vector still reconstructs directions satisfying $A\Delta x = 0$ and $A^\top \Delta y + \Delta s = 0$ exactly. This is the *OSS* introduced by Mohammadisiahroudi et al. [107].

The *modified normal-equation system (MNES)* instead chooses an index set B for which $A_B \in \mathbb{R}^{m \times m}$ is nonsingular. With $D = X^{1/2}S^{-1/2}$, its principal submatrix D_B , $\hat{A} = A_B^{-1}A$, and $E = D_B^{-1}\hat{A}D$, its coefficient and right-hand side are

$$H_{\text{MNES}} = EE^\top = D_B^{-1}A_B^{-1}H_{\text{NE}}(D_B^{-1}A_B^{-1})^\top, \quad h_{\text{MNES}} = D_B^{-1}A_B^{-1}h_{\text{NE}}, \quad (18)$$

where h_{NE} denotes the right-hand side of (16). If an approximate solution has residual $\hat{r}$, the reconstruction uses the correction $v_B = D_B \hat{r}$, $v_{\bar{B}} = 0$; the primal and dual feasibility rows are then exact, while the residual $-Sv$ is confined to the complementarity row [106]. The names OSS and MNES always mean the formulations (17) and (18). Neither name implies that V , $\hat{A}$, or the resulting system is sparse; that property must be established for the instance at hand.

Definition 2.8 (OSS/MNES inexact Newton residual convention). Let $\eta \in [0, 1)$ be the named forcing tolerance. An inexact OSS solve uses the linear-system residual $r_{\text{OSS}} = H_{\text{OSS}}\tilde{z} - r_c$ and the convention $\|r_{\text{OSS}}\|_2 \leq \eta\bar{\mu}$. An inexact MNES solve uses $\hat{r} = H_{\text{MNES}}\tilde{z} - h_{\text{MNES}}$ and the $\sqrt{\bar{\mu}}$ -scaled convention

$$\|\hat{r}\|_2 \leq \eta\sqrt{\bar{\mu}/n}, \quad \text{which ensures} \quad \|Sv\|_\infty \leq \eta\bar{\mu}. \quad (19)$$

These are residuals of the named Newton formulation, not the global RHS-relative feasibility residual of Definition 2.26. The formulation-specific scaling is part of the contract [106, 107].

For primal-barrier geometry, let $x_F(\mu)$ lie in the affine slice $x^0 + \ker A$, and let W have orthonormal columns spanning $\ker A$. Its reduced Hessian is

$$H_{\text{red}}(\mu) = W^\top \nabla^2 F(x_F(\mu))W. \quad (20)$$

Definition 2.9 (Reduced central-path Hessian condition number). The *reduced central-path Hessian condition number* is

$$\kappa_{\text{red}}(\mu; F) = \frac{\lambda_{\max}(H_{\text{red}}(\mu))}{\lambda_{\min}(H_{\text{red}}(\mu))}. \quad (21)$$

The Euclidean metric and an orthonormal tangent basis are part of this definition. The value is invariant under replacing W by another orthonormal basis, but not under an arbitrary nonorthogonal reparametrization. For the LP log barrier, $\nabla^2 F(x) = X^{-2}$.

Definition 2.10 (Ambient and KKT condition numbers). For the named ambient cone barrier F , the *ambient barrier-Hessian condition number* is $\kappa_{\text{amb}}(x; F) = \kappa_2(\nabla^2 F(x))$ before restriction to the feasible tangent space. It is representation-dependent: changing the ambient cone representation or its coordinates changes the named F and may change κ_{amb} . For a named nonsingular Newton or saddle matrix K in a specified scaling, its *KKT condition number* is $\kappa_2(K) = \sigma_{\max}(K)/\sigma_{\min}(K)$. We write κ_H for the condition number of the actual linear-system matrix H being solved.

These quantities are not interchangeable. Eliminating variables, changing row scales, choosing a nonorthogonal null-space basis, or passing from an indefinite KKT matrix to normal equations can change singular values and can square a condition number. In particular, a statement about κ_{red} says nothing by itself about $\kappa_2(H_{\text{NE}})$, $\kappa_2(H_{\text{OSS}})$, or the full saddle system.

2.2 Quantum computation and access models

Algorithms are uniform quantum circuits with arbitrary input-independent gates, intermediate measurements, and classical control. A query and a gate are different resources. Calls to an oracle, its inverse, or a controlled version count as oracle calls unless a result explicitly states otherwise.

Definition 2.11 (Raw sparse coefficient (sign) access). In the *raw sparse coefficient (sign) access model*, the dimensions and a fixed list of nonzero positions of the LP or SDP data are public. The hidden input is a string $\varsigma \in \{-1, +1\}^{n_{\text{bit}}}$ that determines designated coefficient signs (or designated entries from a fixed finite alphabet). The charged oracle is

$$O_{\varsigma}|i, z\rangle = |i, z \oplus \text{enc}(\varsigma_i)\rangle. \quad (22)$$

A sparse row, sparse column, or coefficient query is admissible only when it can be simulated coherently with the stated number of calls to O_{ς} ; a query does not reveal several hidden signs merely because a coefficient is repeated in a derived matrix. The equivalent phase convention $|i\rangle \mapsto \varsigma_i|i\rangle$ and the XOR convention (22) interconvert with one charged query in either direction, using phase kickback and the assumed controlled form of the oracle.

Remark 2.12. Definition 2.11 is a deliberately restrictive, project-specific specialization of the standard quantum query model [20]. Public support and hidden signs make reductions auditable. Supplying a prefix-parity table, an eliminated matrix, or an arbitrary input-dependent block-encoding completion at unit cost is a stronger model and is never implicit.

Definition 2.13 (Block encoding). Let M act on ℓ qubits. An (α, a, ϵ) *block encoding* of M is an $(\ell + a)$ -qubit unitary U such that

$$\|M - \alpha(|0^a\rangle\langle 0^a| \otimes I)U(|0^a\rangle\langle 0^a| \otimes I)\|_2 \leq \epsilon, \quad \alpha > 0. \quad (23)$$

The scalar α is the *block-encoding normalization* (also called subnormalization), a is the ancilla count, and ϵ is absolute operator-norm error [65].

Normalizations multiply under products: composing encodings with normalizations α and β yields normalization $\alpha\beta$ and the corresponding propagated error. Hence the cost of polynomial inversion is governed by the normalized spectral cutoff. For invertible H , the access-normalized condition parameter is

$$\kappa_{\text{BE}}(H; \alpha_H) = \alpha_H \|H^{-1}\|_2, \quad (24)$$

whereas $\kappa_H = \|H\|_2 \|H^{-1}\|_2$. They coincide only when the encoding is normalized at the operator norm. Sparse-access constructions can have α_H controlled by the row/column sparsity of H , not by $\text{nnz}(A)$, and product encodings can have a still larger normalization [65].

Definition 2.14 (State-preparation access). For a nonzero vector $v \in \mathbb{C}^N$, *state-preparation access* supplies a unitary U_v with

$$U_v|0\rangle = |v\rangle := \frac{1}{\|v\|_2} \sum_{i=1}^N v_i |i\rangle, \quad (25)$$

together with controlled access to U_v and $U_v^\dagger$. If $\|v\|_2$ is needed by the task, it must be supplied or estimated separately; when signed amplitudes matter, the oracle promise also fixes a phase convention for the controlled circuit. For a mixed state, the analogous oracle prepares a specified purification. This model is stronger than receiving independent copies, because inverse access enables amplitude-estimation procedures [131].

State preparation exposes one normalized ray. It does not, merely by being callable, provide coherent numerical access to individual v_i , a diagonal operator $\text{diag}(v)$, or a reusable update oracle. Such interfaces are separately defined output contracts below.

Definition 2.15 (Plain block-encoding access and factor access). *Plain block-encoding access* to a Hermitian positive-definite matrix H supplies a controlled $(\alpha_H, a_H, \epsilon_H)$ block encoding of H and its adjoint. *Factor access* supplies instead a controlled $(\alpha_G, a_G, \epsilon_G)$ block encoding of a possibly rectangular G (padded to square dimensions in the standard way) and its adjoint, together with the promise

$$H = G^\dagger G. \quad (26)$$

These are different input models. The standard product construction using two factor-oracle calls gives an $(\alpha_G^2, 2a_G, 2\alpha_G\epsilon_G + \epsilon_G^2)$ block encoding of H (the ϵ_G^2 term is needed because [definition 2.13](#) does not require $\alpha_G \geq \|G\|$). Thus a QLS solver applied to this composed encoding uses normalization $\alpha_H = \alpha_G^2$ and access-normalized inversion parameter $\alpha_G^2 \|H^{-1}\|_2$. A solver that acts on G directly must instead state its dependence on α_G and on the relevant singular-value cutoff of G ; it may not silently substitute plain access to H .

Normal equations and reduced log-barrier Hessians often have natural factors, so later results state explicitly which side of [Definition 2.15](#) they assume. No factorization is free merely because one exists algebraically.

Definition 2.16 (Quantum linear-system solver). A *quantum linear-system (QLS) solver* for a nonsingular Hermitian matrix H is a procedure that, given the stated matrix access and a preparation oracle for $|b\rangle$, produces a state ρ satisfying

$$D_{\text{tr}}\left(\rho, \frac{H^{-1}|b\rangle\langle b|H^{-1}}{\|H^{-1}|b\rangle\|_2^2}\right) \leq \epsilon_{\text{ls}}. \quad (27)$$

Its resource statement must name the linear-system condition number κ_H , the block-encoding normalization α_H (or sparse-oracle normalization), the preparation cost of $|b\rangle$, success probability, and precision ϵ_{ls} .

HHL introduced the state-output linear-systems primitive [76]. The Childs–Kothari–Somma algorithm achieves polylogarithmic precision dependence in sparse access, variable-time techniques improve amplification overhead, and QSVT implements inversion as a singular-value polynomial [7, 43, 65]. Discrete-adiabatic and adiabatic solvers provide other implementations [47, 128]. These algorithm families have different constants and parameter dependences, but all require an access-and-output ledger; none turns (27) into a classical vector for free.

Definition 2.17 (Charged QRAM access). A *QRAM access assumption* supplies a coherent lookup unitary for an explicit classical table, for example $|i, z\rangle \mapsto |i, z \oplus \text{enc}(d_i)\rangle$ [66]. In *charged QRAM access*, raw-input oracle calls and QRAM writes used to build or update that table are charged as separate resources; subsequent lookups have the advertised QRAM query cost.

QRAM can reduce coherent lookup depth once data have been loaded, but it does not erase the information cost of producing an input-dependent table. The lower bounds in this paper therefore charge setup. A QRAM containing all hidden parities, Newton entries, or solution coordinates has already been given information that raw coefficient access was designed to hide.

Tomography illustrates the output bottleneck. For an N -dimensional pure state, a classical amplitude vector with ℓ_2 error ϵ requires $\tilde{\Theta}(N/\epsilon)$ controlled calls in the state-preparation unitary model, while sampling-based approaches have quadratic dependence on $1/\epsilon$; mixed-state costs also depend on rank and the requested matrix norm [131]. Thus a polylogarithmic-dimensional QLS state-generation cost is compatible with a linear-dimensional classical readout cost.

2.3 Output and accounting contracts

An algorithm solves only the contract it is asked to solve. The following names are canonical throughout the paper.

Definition 2.18 (Value-estimation output). For a scalar target v , an *additive ϵ value-estimation output* is a classical $\tilde{v}$ such that $|\tilde{v} - v| \leq \epsilon$. A *target-relative ϵ value-estimation output* satisfies $|\tilde{v} - v| \leq \epsilon|v|$ and is used only under a nonzero-target promise. An unqualified “relative value estimate” always means this target-relative contract. The required success probability is part of either contract.

Definition 2.19 (Reference-scaled value-estimation output). For a stated scale $s_{\text{ref}} > 0$, a *reference-scaled ϵ value-estimation output* satisfies $|\tilde{v} - v| \leq \epsilon s_{\text{ref}}$. This contract is used when zero may be a valid target or when the problem supplies a natural external scale; the scale is always named rather than called a relative error.

Definition 2.20 (Classical vector output). A *classical vector output* for $x \in \mathbb{R}^N$ is an explicit classical list $\tilde{x} \in \mathbb{R}^N$ satisfying a named error guarantee, including its norm and scale. Unless stated otherwise that guarantee is $\|\tilde{x} - x\|_2 \leq \epsilon$. Producing only $\tilde{x}/\|\tilde{x}\|_2$, samples from its coordinates, or an implicit circuit is a different contract.

Definition 2.21 (Compact optimizer output). A *compact (sparse-support) optimizer output* is a classical description of an identified component or subset I together with the nonzero coordinates of a solution supported on I , satisfying a named optimality and feasibility tolerance. It is an explicit sparse solution description, not a full dense vector and not merely a quantum state or sampling oracle.

Definition 2.22 (Normalized solution-state output). For a nonzero solution vector x in the original problem coordinates, an *ϵ normalized solution-state output* (also called an *amplitude-encoded solution state*) is a density operator ρ such that

$$D_{\text{tr}}(\rho, |x/\|x\|_2\rangle\langle x/\|x\|_2|) \leq \epsilon, \quad D_{\text{tr}}(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1. \quad (28)$$

The output does not include $\|x\|_2$, coordinate values, or a preparation unitary unless those items are explicitly requested. When x is an LP primal candidate, we use the synonymous stable name *amplitude-encoded primal-state output*. A state encoding coefficients in a reduced or null-space basis is a different contract; the later original-coordinate state lower bounds do not automatically apply to that reduced-coordinate output.

Definition 2.23 (Trace-normalized density output). For a nonzero PSD solution X , an ϵ *trace-normalized density output* is ρ satisfying

$$D_{\text{tr}}(\rho, X/\text{tr } X) \leq \epsilon. \quad (29)$$

For a product-PSD solution this means the direct-sum density operator $\bigoplus_j X_j/(\sum_j \text{tr } X_j)$, including its block-label register. This contract deliberately discards the total trace, which must be output separately when needed.

Definition 2.24 (Coordinate-oracle output). An ϵ *coordinate-oracle output* for $x \in \mathbb{R}^N$ is a reusable, controlled circuit $O_{\tilde{x}}$ and its adjoint acting as

$$O_{\tilde{x}}|i, z\rangle = |i, z \oplus \text{enc}(\tilde{x}_i)\rangle, \quad |\tilde{x}_i - x_i| \leq \epsilon \quad (30)$$

for every i , with a stated fixed-point encoding and failure convention. The cost of one invocation and any one-time compilation cost are both part of the contract. A normalized solution state is not a coordinate-oracle output.

Definition 2.25 (Unconditional and heralded preparation). An *unconditional preparation* is a channel whose output state itself satisfies the promised error bound after all internal randomness and flags are discarded. A *heralded preparation* outputs a classical or computational-basis success flag; conditioned on that flag, the state obeys the error bound, the success probability is at least a stated $p > 0$ on every allowed input, and repetitions or amplification needed to obtain success are charged. An unobserved or unspecified “good branch” is neither contract.

Definition 2.26 (Global RHS-relative residual). For $b \neq 0$, the *global RHS-relative ℓ_2 primal residual* of an LP candidate x is

$$\text{res}_p(x) = \frac{\|Ax - b\|_2}{\|b\|_2}. \quad (31)$$

For an SDP candidate X , replace Ax by $\mathcal{A}(X)$. A statement that the residual is at most η always refers to this ratio unless it names a different contract. Per-row relative residuals, coefficientwise residuals, normal-equation residuals, local-norm residuals, and backward error under perturbed data are different and do not imply one another without additional scale assumptions.

Definition 2.27 (First-preparation and amortized total-query accounting). For one fixed input oracle and R requested outputs, *setup* is the reusable input-dependent work performed before the first output is produced. Let Q_{setup} and W_{setup} be, respectively, its raw-input query count and QRAM-write count, and let Q_t and W_t be the corresponding nonreusable costs attributable to output t . Raw queries and QRAM writes are distinct columns in the resource ledger and are never added together. The *first-preparation cost* is the pair $(Q_{\text{setup}} + Q_1, W_{\text{setup}} + W_1)$. The total ledgers are

$$Q_{\text{total}}(R) = Q_{\text{setup}} + \sum_{t=1}^R Q_t, \quad W_{\text{total}}(R) = W_{\text{setup}} + \sum_{t=1}^R W_t, \quad (32)$$

with amortized costs obtained by dividing each column by R . If setup caches the complete fixed instance, later Q_t may be zero in a pure query model. Conversely, every raw query or QRAM write used to prepare consumed advice or additional state copies is charged in setup or in the consuming call.

Definition 2.27 prevents two opposite accounting errors: repeatedly charging the same static information as if it were fresh, and declaring an input-dependent reusable data structure free. In the pure query model with a fixed, finite, cacheable hidden input, an additive per-round raw-query lower bound therefore needs a genuine freshness, memory, communication, or consumable-resource assumption.

2.4 Lower-bound toolkit

Parity. For n_{bit} oracle bits, bounded-error quantum computation of their parity needs $\Omega(n_{\text{bit}})$ queries. This follows from the polynomial method because a T -query algorithm has an acceptance probability represented by a real multilinear polynomial of degree at most $2T$, whereas parity has linear approximate degree [20]. Our raw-sign reductions therefore verify that each optimization-data query can be simulated by $O(1)$ sign queries and then decode parity from the contracted output.

Adversary bounds and composition. Ambainis’s adversary method assigns nonnegative weights to pairs of inputs with different answers and bounds the progress one query can make [6]. Its positive-weight spectral formulation composes, and already covers an OR of independent parity gadgets, with the product of the outer and inner adversary scales [77]. The more general negative-weight bound $\text{Adv}^\pm$ composes for Boolean outer functions and characterizes bounded-error Boolean function evaluation up to constants [92, 122]. We do not invoke either composition statement when component queries overlap or an aggregate predicate is supplied.

Polynomial method and symmetrization. More generally, every measurement probability after T standard input queries is a degree- $2T$ polynomial in the input bits. Averaging over inputs of equal Hamming weight turns it into a univariate polynomial without increasing degree. Approximation constraints at the promised weights can then be combined with classical polynomial inequalities to lower-bound T [20]. Later applications state the promise points and error interval explicitly; symmetrization does not justify inserting unpromised inputs.

Trace distance and Helstrom decoding. Trace distance is contractive under channels, and the statistical distance between the outcome distributions of any measurement on ρ and σ is at most $D_{\text{tr}}(\rho, \sigma)$. Conversely, for equal priors the optimal binary discrimination success probability is $\frac{1}{2}(1 + D_{\text{tr}}(\rho, \sigma))$, the Helstrom bound [116]. Hence a fixed measurement that decodes a hidden bit from an ideal solution state with bias β still has positive bias from any output within trace distance less than β .

State conversion. In the state-conversion problem, an input oracle indexed by x must transform a prescribed family of initial states $\{|\rho_x\rangle\}$ into target states $\{|\sigma_x\rangle\}$ to bounded error. The filtered γ_2 query distance compares the two Gram matrices $[\langle\rho_x|\rho_y\rangle]$ and $[\langle\sigma_x|\sigma_y\rangle]$ through the matrices that record which input symbols a query can distinguish. It gives lower bounds for general state conversion, the direction used in this paper, and matching upper and lower bounds up to a gap in their error parameters. In the special case of function evaluation, the resulting general adversary bound characterizes bounded-error query complexity up to constants [92]. Later uses specify the full oracle completion: prescribing only $U_x|0\rangle$ is insufficient when an algorithm may query U_x away from $|0\rangle$.

Table 1: Notation used across the technical sections. Several sections locally rebind G , N , P , α , and η and state those conventions locally.

Symbol	Meaning
m, n	Number of LP equality constraints and primal variables.
q, d_j	Number of PSD blocks and dimension of block j .
$n_{\text{bit}}, \varsigma$	Number of hidden signs and their string in a raw coefficient-oracle reduction.
N	Dimension of a vector requested by an output contract.
N_H	Dimension of the actual Newton, normal-equation, OSS, or MNES system sent to a linear solver.
$A, b, c; \mathcal{A}, b, C$	LP data; product-form SDP data.
X, S	Diagonal LP matrices $\text{diag}(x)$, $\text{diag}(s)$, or primal and dual PSD variables when the context is semidefinite.
F, ν	Named ambient cone barrier and its declared cone parameter.
μ	Central-path parameter.
$\bar{\mu}$	Average LP complementarity $x^\top s/n$ at an iterate.
$\sigma; \sigma_{\min}, \sigma_{\max}$	Newton centering parameter; minimum and maximum singular values.
g	Primal objective gap $c^\top x_F - \text{OPT}$, not automatically the primal-dual gap.
$N_2(\theta)$	Feasible Euclidean complementarity neighborhood in (12).
W, V	Orthonormal tangent basis used in H_{red} ; a general full-rank null-space basis used in OSS.
$H_{\text{red}}, H_{\text{NE}}, H_{\text{OSS}}, H_{\text{MNES}}$	Reduced barrier Hessian and the named Newton-system matrices.
κ_{red}	Condition number of H_{red} in an orthonormal Euclidean tangent basis.
κ_{amb}	Condition number of the named ambient cone-barrier Hessian before tangent restriction.
κ_H	Spectral condition number of the actual named linear-system matrix H ; not $\kappa_2(A)$ unless proved equal.
s_H	Maximum row/column sparsity of H ; not $\text{nnz}(A)$.
$(\alpha_H, a_H, \epsilon_H)$	Normalization, ancilla count, and operator error of a block encoding of H .
$(\alpha_G, a_G, \epsilon_G)$	Corresponding factor block-encoding data for $H = G^\dagger G$.
κ_{BE}	Access-normalized inversion parameter $\alpha_H \ H^{-1}\ _2$.
$\epsilon_{\text{opt}}, \epsilon_{\text{ls}}$	Optimization accuracy and one-linear-solve state accuracy.
η	Named feasibility or inexact-Newton residual tolerance, as specified by the applicable contract.
D_{tr}	Trace distance $\frac{1}{2} \ \rho - \sigma\ _1$.
$Q_{\text{setup}}, Q_{\text{total}}$	Charged setup queries and total raw queries including setup.
$W_{\text{setup}}, W_{\text{total}}$	Charged setup writes and total QRAM writes including setup.
$\text{nnz}(A)$	Number of stored nonzeros of A ; it does not encode fill or derived-system sparsity.

2.5 Recurring notation

3 The Geometry of Central-Path Conditioning

3.1 Question, answer, and relation to prior work

Quantum interior-point methods that invoke a quantum linear-system solver, and classical interior-point methods that solve Newton equations inexactly, pay for the conditioning of a Newton matrix. A familiar estimate is $\kappa = \Theta(\mu^{-2})$, but the classical derivations concern the logarithmic barrier and often a primal-dual Schur complement rather than the reduced primal Hessian. It is therefore natural to ask whether a different self-concordant barrier can remove the ill-conditioning.

For reduced barrier Hessians the answer is no in the relevant minimax sense. The obstruction is the Euclidean geometry of the objective sublevel set. We prove a lower bound for every self-concordant barrier, pin every eigenvalue between directional chord widths, and show that the

canonical log and log-det barriers attain the condition-number floor up to an instance-dependent constant. Error-bound exponents then give a dictionary of conditioning laws.

The universal quantifier over barriers is important. Earlier conditioning analyses concern the log barrier, including Wright’s analysis of logarithmic barrier Hessians and the survey of Forsgren, Gill, and Wright [63, 134]. The μ^{-2} -type estimates in Augustino et al. [14, Section 7.4], for example, are for primal–dual Schur systems. Conversely, Allamigeon et al. [5] prove a lower bound uniform over self-concordant barriers, but for the number and geometry of central-path iterations, not Hessian conditioning. The error-bound consequences below connect to weak sharp minima [33] and to Sturm’s singularity-degree bounds for semidefinite systems [127].

Remark 3.1 (Novelty and textbook ingredients). The Dikin-ellipsoid and asymmetric-containment facts used in the proofs, as well as the centrality identity $\Pi_{\mathcal{V}}\nabla F(x_{\mu}) = -\Pi_{\mathcal{V}}c/\mu$, are standard self-concordance tools [113, 123]. The contribution here is their universal-over-barriers conditioning statement and its geometric consequences, not new containment machinery. In particular, the evidence for novelty is a prior-art audit, not a claim that these proof ingredients are new.

3.2 Sublevel chords and two containment facts

Let $P \subset \mathbb{R}^N$ be compact and convex, with affine hull $x^0 + \mathcal{V}$ and nonempty relative interior P° . Let W have orthonormal columns spanning the d -dimensional tangent space $\mathcal{V}$. We minimize the nonconstant linear functional $c^{\top}x$, so $c_{\mathcal{V}} := \Pi_{\mathcal{V}}c \neq 0$. Put

$$\text{OPT} = \min_{y \in P} c^{\top}y, \quad \Phi = \{y \in P : c^{\top}y = \text{OPT}\}. \quad (33)$$

For $t \geq 0$, define the objective sublevel set, its diameter, and the longest sublevel chord from an interior point by

$$L(t) = \{y \in P : c^{\top}y \leq \text{OPT} + t\}, \quad D(t) = \text{diam } L(t), \quad \ell(x) = \sup_{y \in L(c^{\top}x - \text{OPT})} \|y - x\|_2. \quad (34)$$

Thus D is nondecreasing. If x has objective gap g , then $x \in L(g)$ and the triangle inequality gives

$$\frac{D(g)}{2} \leq \ell(x) \leq D(g). \quad (35)$$

Indeed, for any $y, z \in L(g)$, at least one of $\|x - y\|_2, \|x - z\|_2$ is at least $\|y - z\|_2/2$.

Let F be a ν_F -self-concordant barrier for P in the sense of [definition 2.2](#). We use two standard consequences of self-concordance. For $x \in P^{\circ}$,

$$\begin{aligned} y \in x^0 + \mathcal{V}, \quad \|y - x\|_x < 1 &\implies y \in P^{\circ}, & (\text{Dikin}) \\ y \in P, \quad \langle \nabla F(x), y - x \rangle \geq 0 &\implies \|y - x\|_x \leq C_F, \quad C_F := \nu_F + 2\sqrt{\nu_F}. & (\text{asymmetric}) \end{aligned}$$

The first implication is understood for points in the affine hull; its contrapositive says that a boundary or exterior point has local distance at least one. We also use semiboundedness,

$$\langle \nabla F(x), y - x \rangle \leq \nu_F \quad (y \in P). \quad (36)$$

These facts follow from the standard self-concordant-barrier containment theorems; affine restriction preserves them and does not increase the declared parameter [113, 123].

At an exact central point $x = x_F(\mu)$, first-order optimality on the affine slice is

$$W^{\top}(c + \mu \nabla F(x)) = 0. \quad (37)$$

Consequently, for every $y \in P$,

$$c^{\top}(x - y) = \mu \langle \nabla F(x), y - x \rangle \leq \nu_F \mu. \quad (38)$$

Taking $y \in \Phi$ proves the convention required in [remark 2.5](#):

$$0 < g_F(\mu) \leq \nu_F \mu. \quad (39)$$

The strict inequality holds because a relative-interior minimizer of a nonconstant linear functional cannot exist. Notice that (39) is generally an inequality. The equality $\nu\mu$ in [definition 2.3](#) is the canonical *primal-dual* gap, not necessarily the primal objective gap g used here.

3.3 Rigidity of the Hessian spectrum

We begin with the barrier-independent lower edge. It does not require centrality.

Theorem 3.2 (Universal minimum-eigenvalue rigidity). *Under the setup of [section 3.2](#), for every ν_F -self-concordant barrier F for P and every $x \in P^\circ$,*

$$\lambda_{\min}(W^\top \nabla^2 F(x) W) \geq \frac{1}{\ell(x)^2}. \quad (40)$$

Here $\ell(x)$ is formed from the objective level of that same point, whether or not x is central.

Proof. Fix a Euclidean unit vector $h \in \mathcal{V}$ and replace it by $-h$ if necessary so that $c^\top h \leq 0$. Compactness implies that the ray $x + th$ leaves P° in finite time. More precisely, if

$$\tau = \sup\{t \geq 0 : x + sh \in P^\circ \text{ for } 0 \leq s < t\},$$

then $0 < \tau < \infty$ and $y = x + \tau h \in P$. The objective does not increase along the ray, so $y \in L(c^\top x - \text{OPT})$ and hence $\tau \leq \ell(x)$. Since $y \notin P^\circ$, the contrapositive of ([Dikin](#)) gives

$$1 \leq \|\tau h\|_x^2 = \tau^2 \langle \nabla^2 F(x) h, h \rangle.$$

Thus the Rayleigh quotient is at least $\tau^{-2} \geq \ell(x)^{-2}$. The Hessian form is even in h , so the initial sign choice loses no direction. Minimizing over the Euclidean unit sphere in $\mathcal{V}$ proves (40). $\square$

The opposite spectral edge is forced by the distance in objective value to the optimum. Combining it with [theorem 3.2](#) gives the sublevel chord law.

Theorem 3.3 (Sublevel chord law). *Under the setup of [section 3.2](#), let $x = x_F(\mu)$ be an exact central point of any ν_F -self-concordant barrier, and set $g = g_F(\mu)$. Then*

$$\lambda_{\max}(W^\top \nabla^2 F(x) W) \geq \frac{\|c_{\mathcal{V}}\|_2^2}{g^2}, \quad (41)$$

$$\frac{1}{\ell(x)^2} \leq \lambda_{\min}(W^\top \nabla^2 F(x) W) \leq \frac{C_F^2}{\ell(x)^2}, \quad (42)$$

$$\kappa_{\text{red}}(\mu; F) \geq \left(\frac{\ell(x) \|c_{\mathcal{V}}\|_2}{C_F g} \right)^2 \geq \left(\frac{D(g) \|c_{\mathcal{V}}\|_2}{2 C_F g} \right)^2. \quad (43)$$

In particular, using only the μ parametrization,

$$\kappa_{\text{red}}(\mu; F) \geq \left(\frac{D(g_F(\mu)) \|c_{\mathcal{V}}\|_2}{2 C_F \nu_F \mu} \right)^2. \quad (44)$$

There is no small- μ threshold and no polyhedrality, nondegeneracy, or strict-complementarity hypothesis.

Proof. The gap statement (39) was proved above. For the largest eigenvalue, let

$$\hat{h} = -\frac{c_{\mathcal{V}}}{\|c_{\mathcal{V}}\|_2}, \quad t_* = \frac{g}{\|c_{\mathcal{V}}\|_2}.$$

The affine point $z = x + t_*\hat{h}$ has $c^\top z = \text{OPT}$. If it were in P° , it would be a relative-interior minimizer of a linear functional, forcing $c_{\mathcal{V}} = 0$. Therefore $z \notin P^\circ$, and (Dikin) gives

$$\langle \nabla^2 F(x)\hat{h}, \hat{h} \rangle \geq t_*^{-2} = \frac{\|c_{\mathcal{V}}\|_2^2}{g^2}.$$

This proves (41).

For the upper bound in (42), take any $y \in L(g)$. By (37),

$$\langle \nabla F(x), y - x \rangle = -\frac{c^\top (y - x)}{\mu} \geq 0.$$

Asymmetric containment (asymmetric) therefore gives $\|y - x\|_x \leq C_F$. If $y \neq x$, the Euclidean unit vector $v = (y - x)/\|y - x\|_2$ satisfies

$$\langle \nabla^2 F(x)v, v \rangle \leq \frac{C_F^2}{\|y - x\|_2^2}.$$

The smallest eigenvalue is no larger than this Rayleigh quotient. Taking a sequence of y whose distance from x tends to $\ell(x)$ proves the upper bound; the lower bound is theorem 3.2. Dividing (41) by the upper bound in (42), then using (35), proves (43). Finally $g \leq \nu_F \mu$ proves (44). $\square$

The maximum-eigenvalue constant in (41) can be exact. On the simplex problem $\min x_3$ subject to $x_1 + x_2 + x_3 = 1$, numerical central paths for the two tested barriers give $\lambda_{\max} g^2 / \|c_{\mathcal{V}}\|_2^2 \rightarrow 1$; for the ordinary log barrier, $g/\mu \rightarrow 1$ and $\lambda_{\max} \mu^2 \rightarrow 2/3 = \|c_{\mathcal{V}}\|_2^2$.

The chord idea controls more than the two extreme eigenvalues. For an unoriented one-dimensional subspace $[h] \subset \mathcal{V}$, choose its unit representative $\hat{h}$ with $c^\top \hat{h} < 0$. If $c^\top h = 0$, choose the sign having the longer exit time, breaking a tie arbitrarily. Define

$$\tau_x([h]) = \sup\{t \geq 0 : x + s\hat{h} \in P^\circ \text{ for } 0 \leq s < t\}. \quad (45)$$

The selected exit point is a sublevel point, so $0 < \tau_x([h]) \leq \ell(x)$. The equatorial convention makes τ_x well-defined on unoriented lines; continuity is not needed.

For $j = 1, \dots, d$, define the two width profiles

$$w_j^-(x) = \inf_{\substack{U \subset \mathcal{V} \\ \text{codim } U = j-1}} \sup_{\substack{h \in U \\ \|h\|_2=1}} \tau_x([h]), \quad (46)$$

$$w_j^+(x) = \sup_{\substack{S \subset \mathcal{V} \\ \dim S = j}} \inf_{\substack{h \in S \\ \|h\|_2=1}} \tau_x([h]). \quad (47)$$

We use infima and suprema because the sign choice can make the exit-time function discontinuous on $c^\perp$.

Theorem 3.4 (Width-spectrum rigidity). *In the setting of theorem 3.3, list the eigenvalues of $W^\top \nabla^2 F(x)W$ in nondecreasing order. For every $j = 1, \dots, d$,*

$$\frac{1}{(w_j^-(x))^2} \leq \lambda_j \left(W^\top \nabla^2 F(x)W \right) \leq \frac{C_F^2}{(w_j^+(x))^2}. \quad (48)$$

Moreover $w_j^+(x) \leq w_j^-(x)$ for every j , and $w_1^+(x) = \sup_{[h]} \tau_x([h]) \geq \ell(x)$.

Proof. For every unit h , Dikin containment at the exit point gives

$$\langle \nabla^2 F(x)h, h \rangle \geq \tau_x([h])^{-2}. \quad (49)$$

For the selected orientation, centrality gives

$$\langle \nabla F(x), \tau_x([h])\hat{h} \rangle = -\frac{\tau_x([h])c^\top \hat{h}}{\mu} \geq 0.$$

Asymmetric containment ([asymmetric](#)) applies directly to the exit point and gives

$$\langle \nabla^2 F(x)h, h \rangle \leq C_F^2 \tau_x([h])^{-2}. \quad (50)$$

The Rayleigh quotient is unchanged by orientation.

The two Courant–Fischer formulas for eigenvalues in nondecreasing order are

$$\begin{aligned} \lambda_j &= \sup_{\text{codim } U=j-1} \inf_{h \in U, \|h\|_2=1} R(h), \\ \lambda_j &= \inf_{\text{dim } S=j} \sup_{h \in S, \|h\|_2=1} R(h), \end{aligned}$$

where $R(h) = \langle \nabla^2 F(x)h, h \rangle$. Applying (49) in the first formula yields

$$\lambda_j \geq \sup_U \frac{1}{(\sup_{h \in U} \tau_x([h]))^2} = \frac{1}{(w_j^-(x))^2}.$$

Applying (50) in the second formula gives

$$\lambda_j \leq C_F^2 \inf_S \frac{1}{(\inf_{h \in S} \tau_x([h]))^2} = \frac{C_F^2}{(w_j^+(x))^2}.$$

For any j -dimensional S and codimension- $(j-1)$ subspace U , the dimension formula gives a nonzero line $[h] \subseteq S \cap U$. Hence

$$\inf_{v \in S, \|v\|=1} \tau_x([v]) \leq \tau_x([h]) \leq \sup_{v \in U, \|v\|=1} \tau_x([v]).$$

Taking the supremum over S and infimum over U proves $w_j^+ \leq w_j^-$. Finally, if $y \in L(g)$, the segment $[x, y]$ lies in P° . The line through $y - x$ is oriented toward y when the objective strictly decreases, and our longer-exit convention gives the same conclusion when it is unchanged. Thus $\tau_x([y - x]) \geq \|y - x\|_2$. Taking the supremum over y proves $w_1^+ \geq \ell(x)$. $\square$

Thus even the interior spectrum is constrained by directional sublevel widths, up to the explicit C_F^2 window and the gap between the two minimax widths. The empirical consequences for spectral clustering are deferred to the next section.

3.4 Canonical barriers attain the floor

We next specialize to compact affine slices of the nonnegative orthant and of the product PSD cone in [section 2.1](#). To distinguish parameters when two barriers are compared, write $\nu_{\text{can}} = n$ for the LP log barrier and $\nu_{\text{can}} = \sum_j d_j$ for the product log-det barrier. This is the cone parameter denoted by ν in [section 2.1](#).

Theorem 3.5 (Canonical achievability and gap-matched comparison). *Consider either of the following strictly primal–dual feasible problems with a compact primal feasible region and nonconstant objective.*

- (i) An LP in (1), with canonical path $(x(\mu), y(\mu), s(\mu))$ and $M(\mu) = \|s(\mu)\|_\infty$.
- (ii) A product-form SDP in (3). Its canonical path is $(X(\mu), y(\mu), S(\mu))$, and $M(\mu) = \max_j \|S_j(\mu)\|_2$.

Fix $\bar{\mu} > 0$ and suppose the tail constant

$$M_{\text{tail}} = \sup_{0 < \mu \leq \bar{\mu}} M(\mu) \quad (51)$$

is finite. Use the Euclidean metric for the LP and the product Frobenius metric for the SDP. Write $g_{\text{can}}(\bar{\mu}) = g_{F_{\text{can}}}(\bar{\mu})$ for the canonical primal objective gap at the endpoint of the tail. The notation $\kappa_{\text{red}}^{\text{can}}(g)$ and $\kappa_{\text{red}}^F(g)$ means the reduced condition number at the central point of gap g for the named barrier, using the g -parametrization in definition 2.3 on an interval where the named objective-gap map g_F is one-to-one. For a canonical central point with $0 < \mu \leq \bar{\mu}$ and primal objective gap g ,

$$\lambda_{\min}(H_{\text{red}}) \geq \ell(x)^{-2}, \quad (52)$$

$$\lambda_{\max}(H_{\text{red}}) \leq M_{\text{tail}}^2 \mu^{-2} \leq M_{\text{tail}}^2 \nu_{\text{can}}^2 g^{-2}, \quad (53)$$

$$\kappa_{\text{red}}^{\text{can}}(g) \leq \left(\frac{M_{\text{tail}} \nu_{\text{can}} D(g)}{g} \right)^2. \quad (54)$$

For every gap $0 < g \leq g_{\text{can}}(\bar{\mu})$ attained on the canonical tail, any ν_F -self-concordant barrier F on the same feasible region, and any exact F -central point having that gap,

$$\kappa_{\text{red}}^{\text{can}}(g) \leq \left(\frac{2M_{\text{tail}} \nu_{\text{can}} (\nu_F + 2\sqrt{\nu_F})}{\|c_V\|_2} \right)^2 \kappa_{\text{red}}^F(g). \quad (55)$$

The comparison is between points of equal objective gap; their values of μ need not agree.

Proof. The first bound is theorem 3.2. For the LP, canonical complementarity gives $x_i = \mu/s_i$, and therefore, for every Euclidean unit tangent vector h ,

$$\langle \nabla^2 F_{\log}(x) h, h \rangle = \sum_i \frac{h_i^2}{x_i^2} \leq \frac{M_{\text{tail}}^2}{\mu^2}.$$

For the SDP, $X_j^{-1} = S_j/\mu$, and for a unit product-Frobenius tangent direction H ,

$$\begin{aligned} D^2 F_{\log \det}(X)[H, H] &= \sum_j \|X_j^{-1/2} H_j X_j^{-1/2}\|_F^2 \\ &\leq \max_j \|X_j^{-1}\|_2^2 \sum_j \|H_j\|_F^2 \leq \frac{M_{\text{tail}}^2}{\mu^2}. \end{aligned}$$

This proves the first inequality in (53). The canonical primal–dual gap is $\nu_{\text{can}}\mu$, whereas its primal objective gap satisfies $g \leq \nu_{\text{can}}\mu$ by (39); hence $\mu^{-1} \leq \nu_{\text{can}}/g$. Combining the largest-eigenvalue bound with (52) and $\ell(x) \leq D(g)$ proves (54). Finally divide that upper bound by the universal lower bound (43) for F to obtain (55). $\square$

The tail boundedness in (51) is an explicit instance hypothesis; it holds in the standard convergent canonical-path setting. The restriction to a tail is necessary because the corresponding supremum can diverge as $\mu \rightarrow \infty$. More importantly, (55) says that no alternative barrier can improve the canonical reduced-Hessian condition number by more than a g -independent instance factor. It does not say that an arbitrary barrier cannot be worse.

3.5 A dictionary from error bounds to conditioning

The diameter $D(g)$ is the relevant sharpness modulus. If, for some $\theta \in [0, 1]$ and $\gamma > 0$,

$$\text{dist}(y, \Phi) \leq \gamma^{-1}(c^\top y - \text{OPT})^\theta \quad (y \in P), \quad (56)$$

then choosing nearest optimal points for two members of $L(g)$ gives

$$D(g) \leq \text{diam } \Phi + 2\gamma^{-1}g^\theta. \quad (57)$$

Conversely, $\Phi \subseteq L(g)$ and a central point x of gap g belongs to $L(g)$. For every $x^* \in \Phi$, $g = c_V^\top(x - x^*) \leq \|c_V\|_2 \|x - x^*\|_2$, so

$$D(g) \geq \text{diam } \Phi, \quad D(g) \geq \frac{g}{\|c_V\|_2} \quad \text{at every attained central gap } g. \quad (58)$$

Corollary 3.6 (Conditioning regimes). *Assume the hypotheses of [theorem 3.3](#); for canonical upper bounds also assume [theorem 3.5](#), and restrict those conclusions to the canonical tail.*

- (i) *If $\dim \Phi \geq 1$, then every fixed ν_F -self-concordant barrier satisfies $\kappa_{\text{red}}^F(g) = \Omega(g^{-2})$. The canonical barrier satisfies $\kappa_{\text{red}}^{\text{can}}(g) = \Theta(g^{-2})$.*
- (ii) *If P is a polytope with a unique optimum, including a degenerate vertex, then $D(g) = \Theta(g)$ as $g \downarrow 0$. Consequently the canonical log barrier has $\kappa_{\text{red}}^{\log}(g) = \Theta(1)$.*
- (iii) *If the optimum is unique and $D(g) = \Theta(\sqrt{g})$, as under a two-sided quadratic-growth geometry, every fixed barrier has $\kappa_{\text{red}}^F(g) = \Omega(g^{-1})$ and the canonical barrier has $\kappa_{\text{red}}^{\text{can}}(g) = \Theta(g^{-1})$.*

The constants may depend on the instance and, for a noncanonical barrier, on ν_F . Whenever the canonical objective gap is $\Theta(\mu)$, as in the standard strictly complementary tail, the same three canonical rates are $\Theta(\mu^{-2})$, $\Theta(1)$, and $\Theta(\mu^{-1})$, respectively.

Proof. If $\dim \Phi \geq 1$, the face contains two distinct points and hence $\text{diam } \Phi > 0$; compactness makes this diameter finite and attained. Substitution of $D(g) \geq \text{diam } \Phi$ in (43) proves the universal lower bound. Since $D(g) \leq \text{diam } P$, (54) gives the matching canonical upper bound.

For a polytope, the solution set of a feasible bounded linear program is a weak sharp minimum [33, 99]. With a unique optimum x^* , this gives (56) with $\theta = 1$, hence $D(g) \leq 2g/\gamma$. For the reverse bound, choose a feasible $z \neq x^*$ and set $\Delta = c^\top z - \text{OPT} > 0$. For every $0 < g \leq \Delta$, convexity makes

$$y_g = \left(1 - \frac{g}{\Delta}\right) x^* + \frac{g}{\Delta} z$$

feasible, and $c^\top y_g = \text{OPT} + g$. Thus

$$D(g) \geq \|y_g - x^*\|_2 = \frac{\|z - x^*\|_2}{\Delta} g.$$

Combining these two bounds with [theorems 3.3](#) and [3.5](#) proves (ii). Part (iii) follows by substituting $D(g) = \Theta(\sqrt{g})$ in the same two theorems. The final conversion is immediate under the separately stated relation $g = \Theta(\mu)$. $\square$

Remark 3.7. The statement uniform over barriers in [corollary 3.6](#) is the lower rate. The matching upper rate is established for the canonical barrier. Thus the barrier-optimal rate in the positive-dimensional-face case is $\Theta(g^{-2})$; the results do not supply an upper bound of that order for every conceivable self-concordant barrier.

Part (ii) includes degenerate vertices: nondegeneracy is not needed for weak sharpness. Part (iii) explains why the reduced primal log-det Hessian at a generic curved, unique SDP optimum can have condition number $\Theta(\mu^{-1})$, even though its largest eigenvalue and a primal-dual Schur complement can live at the μ^{-2} scale. This is the conic analogue of the μ^{-1} structured logarithmic-barrier behavior identified for nonlinear programs by Wright [134].

The same substitution produces fractional exponents. Sturm proves an $O(\epsilon^{2^{-d}})$ distance bound when the relevant semidefinite feasibility system has singularity degree d [127]. An upper error bound alone does not force a conditioning lower bound: its exponent must also be attained by sublevel chords. When it is sharp, the result is immediate.

Corollary 3.8 (Attained Hölder and singularity-degree laws). *Suppose that, along central gaps $g \downarrow 0$, $D(g) \geq ag^\theta$ for constants $a > 0$ and $\theta \in [0, 1]$. Then every ν_F -self-concordant barrier satisfies*

$$\kappa_{\text{red}}^F(g) \geq \frac{a^2 \|c_V\|_2^2}{4C_F^2} g^{2\theta-2}. \quad (59)$$

If, in addition, the problem is a canonical LP or product-form SDP satisfying all hypotheses of theorem 3.5 and $D(g) = O(g^\theta)$, then the canonical barrier attains $\Theta(g^{2\theta-2})$ on its tail. In particular, when the Sturm exponent $\theta = 2^{-d}$ is attained, the exponents are

$$d = 0 : 0, \quad d = 1 : -1, \quad d = 2 : -\frac{3}{2}, \quad 2\theta - 2 = 2^{1-d} - 2 \longrightarrow -2. \quad (60)$$

Proof. Insert $D(g) \geq ag^\theta$ into (43). If the matching upper bound on D holds, combine the resulting lower bound with (54). Substituting $\theta = 2^{-d}$ gives (60). $\square$

The following 3×3 problem shows that the $d = 2$ fractional law occurs in a concrete SDP rather than only in the error-bound formalism.

Proposition 3.9 (An explicit $g^{-3/2}$ witness). *Consider*

$$\begin{aligned} \min_{X \in \mathbb{S}^3} \quad & X_{22} \\ \text{subject to} \quad & \text{tr } X = 1, \quad X_{33} = X_{12}, \quad X \succeq 0. \end{aligned} \quad (61)$$

Its unique optimum is $X^ = e_1 e_1^\top$, its optimality system reaches the minimal face in two facial-reduction steps, and*

$$D(g) = \Theta(g^{1/4}). \quad (62)$$

Consequently every self-concordant barrier obeys $\kappa_{\text{red}}(g) = \Omega(g^{-3/2})$, while the log-det barrier, under the hypotheses of theorem 3.5, has $\kappa_{\text{red}}(g) = \Theta(g^{-3/2})$ on its canonical tail.

Proof. Write a feasible matrix as

$$X = \begin{bmatrix} a & b & t \\ b & d & e \\ t & e & b \end{bmatrix}, \quad a + d + b = 1, \quad X \succeq 0.$$

Here $d = X_{22}$ is the objective and $b = X_{12} = X_{33} \geq 0$. If $d = 0$, positive semidefiniteness makes the second row and column zero, hence $b = e = 0$. Then the third diagonal entry is zero, so the third row and column vanish and $t = 0$; the trace constraint gives $a = 1$. The optimum is therefore unique.

For $X \in L(g)$, the 2×2 principal minors give

$$b^2 \leq ad \leq d \leq g, \quad t^2 \leq ab \leq b \leq \sqrt{g}, \quad e^2 \leq db \leq g^{3/2}.$$

Also $|a - 1| = d + b \leq g + \sqrt{g}$. Thus $\|X - X^*\|_F = O(g^{1/4})$, and the triangle inequality gives $D(g) = O(g^{1/4})$.

For the reverse bound, let $0 < g \leq 1/16$, set

$$d = g, \quad b = \frac{\sqrt{g}}{2}, \quad a = 1 - g - \frac{\sqrt{g}}{2}, \quad e = 0, \quad t = \pm \frac{g^{1/4}}{2\sqrt{2}}.$$

Then $a \geq 13/16$. The one- and two-dimensional principal minors are positive, and

$$\det X = agb - b^3 - gt^2 = g^{3/2} \left(\frac{a}{2} - \frac{1}{4} \right) > 0.$$

Both sign choices therefore give positive-definite feasible points in $L(g)$. Their Frobenius distance is $2\sqrt{2}|t| = g^{1/4}$, proving the lower bound in (62).

For completeness, the two facial reductions can be read directly from the optimality equations. The first exposing equation $X_{22} = 0$ removes the e_2 row and column; then $X_{12} = 0$ turns the affine equation $X_{33} = X_{12}$ into $X_{33} = 0$, which removes the e_3 row and column. No one-step positive-semidefinite exposing combination can do both: a PSD linear combination of the trace, $X_{33} - X_{12}$, and X_{22} equations whose $(1, 1)$ entry is zero must have zero first row, forcing the coefficient of $X_{33} - X_{12}$ to vanish. Thus two steps are necessary and sufficient. Finally apply [corollary 3.8](#) with $\theta = 1/4$. $\square$

For the log-det barrier, the leading constant can also be computed exactly. For sufficiently small positive gap g , the center and central parameter are

$$X_{\log}(g) = \begin{pmatrix} a & b & 0 \\ b & g & 0 \\ 0 & 0 & b \end{pmatrix}, \quad b = \frac{-g + \sqrt{3g - 2g^2}}{3}, \quad a = 1 - g - b, \quad \mu = \frac{ag - b^2}{1 - b - 2g}.$$

Indeed, sign symmetry forces the two displayed zero entries, and stationarity of $g/\mu - \log b - \log(g(1 - g - b) - b^2)$ gives $3b^2 + 2gb + g^2 - g = 0$ and the stated μ . Strict convexity makes this interior stationary point the unique center. The spectral calculation in [section 4](#) gives

$$\kappa_{\text{red}}\mu^{3/2} \longrightarrow \frac{2\sqrt{2}}{7}, \quad \kappa_{\text{red}}g^{3/2} \longrightarrow \frac{3\sqrt{3}}{7}.$$

The first constant is approximately 0.40406, refining the earlier numerical value 0.40. The exact center, Frobenius-metric spectrum, and these limits are verified in Lean; the accompanying report `formal/FRACTIONAL_SDP.md` states the verification scope. The geometric and singularity-degree proof above is separate from that formalization.

Near-optimal faces yield a related finite plateau rather than a divergent tail.

Corollary 3.10 (Near-degeneracy plateau). *Suppose a family of compact instances has a scale $\vartheta > 0$ and constants independent of g, ϑ such that*

$$D(g) = \begin{cases} \Theta(g/\vartheta), & 0 < g \lesssim \vartheta, \\ \Theta(1), & \vartheta \lesssim g \leq g_0. \end{cases} \quad (63)$$

Assume also that $\|c_V\|_2$ is bounded below and that the barrier parameters under comparison are bounded above uniformly in ϑ ; for the canonical upper bound, assume the same of $M_{\text{tail}}\nu_{\text{can}}$ and assume $g_0 \leq g_{\text{can}}(\bar{\mu})$ uniformly in ϑ . Then every self-concordant barrier has the corresponding floor

$$\kappa_{\text{red}}^F(g) = \begin{cases} \Omega(\vartheta^{-2}), & 0 < g \lesssim \vartheta, \\ \Omega(g^{-2}), & \vartheta \lesssim g \leq g_0, \end{cases} \quad (64)$$

up to its explicit C_F and instance constants. The canonical barrier matches both orders on the tail under [theorem 3.5](#). Thus the g^{-2} growth stops near $g \asymp \vartheta$ at height $\Theta(\vartheta^{-2})$.

Proof. In the first range, $D(g)/g = \Theta(\vartheta^{-1})$; in the second it is $\Theta(g^{-1})$. Apply the lower bound (43) and the canonical upper bound (54). $\square$

For the simplex family $\min\{x_3 + \vartheta x_2 : x_1 + x_2 + x_3 = 1, x \geq 0\}$, the log-barrier plateau has the following precise two-scale limit. For each fixed $\vartheta > 0$,

$$\lim_{\mu \downarrow 0} \kappa_{\text{red}}(\mu) = \frac{1 + \vartheta^2 + \sqrt{1 - \vartheta^2 + \vartheta^4}}{1 + \vartheta^2 - \sqrt{1 - \vartheta^2 + \vartheta^4}}, \quad \lim_{\vartheta \downarrow 0} \vartheta^2 \left(\lim_{\mu \downarrow 0} \kappa_{\text{red}}(\mu) \right) = \frac{4}{3}. \quad (65)$$

To verify the inner limit, write the equality multiplier as $\alpha_\mu = \mu/x_1$. Centrality gives $x_2 = \mu/(\vartheta + \alpha_\mu)$, $x_3 = \mu/(1 + \alpha_\mu)$, and $\alpha_\mu \rightarrow 0$. On the tangent plane, put $h = (-u - v, u, v)$. After multiplication by μ^2 , the limiting Rayleigh quotient is

$$\frac{\vartheta^2 u^2 + v^2}{2(u^2 + uv + v^2)}.$$

Its generalized eigenvalues are $(1 + \vartheta^2 \pm \sqrt{1 - \vartheta^2 + \vartheta^4})/3$, whose ratio is the first expression in (65); expansion at $\vartheta = 0$ gives the second limit. The computations at $\vartheta = 10^{-2}$ and 10^{-3} round $\vartheta^2 \lim_{\mu \downarrow 0} \kappa_{\text{red}}(\mu)$ to $4/3$.

3.6 Transfer to approximate centrality

The floor is stable in the short-step neighborhood. All norms in the next statement are restricted to the tangent space.

Corollary 3.11 (Approximate-centrality transfer). *Fix $\mu > 0$, let $x_* = x_F(\mu)$, and suppose $x \in P^\circ$ satisfies*

$$\lambda_\mu(x) := \|c/\mu + \nabla F(x)\|_x^* \leq \rho \leq \frac{1}{6}. \quad (66)$$

Then, with $H_ = W^\top \nabla^2 F(x_*) W$ and $H = W^\top \nabla^2 F(x) W$,*

$$\|x - x_*\|_{x_*} \leq \frac{\rho}{(1 - \rho)^2} \leq \frac{1}{4}, \quad \kappa(H) \geq \left(\frac{3}{4}\right)^4 \kappa(H_*). \quad (67)$$

Consequently x inherits the right-hand sides of (43) and (44), evaluated at the exact center x_ , with the additional factor $(3/4)^4$.*

Proof. Restrict $f = F + c^\top(\cdot)/\mu$ to the affine slice. It is standard self-concordant, has Hessian $\nabla^2 F$, and is minimized at x_* . With

$$\omega(t) = t - \log(1 + t), \quad \omega_*(t) = -t - \log(1 - t),$$

the self-concordant Newton estimates give

$$\omega(\|x - x_*\|_{x_*}) \leq f(x) - f(x_*) \leq \omega_*(\lambda_\mu(x)) \leq \omega_*(\rho).$$

The elementary scalar inequality $\omega^{-1}(\omega_*(\rho)) \leq \rho/(1 - \rho)^2$ follows by substituting $t = \rho/(1 - \rho)^2$ and differentiating the difference $\omega(t) - \omega_*(\rho)$ from its value zero at $\rho = 0$. This proves the distance bound, and $\rho \leq 1/6$ gives $\rho/(1 - \rho)^2 \leq 6/25 < 1/4$.

If $r = \|x - x_*\|_{x_*} < 1$, self-concordant Hessian comparison gives

$$(1 - r)^2 H_* \preceq H \preceq (1 - r)^{-2} H_*.$$

It follows that $\lambda_{\max}(H) \geq (1 - r)^2 \lambda_{\max}(H_*)$ and $\lambda_{\min}(H) \leq (1 - r)^{-2} \lambda_{\min}(H_*)$; hence $\kappa(H) \geq (1 - r)^4 \kappa(H_*) \geq (3/4)^4 \kappa(H_*)$. $\square$

3.7 Implications and open geometric questions

For QIPMs that solve barrier-Newton systems, changing the self-concordant barrier cannot remove a sublevel-chord conditioning obstruction. At target primal objective gap ϵ , every barrier pays at least

$$\left(\frac{D(\epsilon) \|c_{\mathcal{V}}\|_2}{2(\nu_F + 2\sqrt{\nu_F})\epsilon} \right)^2$$

in the reduced Hessian, and, for canonical LPs or product-form SDPs with finite M_{tail} and $0 < \epsilon \leq g_{\text{can}}(\bar{\mu})$ attained on the canonical tail, the canonical barrier is within the explicit instance factor in (55). In particular, positive-dimensional optimal faces impose a quadratic inverse-gap floor, curved unique optima can impose a linear inverse-gap floor, and sharp singularity-degree exponents can impose fractional laws.

This conclusion is deliberately scoped to κ_{red} as defined in definition 2.9. It makes no claim about condition-number-free QIPM approaches such as Apers and Gribling [10], about preconditioners outside barrier-Hessian form, or about scalar congruences. Those access-model and system-representation questions are treated in later sections.

Open Problem 3.12 (Degenerate-vertex constants). For a polytope with a unique degenerate optimal vertex, characterize the finite limiting constant (or limiting spectrum) of the canonical reduced log Hessian directly from the active-set geometry.

Open Problem 3.13 (Facial characterization of chord growth). Characterize the exact growth of $D(g)$ for semidefinite programs from facial data beyond the generic strict-complementarity and attained singularity-degree cases. In particular, determine when Sturm’s 2^{-d} exponent is sharp for objective sublevel diameters rather than only an upper error-bound exponent.

4 Clustered Spectra and Solver Cost

The chord law in theorem 3.3 says when the condition number must diverge, but not how the eigenvalues between its two edges are distributed. This section resolves that question for the logarithmic barrier of a degenerate LP and then asks what the resulting spectrum costs. The answer separates solver families. The spectrum eventually consists of two intervals with parameter-independent internal condition numbers, so a classical Krylov method needs only polylogarithmically many matrix–vector products. Quantum polynomial solvers cannot use the empty spectral gap in the same way: in sufficiently large dimension, plain block-encoding access costs $\tilde{\Theta}(\kappa)$ in the worst case, whereas the factor access already defined in definition 2.15 has a $\tilde{\Theta}(\sqrt{\kappa})$ polynomial route under norm-tight encodings. Clustering improves neither quantum model beyond the dependence its access model already provides, while it exposes a classical–quantum separation on the Newton systems themselves. There is an important qualification. At exact central points the path’s own Newton right-hand sides are asymptotically orthogonal to the weak modes, so path-following steps can use a condition-number-free window solve after the gap opens.

4.1 The two-cluster tail

Consider the standard-form LP (1) and suppose that its primal feasible polytope P is compact with $P^\circ \neq \emptyset$. Let $Z \in \mathbb{R}^{n \times p}$ have orthonormal columns spanning $\ker A$, assume as in section 3 that the objective is nonconstant on the affine hull, and follow the exact log-barrier central path $(x(\mu), y(\mu), s(\mu))$. Let B be the maximal primal optimal support and $N = [n] \setminus B$; equivalently, this is the Goldman–Tucker optimal partition. Write P_B and P_N for coordinate projections, and put $D_P = \text{diam } P$.

Choose a fixed central-path tail $0 < \mu \leq \bar{\mu}$ on which

$$S_* = \sup_{0 < \mu \leq \bar{\mu}} \max_j s_j(\mu) < \infty, \quad \beta = \inf_{0 < \mu \leq \bar{\mu}} \min_{j \in B} x_j(\mu) > 0. \quad (68)$$

Strict complementarity of the central-path limit also gives $\underline{s}_N := \inf_{0 < \mu \leq \bar{\mu}} \min_{i \in N} s_i(\mu) > 0$. These are instance constants, not asymptotic variables.

Proposition 4.1 (Two-cluster log-barrier Hessian). *Let*

$$H_\mu = Z^\top X(\mu)^{-2} Z = H_{\text{red}}(\mu), \quad r = \text{rank}(P_N Z), \quad (69)$$

where the equality uses [section 2.1.1](#)'s reduced-Hessian notation with $W = Z$. If $r > 0$, let $\sigma > 0$ be the smallest positive singular value of $P_N Z$. The largest r eigenvalues of H_μ then lie in

$$\left[\frac{\sigma^2 \min_{i \in N} s_i(\mu)^2}{\mu^2}, \frac{\max_{i \in N} s_i(\mu)^2}{\mu^2} + \beta^{-2} \right]. \quad (70)$$

If $p - r > 0$, the remaining $p - r$ eigenvalues lie in

$$[D_P^{-2}, \beta^{-2}]. \quad (71)$$

Each nonempty interval has a μ -independent internal ratio. When both are nonempty, their separation grows as $\Theta(\mu^{-2})$; when $r = 0$ the top-cluster assertion is empty.

Proof. Complementarity gives $x_i(\mu)^{-2} = s_i(\mu)^2/\mu^2$. Split

$$H_\mu = H_{\mu,N} + H_{\mu,B}, \quad H_{\mu,N} = Z^\top P_N^\top X_N^{-2} P_N Z, \quad H_{\mu,B} = Z^\top P_B^\top X_B^{-2} P_B Z. \quad (72)$$

The first summand is positive semidefinite of rank r . When $r > 0$, its nonzero eigenvalues are the squared nonzero singular values of $X_N^{-1} P_N Z$ and therefore lie in

$$\left[\frac{\sigma^2 \min_{i \in N} s_i(\mu)^2}{\mu^2}, \frac{\max_{i \in N} s_i(\mu)^2}{\mu^2} \right].$$

The tail bound on x_B gives $0 \preceq H_{\mu,B} \preceq \beta^{-2} I$. Weyl monotonicity now gives the lower edge in (70), and Weyl's upper inequality gives its upper edge. Since $\ker H_{\mu,N} = \ker(P_N Z)$ has dimension $p - r$, the min-max principle applied to this subspace gives $\lambda_{p-r}^\uparrow(H_\mu) \leq \beta^{-2}$ when $p - r > 0$, where eigenvalues are ordered increasingly. Hence every one of the lowest $p - r$ eigenvalues has the claimed upper bound.

It remains to bound their lower edge. For every Euclidean unit tangent direction, one of its two signs exits the compact polytope within distance D_P . The Dikin containment argument in [theorem 3.2](#) therefore gives $H_\mu \succeq D_P^{-2} I$. Finally, (68) and $\underline{s}_N > 0$ bound each nonempty interval's endpoint ratio independently of μ . If both clusters are nonempty, the lower endpoint of (70) is $\Theta(\mu^{-2})$ while (71) is fixed. $\square$

The extremes agree with the geometry in [section 3](#). At a unique nondegenerate optimum—equivalently here, a singleton optimal face with a nondegenerate optimal basis— $r = p$: if $h \in \ker A$ and $h_N = 0$, then the optimal basis gives $A_B h_B = 0$ and hence $h = 0$. The bottom cluster is empty and the condition number remains bounded. If the optimal face has positive dimension, $\ker(P_N Z)$ contains its tangent space and the bottom cluster is nonempty; its fixed scale and the μ^{-2} top scale recover the chord-law divergence.

The two-window statement is specific to polyhedral log-barrier tails. The log-det central path of (61), the degree-two SDP witness in [proposition 3.9](#), has four increasingly ordered reduced-Hessian eigenvalues in the inherited Frobenius metric:

$$\lambda_1 \sim \sqrt{2} \mu^{-1/2}, \quad \lambda_2 \sim \mu^{-1}, \quad \lambda_3 \sim \sqrt{2} \mu^{-3/2}, \quad \lambda_4 \sim \frac{4}{7} \mu^{-2}.$$

To see the metric normalization, use tangent coordinates (b, g, t, e) for the symmetric matrix with diagonal $(1 - g - b, g, b)$ and upper entries (b, t, e) . At the center $t = e = 0$, write $q = ag - b^2$ and $A_2 = \begin{pmatrix} a & b \\ b & g \end{pmatrix}$. The off-block operator is A_2^{-1}/b , giving the first and third scales because $b \sim \sqrt{\mu/2}$ and $q \sim \mu$. The (b, g) coordinate Hessian K has Gram matrix $G = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}$; its operator is $G^{-1}K$. Direct differentiation gives $\mu K_{bb} \rightarrow 6$, $\mu^{3/2} K_{bg} \rightarrow -\sqrt{2}$, and $\mu^2 K_{gg} \rightarrow 1$. Thus the larger eigenvalue has coefficient $(G^{-1})_{22} = 4/7$, while $\mu^3 \det K \rightarrow 4$ and $\det G = 7$ give the smaller coefficient one. These four distinct powers fix the eventual order. In particular, there is no $O(1)$ window. The complete calculation, including the actual second derivative, the characteristic polynomial with multiplicities, and the full central-path tail, is verified in Lean as documented in `formal/FRACTIONAL_SDP.md`. It replaces the earlier numerical four-window observation with exact asymptotics.

4.2 Why conjugate gradients see the clusters

Cluster-sensitive CG estimates have a long classical lineage [16]. The following elementary construction is enough here and makes the logarithmic dependence on the inter-cluster gap explicit.

Proposition 4.2 (Two intervals are Krylov-benign). *Let $H \succ 0$ have*

$$\text{spec}(H) \subseteq [a_1, b_1] \cup [a_2, b_2], \quad 0 < a_1 \leq b_1 < a_2 \leq b_2,$$

and set $\rho_i = b_i/a_i$ and $\Lambda = b_2/a_1$. Thus $\kappa(H) \leq \Lambda$, with equality only when both enclosure endpoints are attained. For every $0 < \epsilon \leq 1/2$ there is a residual polynomial q , with $q(0) = 1$ and $\max_{\lambda \in \text{spec}(H)} |q(\lambda)| \leq \epsilon$, of degree

$$O(\sqrt{\rho_1 \rho_2} \log \Lambda \log(1/\epsilon) + \sqrt{\rho_2} \log(1/\epsilon)). \quad (73)$$

Consequently CG reaches relative H -norm error ϵ in $O_\rho(\log(\Lambda/\epsilon) \log(1/\epsilon))$ iterations.

Proof. If $a_2 < b_2$, let p_2 be the shifted-Chebyshev residual for $[a_2, b_2]$, normalized by $p_2(0) = 1$, of degree $m_2 = \lceil \sqrt{\rho_2} \rceil$. The standard Chebyshev estimate gives

$$\max_{[a_2, b_2]} |p_2| \leq 2e^{-2m_2/\sqrt{\rho_2}} \leq \frac{1}{2}.$$

On $[0, a_2]$ its magnitude is at most one: after the affine map from $[a_2, b_2]$ to $[-1, 1]$, the normalized Chebyshev quotient is monotone in magnitude between the image of zero and the left endpoint. At the edge $a_2 = b_2$, instead take $p_2(x) = 1 - x/a_2$ and $m_2 = 1$; the same two bounds hold.

If $a_1 = b_1$, take the exact factor $r_1(x) = 1 - x/a_1$ and $m_1 = 1$. Otherwise let r_1 be the shifted-Chebyshev residual for $[a_1, b_1]$ of degree $m_1 = O(\sqrt{\rho_1} \log(1/\epsilon))$, with $r_1(0) = 1$ and $\max_{[a_1, b_1]} |r_1| \leq \epsilon$. In its explicit quotient, the denominator is $T_{m_1}(\eta_1)$, where $\eta_1 = (a_1 + b_1)/(b_1 - a_1) > 1$, and $T_{m_1}(\eta_1) \geq \eta_1^{m_1}$. Combining this denominator bound with $|T_m(t)| \leq (2|t|)^m$ for the numerator outside the approximation interval gives, also for the exact-factor edge case,

$$\max_{[a_2, b_2]} |r_1| \leq (4\Lambda)^{m_1}. \quad (74)$$

Choose

$$j = \lceil m_1 \log_2(4\Lambda) + \log_2(1/\epsilon) \rceil, \quad q = r_1 p_2^j.$$

Then $q(0) = 1$. On the first interval, $|q| \leq \epsilon$ because $|p_2| \leq 1$; on the second, (74) and the choice of j again give $|q| \leq \epsilon$. Its degree is $m_1 + jm_2$, which is (73). CG minimizes the H -norm error over residual polynomials of its degree, so its error is no larger. $\square$

The numerical check used a 40-variable simplex-slice LP with a two-coordinate active set and a 37-dimensional optimal face. From $\mu = 10^{-2}$ to 10^{-8} , κ grew from about 10 to 7×10^{12} , the bottom-cluster ratio stayed below 1.006, and the top ratio approached 4.004 (from 2.44 at the shallowest point). CG solved random right-hand sides to a recursively updated residual tolerance of 10^{-8} in 3–8 iterations. At $\mu = 10^{-8}$, the reported eight-step recursive residual masks a postcomputed true relative residual $\|b - Hx\|_2 / \|b\|_2 \approx 10^{-4}$ and relative H -norm error about 6×10^{-6} , a float64 loss-of-conjugacy effect. A separate 80-bit check, outside the accompanying scripts, used six iterations; the benign iteration count therefore survives the precision check. The unstructured $\sqrt{\kappa}$ estimate is about 2.6×10^6 iterations. The accompanying analysis scripts produce the float64 spectrum and recursive-residual counts.

4.3 Quantum polynomial boundedness and the access-model gap

CG evaluates its residual polynomial only on eigenvalues. QSVT has a different contract: after normalization, its polynomial must be bounded on all of $[-1, 1]$, including the empty spectral gap and the negative half of the interval. This difference rules out transplanting the composite polynomial in [proposition 4.2](#). An order-of-magnitude evaluation of this construction on representative intervals reaches about 10^{85} inside the gap; this illustrative estimate is not used below.

Normalize $\|H\|_2 = 1$ and write

$$\text{spec}(H) \subseteq [1/\Lambda, \rho_1/\Lambda] \cup [1/\rho_2, 1], \quad \Lambda = \kappa(H), \quad (75)$$

where $\rho_1, \rho_2 \geq 1$ are fixed. Cost statements below assume norm-tight encodings, $\alpha_H = \Theta(\|H\|)$ and $\alpha_G = \Theta(\|G\|)$. Without this hypothesis the plain-access parameter is $\kappa_{\text{BE}} = \alpha_H \|H^{-1}\|$ from [equation \(24\)](#), and the factor parameter is $\kappa_G = \alpha_G \|H^{-1}\|^{1/2}$.

Theorem 4.3 (Theorem Q: clustered solves in the two access models). *For the two-cluster spectra (75):*

- (i) *In the one-sided polynomial model, a residual polynomial bounded by M on $[0, 1]$, with $q(0) = 1$ and $|q(1/\Lambda)| \leq \epsilon \leq 1/2$, has degree*

$$d \geq \sqrt{\frac{(1 - \epsilon)\Lambda}{2M}}. \quad (76)$$

Thus its degree-amplitude-normalization metric satisfies $dM = \Omega(\sqrt{\Lambda})$.

- (ii) *In the fixed-polynomial plain-access model, any single polynomial that, for $\rho_1 \geq 2$, relatively inverts a bottom interval containing the points $y = 1, 2$ to accuracy at most $1/8$, and is bounded by M on the QSVT-required symmetric interval $[-\Lambda, \Lambda]$, obeys*

$$dM \geq \frac{5}{16} \sqrt{\Lambda^2 - 4} = \Omega(\Lambda). \quad (77)$$

This is a lower bound on the polynomial metric for a right-hand side supported on the bottom cluster, not a general degree-to-QLS-cost theorem.

- (iii) *With factor access $H = G^\dagger G$, the one-sided construction is an even singular-value polynomial with $dM = \tilde{O}(\sqrt{\Lambda})$. Item (i), applied to the residual of this construction, gives the matching $\Omega(\sqrt{\Lambda})$ degree lower bound.*
- (iv) *Let N_{QLS} denote the Hilbert-space dimension, distinct from the optimal-partition index set N above. In the full query model, plain access has $\tilde{\Theta}(\kappa)$ worst-case QLS cost for $N_{\text{QLS}} \gtrsim \kappa$, even for two-point clusters. More precisely, Orsucci and Dunjko [118, Proposition 6] prove an $\Omega(\min\{\kappa, N_{\text{QLS}}\})$ normalized-block-encoding lower bound for all algorithms. Their Proposition 7 supplies the sparse/Grover lower bound $\Omega(\min\{\kappa, \sqrt{N_{\text{QLS}}}\})$, whose linear-in- κ regime requires $N_{\text{QLS}} \geq \kappa^2$.*

Proof. For (i), the mean-value theorem gives a point $\xi \in (0, 1/\Lambda)$ with $|q'(\xi)| \geq (1-\epsilon)\Lambda$. Markov's inequality, rescaled from $[-1, 1]$ to $[0, 1]$, gives $\max_{[0,1]} |q'| \leq 2d^2M$. Rearranging proves (76). Since $M \geq |q(0)| = 1$, multiplying the bound by M gives $dM = \Omega(\sqrt{\Lambda})$.

For (ii), use the rescaled eigenvalue $y = \Lambda\lambda$. Relative error $\delta' \leq 1/8$ for the target $1/y$ implies

$$p(1) \geq 1 - \delta', \quad p(2) \leq \frac{1 + \delta'}{2}, \quad p(1) - p(2) \geq \frac{5}{16}.$$

The mean-value theorem therefore gives a $\xi \in (1, 2)$ with $|p'(\xi)| \geq 5/16$. Bernstein's inequality on the symmetric interval states

$$|p'(y)| \leq \frac{dM}{\sqrt{\Lambda^2 - y^2}}.$$

Since $|\xi| < 2$, this yields (77).

The metric dM has the stated operational meaning only after fixing the input support. For a right-hand side in the bottom cluster, normalizing a polynomial bounded by M reduces the useful transformed amplitude by $\Theta(M)$, so amplitude amplification gives the dM proxy. For a general right-hand side the success amplitude also depends on its spectral overlap. Indeed, on a two-point spectrum the degree-one separator $p(y) = 1 - (y - 1)/\Lambda$ has $dM = O(1)$ even though the all-algorithms result in item (iv) still gives an $\Omega(\kappa)$ worst case. Thus item (ii) is not the basis for the plain-access row of the cost table; item (iv) is.

For (iii), the matching lower bound is item (i) applied to the residual of the construction below. Writing $q(y) = 1 - yp_y(y)$ in the rescaled variable $y = \Lambda\lambda$, one has $q(0) = 1$, $|q(1)| \leq e^{-T} + O(\epsilon) = O(\epsilon)$, and $|q(y)| \leq e^{-Ty} + y|p_y(y) - f_T(y)| \leq 1 + O(\epsilon)$ on $[0, \max\{L_G, \Lambda\}] \supseteq [0, \Lambda]$, the approximation interval of the construction below. In λ units this is a residual polynomial of degree $d + 1$ bounded by $1 + O(\epsilon)$ on $[0, 1]$, so item (i) forces $d = \Omega(\sqrt{\Lambda})$; since $M \geq |p_y(1)| \geq 1 - O(\epsilon)$, also $dM = \Omega(\sqrt{\Lambda})$. Here M is the amplitude bound of the inverse polynomial in y units, whose target is $1/y$; an inverse approximant in λ units necessarily has bound at least $(1 - \epsilon)\Lambda$, so the two normalizations are not interchangeable.

Now include the encoding normalization explicitly. A block encoding of G presents singular values $s = \sigma/\alpha_G$ in $[-1, 1]$. Put $L_G = \Lambda\alpha_G^2$. If p_y is bounded on $[0, L_G]$, then

$$q_G(s) = p_y(\Lambda\alpha_G^2 s^2) \tag{78}$$

is even and bounded on $[-1, 1]$. On a singular value of G , its argument is $\Lambda\sigma^2$, exactly the one-sided eigenvalue variable for $G^\dagger G$. Norm-tightness gives $\alpha_G = \Theta(\|G\|) = \Theta(1)$ under (75), so $L_G = \Theta(\Lambda)$; it does not set α_G equal to one.

For completeness, a $\tilde{O}(\sqrt{\Lambda})$ one-sided polynomial follows from the endpoint geometry. In $y = \Lambda\lambda$ units define

$$f_T(y) = \frac{1 - e^{-Ty}}{y} = \int_0^T e^{-ty} dt, \quad T = \Theta(\log(1/\epsilon)).$$

For $y \geq 1$, its relative error as an approximation to $1/y$ is at most e^{-T} , and $|f_T(y)| \leq T$ on $[0, L'_G]$, where $L'_G = \max\{L_G, \Lambda\} = \Theta(\Lambda)$. The shifted-Chebyshev expansion of e^{-ty} on this interval, truncated uniformly for $0 \leq t \leq T$, has degree

$$O\left(\sqrt{L'_G T \log(L'_G T/\epsilon)} + \log(L'_G T/\epsilon)\right).$$

Integrating its coefficients over t produces a polynomial $p_y(y)$ approximating f_T to absolute error $O(\epsilon/\Lambda)$, hence to relative error $O(\epsilon)$ for every $y \in [1, \Lambda]$. Its bound is $M = O(T)$, so degree times subnormalization is $\tilde{O}(\sqrt{\Lambda})$ under norm-tight access. The even polynomial q_G in (78) has only twice the degree. This makes the construction admissible in the factor-access polynomial model

and proves item (iii). Algebraically, the corresponding factor route to the linear-system solution is also transparent. For a full-column-rank factor, first form $w = (G^\dagger)^+ b$ and then $x = G^+ w$; an SVD verifies $x = (G^\dagger G)^{-1} b$. Define the two relative output amplitudes

$$\zeta_1 = \frac{\|(G^\dagger)^+ b\|}{\|(G^\dagger)^+ \| \|b\|}, \quad \zeta_2 = \frac{\|G^+ w\|}{\|G^+ \| \|w\|}.$$

Each pseudoinverse application costs $\tilde{O}(\kappa_G)$ to constant relative amplitude. The end-to-end $\tilde{O}(\kappa_G) = \tilde{O}(\sqrt{\kappa})$ solve follows only under the explicit overlap hypothesis $\zeta_1, \zeta_2 = \Omega(1)$; otherwise amplitude amplification pays the corresponding inverse overlaps.

There is no generic factor-access QLS guarantee at $\tilde{O}(\sqrt{\kappa})$. On the Orsucci–Dunjko two-point instance, the square-root factor is diagonal and its oracle is implementable with $O(1)$ hidden-input queries. A generic square-root-cost factor solver would therefore contradict their $\Omega(\kappa)$ query lower bound. The factor statement here is a polynomial-metric result and an overlap-hypothesis solve, not an unrestricted QLS upper bound. Item (iv) is exactly the prior-art all-algorithms grounding for the plain-access worst case. Together with the standard $\tilde{O}(\kappa_{\text{BE}})$ block-encoding QLS upper bound, it establishes the norm-tight $\tilde{\Theta}(\kappa)$ worst-case row below. $\square$

When $\rho_1 = 1$, item (ii) is vacuous because its two interpolation points are absent; this does not make the case algorithmically easy. Orsucci–Dunjko’s all-algorithms lower bound in item (iv) already covers exact two-point clusters.

The symmetric constraint is decisive. The polynomial in the last proof is well behaved on $[0, \Lambda]$ but grows on the negative half; it is not a plain-access QSVT polynomial. The value $f_T(0) = T \neq 0$ also prevents the odd extension used for the usual $1/x$ target. A discretized Remez linear program makes the distinction visible. Requiring a polynomial to separate $p(1/\Lambda) \leq 0.1$ from $p(2/\Lambda) \geq 0.9$ while staying bounded by one gave the following minimum degrees; the computation is included in the accompanying analysis scripts.

Λ	$d_{[-1,1]}$	$d_{[-1,1]}/\Lambda$	$d_{[0,1]}$	$d_{[0,1]}/\sqrt{\Lambda}$
50	52	1.04	12	1.70
100	102	1.02	17	1.70
200	205	1.02	24	1.70
400	376	0.94	33	1.65

Thus the symmetric calculation is 0.94–1.04 times Λ , whereas the one-sided relaxation is about $1.7\sqrt{\Lambda}$.

Suppressing logarithmic precision factors and keeping the access model fixed, the result is the following solve-cost comparison.

solver	matrix access	two-cluster cost
classical CG	matrix–vector products	$\text{polylog}(\kappa)$
factor polynomial route ¹	norm-tight $H = G^\dagger G$	$\tilde{\Theta}(\sqrt{\kappa})$
QLS worst case, $N_{\text{QLS}} \gtrsim \kappa$	norm-tight plain encoding of H	$\tilde{\Theta}(\kappa)$

With [section 2.1.1](#)’s convention $D = X^{1/2} S^{-1/2}$, the normal-equations matrix is $H = AD^2 A^\top$, and the factor $G = DA^\top$ satisfies $H = G^\dagger G$ and is naturally available from the IPM data when the corresponding factor oracle can actually be implemented. One should act on that factor rather than compose it into an encoding of H and inadvertently square the singular condition number.

¹The factor row is a polynomial-metric statement and an end-to-end solve cost only under the overlap hypothesis $\zeta_1, \zeta_2 = \Omega(1)$; it is not a generic factor-access QLS guarantee.

The all-algorithms plain-access construction is prior art due to Orsucci and Dunjko. Somma and de Wolf’s sum-of-squares separation is also prior art, but it is a bound for guided ground-state energy estimation, under their dimension condition $\dim \geq \log(1/\epsilon)/\gamma_{\text{guide}}^2$, where γ_{guide} is their guide overlap, and their stated error assumptions [125]. It distinguishes $\Omega(1/\delta)$ plain resolution from $\Omega(1/\sqrt{\delta})$ sum-of-squares resolution; we do not claim a transfer of that result to QLS. Within the polynomial model, the matching lower bound for the even/one-sided factor class is instead item (i). Cluster-sensitive Krylov convergence is also classical. What appears to be new here is the observation that the hard two-cluster quantum instances are solved by CG in $O(1)$ iterations when the clusters are points, together with the resulting separation on central-path Newton spectra. The novelty claim is this classical-side observation and assembly, not the quantum lower bounds themselves.

Open Problem 4.4 (Exact-subnormalization shift). The sharpest access-model question left by Theorem Q is the following. The shift $H \mapsto I - H$ moves the bottom cluster to an endpoint, where $1/\Lambda$ -scale discrimination would have square-root polynomial degree. Can a subnormalization-one block encoding of $I - H$ be constructed from plain access to H without paying the apparent $\Theta(\Lambda)$ uniform amplification cost near the new spectral edge?

Orsucci and Dunjko raise the closely related normalization problem for $B = I - \eta A$ (in their notation) in their Section 4.3 [118]. They observe that a linear combination of I and a normalized block encoding of A yields only the subnormalized pB , that any black-box amplification of B/β to B is in general inefficient by the lower bound of [65, Theorem 73] applied to $f(x) = \beta x$, and they leave open whether a normalized block encoding of a matrix with operator norm at most one can be built from a sparse-matrix oracle for it, answering it positively for diagonally dominant A and for sums of positive semidefinite local Hamiltonians. [Open Problem 4.4](#) is the promise-sensitive form of their black-box obstruction: the subnormalization to remove is the fixed factor two, the spectrum of H is promised to satisfy (75), and the question is whether that promise lowers the $\Theta(\Lambda)$ uniform-amplification degree at the shifted edge, not whether a general normalization procedure exists.

4.4 The central path does not use a worst-case right-hand side

The preceding obstruction concerns arbitrary right-hand sides. The exact log-barrier path generates a much more special vector. Let $P_{\text{bot}}(\mu)$ denote the spectral projector of H_μ onto the bottom cluster in [proposition 4.1](#).

Proposition 4.5 (Universal Newton right-hand side and weak-mode alignment). *In the setting of [proposition 4.1](#), at an exact central point define $v_\mu = Z^\top X^{-1}e$. For a centering target $\mu_2 = \gamma\mu$ with $0 < \gamma < 1$, the exact reduced Newton equations are*

$$\mu_2 H_\mu d_c = r_c, \quad r_c = -(1 - \gamma)\mu v_\mu, \quad \mu H_\mu d_a = r_a, \quad r_a = -\mu v_\mu. \quad (79)$$

Here d_a is the affine-scaling direction formed with the current local barrier metric; γ avoids collision with the singular value σ in [proposition 4.1](#). If the right-hand side is nonzero, then

$$\frac{\|P_{\text{bot}}(\mu)r\|_2}{\|r\|_2} = O(\mu^2) \quad (r = r_c \text{ or } r_a). \quad (80)$$

For either equation in (79), if the bottom component of its exact solution is dropped, the resulting relative Euclidean residual is exactly the fraction in (80).

Proof. At exact centrality, $Z^\top(c - \mu X^{-1}e) = 0$. Negating the reduced gradient for the target μ_2 gives

$$-Z^\top(c - \mu_2 X^{-1}e) = -(1 - \gamma)\mu Z^\top X^{-1}e,$$

and setting the target barrier term to zero gives $-Z^\top c = -\mu Z^\top X^{-1}e$. This proves the right-hand sides in (79). The reduced Hessian of the target barrier objective is $\mu_2 H_\mu$; the affine direction uses the current local metric μH_μ . Thus the displayed system scalars are also exact.

For the alignment claim, recall the splitting $H_\mu = H_{\mu,N} + H_{\mu,B}$ in (72). The bottom eigenspace of $H_{\mu,N}$ is exactly

$$\mathcal{T}_\Phi = \ker(P_N Z), \quad (81)$$

the reduced tangent space of the primal optimal face. The positive spectrum of $H_{\mu,N}$ begins at $\Theta(\mu^{-2})$, whereas $\|H_{\mu,B}\|_2 = O(1)$. The Davis–Kahan theorem therefore gives

$$\|P_{\text{bot}}(\mu) - P_{\mathcal{T}_\Phi}\|_2 = O(\mu^2) \quad (82)$$

[54].

Let x^a be the analytic center of the primal optimal face. The log-barrier path converges to x^a , and strict complementarity gives the tail expansion $x_B(\mu) = x_B^a + O(\mu)$ [101, 134]. For every reduced unit vector $u \in \mathcal{T}_\Phi$, the ambient direction Zu has $(Zu)_N = 0$, and first-order optimality of x^a on the face says

$$\langle (Zu)_B, (X_B^a)^{-1} e_B \rangle = 0.$$

Since the coordinates of x_B^a are positive, the tail expansion then gives $\langle u, Z^\top X^{-1}e \rangle = O(\mu)$. On the other hand $\|X^{-1}e\|_2 = O(1/\mu)$ because $x_N = \mu S_N^{-1} e_N$. Combining these facts with (82) yields

$$\|P_{\text{bot}}(\mu)v_\mu\|_2 = O(\mu). \quad (83)$$

Centrality also gives $v_\mu = Z^\top c/\mu$. The nonconstant-objective hypothesis makes $Z^\top c \neq 0$, so $\|v_\mu\|_2 = \Theta(1/\mu)$; dividing (83) proves (80).

Finally, write $d = d_{\text{top}} + d_{\text{bot}}$ for either system $\tau H_\mu d = r$, where $\tau = \mu_2$ for centering and $\tau = \mu$ for the affine direction, and use $\tilde{d} = d_{\text{top}}$. Then

$$\tau H_\mu \tilde{d} - r = -\tau H_\mu d_{\text{bot}} = -P_{\text{bot}}(\mu)r.$$

Taking Euclidean norms proves the exact residual identity. $\square$

Corollary 4.6 (Condition-number-free top-window solve). *Suppose an inexact Newton step only requires a relative linear-system residual at most η , with η independent of μ . On the exact log-barrier path, once $\mu = O(\sqrt{\eta})$, it suffices to invert the top cluster. Plain-access QSVT can do so with degree*

$$O(\rho_2 \log(\rho_2/\eta)), \quad (84)$$

independent of $\kappa(H_\mu)$.

Proof. Dropping the bottom solution component contributes $O(\mu^2)$ relative residual by [proposition 4.5](#); choose the tail constant so that this is at most $\eta/2$. After scaling the top interval to $[1/\rho_2, 1]$, the standard bounded QSVT approximation to $1/x$ has degree $O(\rho_2 \log(\rho_2/\eta))$ and can make the remaining residual at most $\eta/2$. The interval stays a fixed distance from zero, so this polynomial is admissible with plain access and has no dependence on the remote bottom cluster. $\square$

The contract in this corollary is the reduced-primal relative residual $\|\tau H_\mu \tilde{d} - r\|/\|r\|$, with $\tau = \mu_2$ for centering and $\tau = \mu$ for the affine direction. Here r_c denotes the reduced primal right-hand side in (79), not the KKT complementarity residual denoted by r_c in [section 2.1.1](#). This contract is distinct from the absolute OSS and MNES conventions in [definition 2.8](#): OSS requires an $O(\bar{\mu})$ residual, while MNES requires an $O(\sqrt{\bar{\mu}/n})$ system residual chosen to produce an $O(\bar{\mu})$ complementarity residual. After identifying residual vectors, a relative tolerance η_{rel} implies an absolute requirement $\|\text{res}\| \leq \tau_{\text{abs}}$ only when $\eta_{\text{rel}}\|r\| \leq \tau_{\text{abs}}$. Here $r_a = -Z^\top c$ and

$r_c = -(1 - \gamma)Z^\top c$ are generally constant in norm, so a constant relative tolerance does not by itself imply either shrinking absolute contract.

The measured law is substantially smaller than its asymptotic bound. On Netlib **afiro**, the dropped-bottom residual was 6.7×10^{-6} , 6.4×10^{-10} , and 6.7×10^{-14} at $\mu = 10^{-1}, 10^{-3}, 10^{-5}$, respectively: an instance constant of about 7×10^{-4} multiplying μ^2 . Both centering and affine columns agree, as (79) predicts; the top-cluster RHS mass rounded to 1.0000, and the solution’s top mass remained at least 0.9929. The symmetric 40-variable family had machine-zero weak-mode mass. The accompanying analysis scripts generate these measurements.

An end-to-end damped-Newton run tested more than exact central points. It used shrink factor 0.85, up to four corrections per parameter value, and fraction-to-boundary steps. A window solver dropped the bottom modes only after detecting a gap larger than two decades and otherwise used the full solve. The window and exact runs agreed in final gap and correction count: the toy family finished at gap 1.758×10^{-8} with 205 corrections in both modes, and **afiro** finished at relative gap 7.4×10^{-10} with 393 corrections in both. This does not establish iterate-by-iterate identity; direct instrumentation of the two runs shows that intermediate iterates can differ. In a separate, independently reproduced experiment, a naive geometric-mean threshold without gap detection stalled on **afiro** at relative gap 0.98 by splitting the early spectral continuum arbitrarily. That negative case is separate from the reported adaptive experiment. The operational rule is therefore: use full solves until a genuine cluster gap opens, then drop the weak modes. The accompanying analysis scripts implement the adaptive experiment.

4.5 A Netlib conditioning survey

To test how early the asymptotic picture becomes visible, we surveyed 14 small presolved standard-form Netlib LPs. For each instance, a damped Newton method followed the log-barrier path in $\ker A$. We fitted $\kappa_{\text{red}}(g) \asymp g^{-\alpha}$ over the last four grid points, spanning about three gap decades, counted clusters using gaps larger than one decade, and ran CG to 10^{-8} tolerance with a 4000-iteration cap at the deepest point. A second computation solved directional LPs in 30 fixed random directions over the objective sublevel polytope. The resulting $\widehat{D}(g)$ is a lower estimate of the diameter, not the diameter itself; fitting $\widehat{D}(g) \asymp g^\theta$ and comparing with $\alpha = 2 - 2\theta$ is therefore a consistency check. It is computationally independent of the path Hessian eigensolves, but is not an exact second measurement of the chord-law diameter.

instance	m	n	α	κ_{end}	clusters	CG its
adlittle	55	136	2.07	1.7×10^{13}	2	4000 cap
afiro	9	18	2.00	3.8×10^{12}	2	20
beaconfd	11	30	0.02	1.3×10^6	3	28
bore3d	36	65	0.03	3.4×10^{11}	3	521
kb2	46	71	0.36	9.6×10^9	4	198
sc105	34	65	0.03	9.9×10^3	1	55
sc205	67	125	0.03	7.2×10^5	1	286
sc50a	17	33	0.01	1.5×10^3	1	20
sc50b	15	29	0.00	5.8×10^2	1	16
scagr7	93	147	1.30	8.1×10^{12}	2	4000 cap
scorpion	77	136	0.12	1.1×10^5	1	160
seba	10	17	0.00	1.3×10^3	2	7
brandy	107	211	numerical failure			
recipelp	63	107	numerical failure			

Table 2: Tail conditioning fits and deepest-point solver diagnostics. The cluster count uses a one-decade heuristic and should not be read as an exact spectral partition.

Two endpoint classes dominate [table 2](#). The positive-dimensional-face instances **afiro** and

adlittle have $\alpha \approx 2$, while the vertex-like **sc** instances, **seba**, **scorpion**, **beaconfd**, and **bore3d** have frozen exponents $\alpha \approx 0$; their condition-number levels can still be large. For **afiro**, $\widehat{D}(g)$ freezes at 126.84, so $\theta \rightarrow 0$ predicts $\alpha = 2.000$, matching the path fit 2.00. For **sc50b**, $\widehat{D}/g$ freezes at 126, so $\theta \rightarrow 1$ predicts $\alpha = 0$, matching its condition-number plateau near 582.

The intermediate **kb2** slope is a crossover, not evidence for a third asymptotic class. Its random-direction estimate of θ climbs from 0.08 to 0.99 across the available gap decades, the in-the-wild version of the near-degeneracy plateau in [corollary 3.10](#); the measured $\alpha = 0.36$ lies in the knee and will tend to zero. The **scagr7** local fitted θ slopes range from 0.1 to 0.6 and may reflect a slower multiscale crossover. We claim no fractional asymptotic exponent for a Netlib instance; the proved fractional example remains the constructed SDP in [proposition 3.9](#).

The instructive counterpoint is **adlittle**. At the first float64-safe probe, its 81 reduced eigenvalues form one cluster with spread ratio $\kappa = 4.3 \times 10^4$ (about 4.6 decades). The middle probe at $\kappa = 1.85 \times 10^7$ is also one segment, with spread ratio 1.9×10^7 (about 7.3 decades), and at the deepest safe probe the spectrum spans nine decades at $\kappa = 1.31 \times 10^9$. No decade gap appears at any of these three depths. CG correspondingly takes 250, 985, and 2935 iterations; counts above 81 reflect finite-precision loss of conjugacy. The deeper survey row heuristically reports two clusters at $\kappa = 1.7 \times 10^{13}$, where float64 spectral diagnostics are already fragile. This sharpens rather than contradicts [proposition 4.1](#): the asymptotic split is real, but the safe probe establishes only that it has not opened through $\kappa \approx 1.3 \times 10^9$. The follow-up calculation is included in the accompanying analysis scripts.

That wall is itself visible in **brandy** and **recipe1p**: at computed condition numbers above roughly 10^{15} , roundoff produces negative eigenvalues for a mathematically positive-definite reduced Hessian. This is consistent with the classical finite-precision analysis of interior methods [\[135\]](#). Presolve, rank reduction, and Phase-I choices affect the face approached by the path, so every exponent in [table 2](#) belongs to the particular presolved representation. The accompanying analysis scripts implement the path survey and sublevel fits; the table reports their deterministic cached-instance run.

4.6 Scope of the escape

The two-cluster theorem and the $O(\mu^2)$ right-hand-side fraction use the LP logarithmic barrier, strict complementarity on a sufficiently small tail, and exact centrality. For a general barrier, it is plausible that the barrier’s face-center first-order condition produces an analogous alignment, perhaps only an $O(\mu)$ fraction, but this is a conjecture: neither the cluster identification nor the required path-limit expansion has been proved in that generality. Off-path iterates can acquire weak-mode components. Nothing here establishes the same rate for predictor–corrector methods, infeasible-start variants, or unreduced primal–dual systems. Accordingly, [corollary 4.6](#) is an exact-path structural statement and the trajectory experiment is evidence for the stated adaptive rule, not a general convergence theorem for those variants.

5 Beyond the Condition Number

The condition number is a worst-case spectral parameter: it charges for a small eigenvalue even when the right-hand side barely sees its eigenspace. Dalzell, Li, and Su (DLS) recently gave two quantum linear-system solvers that replace this worst-case dependence by right-hand-side-dependent truncation and filtering parameters [\[53\]](#); their filtering solver improves an earlier construction of Li [\[94\]](#). The classical idea of effective conditioning is older [\[41\]](#), but the DLS parameters had not been evaluated on IPM Newton systems. We do so here.

Sections [3–4](#) showed that a degenerate LP produces two increasingly separated spectral clusters. The question in this section is finer: when does the Newton right-hand side allow a solver to exploit that separation? Exact block identities give a complete coupling criterion.

They also expose a truncation transition, while a basis-invariant spectral dimension gives more general solver-selection laws. Finally, when the active range is supplied, a directly assembled equilibration removes the central-path divergence while preserving the Newton state exactly.

Throughout the first part, $H \succ 0$ is presented through a block encoding of H/α_H . For a nonzero right-hand side f , set $x = H^{-1}f$ and define

$$\xi(H, f) = \alpha_H \frac{\|H^{-1}f\|}{\|f\|}, \quad \rho(H, f) = \alpha_H \frac{\|H^{-2}f\|}{\|H^{-1}f\|}. \quad (85)$$

The second quantity is the ideal DLS filtering parameter: it is $\|(H/\alpha_H)^{-1}x\|/\|x\|$, not the first-inverse quantity ξ . We avoid ν , which [section 2.1](#) reserves for the barrier parameter. More generally, for a nonsingular, possibly non-Hermitian matrix A encoded as A/α_A , filtering uses

$$\rho(A, f) = \alpha_A \frac{\|A^{-\dagger}x\|}{\|x\|}, \quad x = A^{-1}f. \quad (86)$$

The DLS truncation parameter instead asks how far toward zero one must invert in order to retain all but the prescribed normalized solution mass.

5.1 Exact-center coupling law

Consider a standard-form LP with full-row-rank A , and suppose its exact central path converges to a strictly complementary solution. Let

$$B = \{i : x_i^* > 0\}, \quad N = \{i : s_i^* > 0\}, \quad U = \text{range}(A_B), \quad K = U^\perp,$$

and assume $0 < \dim U < m$. This is the mixed, primal-degenerate case; if $U = \mathbb{R}^m$, the normalized normal matrix has no asymptotically slow cluster. At an exact center, complementarity gives

$$H_\mu = H_{\text{NE}}(\mu) := AXS^{-1}A^\top = \mu^{-1}C_\mu + \mu D_\mu, \quad C_\mu = A_B X_B^2 A_B^\top, \quad D_\mu = A_N S_N^{-2} A_N^\top. \quad (87)$$

Here $C_\mu \rightarrow C_0$, $D_\mu \rightarrow D_0$, and, in $U \oplus K$ blocks,

$$D_\mu = \begin{pmatrix} D_\mu^U & F_\mu^\top \\ F_\mu & E_\mu \end{pmatrix}.$$

The restrictions $C = C_0|_U$ and $E = E_0|_K$ are positive definite. Indeed, C is a positive-weight Gram matrix with range U , and if $z \in K$ obeys $A_N^\top z = 0$, then also $A_B^\top z = 0$; full row rank of A forces $z = 0$, proving positivity of E on K . At the optimal limit, $b = Ax^* = A_B x_B^*$, so the exact-centering right-hand side lies in U . A short step to $\sigma_\mu \mu$, $\sigma_\mu \neq 1$, has $H_\mu \Delta y = (1 - \sigma_\mu)b$. The scalar is immaterial to normalized QLS parameters, so the next theorem uses $b \neq 0$. This nonzero condition also follows in the stated LP setting. Indeed $x^* \neq 0$; if $b = 0$, optimality gives $c^T x^* = 0$, while a strictly feasible dual slack s would give $s^T x^* = c^T x^* > 0$, a contradiction.

Theorem 5.1 (Exact-center coupling criterion). *Define*

$$\begin{aligned} R_\mu &= D_\mu^U - F_\mu^\top E_\mu^{-1} F_\mu, \\ p_\mu &= (C_\mu|_U + \mu^2 R_\mu)^{-1} b, & g_\mu &= E_\mu^{-1} F_\mu p_\mu, \\ q_\mu &= (C_\mu|_U + \mu^2 R_\mu)^{-1} (p_\mu + F_\mu^\top E_\mu^{-1} g_\mu). \end{aligned} \quad (88)$$

Then the following identities are exact:

$$H_\mu^{-1} b = \mu \begin{pmatrix} p_\mu \\ -g_\mu \end{pmatrix}, \quad (89)$$

$$H_\mu^{-2} b = \begin{pmatrix} \mu^2 q_\mu \\ -E_\mu^{-1} g_\mu - \mu^2 E_\mu^{-1} F_\mu q_\mu \end{pmatrix}. \quad (90)$$

For a norm-tight encoding, $\alpha_H = \Theta(\|H_\mu\|) = \Theta(\mu^{-1})$,

$$\xi(H_\mu, b) = \Theta(1), \quad \rho(H_\mu, b) = \Theta\left(1 + \frac{\|g_\mu\|}{\mu^2}\right). \quad (91)$$

In particular, with

$$g = E^{-1}FC^{-1}b, \quad (92)$$

filtering improves asymptotically on the full condition-number scale if and only if $g = 0$. If $g \neq 0$, then $\rho(H_\mu, b) = \Theta(\mu^{-2}) = \Theta(\kappa(H_\mu))$; if $g = 0$, then $\rho = O(\mu^{-1}) = O(\sqrt{\kappa(H_\mu)})$, with bounded cost precisely when $g_\mu = O(\mu^2)$.

Proof. Write a vector in $U \oplus K$ coordinates. Multiplying $H_\mu(u, v)^\top = b$ by μ , the lower block equation gives $v = -E_\mu^{-1}F_\mu u$. Substitution in the upper equation gives $(C_\mu|_U + \mu^2 R_\mu)u = \mu b$, proving (89).

Apply the same elimination to $H_\mu(u, v)^\top = H_\mu^{-1}b$. Its lower equation is $F_\mu u + E_\mu v = -g_\mu$, so $v = -E_\mu^{-1}g_\mu - E_\mu^{-1}F_\mu u$. The upper equation then becomes

$$(C_\mu|_U + \mu^2 R_\mu)u = \mu^2(p_\mu + F_\mu^\top E_\mu^{-1}g_\mu),$$

which proves (90).

Continuity and positive definiteness give $p_\mu \rightarrow p = C^{-1}b \neq 0$. Moreover, $q_\mu \rightarrow C^{-1}(p + F^\top E^{-1}g)$, and the vector in parentheses is nonzero because its inner product with p is $\|p\|^2 + \|E^{-1}Fp\|^2 > 0$. Equations (89)–(90) therefore imply

$$\|H_\mu^{-1}b\| = \Theta(\mu), \quad \|H_\mu^{-2}b\| = \Theta(\|g_\mu\| + \mu^2).$$

For the lower bound, write the second inverse as (a_μ, v_μ) , where $a_\mu = \mu^2 q_\mu$. Its lower block equation gives $g_\mu = -E_\mu v_\mu - F_\mu a_\mu$. Boundedness of E_μ, F_μ controls $\|g_\mu\|$ by $\|(a_\mu, v_\mu)\|$, and $q_\mu \rightarrow q \neq 0$ controls μ^2 by the same norm. This also covers cancellation between the two terms in the lower block. Substitution in (85) proves (91). Finally $g_\mu \rightarrow g$. Thus $g \neq 0$ gives the full μ^{-2} scale. Under strict complementarity the LP central path, and hence $C_\mu, D_\mu, F_\mu, E_\mu$ and g_μ , is analytic at $\mu = 0$. Halická proves this boundary analyticity for LP and the corresponding result for SDP [73, 74]. If $g = 0$, differentiability alone gives $g_\mu = O(\mu)$, hence $\rho = O(\mu^{-1})$. More precisely, let g'_0 denote the derivative of the coupling at zero. If $g'_0 \neq 0$, then $\|g_\mu\| = \Theta(\mu)$ and $\rho = \Theta(\mu^{-1})$. If $g'_0 = 0$, the analytic Taylor remainder gives $g_\mu = O(\mu^2)$ and bounded ρ . Thus boundedness is equivalent to $g_0 = g'_0 = 0$ for this analytic family. The norm law itself also proves boundedness exactly when $g_\mu = O(\mu^2)$. $\square$

Equivalently, the obstruction is $\Pi_K D_0(C|_U)^{-1}b \neq 0$. The exceptional limiting right-hand-side space is $C(\ker F)$, and the obstruction vanishes for every $b \in U$ exactly when $F = 0$, meaning that D_0 reduces $U \oplus K$. Analyticity sharpens the continuous-rate analysis: after $g = 0$, the only divergent intermediate rate is a first-order term $g_\mu = \Theta(\mu)$, which gives $\rho = \Theta(\mu^{-1})$; every higher integer order gives bounded ρ . For a non-tight encoding, the exact scale carried by the proof is

$$\rho(H_\mu, b) = \Theta\left(\alpha_H \left(\mu + \frac{\|g_\mu\|}{\mu}\right)\right). \quad (93)$$

Proposition 5.2 (Sparse sharpness witnesses). *There are two row- and column-two LPs with the same $\kappa(H_\mu) \sim (2\mu^2)^{-1}$ and $\xi = \Theta(1)$, but respectively $\rho = \Theta(\mu^{-2})$ and $\rho = 1$.*

Proof. For the coupled instance, take

$$\min x_2 + x_3 \quad \text{subject to} \quad \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{pmatrix} x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad x \geq 0. \quad (94)$$

Its primal optimum and optimal partition are unique. The optimal duals have $y_1 = 0$, $y_2 \in [-1, 1]$; their strictly complementary members have $y_2 \in (-1, 1)$. The displayed slack $s^* = (0, 1, 1)$ corresponds to $y^* = 0$, the dual analytic center and hence the central-path limit, while $x^* = e_1$. The central point is $x = (1 - t, t, t)$, where

$$t = \frac{2 + 3\mu - \sqrt{4 - 4\mu + 9\mu^2}}{4} = \mu + O(\mu^2).$$

With $\theta_1 = (1 - t)^2/\mu$ and $\theta = t^2/\mu$,

$$H_\mu = \begin{pmatrix} \theta_1 + \theta & \theta \\ \theta & 2\theta \end{pmatrix}, \quad \kappa(H_\mu) \sim \frac{1}{2\mu^2}.$$

Here $C = 1$, $E = 2$, $F = 1$, and the centering right-hand side is $b = (1, 0)^\top$, so $p = 1$ and $g = 1/2$. Direct inversion gives

$$H_\mu^{-1}b = \frac{(2, -1)^\top}{2\theta_1 + \theta}, \quad H_\mu^{-2}b \longrightarrow (0, -1/4)^\top.$$

For $\alpha_H = \|H_\mu\|$, $\xi \rightarrow \sqrt{5}/2$ and $\mu^2\rho \rightarrow 1/(2\sqrt{5})$. The normalized solution is exactly $(2, -1)/\sqrt{5}$. Since the slow eigenspace tends to the second coordinate axis, its slow-cluster amplitude tends to $1/\sqrt{5}$.

For the decoupled instance, replace the constraint matrix by

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix}$$

and keep the same objective and right-hand side. It has the same unique primal optimum, optimal partition, dual-optimal interval, central-path dual limit, and it also satisfies the proposition's row/column-two sparsity bound, but, for $0 < \mu \leq 1/\sqrt{2}$ and $\alpha_H = \|H_\mu\|$,

$$H_\mu = \text{diag}(\mu^{-1}, 2\mu), \quad \kappa(H_\mu) = \frac{1}{2\mu^2}, \quad \xi = \rho = 1.$$

Thus neither κ nor the first-inverse norm distinguishes the two instances; active-to-inactive coupling does. $\square$

Formal verification and scope. The Lean development in `formal/QipmFormal/Coupling/` checks the LP normal-matrix split and centering right-hand side, block elimination and the actual inverse identities, noncancellation, and the exceptional right-hand-side space. It derives the filtering law from continuous block data and checks the differentiable and analytic rate consequences. The sparse witnesses are checked from their primal and dual equations, with Euclidean norms and the displayed normalization qualifications. The claim map and reproduction command are in `formal/COUPLING.md`. The audit permits only `propext`, `Classical.choice`, and `Quot.sound`. The general existence and boundary analyticity of the LP central path remain mathematical inputs; the development verifies the block regularity consequences rather than formalizing those external theorems. It does not verify the quantum solver, construct its block encoding, or establish the later truncation and spectral-dimension claims.

5.2 Truncation at the slow-shell boundary

Let $P_s(\mu)$ project onto the eigenvalues of H_μ of order $\Theta(\mu)$, which become $\Theta(\mu^2)$ after norm normalization. For a general nonzero f , define its slow right-hand-side amplitude and slow solution mass by

$$q_s = \frac{\|P_s f\|}{\|f\|}, \quad p_s = \frac{\|P_s H_\mu^{-1} f\|^2}{\|H_\mu^{-1} f\|^2}.$$

Proposition 5.3 (Two-cluster truncation transition). *If the fast-cluster component of f is bounded below, then*

$$p_s = \Theta\left(\frac{q_s^2}{\mu^4 + q_s^2}\right). \quad (95)$$

Consequently $q_s = \Theta(\mu^2)$ is the critical boundary. There the slow solution mass is constant and fixed-accuracy DLS truncation must retain the full $\Theta(\mu^{-2})$ normalized inverse scale for every target error below that mass. If $q_s = o(\mu^2)$, truncation can eventually discard the slow cluster at fixed accuracy; filtering need not yet be bounded. Under a tight encoding, $q_s = O(\mu^4)$ is sufficient for bounded ideal filtering cost. At exact centering, generic coupling gives $q_s = \Theta(\mu^2)$.

Proof. On the unnormalized fast and slow clusters, H_μ^{-1} has scales $\Theta(\mu)$ and $\Theta(\mu^{-1})$, respectively. The lower bound on the fast part of f therefore gives fast solution norm $\Theta(\mu\|f\|)$, while the slow solution norm is $\Theta(\mu^{-1}q_s\|f\|)$. Dividing the squared slow norm by the sum proves (95).

For the exact-centering vector $b \in U$, standard separated-subspace perturbation, or direct substitution in the block eigenvector equation, gives in the leading K component

$$P_s(\mu)b = -\mu^2 D_{0,KU} C^{-1} b + o(\mu^2). \quad (96)$$

It is therefore $\Theta(\mu^2)$ when the limiting coupling is nonzero. Equation (95) proves the truncation claims. One additional inverse weights the slow part by another μ^{-2} relative to the fast part, so its contribution is bounded only at $q_s = O(\mu^4)$, which is the filtering statement. $\square$

The comparison with [proposition 4.5](#) and [corollary 4.6](#) requires naming the two formulations. This section uses the m -dimensional normal matrix

$$H_{\text{NE}}(\mu) = AXS^{-1}A^\top, \quad \text{spec}(H_{\text{NE}}(\mu)) \subset \Theta(\mu^{-1}) \cup \Theta(\mu),$$

whereas [section 4.4](#) uses the $(n - m)$ -dimensional reduced Hessian

$$H_{\text{red}}(\mu) = Z^\top X^{-2}Z, \quad \text{spec}(H_{\text{red}}(\mu)) \subset \Theta(\mu^{-2}) \cup \Theta(1).$$

When the corresponding clusters are nonempty, both have fast normalized scale $\Theta(1)$ and slow normalized scale $\Theta(\mu^2)$, but they act on different variables and right-hand sides. [Section 4](#) proves an $O(\mu^2)$ slow right-hand-side fraction for $H_{\text{red}}(\mu)$. For $H_{\text{NE}}(\mu)$, the present coupling calculation instead gives $q_s = \Theta(\mu^2)$ only when the limiting coupling is nonzero; the decoupled witness has $P_s b = 0$.

The residual-versus-state distinction can be proved directly for the normal equation. Under nonzero limiting coupling, let $z = H_{\text{NE}}(\mu)^{-1}b$ and discard $z_s = P_s z$. Because P_s is a spectral projector,

$$\frac{\|H_{\text{NE}}(\mu)(z - z_s) - b\|}{\|b\|} = \frac{\|P_s b\|}{\|b\|} = q_s = \Theta(\mu^2), \quad (97)$$

while (95) gives $\|z_s\|^2/\|z\|^2 = \Theta(1)$ under the same nonzero-coupling hypothesis. A state-preparation algorithm below that constant solution mass must retain the slow cluster, but a fixed relative-residual contract may discard it. Thus the two formulations agree on their normalized spectral separation, while both the formulation and the output contract matter; this is the precise sense in which the result is consistent with [section 4](#).

5.3 Effective spectral dimension

Two clusters are not necessary for a sharp solver-selection law. Normalize $\|H\| = 1$ and $\|b\| = 1$, let

$$\mathbf{m}_b(S) = \langle b, \mathbf{1}_S(H)b \rangle, \quad (98)$$

and let λ_* be the smallest eigenvalue visible to b , with $\kappa_* = \lambda_*^{-1}$. We say that (H, b) has effective spectral dimension D if constants $c, C > 0$, independent of λ_* , satisfy

$$ct^{D/2} \leq F_b(t) := \mathbf{m}_b([0, t]) \leq Ct^{D/2} \quad (\lambda_* \leq t \leq 1). \quad (99)$$

This is invariant under simultaneous orthogonal changes $(H, b) \mapsto (VHV^\top, Vb)$; it is not coordinate sparsity in disguise. Unlike an unanchored per-shell condition, (99) controls the mass at the smallest visible eigenvalue and permits fixed gaps between the first few eigenvalues.

Theorem 5.4 (Spectral-dimension phase laws). *Assume the norm-tight unit-normalization convention: after $\|H\| = 1$, the block-encoding normalization is $\alpha_H = \Theta(1)$, and truncation cutoffs are measured in this same normalization. Operationally,*

$$\kappa_{\text{eff}}(\epsilon_{\text{ls}}) := \inf \left\{ K \in [1, \kappa_*] : \frac{\langle b, H^{-2} \mathbf{1}_{[0, 1/K)}(H)b \rangle}{\langle b, H^{-2}b \rangle} \leq \epsilon_{\text{ls}}^2 \right\}. \quad (100)$$

For an encoding of H/α_H , the solver's encoded cutoff scale is $\alpha_H \kappa_{\text{eff}}$; the standing tight convention identifies these up to constants. Under (99), this quantity obeys

$$\kappa_{\text{eff}}(\epsilon_{\text{ls}}) = \begin{cases} \Theta_{\epsilon_{\text{ls}}}(\kappa_*), & D < 4, \ 0 < \epsilon_{\text{ls}} < 1, \\ \kappa_*^{1 - \Theta(\epsilon_{\text{ls}}^2)}, & D = 4, \\ \Theta(\min\{\kappa_*, \epsilon_{\text{ls}}^{-4/(D-4)}\}), & D > 4. \end{cases} \quad (101)$$

For $D < 4$, the hidden constant may depend on the fixed error and on the cumulative-law constants. A sublinear power of κ_* is possible only when the allowed error tends to one as $\kappa_* \rightarrow \infty$. At $D = 4$, a definite leading cumulative coefficient gives exponent $1 - \epsilon_{\text{ls}}^2 + o(1)$; under only (99), $1 - \Theta(\epsilon_{\text{ls}}^2)$ is sharp, and the result saturates at $\Theta(\kappa_*)$ when $\epsilon_{\text{ls}}^2 \log \kappa_* = O(1)$. The ideal filtering parameter is

$$\rho = \begin{cases} \Theta(\kappa_*), & D < 4, \\ \Theta(\kappa_*/\sqrt{\log \kappa_*}), & D = 4, \\ \Theta(\kappa_*^{(8-D)/4}), & 4 < D < 8, \\ \Theta(\sqrt{\log \kappa_*}), & D = 8, \\ \Theta(1), & D > 8. \end{cases} \quad (102)$$

Thus fixed-accuracy truncation becomes independent of mesh conditioning above $D = 4$, whereas filtering does so only above $D = 8$. For a non-tight encoding, both the effective cutoff scale and every filtering branch acquire a factor α_H ; in particular the $D > 8$ filtering cost is $\Theta(\alpha_H)$ rather than $\Theta(1)$.

Proof. For $p > 0$, put $Z_p = \langle b, H^{-p}b \rangle$ and write $s = D/2$. Abel summation, equivalently Stieltjes integration by parts, gives the exact layer-cake identity

$$Z_p = 1 + p \int_{\lambda_*}^1 t^{-p-1} F_b(t) dt. \quad (103)$$

Using $F_b(t) = \Theta(t^s)$ in this integral gives

$$Z_p = \begin{cases} \Theta(\kappa_*^{p-D/2}), & D < 2p, \\ \Theta(\log \kappa_*), & D = 2p, \\ \Theta(1), & D > 2p. \end{cases} \quad (104)$$

The squared norm of $H^{-1}b$ is Z_2 . If a cutoff discards eigenvalues strictly below $u = 1/K$, as in (100), its discarded squared solution norm is

$$T_2(u) = u^{-2} F_b(u^-) + 2 \int_{\lambda_*}^u t^{-3} F_b(t) dt, \quad (105)$$

where $F_b(u^-) = \mathfrak{m}_b([0, u))$ is the left limit. For $D < 4$, fix a sufficiently small constant $u_0 > 0$. Dyadically summing the retained moment on $[u, 1]$ and comparing it with Z_2 gives

$$\frac{\langle b, H^{-2} \mathbf{1}_{[u, 1]}(H)b \rangle}{Z_2} = \Theta\left(\left(\frac{\lambda_*}{u}\right)^{2-D/2}\right) \quad (\lambda_* \leq u \leq u_0). \quad (106)$$

Indeed, the upper bound follows by geometric summation, while a shell $[u, \gamma_{\text{sh}}u]$ with a fixed $\gamma_{\text{sh}} > 1$ chosen from the two cumulative-law constants has mass $\Theta(u^{D/2})$ and gives the matching lower bound. Requiring the retained fraction to be at least $1 - \epsilon_{\text{ls}}^2$ therefore gives

$$u = \Theta\left(\lambda_*(1 - \epsilon_{\text{ls}}^2)^{-2/(4-D)}\right) = \Theta_{\epsilon_{\text{ls}}}(\lambda_*), \quad (107)$$

for every fixed $\epsilon_{\text{ls}} < 1$, proving the first line. A cutoff $u = \lambda_*^{1-\beta}$ for fixed $\beta > 0$ has retained fraction tending to zero, so it is possible only when $\epsilon_{\text{ls}} \rightarrow 1$. For $D = 4$, both (103) and (105) are logarithmic. Discarding an ϵ_{ls}^2 fraction yields $K = \kappa_*^{1-\Theta(\epsilon_{\text{ls}}^2)}$, with the stated refinement when the leading cumulative coefficient exists. For $D > 4$, the discarded tail is $\Theta(K^{-(D-4)/2})$ relative to the constant total. Setting it to ϵ_{ls}^2 and capping at κ_* proves (101).

For filtering, $x = H^{-1}b$ and hence

$$\rho^2 = \alpha_H^2 \frac{\|H^{-1}x\|^2}{\|x\|^2} = \alpha_H^2 \frac{Z_4}{Z_2} = \Theta\left(\frac{Z_4}{Z_2}\right).$$

Applying (104) with $p = 4$ in the numerator and $p = 2$ in the denominator gives all five cases in (102). $\square$

Local conventions in the remaining witnesses are as follows: e_0, e_{e_1}, e_f are coordinate basis vectors, whereas e_k in the marked-diagonal construction is the k -dimensional all-ones vector; the symbols $\delta_c, \delta_Q, \delta_G$ denote, respectively, the corrector perturbation, projector error, and expander contrast.

Proposition 5.5 (Constant-degree torus witnesses). *Let L be even and T_L^d the periodic d -dimensional L -grid. With its combinatorial Laplacian $\Delta_{T_L^d}$, set*

$$H_{L,d} = \frac{L^{-2}I + \Delta_{T_L^d}}{L^{-2} + 4d}. \quad (108)$$

Then $H_{L,d}$ is $(2d+1)$ -sparse, has norm one, and has $\kappa = 1 + 4dL^2 = \Theta(L^2)$. A localized source e_0 realizes $D = d$; the dipole $(e_0 - e_{e_1})/\sqrt{2}$ realizes $D = d + 2$. For the dipole branch, the resulting phases are:

physical d	effective D	κ_{eff}	ρ
3	5	$\Theta(\min\{\kappa, \epsilon_{\text{ls}}^{-4}\})$	$\Theta(\kappa^{3/4})$
4	6	$\Theta(\min\{\kappa, \epsilon_{\text{ls}}^{-2}\})$	$\Theta(\kappa^{1/2})$
5	7	$\Theta(\min\{\kappa, \epsilon_{\text{ls}}^{-4/3}\})$	$\Theta(\kappa^{1/4})$
6	8	$\Theta(\min\{\kappa, \epsilon_{\text{ls}}^{-1}\})$	$\Theta(\sqrt{\log \kappa})$
≥ 7	> 8	<i>mesh-independent at fixed ϵ_{ls} mesh-independent</i>	

Proof. Fourier modes diagonalize the torus Laplacian, with

$$\lambda_k = \frac{L^{-2} + 4 \sum_{j=1}^d \sin^2(\pi k_j / L)}{L^{-2} + 4d}.$$

For the localized source, every squared Fourier overlap is L^{-d} . Lattice-volume bounds, first for $\lambda_k \asymp L^{-2}(1 + \|k\|^2)$ at low frequency and then by the same product-counting bounds on the remaining fixed spectral range, give

$$\mathbf{m}_{e_0}([0, t]) = \Theta(t^{d/2}) \quad (\lambda_* \leq t \leq 1).$$

Thus the cumulative hypothesis, including its bottom anchoring, holds with $D = d$. Equivalently, Abel summation gives

$$\sum_k \lambda_k^{-p} = \begin{cases} \Theta(L^{2p}), & d < 2p, \\ \Theta(L^{2p} \log L), & d = 2p, \\ \Theta(L^d), & d > 2p, \end{cases}$$

The dipole Fourier symbol is $1 - e^{2\pi i k_1/L}$. It vanishes on modes with $k_1 = 0$, so no pointwise comparison with λ_k is valid. Instead sum over permutation-symmetric spectral balls. Symmetry gives

$$\sum_{\lambda_k \leq t} \sin^2(\pi k_1/L) = \frac{1}{d} \sum_{\lambda_k \leq t} \sum_{j=1}^d \sin^2(\pi k_j/L).$$

Together with the lattice-volume estimate, this aggregated identity yields

$$\mathbf{m}_{(e_0 - e_{e_1})/\sqrt{2}}([0, t]) = \Theta(t^{(d+2)/2})$$

from the smallest visible eigenvalue through $t = 1$. At the bottom, the first visible modes have squared overlap $\Theta(L^{-(d+2)})$, exactly $\Theta(\lambda_*^{(d+2)/2})$; the constant mode has zero overlap and $\lambda_* = \Theta(L^{-2})$. Hence the repaired cumulative definition is satisfied with $D = d + 2$. The table follows by substituting $D = d + 2$ in [theorem 5.4](#). $\square$

The two source types occur as actual QIPM correctors, not merely abstract linear systems.

Proposition 5.6 (Positive full-step feasible corrector). *Let B_T be an oriented incidence matrix of T_L^d and set*

$$\mathcal{A} = [B_T, I], \quad W = \text{diag}(I_E, L^{-2}I_V), \quad \mathcal{A}W\mathcal{A}^\top = \Delta_{T_L^d} + L^{-2}I. \quad (109)$$

For $d \geq 2$, choose perpendicular torus edges f, g meeting at a vertex, fix $\tau > 0$ and $0 < \delta_c \leq 1/4$, and retain $x_i/s_i = w_i$, where w_i is the corresponding diagonal entry of W . Set

$$p_i = x_i s_i = \begin{cases} \tau(1 + \delta_c), & i = f, \\ \tau(1 - \delta_c), & i = g, \\ \tau, & \text{otherwise,} \end{cases} \quad x_i = \sqrt{p_i w_i}, \quad s_i = \sqrt{p_i / w_i}. \quad (110)$$

With $b_{\text{LP}} = \mathcal{A}x$, $c = s$, and $y = 0$, this is primal and dual feasible, has average complementarity τ , and has centrality $\|XSe - \tau e\|/\tau = \sqrt{2}\delta_c$. Its full $\sigma = 1$ feasible centering step is positive, has a normal-equation right-hand side of effective dimension $D = d + 2$, preserves average complementarity, and satisfies

$$\frac{\|X^+ S^+ e - \tau e\|}{\tau} \leq \frac{\delta_c^2}{1 - \delta_c^2}. \quad (111)$$

Proof. The definitions make feasibility and the centrality calculation immediate. The feasible corrector equations are

$$\mathcal{A}\Delta x = 0, \quad \mathcal{A}^\top \Delta y + \Delta s = 0, \quad S\Delta x + X\Delta s = \tau e - XSe.$$

Eliminating $\Delta x, \Delta s$ gives

$$(\mathcal{A}W\mathcal{A}^\top)\Delta y = -\mathcal{A}S^{-1}(\tau e - XSe). \quad (112)$$

After normalization, its right-hand side is proportional to

$$\frac{\mathcal{A}_f}{\sqrt{1+\delta_c}} - \frac{\mathcal{A}_g}{\sqrt{1-\delta_c}}.$$

Its low-frequency Fourier symbol vanishes linearly and has nonzero gradient, so its cumulative spectral weight is $\Theta(t^{(d+2)/2})$. Explicitly, for any nonzero linear form ℓ , lattice-ball comparison gives $\sum_{\|k\|\leq r} |\ell(k)|^2 = \Theta(r^{d+2})$; applying this to the linear part of the symbol gives the claimed cumulative spectral weight.

It remains to prove full-step safety. Put $u = X^{-1}\Delta x$, $v = S^{-1}\Delta s$, and $P = \text{diag}(p)$. The Newton equations imply

$$P(u+v) = \tau e - p, \quad u^\top P v = \Delta x^\top \Delta s = 0. \quad (113)$$

Thus $P^{1/2}u$ and $P^{1/2}v$ are orthogonal and sum to $P^{-1/2}(\tau e - p)$, whose squared norm is $2\tau\delta_c^2/(1-\delta_c^2)$. Since $p_i \geq \tau(1-\delta_c)$, for either $z = u, v$,

$$|z_i|^2 \leq \frac{2\delta_c^2}{(1-\delta_c^2)(1-\delta_c)} < 1 \quad (\delta_c \leq 1/4).$$

Hence $x + \Delta x > 0$ and $s + \Delta s > 0$. Moreover

$$p_i^+ = p_i(1+u_i)(1+v_i) = \tau + p_i u_i v_i.$$

Equation (113) gives $\sum_i p_i u_i v_i = 0$, so the average remains τ . Finally,

$$\|P(u \circ v)\| \leq \|P^{1/2}u\| \|P^{1/2}v\| \leq \frac{1}{2} \|P^{-1/2}(\tau e - p)\|^2 = \frac{\tau\delta_c^2}{1-\delta_c^2},$$

which proves (111). $\square$

The localized branch is also QIPM-native, but through an infeasible corrector. At complementarity τ , set $x_i = \sqrt{\tau w_i}$, $s_i = \sqrt{\tau/w_i}$ and $b_{\text{LP}} = \mathcal{A}x$. Give the exactly centered ($x_i s_i = \tau > 0$), primal-feasible point a one-coordinate dual residual $r_d = \eta e_f$. A $\sigma = 1$ infeasible feasibility corrector has

$$(\mathcal{A}W\mathcal{A}^\top)\Delta y = \eta \mathcal{A}W e_f. \quad (114)$$

A ground variable at vertex zero gives normalized right-hand side e_0 and $D = d$; a torus edge gives a dipole and $D = d + 2$.

These witnesses are solver-selection examples, not end-to-end speedups. The torus already admits FFT and multigrid solvers. The constructions describe a single corrector rather than an IPM trajectory, and do not charge block encoding, norm estimation, state or observable readout, or classical iterate recovery.

5.4 Stable active-subspace equilibration

The negative coupling law has a positive structural counterpart. Suppose the active range is available through its orthogonal projector Q , and put $R = I - Q$. The next theorem directly assembles a left-equilibrated operator; this direct assembly, rather than multiplication of normalized encodings, is the source of the gain.

Theorem 5.7 (Stable split equilibration). *Suppose*

$$H_\mu = \mu^{-1}C_\mu + \mu D_\mu, \quad 0 < \mu \leq 1,$$

where, for constants independent of μ ,

$$C_\mu = QC_\mu Q, \quad c_-Q \preceq C_\mu \preceq c_+Q, \quad D_\mu \succeq 0, \quad \|D_\mu\| \leq d_+, \quad RD_\mu R \succeq d_-R.$$

Let $P_\mu = \mu Q + \mu^{-1}R$ and assemble

$$M_\mu = C_\mu + RD_\mu + \mu^2 QD_\mu = C_\mu + D_\mu - (1 - \mu^2)QD_\mu. \quad (115)$$

Then $M_\mu = P_\mu H_\mu$ exactly and $\|M_\mu\| + \|M_\mu^{-1}\| = O(1)$. For every $f \in U = \text{range } Q$,

$$M_\mu^{-1}f = \mu^{-1}H_\mu^{-1}f, \quad (116)$$

so the normalized Newton state is preserved exactly, without recovery or postselection. If C_μ, D_μ, Q have block-encoding normalizations $\alpha_C, \alpha_D, 1$, direct LCU assembly has normalization

$$\Gamma_\mu \leq \alpha_C + (2 - \mu^2)\alpha_D. \quad (117)$$

Thus bounded component normalizations give a uniformly conditioned, uniformly normalized conventional QLS solve through the Hermitian dilation below, and the ideal DLS filtering parameter is $O(\Gamma_\mu)$ as well.

The left-equilibrated M_μ need not be Hermitian. To use the Hermitian QLS contract of [definition 2.16](#), encode its standard dilation

$$\mathcal{H}(M_\mu) = \begin{pmatrix} 0 & M_\mu \\ M_\mu^\dagger & 0 \end{pmatrix}. \quad (118)$$

The dilation has the same block-encoding normalization Γ_μ and the same singular condition number as M_μ . Moreover, $\mathcal{H}(M_\mu)^{-1}(f, 0)^\top = (0, M_\mu^{-1}f)^\top$, so selecting the second register preserves the solution normalization and requires no postselection beyond the fixed register label.

Proof. Since $RC_\mu = 0$,

$$P_\mu H_\mu = C_\mu + \mu^2 QD_\mu + RD_\mu = M_\mu.$$

In $U \oplus U^\perp$ blocks,

$$M_\mu = \begin{pmatrix} C_\mu + \mu^2 D_{UU} & \mu^2 D_{UK} \\ D_{KU} & D_{KK} \end{pmatrix}.$$

The Schur complement of D_{KK} is

$$G_\mu = C_\mu + \mu^2(D_{UU} - D_{UK}D_{KK}^{-1}D_{KU}) \succeq c_-I_U,$$

because the parenthesized Schur complement is positive semidefinite. Also $D_{KK} \succeq d_-I_{U^\perp}$. The block inverse formula therefore bounds every block of M_μ^{-1} using only c_-, d_-, d_+ , while $\|M_\mu\| \leq c_+ + 2d_+$.

For state preservation, let $x_\mu = H_\mu^{-1}f$. Because $f \in U$, $P_\mu f = \mu f$, and hence $M_\mu x_\mu = P_\mu H_\mu x_\mu = \mu f$. This proves (116). Finally, applying LCU to the second form in (115) gives coefficient weight $\alpha_C + \alpha_D + (1 - \mu^2)\alpha_D$, proving (117). Uniform conditioning bounds $\Gamma_\mu \|M_\mu^{-1}\|$ and its restriction to any solution state, which proves the two solver claims. $\square$

The split also supplies a projector-error firewall. If $\|\widehat{Q} - Q\| \leq \delta_Q$ and one assembles

$$\widehat{M}_\mu = C_\mu + D_\mu - (1 - \mu^2)\widehat{Q}D_\mu,$$

then $\|\widehat{M}_\mu - M_\mu\| \leq \delta_Q d_+$. For a sufficiently small constant δ_Q , the Neumann perturbation bound and [theorem 5.7](#) give

$$\frac{\|\widehat{M}_\mu^{-1}f - M_\mu^{-1}f\|}{\|M_\mu^{-1}f\|} = O(\delta_Q), \quad (119)$$

uniformly in μ . Thus $\delta_Q = O(\epsilon_{\text{ls}})$ suffices. In contrast, forming $\widehat{P}_\mu H_\mu$ from the raw normal matrix loses this stability. For $H_\mu = \text{diag}(\mu^{-1}, \mu)$ and a projector rotated by angle θ , its $(2, 1)$ entry is

$(1 - \mu^{-2}) \sin \theta \cos \theta$; uniform normalization requires $\sin \theta = O(\mu^2)$. Given a block encoding of A_B/α_B , QSVT can construct Q to the needed accuracy in

$$O\left(\frac{\alpha_B}{\sigma_{\min}^+(A_B)} \log(1/\epsilon_{\text{ls}})\right) \quad (120)$$

queries, charging active-basis conditioning but no polynomial power of μ .

Proposition 5.8 (Separately normalized composition cannot help). *Let H/α_H and an invertible P/α_P be separately encoded, with $\alpha_H \geq \|H\|$ and $\alpha_P \geq \|P\|$, and compose their encodings. For $x = H^{-1}f$, the DLS parameter of the resulting system $PHx = Pf$ satisfies*

$$\rho_{\text{prod}} = \alpha_H \alpha_P \frac{\|P^{-\dagger} H^{-1} x\|}{\|x\|} \geq \alpha_H \frac{\|H^{-1} x\|}{\|x\|} = \rho_H. \quad (121)$$

Proof. Equation (121) uses the general non-Hermitian parameter (86), because $(PH)^{-\dagger} = P^{-\dagger} H^{-1}$. Since $\alpha_P \geq \|P\|$, every vector y satisfies $\alpha_P \|P^{-\dagger} y\| \geq \|y\|$. Apply this with $y = H^{-1}x$. The displayed expression is precisely the filtering parameter for normalization $\alpha_H \alpha_P$, proving the result. $\square$

This normalization obstruction is known in quantum preconditioning: Lapworth and Sünderhauf show explicitly how multiplying separately normalized block encodings can erase a classical gain, while Nie, Lai, and An show how algebraic regrouping can restore it for Pauli structure [90, 114]. Here separate composition has $\alpha_H, \alpha_P = \Theta(\mu^{-1})$, whereas the direct split has $\Gamma_\mu = O(1)$, a $\Theta(\mu^{-2})$ normalization compression.

Proposition 5.9 (Constant-sparse expander occurrence). *There is a family of constant-sparse standard-form LPs for which the unpreconditioned exact-centering solve has $\rho = \Theta(\mu^{-2})$, while (115) has $O(1)$ singular conditioning and $O(1)$ block-encoding normalization and prepares exactly the same normalized Newton state independently of μ . The uniform bounds hold on $0 < \mu \leq \mu_0$ with μ_0 and all constants depending only on the degree d , the Laplacian gap, and $\delta_{\mathcal{G}}$, not on m .*

Proof. Let $\mathcal{G} = (\mathcal{L}, \mathcal{R}, \mathcal{E})$ be a connected balanced d -regular bipartite expander on m vertices, with constant d and a constant Laplacian gap. Orient each edge from $\mathcal{L}$ to $\mathcal{R}$, let $B_{\mathcal{G}}$ be its signed incidence matrix, and put

$$r = \mathbf{1}_{\mathcal{R}}, \quad \zeta = 2r - e, \quad u_0 = m^{-1/2}e.$$

Here r is the right-part indicator and e is the all-ones vector of the dimension required by context; these symbols are local to this witness and are distinct from the coupling variables above. Then $B_{\mathcal{G}}^\top r = e$ and $B_{\mathcal{G}} e = L_{\mathcal{G}} r = d\zeta$. Fix $0 < \delta_{\mathcal{G}} < 1$, set $\omega_i = 1 + \delta_{\mathcal{G}} \zeta_i$ and $a_i = \sqrt{2/\omega_i}$, and consider

$$\begin{aligned} \min_{x, u, v \geq 0} \quad & \sum_{i=1}^m a_i (u_i + v_i) \\ \text{subject to} \quad & B_{\mathcal{G}} x + u - v = B_{\mathcal{G}} e. \end{aligned} \quad (122)$$

Its constraint matrix $[B_{\mathcal{G}}, I, -I]$ has row and column sparsity $O(d)$.

The point $x = e, u = v = te$ is strictly primal feasible for $t > 0$. Taking $y = -\varepsilon r$ for sufficiently small $\varepsilon > 0$ gives strictly positive dual slacks. Nonnegativity of the objective shows that

$$x^* = e, \quad u^* = v^* = 0, \quad y^* = 0, \quad s_x^* = 0, \quad s_u^* = s_v^* = a$$

is optimal and strictly complementary. Its active face is bounded: if $h \geq 0$ and $B_{\mathcal{G}} h = 0$, any left-vertex row forces all incident components of h to vanish. Also e is the primal face's analytic center because $B_{\mathcal{G}}^\top r = e$.

For completeness, the dual limit is also unique even though the dual optimal face is not a singleton. Dual feasibility and optimality imply $B_G^\top y \leq 0$ and $e^\top B_G^\top y = 0$, hence $B_G^\top y = 0$. Connectedness gives $y = ce$, and the remaining dual inequalities give

$$\{y : y \text{ is dual optimal}\} = \{ce : |c| \leq \min_i a_i\}.$$

On its relative interior the dual analytic-center objective is $\sum_i \log(a_i^2 - c^2)$, which is strictly maximized at $c = 0$. Thus $y(\mu) \rightarrow 0$, and the displayed primal–dual point is the central-path limit.

Along the path,

$$\begin{aligned} H_\mu &= \mu^{-1}C_\mu + \mu D_\mu, \\ C_\mu &= B_G \operatorname{diag}(x_\mathcal{E}(\mu)^2) B_G^\top \longrightarrow L_G, \\ D_\mu &= \operatorname{diag}\left(\frac{1}{(a_i - y_i(\mu))^2} + \frac{1}{(a_i + y_i(\mu))^2}\right) \longrightarrow \operatorname{diag}(\omega_i). \end{aligned} \quad (123)$$

Here $U = \operatorname{range} B_G = e^\perp$ and $Q = I - |u_0\rangle\langle u_0|$. The constants are uniform in m . Write $s_x = -B_G^\top y$, $s_u = a - y$, $s_v = a + y$; the central equations are $x_e s_{x,e} = \mu$, $u_i s_{u,i} = \mu$, $v_i s_{v,i} = \mu$, and $B_G x + u - v = B_G e$. Seek a solution constant on edges and on each side: $x_e = \bar{x}$, $y_i = y_\mathcal{L}$ on $\mathcal{L}$, $y_i = y_\mathcal{R}$ on $\mathcal{R}$, with $\bar{x} = 1 + \mu^2 w$, $y_\mathcal{L} = \mu z_\mathcal{L}$, $y_\mathcal{R} = \mu z_\mathcal{R}$. Since $\omega = 1 \mp \delta_G$ on $\mathcal{L}$ and $\mathcal{R}$, the system reduces to three scalar equations independent of m :

$$dw + \frac{2z_\mathcal{R}}{2/(1 + \delta_G) - \mu^2 z_\mathcal{R}^2} = 0, \quad -dw + \frac{2z_\mathcal{L}}{2/(1 - \delta_G) - \mu^2 z_\mathcal{L}^2} = 0, \quad z_\mathcal{L} - z_\mathcal{R} = \frac{1}{1 + \mu^2 w}.$$

At $\mu = 0$ they have the solution $w = (1 - \delta_G^2)/(2d)$, $z_\mathcal{L} = (1 + \delta_G)/2$, $z_\mathcal{R} = -(1 - \delta_G)/2$, with Jacobian determinant $-2d \neq 0$. The implicit function theorem gives a branch on $0 < \mu \leq \mu_0(d, \delta_G)$ with $\bar{x} > 0$, $y_\mathcal{L} - y_\mathcal{R} = \mu/\bar{x} > 0$, and $|y_i| < a_i$; the ansatz therefore satisfies the full central-path system, and uniqueness of the central point identifies it as the central path. Hence $C_\mu = \bar{x}^2 L_G$ exactly, D_μ takes two side-constant values, and the expander gap together with $\omega_i \in [1 - \delta_G, 1 + \delta_G]$ gives the hypotheses of [theorem 5.7](#) with constants depending only on d , the Laplacian gap, and δ_G .

For the exact-centering right-hand side $f = b = B_G e$, $L_G(Qr) = b$ and $Qr = \zeta/2$. Its limiting active-to-null response is nonzero:

$$u_0^\top \operatorname{diag}(\omega) Qr = \frac{\delta_G \sqrt{m}}{2} = \delta_G \|Qr\|. \quad (124)$$

The coupling theorem therefore gives $\rho(H_\mu, f) = \Theta(\mu^{-2})$. In contrast, direct assembly gives

$$M_\mu = C_\mu + \mu^2 D_\mu + (1 - \mu^2)|u_0\rangle\langle u_0| D_\mu. \quad (125)$$

It has uniform singular conditioning and $O(d)$ normalization: C_μ is a bounded-weight degree- d Laplacian, D_μ is diagonal, u_0 is the uniform state, and the last term is rank one. Since $b \in U$, $M_\mu^{-1}b = \mu^{-1}H_\mu^{-1}b$ at every μ , completing the witness. $\square$

The theorem assumes a supplied active partition, coherent current weights, and direct access to C_μ, D_μ, Q . It also charges $\sigma_{\min}^+(A_B)$, right-hand-side preparation, norm estimation, and the requested output. [Section 13](#) gives a partition-free construction on the strict-complementarity tail and thereby removes the supplied-partition assumption; that construction is not needed here. Classical application of [\(125\)](#) is already linear-time per Krylov iteration. The quantum statement is therefore meaningful only for a state or a few observables; tomography or a classical iterate restores dimension-dependent cost.

5.5 An exact-center marked-diagonal specialization

The sparse search lower bound of Orsucci and Dunjko [118, Proposition 7] can also be realized at an exact LP center. This gives a useful QIPM specialization, but its marked-search core is prior art.

Choose an integer $T \geq 2$, set the local dimension $N_{\text{md}} = T^8$ and $\mu = T^{-2} = N_{\text{md}}^{-1/4}$, and hide $j \in [N_{\text{md}}]$. The subscript distinguishes this dimension from the optimal-partition set N above. Let e_k denote the k -dimensional all-ones vector in this construction. Row i of $A_j \in \mathbb{R}^{N_{\text{md}} \times 2N_{\text{md}}}$ has disjoint support $(2i - 1, 2i)$ and values

$$(\mu/2, \mu/2) \quad (i = j), \quad (1, \mu - 1) \quad (i \neq j). \quad (126)$$

Because μ , $\sqrt{\mu}$, and $\mu^{3/2}$ are rational, all entries have magnitude at most one and $O(\log N_{\text{md}})$ -bit rational descriptions; row sparsity is two and column sparsity one. Set

$$b = \mu^{3/2} e_{N_{\text{md}}}, \quad c = \sqrt{\mu} e_{2N_{\text{md}}}, \quad x^0 = s^0 = \sqrt{\mu} e_{2N_{\text{md}}}, \quad y^0 = 0.$$

Then $A_j x^0 = b$, $(A_j)^\top y^0 + s^0 = c$, and $X^0 S^0 e = \mu e$, so this is an exact μ -center. Since $X^0 (S^0)^{-1} = I$,

$$H_j = A_j A_j^\top = \text{diag}(h, \dots, \mu^2/2, \dots, h), \quad h = 1 + (1 - \mu)^2, \quad (127)$$

and the exact short-step right-hand side is $f = (1 - \sigma)\mu^{3/2} e_{N_{\text{md}}}$. Its normalized preparation is uniform and independent of j . The squared solution weight on the marked coordinate is at least $4/5$, while

$$\begin{aligned} \kappa(H_j) &= \Theta(\mu^{-2}), \quad \xi(H_j, f) = \Theta(1), \quad \rho(H_j, f) = \Theta(\mu^{-2}), \\ \kappa_{\text{eff}}(\epsilon_{\text{ls}}) &= \frac{2h}{\mu^2} = \Theta(\mu^{-2}) \quad \text{for every fixed } \epsilon_{\text{ls}}^2 < 4/5. \end{aligned}$$

After norm-tight normalization the only eigenvalues are 1 and $\mu^2/(2h)$, so κ_{eff} drops to 1 only once ϵ_{ls}^2 reaches the marked weight, which is at least $4/5$. Consequently, under the stated sparse-value oracle, or the canonical diagonal block encoding reducible to it, preparing the conventional exact normalized multiplier state to trace error $1/10$ (so that measuring the marked coordinate succeeds with probability at least $4/5 - 1/10 = 7/10$) requires $\Omega(\sqrt{N_{\text{md}}}) = \Omega(\mu^{-2}) = \Omega(\kappa)$ queries.

Taking $1 - \sigma = \mu^2/4 = \Theta(N_{\text{md}}^{-1/2})$ gives a positive feasible short step whose endpoint remains in a fixed two-norm central neighborhood. This makes the occurrence native to an IPM, subject to three essential caveats. First, it is not representation invariant: row equilibration turns (127) into the identity and makes the scaled multiplier state easy (the primal/slack direction itself is invariant). Second, simply omitting the marked component leaves relative normal residual only $\Theta(\mu^2)$, so ordinary residual-based inexact IPMs can bypass the state-preparation contract. Third, the hard scale occurs at one point of each fixed instance's path; it does not persist as that fixed instance's barrier parameter tends to zero. The lower bound is therefore for the original exact normalized multiplier-state contract, not for every QIPM or every equivalent representation. This distinction also accords with the representation and finite-precision cautions in classical IPM analysis [135]. This $\Omega(\kappa)$ statement is a full-query-model bound and survives factor access. It is consistent with theorem 4.3(iii), which is a polynomial-metric statement and becomes an end-to-end factor-access solve only under its explicit overlap hypothesis.

5.6 Novelty and scope

The application of the 2026 DLS solvers to IPM Newton systems, the exact coupling law, the $D = 4$ versus $D = 8$ selection rule and its QIPM correctors, and the state-preserving stable split appear to be new. This is deliberately not an unconditional priority claim. Effective conditioning,

active-set preconditioning, Green-function moment thresholds, and the generic marked-diagonal search bound are prior art. None of the results in this section proves an end-to-end quantum LP speedup or a new universal QLS lower bound. Access construction, active-set discovery where needed, precision, iterate maintenance, and output costs remain part of any such claim.

6 Block Log-Det Condensation

The trace constraint in a product-PSD log-det center is a shared budget. If one block has a lower spectral edge than all the others, the corresponding resolvent develops a pole as the barrier parameter decreases. The losing blocks can absorb only a finite amount of trace at that edge, and the excess therefore condenses onto the winner. This one mass parameter controls three apparently different effects: the conditioning of the equality-eliminated Hessian, the trace-distance visibility of the winning block, and the query cost of preparing or extracting it.

This section develops that relationship for an abstract two-cost block model. The sparse SDP realizing the model is deferred to [section 7](#), and the stronger mixed-output conversion argument is deferred to [section 8](#). Throughout this section only, G is the number of blocks, d their common dimension, m the winner ground-space multiplicity, τ the barrier parameter, $\alpha = \tau G$, and p the total winner trace mass. Thus m does not denote the number of equality constraints, and τ plays the role called μ in [definition 2.3](#). The C - and Γ -symbols below, as well as d, b, c , are also local; they do not reuse the homonymous data, dimensions, or constants of another section. The spectral resolvent $A(x)$ below is unrelated to the constraint matrix A of [section 2.1](#).

6.1 Two-cost model and exact center

Fix real symmetric costs $C_*, C_L \in \mathbb{S}^d$. There is one winning block with cost C_* and $G - 1$ identical losing blocks with cost C_L . After subtracting the winning spectral minimum v_* from both costs, suppose

$$\text{spec}(C_* - v_* I) = \{\underbrace{0, \dots, 0}_m, \gamma_{m+1}, \dots, \gamma_d\}, \quad \gamma_i > 0, \quad (128)$$

$$\text{spec}(C_L - v_* I) = \{\delta_1, \dots, \delta_d\}, \quad \delta_j > 0. \quad (129)$$

Here $1 \leq m \leq d$, and the block-level winner is unique. Unless otherwise stated, d, m , and all positive gaps are fixed while G grows.

At barrier parameter $\tau > 0$, consider

$$\min_{\substack{Z_g > 0 \\ \sum_g \text{tr } Z_g = 1}} \left\{ \langle C_*, Z_* \rangle + \sum_{g=2}^G \langle C_L, Z_g \rangle - \tau \sum_{g=1}^G \log \det Z_g \right\}. \quad (130)$$

For $x > 0$, define the winner and loser resolvent traces

$$A(x) = \frac{m}{x} + \sum_{i=m+1}^d \frac{1}{x + \gamma_i}, \quad B(x) = \sum_{j=1}^d \frac{1}{x + \delta_j}. \quad (131)$$

An empty regular sum is zero. Also put

$$B_0 = B(0) = \sum_{j=1}^d \delta_j^{-1}, \quad \beta = -B'(0) = \sum_{j=1}^d \delta_j^{-2}, \quad \alpha_* = B_0^{-1}. \quad (132)$$

Proposition 6.1 (Exact block center and finite concentration bounds). *The unique minimizer of (130) is*

$$Z_* = \tau(C_* - \lambda I)^{-1}, \quad Z_g = \tau(C_L - \lambda I)^{-1} \quad (g > 1), \quad (133)$$

where $\lambda = v_* - x$ and x is the unique positive solution of

$$1 = \tau\{A(x) + (G-1)B(x)\}. \quad (134)$$

The winner mass $p_G = \text{tr } Z_* = \tau A(x)$ satisfies, with $\delta_{\max} = \max_j \delta_j$,

$$p_G \geq 1 - \tau(G-1)B_0, \quad (135)$$

$$p_G \geq p_0 > 0 \implies x \leq \frac{d\tau}{p_0}, \quad 1 - p_G \geq \frac{d\tau(G-1)}{\delta_{\max} + d\tau/p_0}. \quad (136)$$

Consequently, for $G \geq 2$ and $0 < \varepsilon < 1$,

$$\tau \leq \frac{\varepsilon}{B_0(G-1)} \implies p_G \geq 1 - \varepsilon, \quad (137)$$

$$p_G \geq 1 - \varepsilon \implies \tau \leq \frac{\varepsilon \delta_{\max}}{d\{G-1-\varepsilon/(1-\varepsilon)\}}, \quad (138)$$

whenever the denominator in (138) is positive. For fixed G and $\tau \downarrow 0$, if $A_{\text{reg}}(0) = \sum_{i=m+1}^d \gamma_i^{-1}$, then

$$x = m\tau + m\{A_{\text{reg}}(0) + (G-1)B_0\}\tau^2 + O_G(\tau^3), \quad (139)$$

$$p_G = 1 - (G-1)B_0\tau + m(G-1)\beta\tau^2 + O_G(\tau^3). \quad (140)$$

Proof. The objective is strictly convex on the affine slice and diverges at its relative boundary, so it has a unique minimizer. With the multiplier sign from [definition 2.4](#), stationarity gives $C_g - \lambda I = \tau Z_g^{-1}$. Positive definiteness requires $\lambda < v_*$, and taking traces gives (134). Its right-hand side is continuous and strictly decreasing from infinity to zero, which proves existence and uniqueness of x .

The loser mass is $1 - p_G = \tau(G-1)B(x)$. Since $B(x) \leq B_0$, this proves (135). Since $A(x) \leq d/x$, $p_G \geq p_0$ implies $x \leq d\tau/p_0$. The bound $B(x) \geq d/(x + \delta_{\max})$ then proves (136). Equations (137)–(138) follow by substituting $p_0 = 1 - \varepsilon$ and rearranging.

Finally, Taylor expansion gives $A(x) = m/x + A_{\text{reg}}(0) + O(x)$ and $B(x) = B_0 - \beta x + O(x^2)$. Substitute the ansatz $x = m\tau + c\tau^2 + O_G(\tau^3)$ into (134). Subtracting one, dividing by τ , and taking the constant term gives $c = m\{A_{\text{reg}}(0) + (G-1)B_0\}$. Substitution into $1 - p_G = \tau(G-1)B(x)$ gives (140). $\square$

The finite bounds already show that constant concentration occurs precisely on a scale no larger than $1/G$. The next result resolves that scale.

6.2 Capacity transition

Theorem 6.2 (Block log-det condensation). *Set $\tau = \alpha/G$ for fixed $\alpha > 0$. Under (128)–(129), the following three-phase law holds as $G \rightarrow \infty$:*

$$0 < \alpha < \alpha_* : \quad \frac{x}{\tau} \longrightarrow \frac{m}{1 - \alpha B_0}, \quad p_G \longrightarrow 1 - \alpha B_0; \quad (141)$$

$$\alpha = \alpha_* : \quad \sqrt{G}x \longrightarrow \sqrt{m/\beta}, \quad \sqrt{G}p_G \longrightarrow \alpha_* \sqrt{m\beta}; \quad (142)$$

$$\alpha > \alpha_* : \quad x \longrightarrow x_\alpha, \quad Gp_G \longrightarrow \alpha A(x_\alpha), \quad (143)$$

where $x_\alpha > 0$ is the unique solution of $B(x_\alpha) = 1/\alpha$.

Proof. Write the secular equation as

$$1 = \frac{\alpha}{G}A(x) + \alpha(1 - G^{-1})B(x). \quad (144)$$

If $\alpha < \alpha_*$, x must tend to zero: along any subsequence bounded away from zero, the first term vanishes and the second has limit at most $\alpha B_0 < 1$. Using the expansions following (140) in (144) gives

$$1 = \frac{\alpha m}{Gx} + \alpha B_0 + o(1),$$

so $Gx \rightarrow \alpha m / (1 - \alpha B_0)$ and $p_G = (\alpha/G)A(x) \rightarrow 1 - \alpha B_0$.

At $\alpha = \alpha_*$ the same compactness argument gives $x \rightarrow 0$. After cancelling $\alpha_* B_0 = 1$, the Taylor expansions give

$$0 = \frac{\alpha_* m}{Gx} - \alpha_* \beta x + O(G^{-1}) + O(x^2), \quad (145)$$

where the bounded regular part of A is absorbed in $O(G^{-1})$. Monotonicity first sandwiches x between constant multiples of $G^{-1/2}$; dividing (145) by $G^{-1/2}$ then yields $\sqrt{G}x \rightarrow \sqrt{m/\beta}$. Since $A(x) = m/x + O(1)$, the mass limit in (142) follows.

If $\alpha > \alpha_*$, x cannot tend to zero because the loser term alone would tend to $\alpha B_0 > 1$. It also cannot diverge. Every convergent subsequence therefore has a positive limit, at which the winner term in (144) vanishes and $B(x) = 1/\alpha$. Strict monotonicity of B makes that limit unique. Multiplying $p_G = (\alpha/G)A(x)$ by G proves (143). $\square$

Below capacity, the pole supplies the trace missing from the losing bath; at capacity, the pole balances the linear response βx ; above capacity, the bath equation alone fixes a positive x_α . The requirement $\delta_j > 0$ is essential: a zero loser gap makes B_0 infinite and destroys this finite-capacity transition.

6.3 Equality-reduced Hessian and conditioning

All conditioning statements below use exactly [definition 2.9](#): the equality-eliminated block Hessian in Frobenius coordinates. They are neither preconditioner-invariant statements nor statements about a saddle KKT matrix. At the center put $D_* = C_* - \lambda I$ and $D_L = C_L - \lambda I$. On the tangent space

$$\mathcal{T} = \left\{ Y = (Y_1, \dots, Y_G) : \sum_{g=1}^G \text{tr } Y_g = 0 \right\}, \quad (146)$$

the barrier-scaled log-det Hessian has quadratic form

$$\mathcal{Q}(Y) = \frac{1}{\tau} \left\{ \text{tr}(D_* Y_* D_* Y_*) + \sum_{g=2}^G \text{tr}(D_L Y_g D_L Y_g) \right\}. \quad (147)$$

Indeed, $\mathcal{Q}$ is the quadratic form of $\tau \nabla^2 F_{\log \det}$ at the center. Therefore $\kappa(\mathcal{Q}|\mathcal{T}) = \kappa_{\text{red}}$ by [definition 2.9](#): the scalar factor τ changes the absolute curvatures but not their ratio.

Let $\delta_{\min} = \min_j \delta_j$, let r be its multiplicity, and, when $m < d$, let $\gamma_{\min} = \min_{i>m} \gamma_i$. Define

$$b(x) = \min(\{x + \delta_{\min}\} \cup \{x + \gamma_{\min} : m < d\}), \quad \Lambda = \max(\{\delta_j\} \cup \{\gamma_i : i > m\}), \quad (148)$$

with the second set omitted when $m = d$.

Lemma 6.3 (Finite Hessian sandwich). *For $m \geq 2$,*

$$\lambda_{\min}(\mathcal{Q}|\mathcal{T}) = \frac{x^2}{\tau}. \quad (149)$$

For $m = 1$, if λ_{diag} is the least curvature on the diagonal trace hyperplane, then

$$\frac{(dG - 1)x^2 + b(x)^2}{dG\tau} \leq \lambda_{\text{diag}} \leq \frac{r(G - 1)x^2 + (x + \delta_{\min})^2}{\tau\{r(G - 1) + 1\}}. \quad (150)$$

If $m = 1 < d$, a winner ground-excited direction has curvature $x(x + \gamma_{\min})/\tau$. Consequently, with constants depending only on the fixed spectral data,

$$\lambda_{\min}(\mathcal{Q}|\tau) \asymp \frac{1}{\tau} \begin{cases} \min\{x(x + \gamma_{\min}), x^2 + (x + \delta_{\min})^2/G\}, & m = 1 < d, \\ x^2 + (x + \delta_1)^2/G, & m = d = 1. \end{cases} \quad (151)$$

For $G \geq 3$,

$$\frac{(x + \delta_{\max})^2}{\tau} \leq \lambda_{\max}(\mathcal{Q}|\tau) \leq \frac{(x + \Lambda)^2}{\tau}. \quad (152)$$

Proof. Use eigenbases of the two costs. Diagonal and symmetric off-diagonal coordinates are orthogonal invariant subspaces. An off-diagonal (i, j) coordinate in a block with shifted eigenvalues d_i, d_j has curvature $d_i d_j / \tau$, and a diagonal coordinate has curvature d_i^2 / τ . Only diagonal coordinates are coupled by (146).

If $m \geq 2$, the difference of two winner ground diagonals, or their off-diagonal coordinate, is tangent and has curvature x^2 / τ . No ambient coordinate has smaller weight, proving (149). For $m = 1$, write the soft winner diagonal coefficient as a and all other $dG - 1$ diagonal coefficients as z . The trace constraint gives $\mathbf{1}^T z = -a$, hence $\|z\|_2^2 \geq a^2 / (dG - 1)$; every corresponding weight is at least $b(x)^2 / \tau$. Minimizing the resulting two-term quotient gives the lower bound in (150). For the upper bound, cancel a equally among the $r(G - 1)$ loser coordinates with gap $\delta_{\min}$ and calculate its Rayleigh quotient.

The stated winner off-diagonal coordinate supplies the other upper bound in (151); all remaining off-diagonal weights are bounded below by the same expression up to fixed spectral constants. Together with the diagonal sandwich this proves (151), including the scalar case where no within-block off-diagonal exists. Finally, opposite maximum-gap diagonal directions in two loser blocks are tangent and give the lower bound in (152); its upper bound is the largest ambient coordinate weight. $\square$

Theorem 6.4 (Multiplicity-dependent conditioning phases). *For fixed $\alpha > 0$, $\tau = \alpha/G$, and fixed spectral data,*

$$\kappa_{\text{red}} = \begin{cases} \Theta(G^2), & m \geq 2, 0 < \alpha < \alpha_*, \\ \Theta(G), & m = 1, 0 < \alpha \leq \alpha_*, \\ \Theta(G), & m \geq 2, \alpha = \alpha_*, \\ \Theta(1), & \alpha > \alpha_* \text{ for every } m. \end{cases} \quad (153)$$

Proof. In the subcritical phase, theorem 6.2 gives $x = \Theta(G^{-1})$. Hard loser curvature is $\Theta(G)$. For $m \geq 2$, (149) gives minimum curvature $\Theta(G^{-1})$; for $m = 1$, (151) gives $\Theta(1)$. At criticality $x = \Theta(G^{-1/2})$, and either minimum formula gives $\Theta(1)$, while the maximum remains $\Theta(G)$. Above capacity x is bounded away from zero, so every curvature is $\Theta(G)$. $\square$

The assumption $m < d$ is needed only for the particular soft-stiff direction used when $m = 1 < d$; trace spread across losers supplies the scalar direction.

6.4 Finite mass-conditioning laws

The phase law fixes a schedule. The sharper statement below instead starts from the observable winner mass.

Theorem 6.5 (Finite mass-conditioning law). *Let $G \geq 3$ and let $p = p_G \in (0, 1)$ be the winner mass at an exact center. Define*

$$R_G(p) = \frac{(G - 1)p}{1 - p}. \quad (\text{MC1})$$

Then

$$m \geq 2 : \quad \kappa_{\text{red}} \geq R_G(p)^2, \quad (\text{MC2})$$

$$m = 1 < d : \quad \kappa_{\text{red}} \geq \min \left\{ 1, \frac{\delta_{\text{max}}}{\gamma_{\text{min}}} \right\} R_G(p). \quad (\text{MC3})$$

If $m = d = 1$, writing $\delta = \delta_1$, the exact closed form is

$$\kappa_{\text{red}} = \frac{G}{1 + (1-p)^2 / \{(G-1)p^2\}}. \quad (\text{MC4})$$

For scalar blocks with $G = 2$, the tangent space is one-dimensional and $\kappa_{\text{red}} = 1$, so (MC4) is asserted only for $G \geq 3$.

Proof. The exact mass equations are

$$p = \tau A(x), \quad 1 - p = \tau(G-1)B(x). \quad (154)$$

Because $A(x) \leq d/x$ and $B(x) \geq d/(x + \delta_{\text{max}})$,

$$\frac{1-p}{p} = (G-1) \frac{B(x)}{A(x)} \geq (G-1) \frac{x}{x + \delta_{\text{max}}}, \quad \frac{x + \delta_{\text{max}}}{x} \geq R_G(p). \quad (155)$$

Two opposite δ_{max} loser diagonals give $\lambda_{\text{max}} \geq (x + \delta_{\text{max}})^2/\tau$. If $m \geq 2$, divide this by (149) and use (155).

If $m = 1 < d$, the winner ground-excited direction gives $\lambda_{\text{min}} \leq x(x + \gamma_{\text{min}})/\tau$, while

$$\frac{x + \delta_{\text{max}}}{x + \gamma_{\text{min}}} \geq \min \left\{ 1, \frac{\delta_{\text{max}}}{\gamma_{\text{min}}} \right\}.$$

Factoring the curvature ratio proves (MC3).

For $d = m = 1$, put $a = x^2/\tau$ and $b = (x + \delta)^2/\tau$. On the scalar trace-zero space, $G - 2$ loser-difference directions have curvature b , and the direction balancing the winner against equal loser changes has curvature $\varrho = \{(G-1)a + b\}/G$. Since $b > a$, these are the largest and smallest eigenvalues. The mass equations reduce to

$$\frac{x}{x + \delta} = \frac{1-p}{(G-1)p}.$$

Thus $\kappa_{\text{red}} = b/\varrho = G/\{1 + (G-1)a/b\}$, which is (MC4). $\square$

In particular, fixed $p \geq p_0 > 0$ forces $\kappa_{\text{red}} = \Omega(G^2)$ when $m \geq 2$ and $\kappa_{\text{red}} = \Omega(G)$ when $m = 1$. A direct finite proof also follows from $x \leq d\tau/p_0$, the loser mass bound $\tau = O(G^{-1})$, and the test directions above; the exact laws identify the dependence on p without first specifying τ .

6.5 Visibility, collision, and the conditioning dichotomy

The next result needs only the two-cost symmetry, not the spectral asymptotics. Let $C_W, C_L \in \mathbb{S}^d$, $G \geq 2$, and $\tau > 0$. Compare the all-loser and unique-winner trace-one centers

$$\rho_0 = \arg \min_{\substack{Z_g > 0 \\ \sum_g \text{tr } Z_g = 1}} \left\{ \sum_{g=1}^G \langle C_L, Z_g \rangle - \tau \sum_g \log \det Z_g \right\}, \quad (156)$$

$$\rho_1 = \arg \min_{\substack{Z_g > 0 \\ \sum_g \text{tr } Z_g = 1}} \left\{ \langle C_W, Z_1 \rangle + \sum_{g=2}^G \langle C_L, Z_g \rangle - \tau \sum_g \log \det Z_g \right\}. \quad (157)$$

Identify each tuple with its block-diagonal density matrix and write

$$\rho_0 = \bigoplus_{g=1}^G L_0, \quad \rho_1 = W_1 \oplus \bigoplus_{g=2}^G L_1, \quad p = \text{tr } W_1. \quad (\text{O18})$$

Permutation symmetry gives $\text{tr } L_0 = 1/G$ and $\text{tr } L_1 = (1-p)/(G-1)$.

Theorem 6.6 (Visibility and label collision (Theorem O)). *For the centers above,*

$$\left| p - \frac{1}{G} \right| \leq D_{\text{tr}}(\rho_0, \rho_1) \leq \max \left\{ p, \frac{1}{G} \right\}. \quad (\text{O1})$$

If two independent component labels are measured and equality is tested, then

$$c_0 = \frac{1}{G}, \quad c_1 = p^2 + \frac{(1-p)^2}{G-1}, \quad \boxed{c_1 - c_0 = \frac{(Gp-1)^2}{G(G-1)}}. \quad (\text{O2})$$

The measurement is public and acts only on the two label registers, so the identity is unchanged by internal degeneracy or by purifying and then discarding auxiliary registers.

Proof. Measuring whether the label is 1 gives the lower bound in (O1). Let λ_0, λ_1 be the scalar trace multipliers of the two centers. Their loser blocks satisfy

$$L_b = \tau(C_L - \lambda_b I)^{-1} \quad (b = 0, 1),$$

and are therefore Loewner ordered. If $\lambda_1 \leq \lambda_0$, then $L_1 \preceq L_0$ and the trace identities imply $p \geq 1/G$. Hence

$$(G-1)\|L_1 - L_0\|_1 = p - \frac{1}{G}, \quad \|W_1 - L_0\|_1 \leq p + \frac{1}{G}.$$

Adding the trace norms of the direct-sum blocks and dividing by two gives $D_{\text{tr}}(\rho_0, \rho_1) \leq p$. If $\lambda_1 \geq \lambda_0$, the order and inequality reverse, and the same calculation gives the upper bound $1/G$. Finally the two label distributions are

$$(1/G, \dots, 1/G), \quad \left(p, \frac{1-p}{G-1}, \dots, \frac{1-p}{G-1} \right).$$

Summing their squared entries and subtracting proves (O2). $\square$

Only the upper trace-distance bound used the central resolvent form. The lower bound and collision identity require only a uniform all-loser marginal and exchangeable losers. When $p \geq 1/G$ (which holds whenever $C_W \preceq C_L$, since $X \mapsto (X - \lambda I)^{-1}$ is operator antitone on $X \succ \lambda I$, and for the family of section 7.6 by (225)), the trace-distance bounds reduce to $p - 1/G \leq D_{\text{tr}} \leq p$.

For the conditioning consequences, now specialize $C_W = C_*$ and retain the spectral hypotheses of section 6.1. For $G \geq 3$, combining theorems 6.5 and 6.6 gives the finite conditioning-visibility laws. With $c = \min\{1, \delta_{\max}/\gamma_{\min}\}$,

$$p \leq \frac{\sqrt{\kappa_{\text{red}}}}{G-1 + \sqrt{\kappa_{\text{red}}}} \quad (m \geq 2), \quad (\text{CV1})$$

$$p \leq \frac{\kappa_{\text{red}}}{c(G-1) + \kappa_{\text{red}}} \quad (m = 1 < d), \quad (\text{CV2})$$

$$p = \frac{1}{1 + \sqrt{G-1} \sqrt{G/\kappa_{\text{red}} - 1}} \quad (m = d = 1). \quad (\text{CV3})$$

Each right-hand side, maximized with $1/G$, bounds the trace distance in (O1). Conversely, if $D_{\text{tr}}(\rho_0, \rho_1) \geq v > 1/G$, then $p \geq v$ and

$$\kappa_{\text{red}} \geq \left(\frac{(G-1)v}{1-v} \right)^2 \quad (m \geq 2), \quad \kappa_{\text{red}} \geq c \frac{(G-1)v}{1-v} \quad (m = 1 < d). \quad (\text{CV4})$$

The scalar formula gives the same linear conclusion. Thus constant visibility forces $\Omega(G^2)$ conditioning for a degenerate winner ground space and $\Omega(G)$ for a simple one. A constant collision gap also forces $p = \Omega(1)$ and hence the same dichotomy. These conclusions retain the coordinate scope stated at the start of [section 6.3](#).

Approximate outputs and reduced symmetry. Let $\tilde{\rho}_b$ satisfy $D_{\text{tr}}(\tilde{\rho}_b, \rho_b) \leq \epsilon$. Contractivity and the tensor-product triangle inequality give $D_{\text{tr}}(\tilde{\rho}_b^{\otimes 2}, \rho_b^{\otimes 2}) \leq 2\epsilon$. Thus the observed two-copy collision gap is at least

$$\Delta_{\text{out}} \geq d(p, G) - 4\epsilon, \quad d(p, G) := \frac{(Gp-1)^2}{G(G-1)}. \quad (159)$$

Ordinary concentration uses $O(\Delta_{\text{out}}^{-2})$ independent experiments for constant success, or an additional $\log(1/\delta)$ factor for failure probability δ .

Equal loser mass is not necessary for the lower bound. Suppose a public label measurement is uniform on the no-winner input and, on every unique-winner input, the winner label has mass $q \geq p \geq 1/G$. If the other masses are arbitrary, Cauchy–Schwarz gives

$$\sum_{g=1}^G w_g^2 \geq q^2 + \frac{(1-q)^2}{G-1} \geq p^2 + \frac{(1-p)^2}{G-1}, \quad (160)$$

where the last inequality holds because the displayed function is increasing for $q \geq 1/G$. The collision gap is therefore at least $d(p, G)$, with equality exactly at winner mass p and uniform loser mass. If the no-winner collision probability is only bounded above by b , replace the gap by $p^2 + (1-p)^2/(G-1) - b$. In particular, if $p \geq p_0 > 0$ and $G \geq 2/p_0$, then $d(p, G) \geq p_0^2/4$.

6.6 Unique-OR query lower bounds

The visibility theorem becomes a raw-query statement only after specifying an independently composed oracle. Let $G \geq 2$ and let a possibly partial Boolean predicate $f : \mathcal{X}_0 \sqcup \mathcal{X}_1 \rightarrow \{0, 1\}$ have general adversary value $\text{Adv}_f = \text{Adv}^\pm(f)$. For G independently queried blocks, set

$$\mathcal{P}_0 = \mathcal{X}_0^G, \quad \mathcal{P}_1 = \bigsqcup_{a=1}^G \mathcal{X}_0^{a-1} \times \mathcal{X}_1 \times \mathcal{X}_0^{G-a}, \quad (161)$$

and let F distinguish $\mathcal{P}_0$ from $\mathcal{P}_1$.

Theorem 6.7 (Unique-OR adversary and preparation (Theorem P)). *In the raw-query model where one query addresses one coordinate of one block,*

$$\text{Adv}^\pm(F) \geq \text{Adv}_f \sqrt{G}. \quad (\text{O3})$$

Consequently, suppose public fixed component projectors give the uniform component marginal on every no-winner target and give its winning component mass at least $p \geq 1/G$ on every unique-winner target; loser masses may be arbitrary. Any circuit that prepares, on every promised input, a state within trace distance ϵ of such a target uses $\Omega(\text{Adv}_f \sqrt{G})$ raw queries whenever $d(p, G) - 4\epsilon = \Omega(1)$. By tightness of the general adversary bound, this is equivalently $\Omega(Q(f) \sqrt{G})$ up to universal constants.

Proof. Take the rectangular zero-to-one block B , with rows indexed by $\mathcal{X}_0$ and columns by $\mathcal{X}_1$, of a feasible adversary matrix for f , normalized so that

$$\|B\| = \text{Adv}_f, \quad \max_j \|B \circ \Delta_j\| \leq 1, \quad (162)$$

where $\Delta_j[x, y] = 1$ exactly when query j has different answers on x, y , and let I_0 be the identity on $\ell_2(\mathcal{X}_0)$. For the sector of $\mathcal{P}_1$ whose winner is block a , define

$$C_a = I_0^{\otimes(a-1)} \otimes B \otimes I_0^{\otimes(G-a)}, \quad C = [C_1 \ C_2 \ \cdots \ C_G],$$

and use the Hermitian adversary matrix

$$\Gamma = \begin{pmatrix} 0 & C \\ C^* & 0 \end{pmatrix}.$$

The winner sectors are disjoint, and

$$CC^* = \sum_{a=1}^G I_0^{\otimes(a-1)} \otimes BB^* \otimes I_0^{\otimes(G-a)}.$$

The summands commute. Tensoring a top eigenvector of BB^* across all G blocks gives their common top eigenvector, so $\|\Gamma\| = \text{Adv}_f \sqrt{G}$. Filtering by raw query (a, j) kills every winner sector except a , since row and column inputs agree in block a in all other sectors. The remaining block is $B \circ \Delta_j$ and has norm at most one by (162). Thus Γ is feasible and proves (O3).

If $d(p, G) - 4\epsilon$ is a positive constant, (159) turns a constant number of fresh preparations and public collision measurements into a bounded-error algorithm for F . The adversary lower bound therefore applies to the preparation circuit. Tightness and composition of the negative-weight adversary bound give the equivalent $Q(f)$ statement [77, 92, 122]. $\square$

The theorem assumes that all Cartesian-product inputs in (161) are valid, the two fibers of f are nonempty, one raw query addresses only one block coordinate, and the component projectors are public. It requires no clean evaluator, candidate verification, inverse preparation oracle, or restriction on internal conditional states. If one coefficient-oracle call can be simulated with at most s raw queries, it yields the transferred lower bound

$$\Omega\left(\frac{\text{Adv}_f \sqrt{G}}{s}\right) \quad (163)$$

coefficient calls. An oracle supplying a winner bit, aggregate flag, or input-dependent projector is stronger and can remove the inner factor Adv_f .

There is also a sharp warning at low visibility. The public all-loser center ρ_0 is a valid zero-query approximation on the zero-or-one promise when

$$\epsilon \geq \max\{p, 1/G\}, \quad (164)$$

by theorem 6.6. Hence a constant objective gap need not imply a constant-error lower bound for trace-normalized primal-density output. The argument concerns the direct-sum density of definition 2.23, not a normalized vectorization whose label probabilities are Frobenius rather than trace weights.

6.7 Preparation, extraction, and the conservation law

For upper bounds, impose the stronger exactly-one-winner promise and assume a clean coherent evaluator

$$V_f|g, 0\rangle = |g, f_g\rangle, \quad f_g = 1 \iff g = w, \quad (\text{A1})$$

costing T_{eval} raw component queries; its inverse has the same cost. The two costs, G , and τ are public. Thus the center has the form

$$\rho_p = p|w\rangle\langle w| \otimes \rho_W + \frac{1-p}{G-1} \sum_{g \neq w} |g\rangle\langle g| \otimes \rho_L, \quad (\text{A2})$$

where the normalized fixed-block resolvents ρ_W, ρ_L are public. Put

$$a_W = p, \quad a_L = \frac{1-p}{G-1}, \quad a_{\max} = \max\{a_W, a_L\}. \quad (\text{A3})$$

Proposition 6.8 (Coherent rejection preparation). *There is a purification circuit for (A2) using*

$$\tilde{O}\left(T_{\text{eval}}\sqrt{Ga_{\max}} + T_{\text{loc}}\right) \quad (\text{A4})$$

operations, where T_{loc} is the gate cost, not raw-query cost, of the one public conditional local preparation. When $p \geq 1/G$, its raw-query part is

$$\tilde{O}(T_{\text{eval}}\sqrt{Gp}). \quad (\text{A5})$$

Proof. Prepare $G^{-1/2} \sum_g |g\rangle$, compute f_g , and rotate an acceptance qubit with amplitude $\sqrt{a_{f_g}/a_{\max}}$, but do not yet copy the label or prepare the internal state. This trial succeeds with probability

$$P_{\text{acc}} = \frac{1}{G} \left(\frac{a_W}{a_{\max}} + (G-1) \frac{a_L}{a_{\max}} \right) = \frac{1}{Ga_{\max}}. \quad (\text{A6})$$

Because this probability is known from the classically solved secular equation, exact phase-matched amplitude amplification of this trial acts on the label, predicate, and acceptance registers [31]. After it succeeds, the label has probability a_{f_g} ; only then coherently copy the label to an environment register and use the retained predicate once to prepare a purification of ρ_W or ρ_L . The one final local preparation supplies the right conditional internal state. Tracing out the copied label, predicate, acceptance, and local purification registers gives exactly (A2). The evaluator is therefore amplified but the T_{loc} work is not, proving both stated bounds; the tilde accounts only for requested implementation accuracy. $\square$

The scalar secular equation and fixed-size resolvents can be computed classically. If d varies, one must add $\text{poly}(d, \log(1/\epsilon))$ arithmetic and state-synthesis costs. A bounded-error evaluator similarly requires coherent error reduction below the final state error.

Suppose independent prepared copies satisfy $D_{\text{tr}}(\tilde{\rho}, \rho_p) \leq \epsilon < p$. Measuring the label proposes the winner with probability at least $p - \epsilon$, so verification and repetition extract it with failure δ using

$$O\left(\frac{Q_\rho + T_{\text{eval}}}{p - \epsilon} \log \frac{1}{\delta}\right) \quad (\text{A7})$$

raw queries. For exact copies take $\epsilon = 0$; if $\epsilon \leq p/2$, this has the usual $O(1/p)$ repetition factor. Given instead a coherent preparation unitary U_ρ , its inverse, and a clean winner reflection, amplification gives

$$O\left(\frac{Q_U + T_{\text{eval}}}{\sqrt{p}} \log \frac{1}{\delta}\right). \quad (\text{A8})$$

The coherent bound needs an operator-level implementation of U_ρ and its inverse; closeness of the one output density matrix says nothing about the unitary completion away from $|0\rangle$.

Combining (A5) and (A8) yields the preparation–extraction conservation law

$$\tilde{O}(T_{\text{eval}}\sqrt{Gp}) O(p^{-1/2}) = \tilde{O}(T_{\text{eval}}\sqrt{G}). \quad (\text{A9})$$

Condensation moves the work from extraction to preparation without changing the end-to-end component-query scale. For exact output, or for $\epsilon \leq p/2$ in the copy-extraction column, explicitly,

regime	central preparation	copy extraction	coherent extraction	
subcritical	$\tilde{O}(T_{\text{eval}}G^{1/2})$	$O(1)$	$O(1)$	(A10)
critical	$\tilde{O}(T_{\text{eval}}G^{1/4})$	$O(G^{1/2})$	$O(G^{1/4})$	
supercritical	$\tilde{O}(T_{\text{eval}})$	$O(G)$	$O(G^{1/2})$	

The supercritical preparation statement concerns exact output, or accuracy fine enough to resolve an $O(1/G)$ exceptional mass. At fixed error the zero-query boundary (164) can apply.

Direct quantum search costs

$$\tilde{O}(T_{\text{eval}}\sqrt{G}) \quad (\text{A11})$$

and, after solving and recovering the selected component, the honest end-to-end bound is

$$\tilde{O}(T_{\text{eval}}\sqrt{G}) + T_{\text{solve}} + T_{\text{recover}}. \quad (\text{A12})$$

These bounds use standard quantum search and minimum finding [29, 57]. This is search-then-crossover, not a new interior-point method. For a local predicate of bounded-error raw-query complexity q , independent general-adversary composition gives the total bounded-error raw-query cost

$$Q(\text{UniqueOR}_G \circ f) = \Theta(q\sqrt{G}). \quad (\text{A13})$$

This total-complexity composition follows from the general adversary bound [77, 92, 122]. Composing that lower bound with (A8) gives

$$Q_U + T_{\text{eval}} = \Omega(q\sqrt{Gp}), \quad (\text{A14})$$

and an ϵ -accurate independent-copy preparation with $\epsilon < p$ gives the weaker

$$Q_\rho + T_{\text{eval}} = \Omega((p - \epsilon)q\sqrt{G}). \quad (\text{A15})$$

For exact output this is $\Omega(pq\sqrt{G})$, and the same asymptotic bound holds when $\epsilon \leq p/2$. These statements require independent component oracles; overlapping queries or an aggregate winner flag invalidate the composition. The new conclusion here is the optimization-specific conservation law, not the standard search, amplification, minimum-finding, or adversary primitives.

For comparison, the block product barrier has parameter dG , so a short-step continuation from constant τ to the $\Theta(1/G)$ scale has $O(\sqrt{dG} \log G)$ Newton iterations. This count includes neither a linear-solve condition number nor state preparation or recovery, and it must not be multiplied by (153) without choosing and analyzing an actual solver and preconditioner.

6.8 Sublinear multiwinner capacity

Now let exactly k blocks have cost C_W , with $v = \lambda_{\min}(C_W)$ of multiplicity m , and suppose $C_L - vI \succ 0$. Redefine locally

$$A(x) = \text{tr}(C_W - vI + xI)^{-1}, \quad B(x) = \text{tr}(C_L - vI + xI)^{-1}. \quad (\text{165})$$

Then $A(x) = m/x + O(1)$, $B_0 = B(0) < \infty$, and $\beta = -B'(0) > 0$. The exact equations are

$$1 = \tau\{kA(x) + (G - k)B(x)\}, \quad p_{G,k} = \tau kA(x), \quad 1 - p_{G,k} = \tau(G - k)B(x). \quad (\text{166})$$

Theorem 6.9 (Multiwinner capacity law (Theorem M)). *Let $1 \leq k = k_G = o(G)$, put $\tau = \alpha/G$, and keep $\alpha > 0$ fixed. With $\alpha_* = 1/B_0$,*

$$p_{G,k} \longrightarrow 1 - \alpha B_0, \quad x \sim \frac{\alpha m k}{G(1 - \alpha B_0)} \quad (0 < \alpha < \alpha_*), \quad (167)$$

$$p_{G,k} \sim \alpha_* \sqrt{m \beta k / G}, \quad x \sim \sqrt{m k / (\beta G)} \quad (\alpha = \alpha_*), \quad (168)$$

$$p_{G,k} \sim \alpha A(x_\alpha) k / G, \quad B(x_\alpha) = 1/\alpha \quad (\alpha > \alpha_*). \quad (169)$$

Proof. Divide the first equation in (166) by the G already absorbed in $\tau = \alpha/G$. Below criticality $x \rightarrow 0$: otherwise the winner term vanishes because $k/G \rightarrow 0$, while the loser term is at most $\alpha B_0 < 1$. The loser mass then tends to αB_0 , proving the first mass law, and the pole $A(x) \sim m/x$ gives the formula for x .

At criticality put $q = k/G$. Taylor expansion of the loser equation and the winner pole give two expressions for the total winner mass:

$$p_{G,k} = q + \frac{\beta}{B_0} x + o(q + x) = \frac{qm}{B_0 x} + O(q). \quad (170)$$

As in the one-winner proof, monotonicity first bounds x by constant multiples of $\sqrt{q}$; balancing the leading terms then gives $x \sim \sqrt{mq/\beta}$ and the critical mass. Above criticality the same compactness and strict-monotonicity argument as in [theorem 6.2](#) gives $x \rightarrow x_\alpha > 0$; the winner equation gives the final law. $\square$

The assumption $k = o(G)$ is substantive. If $k/G \rightarrow q \in (0, 1]$, then the limit instead solves $1 = \alpha\{qA(x) + (1 - q)B(x)\}$ with $x > 0$; the winner pole is already extensive and there is no singular capacity transition at fixed α .

Nonidentical losers. Let every losing cost satisfy $C_{\ell,G} - vI \succ 0$, and define

$$B_{\ell,G}(x) = \text{tr}(C_{\ell,G} - vI + xI)^{-1}.$$

Then

$$\overline{B}_G(x) = \frac{1}{G - k} \sum_{\ell \text{ losing}} B_{\ell,G}(x).$$

If $k = o(G)$ and $\overline{B}_G \rightarrow B$ locally uniformly on $x \geq 0$, with $B(0) = B_0 \in (0, \infty)$, the subcritical and supercritical conclusions and the subcritical formula for x remain valid. At criticality it is enough additionally that

$$\overline{B}_G(0) - B_0 = o(\sqrt{k/G}), \quad \overline{B}_G(x) = \overline{B}_G(0) - \beta x + r_G(x), \quad (171)$$

where $\beta > 0$ and, for every fixed $C > 0$,

$$\sup_{0 < x \leq C\sqrt{k/G}} \frac{|r_G(x)|}{x} \longrightarrow 0.$$

Indeed, replacing B by $\overline{B}_G$ in (166) leaves the subcritical and supercritical limit arguments unchanged. Under (171), its finite-size error is lower order than both leading terms in (170), proving the critical law. These hypotheses control mass only; they do not restore exchangeable losers, a public common internal state, or the exact visibility identity.

For identical blocks and a fixed winner set S of size $k < G$, let $q = k/G$. Repeating [theorem 6.6](#)'s Loewner proof gives

$$|p_{G,k} - q| \leq D_{\text{tr}}(\rho_0, \rho_S) \leq \max\{p_{G,k}, q\}, \quad (172)$$

and direct calculation gives the two-copy label-collision gap

$$c_S - c_0 = \frac{p_{G,k}^2}{k} + \frac{(1 - p_{G,k})^2}{G - k} - \frac{1}{G} = \boxed{\frac{(Gp_{G,k} - k)^2}{Gk(G - k)}}. \quad (173)$$

Thus constant total winner mass gives only an $O(1/k)$ collision gap when k grows. Pairwise trace distance does not furnish one common measurement for an unknown S . [Section 8](#) bypasses this $1/k$ loss for the structured family by a composed small-success theorem; that stronger conclusion is not available in the abstract two-matrix model.

6.9 Exactly- k coherent preparation

Assume now that $1 \leq k < G$ is known, the promise is exactly k winners, a clean coherent membership marker costs T_{mark} raw queries, and the two costs, G, k, τ , multiplier, and normalized internal resolvents are public. Controlled marker calls and the inverse preparation circuit are available. Put, for $p = p_{G,k}$,

$$a_W = \frac{p}{k}, \quad a_L = \frac{1 - p}{G - k}, \quad a_{\max} = \max\{a_W, a_L\}. \quad (\text{AP0})$$

The construction in [proposition 6.8](#), with the membership bit in place of the unique-winner bit, succeeds in one trial with probability $1/(Ga_{\max})$. This probability is public and exactly known, so phase-matched exact amplitude amplification uses

$$O(\sqrt{Ga_{\max}} + 1) \quad (\text{AP1})$$

marker calls [\[31\]](#). After amplification succeeds, copying the label to the environment and controlling public purifications of the normalized winner and loser blocks gives the required internal state once. Specifically, let W and L denote the common unnormalized winner and loser blocks of the center, and put

$$\omega_W = \frac{W}{p/k}, \quad \omega_L = \frac{L}{(1 - p)/(G - k)} \quad (\text{AP2})$$

The resulting circuit satisfies

$$\boxed{Q_U = O\left(T_{\text{mark}}[\sqrt{Ga_{\max}} + 1]\right)}. \quad (\text{AP3})$$

When $p \geq k/G$, this is $O(T_{\text{mark}}[\sqrt{Gp/k} + 1])$; when $p < k/G$, it is $O(T_{\text{mark}}[\sqrt{G(1 - p)/(G - k)} + 1])$. Variable error adds the standard logarithmic factor. Costs for loading nonpublic eigenvectors, resolvents, or multipliers must be added.

Conversely, amplification from the prepared purification finds a winner in

$$O\left(\frac{Q_U + T_{\text{mark}}}{\sqrt{p}}\right) \quad (\text{AP4})$$

raw queries. If the local predicate has bounded-error raw-query complexity q_f and independent composition gives winner-finding lower bound $\Omega(q_f \sqrt{G/k})$, then

$$\boxed{Q_U + T_{\text{mark}} = \Omega\left(q_f \sqrt{Gp/k}\right)}. \quad (\text{AP5})$$

This follows by composing [\(AP4\)](#) with that lower bound. It requires an operator-level unitary and inverse. For mixed-state output, independent copies yield only the weaker repetition tradeoff; [section 8](#)'s upgrade instead comes from the composed small-success theorem.

For a parity marker, $T_{\text{mark}} = \Theta(N)$ and adversary composition gives $\Theta(N\sqrt{G/k})$ winner-finding complexity. For $k = o(G)$, [theorem 6.9](#) therefore yields the parity profile

$$Q_U = \begin{cases} \Theta(N\sqrt{G/k}), & 0 < \alpha < \alpha_*, \\ \Theta(N(G/k)^{1/4}), & \alpha = \alpha_*, \\ O(N), & \alpha > \alpha_*. \end{cases} \quad (\text{AP6})$$

The first two lines match [\(AP5\)](#) when the marker is optimal and the amplification factor diverges. The supercritical upper bound follows from $a_{\text{max}} = \Theta(1/G)$ even if $p < k/G$. At criticality $p \rightarrow 0$, so a fixed-error density-output contract need not inherit this operator-level lower bound.

This circuit is not a zero-versus- k algorithm: the all-loser branch has a different trace multiplier and loser resolvent. Under that different promise, exact existence search followed by conditional preparation costs $O(T_{\text{mark}}\sqrt{G/k})$, while always returning the public all-loser center costs zero queries and has error at most $\max\{p_{G,k}, k/G\}$ by [\(172\)](#). The collision decoder matches the former only when [\(173\)](#), after output error, retains a constant gap.

6.10 Robust approximate centrality

The mass and conditioning obstruction persists without exact stationarity or commutation. Retain the one-winner notation and, for positive definite $\tilde{Z}_g$ and $\tilde{\lambda} < v_*$, put

$$x = v_* - \tilde{\lambda}, \quad D_* = C_* - \tilde{\lambda}I, \quad D_L = C_L - \tilde{\lambda}I, \quad \tilde{T} = \sum_g \text{tr} \tilde{Z}_g, \quad \tilde{p} = \frac{\text{tr} \tilde{Z}_*}{\tilde{T}}. \quad (174)$$

Write $D_g = D_*$ for the winner and $D_g = D_L$ for every loser. Assume a dimensionless barrier-norm stationarity residual and a trace residual:

$$E_g = \tau^{-1} \tilde{Z}_g^{1/2} D_g \tilde{Z}_g^{1/2} - I, \quad \max_g \|E_g\|_{\text{op}} \leq \eta < 1, \quad |\tilde{T} - 1| \leq \zeta < 1. \quad (175)$$

Theorem 6.10 (Robust capacity, mass, and conditioning (Theorem R)). *Every point satisfying [\(174\)](#)–[\(175\)](#) obeys the noncommutative sandwich*

$$(1 - \eta)\tau D_g^{-1} \preceq \tilde{Z}_g \preceq (1 + \eta)\tau D_g^{-1}. \quad (\text{R1})$$

Consequently,

$$\boxed{\tilde{p} \geq 1 - \frac{(1 + \eta)\tau(G - 1)B_0}{1 - \zeta} \geq 1 - \frac{(1 + \eta)\alpha B_0}{1 - \zeta}}. \quad (\text{R2})$$

A uniform margin $(1 + \eta)\alpha B_0 \leq 1 - \zeta - c$ therefore forces positive G -independent winner mass. Conversely, if $G \rightarrow \infty$, $\alpha \rightarrow \bar{\alpha}$, and $x \rightarrow 0$, then

$$(1 - \eta)\bar{\alpha}B_0 \leq 1 + \zeta. \quad (\text{R3})$$

Thus $(1 + \eta)\alpha B_0 = 1 - \zeta$ and $(1 - \eta)\alpha B_0 = 1 + \zeta$ bracket the critical surface allowed by this residual contract.

For the conditioning statement suppose $G \geq 2$, $d \geq 2$, and $\tilde{p} \geq p_0 \in (0, 1)$. Set

$$q_0 = p_0(1 - \zeta), \quad \Gamma_* = \lambda_{\text{max}}(C_* - v_*I), \quad \delta_{\text{max}} = \lambda_{\text{max}}(C_L - v_*I). \quad (176)$$

Then

$$x \leq \frac{d(1 + \eta)\tau}{q_0}, \quad (\text{R4})$$

and, whenever the denominator is positive,

$$\tau \leq \frac{(1-p_0)(1+\zeta)\delta_{\max}}{d\{(1-\eta)(G-1) - (1-p_0)(1+\zeta)(1+\eta)/q_0\}}. \quad (\text{R5})$$

For the actual log-det Hessian

$$\tilde{\mathcal{H}}[Y]_g = \tau \tilde{Z}_g^{-1} Y_g \tilde{Z}_g^{-1} \quad \text{on } \mathcal{T}, \quad (177)$$

one has

$$\kappa(\tilde{\mathcal{H}}|_{\mathcal{T}}) \geq \frac{\delta_{\min}^2(1-\eta)q_0}{d(1+\eta)^2\tau(x+\Gamma_*)} = \Omega(G). \quad (\text{R6})$$

If $m \geq 2$, the stronger bound is

$$\kappa(\tilde{\mathcal{H}}|_{\mathcal{T}}) \geq \frac{\delta_{\min}^2(1-\eta)^2}{(1+\eta)^2x^2} = \Omega(G^2). \quad (\text{R7})$$

For fixed $p_0, \eta, \zeta, C_*, C_L$, all asymptotic constants are uniform.

Proof. The residual bound is equivalent to

$$(1-\eta)I \preceq \tau^{-1} \tilde{Z}_g^{1/2} D_g \tilde{Z}_g^{1/2} \preceq (1+\eta)I.$$

Congruence by $\tilde{Z}_g^{-1/2}$, inversion, and multiplication by τ prove (R1). Since $D_L \succeq C_L - v_*I$,

$$\sum_{g=2}^G \text{tr } \tilde{Z}_g \leq (1+\eta)\tau(G-1) \text{tr } D_L^{-1} \leq (1+\eta)\tau(G-1)B_0.$$

Divide by $\tilde{T} \geq 1-\zeta$ to prove (R2). The lower sandwich similarly gives $\tilde{T} \geq (1-\eta)\tau(G-1) \text{tr } D_L^{-1}$; taking the limit in the theorem proves (R3).

Winner mass and the upper sandwich give

$$q_0 \leq \text{tr } \tilde{Z}_* \leq (1+\eta)\tau \text{tr } D_*^{-1} \leq \frac{d(1+\eta)\tau}{x},$$

which is (R4). Every eigenvalue of D_L is at most $x + \delta_{\max}$. The lower sandwich on loser blocks and $(1-\tilde{p})\tilde{T} \leq (1-p_0)(1+\zeta)$ give

$$(1-p_0)(1+\zeta) \geq \frac{(1-\eta)\tau(G-1)d}{x + \delta_{\max}}.$$

Substitute (R4) and rearrange to obtain (R5), hence $\tau = O(G^{-1})$.

No commutation is needed for the curvature tests. In an eigenbasis of any loser block $\tilde{Z}_g$, a Frobenius-unit symmetric off-diagonal matrix is tangent. The upper sandwich and $D_L \succeq \delta_{\min}I$ bound every eigenvalue of this block by $(1+\eta)\tau/\delta_{\min}$, so this direction has curvature at least $\delta_{\min}^2/\{(1+\eta)^2\tau\}$. The largest eigenvalue of the winner block is at least q_0/d , while its other eigenvalues are at least $(1-\eta)\tau/(x+\Gamma_*)$. An off-diagonal direction between a largest eigenvector and an orthogonal eigenvector therefore has curvature at most $d(x+\Gamma_*)/\{(1-\eta)q_0\}$. Taking the quotient proves the finite bound in (R6); (R4)–(R5) give its asymptotic form.

If $m \geq 2$, the min–max principle and the lower sandwich show that the two largest winner eigenvalues are at least $(1-\eta)\tau/x$. Their off-diagonal direction has curvature at most $x^2/\{(1-\eta)^2\tau\}$. Comparing with the hard loser direction proves (R7). $\square$

For completeness, scalar blocks also have a robust linear obstruction. When $d = 1$ and $G \geq 3$, (R4)–(R5) remain valid. A difference of two loser coordinates has curvature at least $\delta_1^2/\{(1+\eta)^2\tau\}$. Balancing a unit winner coordinate by $-1/(G-1)$ in every loser yields a tangent Rayleigh quotient at most

$$\frac{\tau}{q_0^2} + \frac{(x + \delta_1)^2}{(1 - \eta)^2 \tau (G - 1)}. \quad (178)$$

Their quotient, together with (R4)–(R5), is $\Omega(G)$. The dimensionless relative residual in (175) is essential: an unscaled small stationarity residual need not imply a Loewner sandwich near a singular spectral edge.

6.11 Relation to prior work and open boundaries

The affine-resolvent density in (133) is classical. It is the density associated with matrix Burg entropy, and Ishihara derives this form from a trace-and-energy maximum-entropy problem [78]. At mode level, the precise statistical-mechanical analogue is Rayleigh–Jeans condensation: reciprocal-gap occupancies, saturation of excited modes, and macroscopic ground-mode mass are established phenomena [18, 143]. Inverse-gap capacity and multi-state condensation also appear in spherical models [48, 97]. We therefore use “matrix-Burg / Rayleigh–Jeans analogue” and make no claim of a Bose–Einstein distribution or a new resolvent center.

Square-root rates on degenerate SDP central paths and generic Newton-system ill-conditioning near low-rank optima are likewise known [4, 49, 147]. Those works do not concern this growing- G equality-reduced Frobenius Hessian. The contributions isolated here are the growing- G critical coefficients, the G versus G^2 ground-multiplicity split, the finite mass-conditioning laws, their conjunction with visibility and raw-query bounds, and the preparation–extraction conservation law. The quantum search, amplification, and adversary tools themselves are standard.

Identical losers are needed for a sharp common capacity, exact collision identity, and the coherent public-resolvent preparation. The empirical resolvent hypotheses in (171) extend the mass law, but visibility for nonidentical losers requires a separate public statistic such as (160). A further open direction is the η -channel between the robust barrier-norm neighborhood and the algorithm-specific inexact Newton residuals of definition 2.8; no conversion is asserted here. All conditioning conclusions remain tied to the unpreconditioned block Hessian in Frobenius coordinates, and all query conclusions remain tied to trace-normalized primal-density output with a public component decoder.

7 Winner-Take-All Global-Trace Holonomy SDP

This section realizes the condensation theory of section 6 with bounded scalar incidence. Each of G qutrit-holonomy components hides a parity of N private signs. This is the same component as in section 11.1, with $N = n_{\text{bit}}$. A single trace budget turns the component spectra into an exact minimum, while a chain of free scalar accumulators keeps the input rows and columns sparse. The qutrit component itself is developed in section 11.1; here we specify only the data needed to couple and analyze the components.

quantity	conclusion
value queries	$Q = \Theta(N\sqrt{G})$, $R = \Theta(NG)$ for error $< 1/20$
one-odd central path	$p = \Theta(1), \Theta(G^{-1/2}), \Theta(G^{-1})$ below, at, and above $\alpha_* = 2/45$
reduced-Hessian conditioning	$\Theta(G), \Theta(G), \Theta(1)$ in the same phases
coherent central-state preparation	$\Theta(N\sqrt{G}), \Theta(NG^{1/4}), O(N)$
end-to-end architecture	search, then cross over to the selected component

The preparation row is phase-dependent and uses the exactly-one-winner promise. In particular, the well-conditioned public-start Newton state is not a winner-revealing output.

7.1 A bounded-incidence product SDP

Put

$$K = 16N, \quad P = 2K + N = 33N, \quad u = \frac{1}{10}, \quad q = \frac{K}{P} = \frac{16}{33}, \quad \gamma = \frac{u}{q} = \frac{33}{160}. \quad (179)$$

For $a < b$, write $H_{ab} = e_a e_b^T + e_b e_a^T$, and put $D_s = \text{diag}(s, 1, 1)$ for $s \in \{-1, +1\}$. Group g has private signs $\sigma_{g,1}, \dots, \sigma_{g,N}$ and holonomy

$$h_g = \prod_{j=1}^N \sigma_{g,j}. \quad (180)$$

It contains P blocks $X_{g,0}, \dots, X_{g,P-1} \in \mathbb{S}_+^3$. For the edge from $i-1$ to i , let $G_{g,i} = I$ except that $G_{g,K+j-1} = D_{\sigma_{g,j}}$ for $1 \leq j \leq N$, and impose the six isometric-svec scalar equations

$$X_{g,i} = G_{g,i} X_{g,i-1} G_{g,i}^T \quad (1 \leq i < P). \quad (181)$$

If $R_{g,0} = I$ and $R_{g,i} = G_{g,i} \cdots G_{g,1}$, these equations say

$$X_{g,i} = R_{g,i} Z_g R_{g,i}^T, \quad Z_g = X_{g,0} \succeq 0. \quad (182)$$

The first K rotations are I , and the final K are D_{h_g} .

The public block costs are

$$C_{g,i} = \begin{cases} 3I + uH_{23} + \gamma H_{12}, & 0 \leq i < K, \\ 3I + uH_{23}, & K \leq i < K + N, \\ 3I + uH_{23} + \gamma H_{13}, & K + N \leq i < P. \end{cases} \quad (183)$$

Because $\sqrt{u^2 + \gamma^2} < 1/4$, every $C_{g,i}$ has spectrum in $(11/4, 13/4)$. Introduce free public scalars $t_1, \dots, t_G$ and add

$$\begin{aligned} t_1 - \text{tr } X_{1,0} &= 0, \\ t_g - t_{g-1} - \text{tr } X_{g,0} &= 0 \quad (2 \leq g \leq G), \\ t_G &= 1. \end{aligned} \quad (184)$$

For $G = 1$, the first and last rows impose $t_1 = \text{tr } X_{1,0} = 1$. The SDP minimizes

$$f(X) = \frac{1}{P} \sum_{g=1}^G \sum_{i=0}^{P-1} \langle C_{g,i}, X_{g,i} \rangle \quad (185)$$

subject to (181)–(184) and $X_{g,i} \succeq 0$; the t_g are unconstrained conic bookkeeping variables.

There are

$$L = GP = 33GN \quad (186)$$

blocks in $\mathbb{S}_+^3$. A copy row has two scalar nonzeros. The first accumulator row has four, an interior row has five, and the anchor has one. Every matrix coordinate and every free-scalar column occurs in at most two rows. Thus the equality matrix has scalar row sparsity at most five and scalar column sparsity at most two. Its support, right-hand side, and all costs are public; only the 12 and 13 copy coefficients on signed edges depend on the input.

Eliminating the path and accumulator variables gives exactly

$$\mathcal{B}_G = \left\{ (Z_1, \dots, Z_G) : Z_g \succeq 0, \sum_{g=1}^G \text{tr } Z_g = 1 \right\}. \quad (187)$$

The public point

$$X_{g,i}^0 = \frac{1}{3G}I, \quad t_g^0 = \frac{g}{G} \quad (188)$$

is strictly primal feasible. With all equality multipliers zero, the dual PSD slacks are $C_{g,i}/P \succ 0$ and stationarity in every free scalar is zero. Hence the formulation is strictly primal–dual feasible in the sense of [definition 2.1](#).

7.2 Exact spectral minimum

Pulling (185) to the roots gives

$$\bar{C}_h = 3I + u(H_{12} + H_{23} + hH_{13}), \quad \min_{Z \in \mathcal{B}_G} \sum_g \langle \bar{C}_{h_g}, Z_g \rangle. \quad (189)$$

Indeed, the first and last plateaus each contribute $K\gamma/P = u$, and conjugation by D_h changes H_{13} to hH_{13} . The signed triangle has spectra

$$\text{spec}(\bar{C}_+) = \left\{ \frac{16}{5}, \frac{29}{10}, \frac{29}{10} \right\}, \quad \text{spec}(\bar{C}_-) = \left\{ \frac{31}{10}, \frac{31}{10}, \frac{14}{5} \right\}. \quad (190)$$

These follow either by diagonalizing the two signed triangle adjacency matrices, whose eigenvalues are $(2, -1, -1)$ and $(1, 1, -2)$, or from the intrinsic component analysis in [section 11.1](#).

For any feasible roots, Rayleigh–Ritz gives

$$\sum_g \langle \bar{C}_{h_g}, Z_g \rangle \geq \sum_g \lambda_{\min}(\bar{C}_{h_g}) \text{tr } Z_g \geq \min_g \lambda_{\min}(\bar{C}_{h_g}). \quad (191)$$

Equality is obtained by placing all trace on a minimum eigenvector of a minimizing group. Therefore

$$\text{OPT} = \begin{cases} 29/10, & h_g = +1 \text{ for every } g, \\ 14/5, & h_g = -1 \text{ for at least one } g. \end{cases} \quad (192)$$

The gap is $1/10$, so the value encodes $\text{OR}_G \circ \text{PARITY}_N$.

The spectrahedron is genuinely semidefinite. Its conic hull is $(\mathbb{S}_+^3)^G$. If (187) had a lift over a finite product of second-order cones, homogenizing the compact lift and projecting to one factor would give such a lift of $\mathbb{S}_+^3$, contrary to Fawzi’s theorem [61]. The path-and-accumulator formulation projects linearly onto the root set, so it has no finite SOC lift either.

7.3 Sharp value-query complexity

Use the raw sparse coefficient access of [definition 2.11](#). In the canonical fixed-position isometric-svec oracle, a public reversible address map sends a requested hidden slot to (g, j) , or sends a public slot to a dummy zero. The value oracle XORs the sign of one of the two hidden coefficients $\pm\sigma_{g,j}$ directly into its target. Thus one raw sign query simulates one coefficient query, coherently and also under inverse or external control. Conversely, one fixed hidden copy slot reveals $\sigma_{g,j}$. The raw-sign and canonical coefficient models therefore have exact unit query normalization. A clean arithmetic-value convention may instead use compute–phase–uncompute and cost two calls.

Here is the promised instantiation of Theorem P in [theorem 6.7](#); its tensor-composition proof is not repeated. Let $E, O \subseteq \{0, 1\}^N$ be the even- and odd-parity strings and let

$$B_{xy} = \mathbf{1}[|x - y|_1 = 1] \quad (x \in E, y \in O). \quad (193)$$

The uniform vectors show $\|B\| = N$, while $B \circ \Delta_j$ is a permutation matrix and has norm one. Hence the inner adversary value in (162) is exactly N . Applying (O3) on the all-even versus exactly-one-odd promise gives

$$Q_{1/3} = \Omega(N\sqrt{G}). \quad (194)$$

This is the explicit even-to-odd hypercube witness inside the general composition machinery of [section 6.6](#).

If $|\tilde{v} - \text{OPT}| \leq \epsilon < 1/20$, thresholding at $57/20$ distinguishes the two cases in (192). The same threshold works for target-relative error $1/100$, since

$$\frac{101}{100} \frac{14}{5} < \frac{57}{20} < \frac{99}{100} \frac{29}{10}. \quad (195)$$

At the non-strict endpoint $\epsilon = 1/20$, the public midpoint $57/20$ is valid for both values and uses no query. Thus the additive threshold is sharp.

For the randomized lower bound, let $\mathcal{D}_0$ make all groups independent and uniform conditional on even parity; let $\mathcal{D}_1$ choose a uniform group J , make it uniform conditional on odd parity, and leave the others as in $\mathcal{D}_0$. Couple adaptive decision trees by returning common fair answers to the first $N - 1$ distinct positions in each group and using the final unseen sign to enforce the promised parity. The transcripts can differ only if the algorithm completes group J . A depth- r tree completes at most r/N groups, so

$$\text{TV}(\text{Tr}_{\mathcal{D}_0}, \text{Tr}_{\mathcal{D}_1}) \leq \frac{r}{NG}. \quad (196)$$

Yao's principle therefore gives $r \geq NG/3$ for worst-case success $2/3$; reading every sign is a matching upper bound. Consequently, for every $\epsilon < 1/20$,

$$\boxed{Q_\epsilon(\text{OPT}) = \Theta(N\sqrt{G}), \quad R_\epsilon(\text{OPT}) = \Theta(NG).} \quad (197)$$

The general adversary and randomized composition frameworks are due to Høyer–Lee–Špalek, Belovs–Lee, Reichardt, and Ben-David–Kothari [22, 24, 77, 122].

7.4 The public start and its scope

Use the log-det barrier on the PSD blocks, but no barrier on the free accumulators:

$$\Phi_\mu(X) = f(X) - \mu \sum_{g,i} \log \det X_{g,i}, \quad \mu = \frac{1}{GP}. \quad (198)$$

At (188), the PSD-block gradient and Hessian in isometric-svec coordinates are

$$g_{g,i} = \frac{1}{P}(C_{g,i} - 3I), \quad \nabla_X^2 \Phi_\mu(X^0) = \frac{9G}{P}I. \quad (199)$$

These right-hand sides and all equality residuals are public. The input appears only in the signed-copy operator on the left of the unreduced Newton system.

After exact elimination of the copies and accumulators, a root tangent is $Y = (Y_1, \dots, Y_G)$ with $\sum_g \text{tr } Y_g = 0$. Orthogonal congruence and the P path copies give

$$\boxed{\nabla_{\text{root}}^2 \Phi_\mu(X^0) = 9GI_{6G-1}.} \quad (200)$$

Thus this root-coordinate Hessian has condition number one, including the directions that redistribute trace among groups. With

$$A_h = H_{12} + H_{23} + hH_{13}, \quad (201)$$

the pulled-back gradient is uA_{h_g} , which is trace zero, and the exact Newton directions are

$$\Delta Z_g = -\frac{u}{9G}A_{h_g}, \quad \Delta t_g = 0. \quad (202)$$

Since $\|A_h\|_F^2 = 6$, the decrement is

$$\lambda^2 = \sum_g \langle uA_{h_g}, (9G)^{-1}uA_{h_g} \rangle = \frac{1}{150}, \quad \lambda = \frac{1}{\sqrt{150}}. \quad (203)$$

The full step remains strictly feasible. The eigenvalues inside the parentheses below are 14/45, 31/90, 31/90 for $h = +1$ and 29/90, 29/90, 16/45 for $h = -1$. Consequently,

$$Z_g^0 + \Delta Z_g = \frac{1}{G} \left(\frac{1}{3}I - \frac{u}{9}A_{h_g} \right) \succeq \frac{14}{45G}I. \quad (204)$$

On the root slice the barrier weight is $\tau = \mu P = 1/G$ (see (219)), which is below one for every $G \geq 2$; Φ_μ is then not standard self-concordant there, and the usual full-step decrement estimate does not apply to λ . We instead bound the next decrement directly. With $B_h = \frac{1}{3}I - \frac{u}{9}A_h$, the full step is $\hat{Z}_g = B_{h_g}/G$ and its slice gradient is $\bar{C}_{h_g} - B_{h_g}^{-1}$. On the trace slice the Newton decrement is the minimum over the trace multiplier ν of the Hessian-inverse norm of the shifted gradient, so the choice $\nu = 0$ gives an upper bound:

$$\lambda_{\text{next}}^2 \leq \frac{1}{G} \sum_g \text{tr}[(B_{h_g}\bar{C}_{h_g} - I)^2] = \frac{u^4}{81} \text{tr} A_h^4 = \frac{2u^4}{9}, \quad \lambda_{\text{next}} \leq \frac{\sqrt{2}}{300}, \quad (205)$$

because $\bar{C}_h = 3I + uA_h$ gives $B_h\bar{C}_h - I = -\frac{u^2}{9}A_h^2$ and $\text{tr} A_h^4 = 18$ for both parities. This bound is uniform in G and in the parity pattern.

Remark 7.1 (Dilution boundary). The normalized root-direction state is

$$|\Delta(h)\rangle = \frac{1}{\sqrt{G}} \sum_{g=1}^G |g\rangle |a_{h_g}\rangle, \quad |a_h\rangle = \frac{|\text{svec}(A_h)\rangle}{\sqrt{6}}. \quad (206)$$

Looping once over the N raw signs and applying phase kickback only to the 13 coordinate prepares this state exactly with N queries. Conversely, the density-matrix entry between group 1's 12 and 13 coordinates is $h_1/(3G)$. Since an entry produced by a q -query circuit has degree at most $2q$, exact preparation needs $q \geq N/2$; the same holds at trace error $\delta < 1/(6G)$. Hence exact preparation costs $\Theta(N)$.

Moreover, the all-even and one-odd states have overlap $1 - 2/(3G)$ and trace distance

$$\frac{2\sqrt{3G-1}}{3G} = \Theta(G^{-1/2}). \quad (207)$$

A constant-error Newton state therefore cannot reveal the unique-winner promise with constant bias. The $\sqrt{G}$ value hardness applies to scalar value and winner-revealing optimizer outputs, not to this normalized public-start direction state.

The qualification ‘‘root-coordinate’’ is essential. A feasible root direction lifts to accumulator changes equal to prefix trace sums; that lift is not an isometry. The free scalars have zero barrier Hessian, so neither the ambient augmented Hessian nor the full saddle KKT matrix is claimed to have condition number one.

7.5 Approximate optimizers

For exactly feasible roots on an instance with at least one odd group, put $m_- = \sum_{g:h_g=-1} \text{tr } Z_g$. If the objective is at most $14/5 + \delta_{\text{obj}}$, (190) and (191) give

$$\frac{14}{5} + \frac{1-m_-}{10} \leq \sum_g \langle \bar{C}_{h_g}, Z_g \rangle \leq \frac{14}{5} + \delta_{\text{obj}}, \quad \boxed{m_- \geq 1 - 10\delta_{\text{obj}}}. \quad (208)$$

When the instance contains both an odd and an even group and $0 \leq \delta_{\text{obj}} \leq 1/10$, the constant is sharp: split trace $1 - 10\delta_{\text{obj}}$ and $10\delta_{\text{obj}}$ between odd and even minimum eigenspaces.

The original sparse formulation has a correspondingly scaled robust contract. Define copy residuals

$$E_{g,i} = X_{g,i} - G_{g,i} X_{g,i-1} G_{g,i}^T \quad (209)$$

and let $r_1, \dots, r_G, r_{\text{top}}$ be the residuals in (184). Suppose $X_{g,i} \succeq 0$ and

$$\sum_{g,i \geq 1} \|E_{g,i}\|_F^2 + \sum_{g=1}^G r_g^2 + r_{\text{top}}^2 \leq \epsilon^2, \quad f(X) \leq \text{OPT} + \delta_{\text{obj}}. \quad (210)$$

Put $Z_{g,i} = R_{g,i}^T X_{g,i} R_{g,i}$, gauge each block into its root frame, and telescope the paths and the accumulator chain. Cauchy–Schwarz and the local cost bound give

$$\left| \sum_g \text{tr } Z_{g,0} - 1 \right| \leq \sqrt{G+1} \epsilon, \quad (211)$$

$$\left| \frac{1}{P} \sum_{g,i} \text{tr } X_{g,i} - 1 \right| \leq \sqrt{GP+1} \epsilon, \quad (212)$$

$$\left| f(X) - \sum_g \langle \bar{C}_{h_g}, Z_{g,0} \rangle \right| < \frac{13}{4} \sqrt{GP} \epsilon. \quad (213)$$

Combining these estimates with (190) yields, for the even root mass m_+ on a yes-instance,

$$m_+ \leq 10\delta_{\text{obj}} + 28\sqrt{G+1} \epsilon + \frac{65}{2} \sqrt{GP} \epsilon. \quad (214)$$

Theorem 7.2 (Robust winner recovery). *Suppose an algorithm prepares a state within trace distance $1/100$ of the trace-normalized physical density*

$$\rho(X) = \frac{\bigoplus_{g,i} X_{g,i}}{\sum_{g,i} \text{tr } X_{g,i}} \quad (215)$$

for PSD blocks and free accumulators satisfying (210), where

$$\delta_{\text{obj}} \leq \frac{1}{100}, \quad \epsilon \leq \frac{1}{1000\sqrt{GP}}. \quad (216)$$

Then a public fixed measurement decides whether any group is odd with constant bias. On every yes-instance, measuring the group register returns an actual odd group with probability greater than $5/6$. Consequently any such first preparation uses $\Omega(N\sqrt{G})$ raw or canonical coefficient queries.

Proof. Set $s = \sqrt{GP} \epsilon \leq 1/1000$. Since $P \geq 33$, $\sqrt{G+1} \epsilon < s/4$. Equations (212) and (214) put the physical even-group numerator below $1/10 + (81/2)s$ and the total trace above $1 - (33/32)s$.

Their ratio is less than $1/7$; trace error $1/100$ leaves odd-group probability above $5/6$. For existence alone, the public contraction

$$M = \frac{1}{2/5} \left(\bigoplus_{g,i} C_{g,i} - \frac{57}{20} I \right) \quad (217)$$

has norm at most one. The same three telescoping estimates put $f(X)/\bar{T} < 2.82$ on yes-instances and > 2.89 on no-instances, where $\bar{T} = P^{-1} \sum_{g,i} \text{tr } X_{g,i}$. Equivalently, $\text{tr}(M\rho(X)) < -3/40$ and $> 1/10$, respectively. The POVM $(I \pm M)/2$, repeated a constant number of times, therefore decides the promise. Now apply (194). $\square$

The $1/\sqrt{GP}$ residual order is necessary. On an all-even base input, flip each of the GN raw signs in turn, take the neighboring instance's exact odd optimizer, and average the resulting physical solutions. This PSD mixture has objective $14/5$, exact accumulator rows, and only the GN flipped-edge copy residuals. With $r_- = (1, -1, 1)^T/\sqrt{3}$ and $Z_-^* = r_- r_-^T$, each nonzero residual before averaging has Frobenius norm $\|D_- Z_-^* D_- - Z_-^*\|_F = 4/3$. The total Euclidean residual is therefore

$$\frac{4}{3\sqrt{GN}} = \frac{4\sqrt{33}}{3\sqrt{GP}}. \quad (218)$$

Yet its density makes (217) report a winner on the all-even input. Approximate feasibility alone is also insufficient: the public point (188) is exactly feasible for every input. These facts make the output and objective clauses of theorem 7.2 substantive.

7.6 Exact central path and condensation

Eliminating a length- P path multiplies the physical barrier coefficient by P . Thus, with $\tau = \mu P$, the root problem is

$$\min_{Z_g \succ 0, \sum_g \text{tr } Z_g = 1} \left\{ \sum_g \langle \bar{C}_{h_g}, Z_g \rangle - \tau \sum_g \log \det Z_g \right\}. \quad (219)$$

Its exact center is

$$Z_g(\tau) = \tau(\bar{C}_{h_g} - \lambda I)^{-1}, \quad 1 = \tau \sum_g \text{tr}(\bar{C}_{h_g} - \lambda I)^{-1}, \quad (220)$$

with the unique $\lambda < \min_g \lambda_{\min}(\bar{C}_{h_g})$. This instantiates proposition 6.1.

On the all-even input, every group has trace exactly $1/G$. With $y = 29/10 - \lambda$ and $\alpha = \tau G$, the scalar equation and its positive solution are

$$1 = \alpha \left(\frac{2}{y} + \frac{1}{y + 3/10} \right), \quad (221)$$

$$y_\alpha = \frac{3\alpha - 3/10 + \sqrt{(3/10 - 3\alpha)^2 + (12/5)\alpha}}{2}. \quad (222)$$

Now suppose group 1 is the unique odd group and put $x = 14/5 - \lambda$. The winner and loser resolvents are

$$A(x) = \frac{1}{x} + \frac{2}{x + 3/10}, \quad B(x) = \frac{1}{x + 2/5} + \frac{2}{x + 1/10}. \quad (223)$$

The exact central path is determined by

$$\boxed{1 = \tau \{A(x) + (G - 1)B(x)\}}, \quad \boxed{p_G = \tau A(x), \quad 1 - p_G = \tau(G - 1)B(x)}. \quad (224)$$

Direct algebra also gives the family-specific ordering

$$A(x) - B(x) = \frac{3/250}{x(x+1/10)(x+3/10)(x+2/5)} > 0 \quad (x > 0). \quad (225)$$

Together with (224), this yields $p_G/(1-p_G) > 1/(G-1)$, and hence $p_G \geq 1/G$. This implication is special to the present spectra and is not part of the general two-cost model of [section 6.1](#). In particular,

$$p_G \geq 1 - \frac{45}{2}\tau(G-1). \quad (226)$$

Here

$$B_0 = B(0) = \frac{45}{2}, \quad \beta = -B'(0) = \frac{825}{4}, \quad \boxed{\alpha_* = B_0^{-1} = \frac{2}{45}}. \quad (227)$$

Applying [theorem 6.2](#) gives, for fixed $\alpha > 0$ and $G \rightarrow \infty$,

$$0 < \alpha < 2/45 : \quad p_G \longrightarrow 1 - \frac{45}{2}\alpha, \quad \frac{x}{\tau} \longrightarrow \frac{1}{1 - (45/2)\alpha}, \quad (228)$$

$$\alpha = 2/45 : \quad \sqrt{G} p_G \longrightarrow \frac{\sqrt{33}}{9}, \quad \sqrt{G} x \longrightarrow \frac{2}{\sqrt{825}}, \quad (229)$$

$$\alpha > 2/45 : \quad G p_G \longrightarrow \alpha A(x_\alpha), \quad B(x_\alpha) = \frac{1}{\alpha}. \quad (230)$$

In the supercritical line, (225) implies the strict coefficient inequality $\alpha A(x_\alpha) > \alpha B(x_\alpha) = 1$. The coefficient $\sqrt{33}/9$ follows from $\alpha_* \sqrt{\beta} = (2/45)(\sqrt{825}/2)$.

At the center the equality-reduced Frobenius Hessian is

$$\mathcal{H}_\tau[Y]_g = \frac{1}{\tau} D_g Y_g D_g, \quad D_g = \overline{C}_{h_g} - \lambda I, \quad \sum_g \text{tr } Y_g = 0. \quad (231)$$

For one odd group, the spectral gaps are $(x, x+3/10, x+3/10)$ in the winner and $(x+1/10, x+1/10, x+2/5)$ in every loser. For $G \geq 3$, the specialized finite sandwich from [lemma 6.3](#) is

$$\kappa_{\text{red}} \asymp \frac{(x+2/5)^2}{\min\{x(x+3/10), x^2 + (x+1/10)^2/G\}}. \quad (232)$$

Therefore [theorem 6.4](#) gives

$$\kappa_{\text{red}} = \begin{cases} \Theta(G), & 0 < \alpha \leq 2/45, \\ \Theta(1), & \alpha > 2/45. \end{cases} \quad (233)$$

On the all-even input and for $G \geq 2$, by contrast,

$$\kappa_{\text{even}} = \left(\frac{y_\alpha + 3/10}{y_\alpha} \right)^2 = \Theta_\alpha(1). \quad (234)$$

The finite mass law ([MC3](#)) is especially transparent here because $\min\{1, (2/5)/(3/10)\} = 1$: for $G \geq 3$,

$$\kappa_{\text{red}} \geq \frac{(G-1)p_G}{1-p_G}. \quad (235)$$

Thus constant winner mass forces $\Omega(G)$ unpreconditioned Euclidean conditioning. This is a trajectory statement, not a lower bound against an input-dependent preconditioner. The associated visibility-conditioning bounds are [equations \(CV2\)](#) and [\(CV4\)](#).

The robust statements require the dimensionless local residual used in Theorem R, not an unscaled stationarity error. If positive definite $\tilde{Z}_g$ and a multiplier $\tilde{\lambda} < 14/5$ satisfy

$$\left\| \tau^{-1} \tilde{Z}_g^{1/2} (\bar{C}_{h_g} - \tilde{\lambda} I) \tilde{Z}_g^{1/2} - I \right\|_{\text{op}} \leq \eta < 1, \quad \left| \sum_g \text{tr} \tilde{Z}_g - 1 \right| \leq \zeta < 1, \quad (236)$$

put $\tilde{p}_G = \text{tr}(\tilde{Z}_1) / \sum_g \text{tr}(\tilde{Z}_g)$. The denominator need not equal one under (236). Then (R1)–(R3) specialize to

$$\tilde{p}_G \geq 1 - \frac{(1+\eta)(45/2)\alpha}{1-\zeta}. \quad (237)$$

The two surfaces $(1+\eta)(45/2)\alpha = 1-\zeta$ and $(1-\eta)(45/2)\alpha = 1+\zeta$ bracket the residual-dependent transition. Equation (R6) also gives $\kappa(\tilde{\mathcal{H}}|\mathcal{T}) = \Omega(G)$ whenever $\tilde{p}_G$ is bounded below. Trace error δ merely subtracts δ from the probability of observing the winner label.

7.7 Preparation, visibility, and extraction

The root and physical central densities are

$$\rho_\tau = \bigoplus_g Z_g(\tau), \quad \rho_\tau^{\text{phys}} = \frac{1}{P} \bigoplus_{g,i} R_{g,i} Z_g(\tau) R_{g,i}^T. \quad (238)$$

They have the same public group-label marginal, so the unique winner is observed with probability p_G . The ordering (225) proves $p_G \geq 1/G$, so O1's general upper bound $\max\{p_G, 1/G\}$ specializes to p_G . Thus [theorem 6.6](#) gives

$$p_G - \frac{1}{G} \leq D_{\text{tr}}(\rho_0, \rho_1) \leq p_G, \quad (239)$$

$$c_1 - c_0 = \frac{(Gp_G - 1)^2}{G(G-1)}, \quad c_0 = \frac{1}{G}. \quad (240)$$

These formulas give a first-use lower bound with no candidate-verification query. If $\tau \leq 1/(90G)$, (226) gives $p_G \geq 3/4$, and (240) gives $c_1 - c_0 \geq 1/8$ for every $G \geq 2$. Two independently prepared states of trace error ϵ change the promise gap by at most 4ϵ . Thus every fixed $\epsilon < 1/32$ leaves a constant gap. At $\tau \leq 1/(135G)$, one has $p_G \geq 5/6$; trace error $1/20$ leaves gap at least $2/9 - 1/5 = 1/45$. Theorem P then implies

$$\boxed{Q_\rho = \Omega(N\sqrt{G})} \quad (241)$$

for a first root or physical central-density preparation at either stated schedule and tolerance. At the first schedule, the physical product-cone duality gap is $3G\tau \leq 1/30$. Setup is included in (241) under the accounting convention of [definition 2.27](#); a parity table or supplied central-state oracle has already paid or bypassed the relevant aggregation cost.

There is a matching coherent upper bound on the exactly-one-winner promise. A clean parity evaluator costs $T_{\text{eval}} = N$ raw queries. Instantiating A4–A5 in [proposition 6.8](#) gives

$$Q_U = O\left(N(\sqrt{Gp_G} + 1)\right), \quad (242)$$

because the normalized 3-dimensional winner and loser resolvents are public functions of (G, τ) . The acceptance probability is classically known, so exact phase-matched amplitude amplification removes any logarithmic overhead from the raw-query count; the conditional local preparation

cost T_{loc} is gate cost only. Together with (A14), this is tight when the amplification factor diverges and the error resolves the winner mass. Hence

$$Q_U = \begin{cases} \Theta(N\sqrt{G}), & 0 < \alpha < 2/45, \\ \Theta(NG^{1/4}), & \alpha = 2/45, \\ O(N), & \alpha > 2/45. \end{cases} \quad (243)$$

For multiple known winners the corresponding extension is AP3–AP6 in section 6.9; it is not needed here.

For an already available exact central-state primitive, (A10) gives the extraction ledger

regime	p_G	verified copies	coherent calls
$0 < \alpha < 2/45$	$\Theta(1)$	$\Theta(1)$	$\Theta(1)$
$\alpha = 2/45$	$\Theta(G^{-1/2})$	$\Theta(G^{1/2})$	$\Theta(G^{1/4})$
$\alpha > 2/45$	$\Theta(G^{-1})$	$\Theta(G)$	$\Theta(G^{1/2})$

Verification costs another N raw queries per proposed label. Coherent amplification needs a preparation unitary, its inverse, and a clean marker; trace-distance access to one output state alone is insufficient. The stronger mixed-output conversion analysis is deferred to section 8. The table counts uses after the state primitive exists, whereas (241) charges its first construction.

Visibility also identifies the fixed-error boundary. If the allowed root trace error is at least p_G , the public all-even root center is a valid zero-query output on the zero-or-one promise by (225) and O1. At criticality this happens for $G = \Omega(\epsilon^{-2})$, and for fixed supercritical α it happens for $G = \Omega_\alpha(\epsilon^{-1})$. In physical coordinates, lifting that reference through the input’s actual prefixes costs exactly N phase queries and preserves trace distance, so

$$p_G \leq \epsilon \implies Q_\epsilon(\rho_{\text{central}}^{\text{root}}) = 0, \quad Q_\epsilon(\rho_{\text{central}}^{\text{phys}}) \leq N. \quad (244)$$

For sufficiently large fixed α , even the public maximally mixed physical state is zero-query: with y_α from (222), it suffices that

$$\frac{1}{3} - \frac{\alpha}{y_\alpha + 3/10} + p_G \leq \epsilon. \quad (245)$$

No $\Omega(N)$ physical-state lower bound is claimed above the transition.

The conclusion is architectural. Condensation makes the winner visible only after preparation has absorbed the search cost, and constant mass also forces linear reduced-Hessian conditioning. The conservation law (A9) and direct-search bounds equations (A11) and (A12) therefore select search-then-crossover:

$$O(N\sqrt{G}) + T_{\text{component solve}} + O(N)_{\text{recovery}}. \quad (246)$$

This family does not exhibit a QIPM speedup.

7.8 Search then solve

Write $\sigma_{g,j} = (-1)^{z_{g,j}}$. With the XOR-oracle answer register in $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, query the public indices $j = 1, \dots, N$ in order while retaining the group address. Phase kickback implements

$$|g\rangle \mapsto \prod_{j=1}^N (-1)^{z_{g,j}} |g\rangle = h_g |g\rangle \quad (247)$$

with exactly N raw queries and no parity or prefix garbage.

Theorem 7.3 (Search-then-solve upper bound). *For failure probability δ , the value of the SDP can be returned using $O(N\sqrt{G}\log(1/\delta))$ raw queries. The same procedure returns either a verified winning label or ‘NONE’ on an all-even input, which has no winning label, together with the corresponding compact exact root optimizer in either branch. It can instead prepare that optimizer’s physical state with another $O(N)$ queries. It needs no QRAM.*

Proof. Apply bounded-error quantum search to the clean marker (247), and verify a returned label by evaluating its parity. Output 14/5 after successful verification and 29/10 when the search reports ‘NONE’. Standard reversible addressing uses $O(N\sqrt{G}\log(GP)\log(1/\delta))$ elementary routing gates and $O(\log G + \log P)$ work qubits, apart from raw-query implementation.

If g_* is odd, take $Z_{g_*} = r_- r_-^T$ with $r_- = (1, -1, 1)^T/\sqrt{3}$; if no odd group exists, take $g_* = 1$ and $Z_{g_*} = (I - r_+ r_+^T)/2$ with $r_+ = (1, 1, 1)^T/\sqrt{3}$. Set every other root to zero. This is the compact optimizer contract of definition 2.21. Streaming the N prefix signs of g_* lifts it through (182) and prepares the normalized physical direct-sum state. Listing its $P = 33N$ nonzero blocks instead costs $\Theta(P)$ output work. $\square$

Together, theorem 7.3 and (194) show directly that bounded-error value estimation and winner recovery cost $\Theta(N\sqrt{G})$. The matching search primitives are standard [29, 57].

The exact-query scope is different. On the promise of zero or one winner, phase-matched exact amplification followed by exact parity verification gives

$$Q_E(\text{OR}_G^{0,1} \circ \text{PARITY}_N) = \Theta(N\sqrt{G}). \quad (248)$$

Without that promise, the unique multilinear output polynomial has a nonzero degree- NG monomial, so the polynomial method gives an $NG/2$ lower bound; computing every parity gives

$$Q_E(\text{OR}_G \circ \text{PARITY}_N) = \Theta(NG). \quad (249)$$

If exactly one winner is promised, its label still costs $\Theta(N\sqrt{G})$ for $G \geq 2$, although the scalar decision is public.

Remark 7.4 (Access and setup scope). A supplied h_g oracle removes the inner N factor only because it is a stronger input. A QRAM parity table has already paid its setup cost. Parallel ports may reduce depth but not total queries. The accumulator chain is public and carries no hidden query information. A compact optimizer is required for the stated end-to-end bound; materializing G arrays incurs their output size. If preprocessing uses S raw queries and later calls use q_t , any value or decodable-winner procedure obeys

$$S + \sum_t q_t = \Omega(N\sqrt{G}). \quad (250)$$

An arbitrary input-dependent block encoding, prefix table, parity table, null-space basis, or eliminated cost is not free raw-coefficient access.

7.9 Formulation tradeoff and novelty scope

The direct root trace row has $3G$ nonzeros but gives an isometric tangent model in the matrix variables. The accumulator chain (184) reduces row sparsity to five and column sparsity to two, while preserving the projected SDP and root-coordinate geometry. Its lift map is not an isometry and its free scalars have no barrier Hessian. Replacing them by differences of nonnegative variables, or adding scalar log barriers, would change the geometry. Thus bounded incidence and an ambient condition-one formulation are not simultaneously claimed.

The query machinery is prior art: adversary composition is due to Høyer–Lee–Špalek, Belovs–Lee, and Reichardt; the randomized composition framework includes Ben-David–Kothari; search and minimum finding are due to BBHT and Dürr–Høyer; and fixed-point search is standard

[22, 24, 29, 57, 77, 122, 141]. Boolean-in-optimum embeddings and sparsity-aware coefficient-query lower bounds already appear in van Apeldoorn et al., Theorem 29 and Corollary 30, and Apers–Gribling, Theorem 8.4 [10, 130]. Fawzi’s non-SOC theorem and the Rayleigh–Ritz trace identity are likewise established tools.

The candidate new contribution is only their conjunction here: a bounded-incidence public normalization chain, an exact spectral minimum, a constant-gap $\text{OR} \circ \text{PARITY}$ value, a condition-one public root start, the exact constants in the condensation transition, and the phase-dependent preparation profile. None of Grover search, parity, composition, the trace-allocation identity, generic SDP ill-conditioning, or log-det resolvents is claimed separately as new.

8 Hidden-Block State Conversion

The central paths in [section 7.6](#) concentrate trace on a hidden set of winning blocks. This section asks for the query cost of preparing that block center when the number of winners is promised to be either zero or k . The endpoints were already visible in [section 6.9](#): sufficiently coarse error permits the public all-loser output, whereas sufficiently accurate coherent preparation costs search complexity. Between them, however, the collision statistic in (173) loses a factor $1/k$, and pairwise trace distance supplies no single measurement that works for every unknown winner set. The remedy is to measure a proposed label and verify it with fresh queries. This verified-sample decoder is one-sided on the no-winner branch and works directly on trace-close mixed outputs.

For parity blocks, a polynomial argument then gives the entire nontrivial accuracy curve. The key step is that averaging over inputs of fixed block parity kills every proper sub-block character, dividing polynomial degree by the block length. We also contrast the coupled centers with a batch of uncoupled triangle-holonomy SDPs: vector output admits no amortization, while one averaged scalar objective has the usual approximate-counting frontier.

8.1 The zero-versus- k conversion problem

Let $f : \mathcal{X}_0 \sqcup \mathcal{X}_1 \rightarrow \{0, 1\}$ be a possibly partial Boolean predicate on one block. Its bounded-error query complexity and negative-weight adversary value satisfy

$$Q(f) = \Theta(A), \quad A = \text{Adv}^\pm(f).$$

There are G independently queried blocks, and $S = \{g : f(x_g) = 1\}$ is promised to have size zero or k , where $1 \leq k < G$. One raw query addresses one coordinate of one block. The target on a k -winner input is

$$\rho_S = \frac{p}{k} \sum_{g \in S} |g\rangle\langle g| \otimes \rho_W + \frac{1-p}{G-k} \sum_{g \notin S} |g\rangle\langle g| \otimes \rho_L, \quad \rho_0 = \frac{1}{G} \sum_{g=1}^G |g\rangle\langle g| \otimes \rho_L^0, \quad (251)$$

where $p = p_{G,k}$ and the constant-dimensional internal states are public. Write

$$\eta_{\text{int}} = D_{\text{tr}}(\rho_L, \rho_L^0), \quad \delta = p - \varepsilon. \quad (252)$$

Here η_{int} is the local internal-state discrepancy, not the residual tolerance η used elsewhere. The task is to output, on every promised input, a state within trace distance ε of its target. The upper bounds below use a clean coherent evaluator as in (A1), with the membership bit for S , at cost T_{eval} per call. An exact evaluator costs $\Theta(Q_E(f))$ raw queries, Q_E denoting exact quantum query complexity; a bounded-error evaluator may be substituted after coherent error reduction to operator-norm error below the per-call budget of the amplification it enters, at cost $O(Q(f) \log(G/\min\{\delta, \varepsilon\}))$ per call, which the stated tildes absorb. For $f = \text{PARITY}_N$, $Q_E(f) = \lceil N/2 \rceil$, and $Q(f)$ and $\text{Adv}^\pm(f)$ are both $\Theta(N)$. As in the condensation applications,

we assume $p \geq k/G$ when invoking rejection preparation. For classical algorithms, queries are charged but state synthesis from the classical transcript is free.

Measuring the label of an ε -accurate output and then evaluating f on the sampled block accepts with probability at least $p - \varepsilon = \delta$ on the k -winner branch, before verifier error. On the zero-winner branch, every proposed block is a loser, so only verifier error can cause acceptance. This conclusion is independent of the state produced by the circuit on that branch. It is the common measurement that the collision approach lacks.

Theorem 8.1 (Verified-sample lower bound (M-1)). *Fix $\delta \in (0, 1)$. Suppose a T -query circuit solves (251) with $\varepsilon \leq p - \delta$. If*

$$\sqrt{G/k} \geq C\delta^{-1} \log(e/\delta) \quad (253)$$

for a sufficiently large universal constant C , then

$$T = \Omega\left(\delta A \sqrt{G/k}\right).$$

For $f = \text{PARITY}_N$, this is $T = \Omega(\delta N \sqrt{G/k})$.

Proof. First reduce the zero-versus-one problem of theorem 6.7 to zero-versus- k . Put $G' = \lfloor G/k \rfloor$. Replace each of the G' input blocks by k addressable copies and pin the remaining $G - kG' < k$ blocks to a fixed public element of $\mathcal{X}_0$. A query to a copy is simulated by one query to its original. Zero winners remain zero, and a unique winner produces exactly k winners. The Cartesian-product promise is preserved. Since (253) places us away from the $G/k = 1$ edge, $G' = \Theta(G/k)$, and (O3) gives a decision lower bound $\Omega(A\sqrt{G/k})$. This copies-plus-public-loser recipe is also valid when k does not divide G .

Now repeat the following trial $R = \Theta(1/\delta)$ times. Run the preparation circuit afresh, measure its label g , and evaluate $f(x_g)$ with error at most $\beta = \delta/8$. Standard error reduction costs $O(Q(f) \log(e/\delta))$ queries per verification. On a k -winner input, contractivity of trace distance under the two-outcome measurement $\{\sum_{g \in S} |g\rangle\langle g| \otimes I, I - \dots\}$ gives

$$\Pr[g \in S] \geq p - \varepsilon \geq \delta, \quad \Pr[\text{accept}] \geq \delta(1 - \beta) \geq \delta/2.$$

On a zero-winner input, every label fails the true predicate, regardless of the prepared state, and $\Pr[\text{accept}] \leq \beta = \delta/8$. A multiplicative Chernoff bound distinguishes these two rates with constant error using $R = \Theta(1/\delta)$ trials. Its total cost

$$O(\delta^{-1}[T + Q(f) \log(e/\delta)])$$

is therefore $\Omega(A\sqrt{G/k})$. Since $Q(f) = \Theta(A)$, (253) absorbs the verification term and proves the claim. $\square$

The restriction (253) is substantive. At constant δ it requires only $G/k = \Omega(1)$. For corollary 8.3 we adopt the uniform hypothesis $k = O(G/\log^2 G)$, which is stronger than the displayed domain requires at constant δ . At variable δ , the domain contains the scale $G/k = \Omega(\delta^{-2})$ up to logarithms. The advantage of theorem 8.1 is its general inner predicate and its complete lack of a purification or inverse-preparation assumption.

Theorem 8.2 (Detuned rejection preparation (M-2)). *Assume $p \geq k/G$, $1 - p \geq c_0 > 0$, and $0 < \varepsilon < p$. With $\delta = p - \varepsilon$, if $4\eta_{\text{int}} \leq \varepsilon$, then*

$$Q_\varepsilon = O\left(T_{\text{eval}} \left[1 + \sqrt{\delta G/k}\right] c_0^{-1/2} \log \frac{e}{\min\{\delta, \varepsilon\}}\right). \quad (254)$$

If $0 < \varepsilon < 4\eta_{\text{int}}$, then

$$Q_\varepsilon = O\left(T_{\text{eval}} \sqrt{G/k} \log(e/\varepsilon)\right). \quad (255)$$

In the accuracy-frontier regime $\varepsilon = \Omega(\delta)$, the logarithm in (254) is simply $O(\log(e/\delta))$.

Proof. We use two detuning regimes. Define

$$m = \begin{cases} \max\{3\delta/2, k/G\}, & \delta \leq 2p/3, \\ \max\{p - \varepsilon/2, k/G\}, & \delta > 2p/3, \end{cases} \quad (256)$$

$$m \leq p : \begin{cases} 3\delta/2 \leq p \text{ and } k/G \leq p, & \delta \leq 2p/3, \\ p - \varepsilon/2 \leq p \text{ and } k/G \leq p, & \delta > 2p/3. \end{cases}$$

Then $k/G \leq m \leq p$, $m - \delta \geq \delta/2$ in the first regime, and $m - \delta \geq \varepsilon/2$ in the second. Moreover $\sqrt{Gm/k} = O(1 + \sqrt{\delta G/k})$: this is immediate in the first regime, while $m = \Theta(\delta)$ in the second. The upper cap $m \leq p$ is essential: together with $1 - p \geq c_0$, it keeps the empty-branch acceptance probability from acquiring an additional $(1 - m)^{-1/2}$ amplification cost.

Run the rejection construction of [proposition 6.8](#), here with the label copied before amplification and the copy included in every reflection, with detuned weights

$$a'_W = m/k, \quad a'_L = (1 - m)/(G - k), \quad a'_{\max} = m/k.$$

Starting from the uniform label superposition, copy the label to an environment register, compute the membership bit, rotate an acceptance qubit with amplitude $\sqrt{a'_b/a'_{\max}}$, and conditionally attach public purifications of ρ_W and ρ_L . A trial has acceptance probability

$$P_k = \frac{k}{Gm}, \quad P_0 = \frac{k(1 - m)}{m(G - k)} \quad (257)$$

on the k - and zero-winner branches, respectively. Both are at least a constant times $c_0 k/(Gm)$, because $m \leq p$ implies $1 - m \geq 1 - p \geq c_0$.

Apply fixed-point amplification simultaneously to the two possible acceptance angles [\[141\]](#). The fixed-point reflections act on every register the trial touches, including the environment copy of the label, which is uncomputed and recomputed inside each reflection. With

$$r = \frac{1}{8} \min\{\delta, \varepsilon\},$$

the acceptance-subspace residual can be made at most r^2 on both branches using

$$O\left(\sqrt{Gm/k} c_0^{-1/2} \log(e/r)\right)$$

evaluator calls. The quadratic residual is necessary: overlap error of a purification becomes its square root in trace distance. Writing $q = k/G$, [\(257\)](#) gives $P_0/P_k = (1 - m)/(1 - q)$. Thus $m = q$ has no branch mismatch. When $q = o(m)$, however, an exact schedule tuned to the larger probability P_k under-rotates the empty branch and can leave $\Theta(m^2)$ junk probability, which is unaffordable for constant p . Fixed-point amplification avoids this mismatch.

Conditioned on acceptance, the k -branch output is exactly [\(251\)](#) with winner mass m in place of p ; the zero-winner output is exactly $G^{-1} \sum_g |g\rangle\langle g| \otimes \rho_L$. Hence the respective errors are at most

$$(p - m) + r, \quad \eta_{\text{int}} + r.$$

If $\delta \leq 2p/3$, then $\varepsilon = p - \delta \geq \delta/2$ and $p - m \leq \varepsilon - \delta/2$, so both errors are below ε . If $\delta > 2p/3$, then $p - m \leq \varepsilon/2$ and $r \leq \varepsilon/8$, again proving the contract. Substituting the bound on m gives [\(254\)](#).

For [\(255\)](#), first decide whether there are zero or k winners by fixed-point search, reducing its error to $O(\varepsilon)$. On the k branch use the exactly- k circuit [\(AP3\)](#); on the empty branch prepare ρ_0 directly. The decision error contributes at most $O(\varepsilon)$ in trace distance, and the stated cost dominates both steps. Deciding the branch first avoids any circular use of an exactly- k preparation promise. $\square$

The precision dependence in (254) is written explicitly. In the second line of (256), suppressing the fixed-point residual only to $O(\delta)$ would not suffice when $\varepsilon \ll \delta$; spending an $O(\varepsilon)$ residual is necessary.

Corollary 8.3 (Constant-visibility regime (M-3)). *Let $f = \text{PARITY}_N$, let p be bounded away from zero and one, and suppose*

$$k = O(G/\log^2 G), \quad 4\eta_{\text{int}} \leq \varepsilon \leq (1-c)p$$

for a constant $c > 0$. Then, up to fixed-point precision logarithms,

$$Q_\varepsilon = \tilde{\Theta}\left(N\sqrt{G/k}\right).$$

The same scale covers $\varepsilon < 4\eta_{\text{int}}$ when $\delta = \Omega(1)$ via (255). Above the minimax radius of the target family, zero queries suffice; (172) gives the useful public-output bound $\max\{p, k/G\}$ when the internal loser states agree.

Proof. Here $\delta = p - \varepsilon = \Omega(1)$ and $A = \Theta(N)$. The domain assumption makes (253) valid, so theorem 8.1 supplies the lower bound. The upper bound is theorem 8.2. The zero-query statement follows by returning the public all-loser target and using (172). $\square$

Theorem 8.4 (Classical accuracy law (M-4)). *For $f = \text{PARITY}_N$, suppose $p \geq k/G$, $1 - p = \Omega(1)$, and*

$$2\eta_{\text{int}} \leq \varepsilon, \quad Ck/G \leq \delta = p - \varepsilon \leq 1/2. \quad (258)$$

Here $C > 0$ is a sufficiently large universal constant. Then the randomized classical raw-query complexity is

$$R_\varepsilon = \Theta(N\delta G/k).$$

Proof. For the upper bound, inspect $t = \Theta(\delta G/k)$ uniformly random blocks without replacement and compute each parity exactly using N queries. Put $q = k/G$. The probability of finding at least one winner is

$$P_{\text{find}} = 1 - \frac{\binom{G-k}{t}}{\binom{G}{t}}.$$

The constant in t can be chosen so that $P_{\text{find}} \geq \delta$ throughout (258). This use of a linear budget relies on $\delta \leq 1/2$; as $\delta \rightarrow 1$, the correct sampling budget contains $(G/k) \log(1/(1-\delta))$.

Let E_{all} be the event that every sampled block is a winner. Its public probability is at most $q^t \leq q \leq \delta$. On E_{all} output a uniformly selected sampled winner. On $E_{\text{find}} \setminus E_{\text{all}}$, do the same with a public probability chosen so that the total probability of winner output is exactly δ ; this is possible because $\Pr(E_{\text{all}}) \leq \delta \leq \Pr(E_{\text{find}})$. In all other cases select uniformly among sampled losers. A loser is available whenever the latter rule is used. Exchangeability makes the selected winner uniform on S and the selected loser uniform on its complement. Attaching the corresponding public internal state therefore prepares exactly the target shape with winner mass δ in place of p . Its k -branch trace error is $p - \delta = \varepsilon$. On the empty branch the output is the uniform-label ρ_L state and has error $\eta_{\text{int}} \leq \varepsilon/2$. Thus $O(N\delta G/k)$ queries suffice.

For the lower bound, append exact N -query parity verification to a sampled output label and repeat the preparation $\Theta(1/\delta)$ times. This decides zero versus k winners with one-sided error and cost $O(\delta^{-1}(T + N))$. For the decision lower bound, use the k -copy embedding from theorem 8.1 and draw each original block uniformly conditioned on its parity. Every proper subset of a block is uniform, so a classical transcript is independent of the planted parity until all N positions of that block have been queried. A q -query tree completes at most q/N original blocks, and it hits the uniformly planted winner with probability at most $O(qk/(NG))$. Yao's principle therefore gives $\Omega(NG/k)$ decision queries. Consequently

$$\delta^{-1}(T + N) = \Omega(NG/k).$$

The lower restriction $\delta \geq Ck/G$ absorbs the additive N term. $\square$

No lower bound is asserted in the sliver $\delta = o(k/G)$: a public uniform-label output may already meet the state contract there. The upper restriction in (258) is equally real, because finding probability ceases to be linear in the number of sampled blocks near one.

8.2 Composed small-success search

The verified-sample argument above loses a factor $\sqrt{\delta}$ through repetition. For parity blocks the polynomial method recovers it. We state the degree-division mechanism separately because it will also be used for the uncoupled batch.

Lemma 8.5 (Parity collapse). *Let a T -query quantum algorithm access $x \in \{\pm 1\}^{G \times N}$, and let $P(x)$ be the probability of a fixed final measurement event. For block parities $z_g = \prod_{j=1}^N x_{g,j}$, define*

$$q(z) = \mathbb{E}[P(x) \mid \prod_j x_{g,j} = z_g \text{ for every } g].$$

The conditional expectation is uniform over each parity fiber, independently across blocks. Then q is a real multilinear polynomial in $z \in \{\pm 1\}^G$ of degree at most $\lfloor 2T/N \rfloor$. If $0 \leq P \leq 1$, then $0 \leq q \leq 1$ on the Boolean cube.

Proof. The polynomial method writes P as a real multilinear polynomial of degree at most $2T$ [20]. Consider a monomial x_R . Conditional expectation factors over blocks. In one block, the average of a character on a parity coset is one when its share of R is empty, is z_g when that share is the full block, and is zero for every proper nonempty share. The last assertion follows by pairing inputs that differ in two coordinates, one inside and one outside the proper share. Thus a surviving raw monomial becomes a monomial in z whose degree is the number of full blocks it contains, at most $|R|/N \leq 2T/N$. Averaging all monomials proves the claim. $\square$

Remark 8.6 (Parity-collapse audit). As a check on the character calculation in lemma 8.5, the identity was verified by exhaustive enumeration for block lengths $N = 2, 3, 4, 5$ and end-to-end on random multilinear polynomials.

Theorem 8.7 (One-sided composed small-success bound (S)). *Suppose a T -query quantum algorithm accepts with probability at most $\beta \leq \delta/2$ on every all-even input and at least δ on every input with exactly k odd-parity blocks. Then absolute constants $c_1, c_2 > 0$ satisfy*

$$T \geq c_1 N \sqrt{\delta G/k} - c_2 N. \quad (259)$$

Proof. Let $P(x)$ be the acceptance probability. By lemma 8.5, conditioning on all block parities gives a polynomial $q(z)$ of degree at most $D = \lfloor 2T/N \rfloor$, bounded between zero and one at every Boolean input. It is at most β at the all-even point and at least δ at every parity vector of odd-block weight k .

Average q over all block permutations. Minsky–Papert symmetrization [103] gives a univariate polynomial h of degree at most D such that

$$0 \leq h(w) \leq 1 \quad (w = 0, 1, \dots, G), \quad h(0) \leq \beta, \quad h(k) \geq \delta.$$

Fix a sufficiently small absolute $\gamma > 0$. First suppose $D \leq \gamma\sqrt{G}$. The Coppersmith–Rivlin inequality bounds a degree- D polynomial bounded at the $G+1$ equally spaced integer points by

$$\sup_{t \in [0, G]} |h(t)| \leq C(\gamma),$$

where $C(\gamma)$ is an absolute constant in this degree range [46, 86]. Only existence of the universal Coppersmith–Rivlin constants is used; the original paper does not specify numerical values; this is the form quoted in Theorem 8 of Klauck–Špalek–de Wolf. Erdélyi proves a related explicit-constant inequality in a different, single-point-versus-supremum parametrization [59].

Markov's inequality, rescaled from $[-1, 1]$ to $[0, G]$, now gives

$$\sup_{[0, G]} |h'| \leq \frac{2D^2 C(\gamma)}{G}.$$

The mean value theorem on $[0, k]$ gives a point ξ with

$$h'(\xi) \geq \frac{h(k) - h(0)}{k} \geq \frac{\delta - \beta}{k} \geq \frac{\delta}{2k}.$$

Therefore $D = \Omega(\sqrt{\delta G/k})$. In the complementary case $D > \gamma\sqrt{G}$, the same conclusion follows directly from $\delta \leq 1 \leq k$. Finally $D \leq 2T/N$, with an optional additive $O(N)$ allowance for integer bookkeeping, proves (259). $\square$

The theorem is new as stated to the best of our knowledge, but its ingredients are standard and the result is plausibly derivable by experts. In particular, it should not be confused with a new general composition theorem.

Corollary 8.8 (Composed search (S1)). *If a quantum algorithm outputs an odd-parity block with probability at least δ whenever exactly k blocks are odd, then*

$$T = \Omega\left(\sqrt{\delta} N \sqrt{G/k}\right) - O(N).$$

This is tight up to constants and the additive N term.

Proof. Verify the output block by computing its parity exactly with N fresh queries. Acceptance is then zero on an all-even input and at least δ on an exactly- k input, so [theorem 8.7](#) applies to $T + N$ queries. Conversely, if $\delta \leq 4k/G$ a uniformly random label is odd with probability $k/G \geq \delta/4$ at zero queries. Otherwise $k/G < 1/4$, so the initial Grover angle $\theta = \arcsin \sqrt{k/G}$ is below $\pi/6$, and $t = \lceil c\sqrt{\delta G/k} \rceil \geq 1$ Grover iterations with $c = 1/6$ keep $(2t + 1)\theta \leq \pi/2$; each uses a $\Theta(N)$ -query parity marker, and together they produce a winner with probability $\sin^2((2t + 1)\theta) \geq \sin^2(2c\sqrt{\delta}) = \Theta(\delta)$. Direct exact verification costs another N queries. For one marked item, Zalka proved exact optimality [142]. The small-success scaling for k marked items follows from the angle bound in Theorem 2 of the published Dohotaru–Høyer paper (Theorem 9 of the arXiv version) plus the standard k -copy embedding. Carolan–Poremba Corollary 2.4 records the packaged inequality $\delta \leq 8(T + 1)^2 k/G$; our attribution uses the angle theorem and embedding, not their intermediate distance-based derivation [35, 55]. Zalka's result alone does not establish the k -marked statement. Aaronson's direct-product Theorem 4.6 instead concerns the find-all- k analogue [1]. $\square$

Corollary 8.9 (Full conversion curve (S2)). *Let $f = \text{PARITY}_N$, $p \geq k/G$, $1 - p = \Omega(1)$, $4\eta_{\text{int}} \leq \varepsilon < p$, and $\delta = p - \varepsilon \geq Ck/G$ for a sufficiently large constant C . Then*

$$Q_\varepsilon = \tilde{\Theta}\left(N\sqrt{(p - \varepsilon)G/k}\right). \quad (260)$$

The tilde hides only fixed-point precision logarithms: the hidden factor is $\log(e/\min\{\delta, \varepsilon\}) = \log(1/\varepsilon) + O(1)$ when $\varepsilon \ll \delta$, and can be unbounded relative to the main term for superpolynomially small ε . The lower bound is nonvacuous once $\delta G/k$ is a sufficiently large constant.

Proof. One preparation followed by label measurement and exact parity verification uses $T + N$ queries. It accepts with probability zero on the empty branch and at least $p - \varepsilon = \delta$ on the k -winner branch. Applying [theorem 8.7](#) and absorbing its additive $O(N)$ term under the domain assumption gives the lower bound. The upper bound is [theorem 8.2](#); its additive one is absorbed on the same domain. $\square$

Unlike [theorem 8.1](#), this conclusion needs neither $k = O(G/\log^2 G)$ nor $G/k = \tilde{\Omega}(\delta^{-2})$. It is parity-specific: the gain comes from [lemma 8.5](#).

Corollary 8.10 (Mixed-output preparation profile (S3)). *Suppose a T -query circuit, on every exactly- k parity input, outputs a state within trace distance $\varepsilon \leq p/2$ of the winner-mass- p target. Then*

$$T = \Omega\left(\sqrt{p} N \sqrt{G/k}\right) - O(N). \quad (261)$$

Thus the first two lines of the AP6 profile ([AP6](#)) are tight for trace-close mixed-state output contracts: the lower bounds are $\Omega(N\sqrt{G/k})$ in the subcritical phase and $\Omega(N(G/k)^{1/4})$ at criticality. In the supercritical phase the displayed lower bound is vacuous and does not certify the $O(N)$ upper bound.

Proof. Run the circuit also on an all-even input, which is outside its preparation promise but on which the physical circuit remains defined. Measure a label and verify its parity exactly. Acceptance is zero on this input and is at least $p - \varepsilon \geq p/2$ on every exactly- k input. Apply [theorem 8.7](#) with $\delta = p/2$. The three phase statements follow by substituting the winner masses from [theorem 6.9](#): $p = \Theta(1)$, $p = \Theta(\sqrt{k/G})$, and $p = \Theta(k/G)$. $\square$

This changes the hypothesis, not the upper bounds, in the preparation ledger of [section 6.9](#). The AP5 lower bound ([AP5](#)) was obtained by coherent extraction through AP4 ([AP4](#)), and therefore assumed access to a preparation unitary and its inverse. For parity blocks and $\varepsilon \leq p/2$, [corollary 8.10](#) replaces that operator-level hypothesis by the weaker contract that the circuit output is a trace-close mixed state. The AP3 upper bound ([AP3](#)) and the AP6 supercritical caveat are unchanged. In the concrete family of [section 7.7](#), this explains which condensed central states reveal hidden holonomy without changing the search-then-crossover architecture of [\(246\)](#).

Combining [corollary 8.9](#) and [theorem 8.4](#) gives, on their common domain,

model	query complexity	dependence on $\delta = p - \varepsilon$
quantum	$\tilde{\Theta}(N\sqrt{\delta G/k})$	square root
randomized classical	$\Theta(N\delta G/k)$	linear

Thus every nontrivial accuracy scale between $\delta = \Theta(k/G)$ and constant δ has the corresponding quadratic search separation: the classical-to-quantum ratio is $\tilde{\Theta}(\sqrt{\delta G/k})$. It approaches one at the public-output boundary and grows to $\tilde{\Theta}(\sqrt{G/k})$ at fixed relative accuracy.

Open Problem 8.11 (General-inner composition). Does every predicate f satisfy a composed small-success lower bound

$$\Omega\left(\sqrt{\delta} Q(f) \sqrt{G/k}\right)$$

for finding a winner with probability δ ? The adversary reduction in [theorem 8.1](#) gives only linear dependence on δ , while the parity proof uses the special disappearance of every proper sub-block character.

8.3 Uncoupled holonomy batches

The global trace row in [section 7.1](#) couples all components. For contrast, take G disjoint copies of the triangle-holonomy component, including one public trace-one row per component and no equality that joins two components. The path length and local costs are those of [equations \(179\)](#) and [\(183\)](#). Consequently the product has $L = 33GN$ constant-size PSD blocks, scalar row sparsity at most three and column sparsity at most two. If

$$h_g = \prod_{j=1}^N \sigma_{g,j}, \quad y_g = (1 + h_g)/2,$$

then [\(190\)](#) gives component optima $v_+ = 29/10$ and $v_- = 14/5$.

Theorem 8.12 (No amortization for vector output). *Returning estimates $\hat{v}_1, \dots, \hat{v}_G$ satisfying $|\hat{v}_g - v_{h_g}| \leq 1/25$ for every g , with worst-case joint success probability at least $2/3$, has quantum raw-query complexity*

$$\Theta(GN) = \Theta(L).$$

Moreover, there are constants $a, b > 0$ such that at most aGN queries give worst-case joint success probability at most 2^{-bG} .

Proof. Threshold every estimate at $57/20$ to recover all h_g . Their product is the parity of all GN raw signs, which has quantum query complexity $\Omega(GN)$; reading all signs is a matching upper bound. For the stronger claim, specialize Lee–Roland’s strong direct-product theorem to G copies of PARITY_N [91]. Its Theorem 1.1 with success parameter $3/4$ gives

$$Q_{1-(3/4)^{G/2}}(f^{(G)}) \geq \frac{G \ln(9/8)}{8000} Q_{1/4}(f).$$

Since $Q_{1/4}(\text{PARITY}_N) = \Theta(N)$, fewer than a fixed constant times GN queries makes simultaneous decoding succeed with worst-case probability at most $(3/4)^{G/2} = 2^{-\Omega(G)}$. $\square$

Now average the G component objectives. Separability gives the single optimum

$$V(\sigma) = \frac{1}{G} \sum_{g=1}^G v_{h_g} = \frac{14}{5} + \frac{1}{10G} \sum_{g=1}^G y_g. \quad (262)$$

Theorem 8.13 (Scalar batch frontier). *For $0 < \varepsilon \leq 1/100$,*

$$Q_\varepsilon(V) = \Theta(N \min\{G, 1/\varepsilon\}), \quad (263)$$

$$R_\varepsilon(V) = \Theta(N \min\{G, 1/\varepsilon^2\}). \quad (264)$$

Proof. An exact coherent query to y_g costs $\Theta(N)$ raw sign queries. Amplitude estimation estimates its mean to error 10ε using $O(1/\varepsilon)$ such calls [31]; when $1/\varepsilon > G$, compute all parities. This proves the quantum upper bound. Classically, sample $O(\min\{G, 1/\varepsilon^2\})$ groups without replacement, compute their parities, and use the empirical mean, scanning all groups when the minimum is G .

For every G , restricting to inputs in which all G groups carry the same N -bit string x makes V change by $1/10 > 2\varepsilon$ when the parity of x flips, so an ε -estimator computes PARITY_N and $T \geq N/2$; this settles $G < 6$, where $\min\{G, 1/\varepsilon\} = O(1)$. For $G \geq 6$, choose Hamming weights $\ell, \ell' \in [G/4, 3G/4]$ separated by $\Delta = \max\{1, \lceil 21\varepsilon G \rceil\}$. Their values in (262) differ by more than 2ε , so thresholding an estimator gives a bounded-error polynomial distinguishing the two weights. Nayak–Wu’s symmetric approximate-counting lower bound gives degree

$$\Omega\left(\frac{\sqrt{m(G-m)}}{\Delta}\right) = \Omega(\min\{G, 1/\varepsilon\}),$$

where m is the selected weight farther from $G/2$ [112]. Applying the already proved degree division in lemma 8.5 forces $\lfloor 2T/N \rfloor = \Omega(\min\{G, 1/\varepsilon\})$.

The same lower bound also follows from perfect negative-adversary composition: compose the partial symmetric outer gap-counting problem with G independent parity blocks and combine Nayak–Wu’s outer bound with the Belovs–Lee composition theorem [22]. Thus the parity-collapse argument here is a construction-specific proof, not a new composition theorem.

For completeness, the classical reduction also charges the entire parity block even to an adaptive algorithm. Fix $h \in \{\pm 1\}^G$ and draw the raw signs of each group uniformly conditional on their product being h_g . A simulator answers every first-time query while at least two signs remain unseen by an independent fair sign. Only when the final unseen sign in a group is

queried does it query h_g and force the required product. This exactly reproduces the conditional distribution under arbitrary interleaving. Each parity-oracle query consumes N distinct raw queries, so a depth- q raw tree induces a parity-bit tree of depth at most $\lfloor q/N \rfloor$. The classical bounded-error complexity of estimating a G -bit mean to error 10ε is $\Theta(\min\{G, 1/\varepsilon^2\})$, by the standard Yao distribution on two nearby central Hamming weights. Multiplication by N proves (264). $\square$

The two frontiers form three regimes:

accuracy	$Q_\varepsilon(V)$	$R_\varepsilon(V)$
$\varepsilon = \Omega(G^{-1/2})$	$\Theta(N/\varepsilon)$	$\Theta(N/\varepsilon^2)$
$G^{-1} = O(\varepsilon) = O(G^{-1/2})$	$\Theta(N/\varepsilon)$	$\Theta(NG)$
$\varepsilon = O(G^{-1})$	$\Theta(NG)$	$\Theta(NG)$

At $\varepsilon = \Theta(G^{-1/2})$, the costs are $\Theta(N\sqrt{G})$ and $\Theta(NG)$, exactly the quantum and classical scales of the coupled winner-take-all value bound (197). This is a dimension–accuracy crossover, not an unqualified exponential advantage.

Corollary 8.14 (Parallel Newton materialization). *At the public point $X_{g,i} = I/3$ and product-barrier parameter $\mu = 1/(GP)$, the uncoupled batch has reduced root Hessian $(9/G)I$ and condition number one. Nevertheless, producing classical descriptions of all G root Newton directions to entrywise error less than $1/200$ costs $\Theta(GN)$ queries, and sublinear constant-factor query budgets have $2^{-\Omega(G)}$ worst-case joint success.*

Proof. Using A_h from (201), direct elimination gives

$$\Delta Z_g = -\frac{1}{90}(H_{12} + H_{23} + h_g H_{13}).$$

Every full physical step remains positive definite, with minimum eigenvalue at least $14/45$. The hidden 13 entry has magnitude $1/90 > 1/200$, so its sign reveals h_g . Apply theorem 8.12. $\square$

These batch results concern raw queries to the displayed signed copy rows. An oracle supplying block parities, aggregate gradients, or component optima has already performed the hard composition. The vector theorem concerns simultaneous classical materialization, not one normalized global quantum direction state; the latter can hide individual entries by amplitude dilution, as remark 7.1 shows. Finally, the claims are for the exact equalities: the path operator has small singular values, so no constant-residual robustness statement is implicit.

8.4 Remaining boundaries

Three boundaries remain. First, the winner-mass decoder is insensitive to the internal-state discrepancy η_{int} . Determining the minimax radius and query complexity of this internal-state channel, especially when p is below the condensation threshold, is open. Second, open problem 8.11 asks for $\sqrt{\delta}$ composition beyond parity. Third, the sliver $\delta = o(k/G)$ is deliberately excluded from the tight laws: public uniform-label outputs can already satisfy some such contracts, and the exact boundary depends on the internal states rather than winner mass alone.

9 LP Loading and Recovery Lower Bounds

This section gives the LP thread of the paper. Its endpoint is a linear-size, bounded-degree standard-form LP whose reduced primal Newton matrix is a scalar identity at a public infeasible start, whose reduced log-barrier Hessian is a scalar identity along the entire central path, and

whose original-primal Newton-direction state nevertheless needs a linear number of raw coefficient queries. The obstruction is not the scalar solve: it is construction of the hidden affine translation and recovery in the original coordinates.

All state, access, residual, and success conventions are those of [sections 2.2](#) and [2.3](#). Throughout this section $N \geq 2$. In particular, a residual without another qualifier is the global RHS-relative residual $\|Ax - b\|_2/\|b\|_2$, and an output state is unconditional. The same results hold for a heralded branch of input-independent constant success probability when repetitions are charged. A reduced-coordinate state is not an original-coordinate output, and an unobserved “good branch” is not an output contract. We count all raw queries used for setup, preconditioning, right-hand-side formation, solution, and recovery.

The underlying amplification motifs are established. Parity and the polynomial method are classical query-complexity tools [\[20\]](#); endpoint padding and biased clocks occur in HHL and Feynman–Kitaev constructions [\[19, 34, 76\]](#); and sparse QLS states carrying an endpoint bit occur in Wang–Zhang and Mori et al. [\[109, 133\]](#). Apers–Gribbling Theorem 8.4 already gives a linear exponent for sparse-LP *value* estimation in balanced dimensions [\[10\]](#). None of these results is a constant-condition QLS lower bound of the kind sometimes informally inferred from the statements below. Our contribution is the LP-native conjunction of regularity, reduced geometry, original-coordinate output contracts, sharp residual thresholds, and complete access accounting.

9.1 Parity amplification and feasible-state hardness

Let $\sigma_1, \dots, \sigma_N \in \{\pm 1\}$, and put

$$p_0 = 1, \quad p_i = \prod_{j=1}^i \sigma_j.$$

Represent a signed scalar at node j by a nonnegative pair (u_j, v_j) , with $d_j = u_j - v_j$ and $q_j = u_j + v_j$. A chain fixes

$$d_0 = 1, \quad d_i - \sigma_i d_{i-1} = 0 \quad (1 \leq i \leq N). \quad (265)$$

At node N , attach a complete binary copy tree with

$$M = 2^{\lceil \log_2(8(N+1)) \rceil}, \quad K = 2M - 2, \quad P = N + 1 + K, \quad (266)$$

and impose $d_j - d_{\pi(j)} = 0$ on every new node. Thus all K copy nodes have difference p_N , while row and column sparsity remain at most four and three.

We use the following elementary decoder throughout the section.

Lemma 9.1 (Fixed-observable parity decoder). *Let O be an input-independent Hermitian observable with $\|O\|_2 \leq 1$. If a family of normalized ideal states obeys*

$$p_N \langle \psi_\sigma | O | \psi_\sigma \rangle \geq \delta > 0,$$

then the POVM $(I \pm O)/2$ guesses p_N with bias at least $\delta/2$. From any ρ_σ with $D_{\text{tr}}(\rho_\sigma, |\psi_\sigma\rangle\langle\psi_\sigma|) \leq \epsilon$, the bias is at least $(\delta - 2\epsilon)/2$. If this is positive by an absolute constant, preparing ρ_σ requires $\Omega(N)$ raw coefficient queries.

Proof. Trace duality changes the expectation of O by at most 2ϵ . The stated POVM has signed expectation $\text{Tr}(O\rho)$. Constant repetition turns any constant positive bias into bounded error. Every sparse row, column, position, or value query in the families below exposes at most one σ_i , even coherently, and is simulated by one call to the sign oracle. The parity lower bound completes the reduction. $\square$

First minimize $\sum_j q_j$ subject to the chain and copy equations. Exact feasibility fixes every $d_j \in \{\pm 1\}$, and $q_j \geq 1$. Hence the unique optimum has $q_j = 1$ and value P . The nullspace has the public orthonormal basis

$$V_j = \frac{e_{u_j} + e_{v_j}}{\sqrt{2}}. \quad (267)$$

On the feasible affine space the barrier objective is

$$\Phi_\mu(q) = \sum_j \left[q_j - \mu \log \frac{q_j^2 - 1}{4} \right]. \quad (268)$$

Thus every central coordinate is $q(\mu) = \mu + \sqrt{\mu^2 + 1}$, and

$$V^T \nabla^2 \Phi_\mu V = 2\mu [(q+1)^{-2} + (q-1)^{-2}] I. \quad (269)$$

At $\mu = 3/4$, $q = 2$, so every oriented pair is $(3/2, 1/2)$. If instead an exactly feasible point has objective gap at most one, writing $q_j = 1 + \delta_j$ gives $\sum_j \delta_j \leq 1$ and

$$\|x\|_2^2 = \sum_j \frac{q_j^2 + 1}{2} \leq P + \frac{3}{2}. \quad (270)$$

Since $K \geq 16(N+1) - 2$, more than 9/10 of this mass lies in the copy tree, and within an oriented copy pair the correct coordinate has at least 9/10 of the pair mass. The computational-basis decoder therefore succeeds with probability greater than 43/50; trace error 1/10 leaves success greater than 19/25. This proves an $\Omega(N)$ lower bound both for every such near-optimal primal state and for the exact $\mu = 3/4$ central state, despite $\kappa_{\text{red}} = 1$.

There is a stronger any-feasible-point contract. Add $t_j \geq 0$ and

$$2u_j + 2v_j + 2t_j = 3 \quad (271)$$

at every node, and retain objective $\sum_j q_j$. Exact feasibility now gives

$$1 \leq q_j \leq \frac{3}{2}, \quad t_j = \frac{3}{2} - q_j. \quad (272)$$

The equality matrix has rank $2P$, and its public orthonormal null basis is

$$\bar{V}_j = \frac{e_{u_j} + e_{v_j} - 2e_{t_j}}{\sqrt{6}}. \quad (273)$$

The positive optimal columns form a basis; the opposite pair coordinate has dual slack two. Thus the LP is bounded and strictly primal–dual feasible, with a unique nondegenerate strictly complementary optimum. Its reduced central Hessian is again a scalar identity, because the reduced barrier is a sum of identical functions of q_j .

For $q \in [1, 3/2]$, one triple has squared norm

$$\bar{w}(q) = \frac{q^2 + 1}{2} + \left(\frac{3}{2} - q \right)^2 \in \left[\frac{5}{4}, \frac{13}{8} \right]. \quad (274)$$

The copy triples therefore carry more than 9/10 of the total mass, and the plus/minus decoder, randomized on t , succeeds conditionally with probability at least 9/10. Consequently *every* exact feasible amplitude state, without an objective promise, has the same $\Omega(N)$ query lower bound at trace error 1/10. The proof also survives $x \geq 0$ and the absolute global ℓ_1 residual $\|\bar{A}x - \bar{b}\|_1 \leq 1/50$: telescoping gives $p_N d_j \geq 49/50$ at every copy node, while the cap gives

$q_j + t_j \leq 151/100$. The resulting copy mass is greater than $43/50$, and the signed conditional expectation exceeds $63/100$. Before trace error the total signed expectation is therefore

$$\frac{43}{50} \frac{63}{100} = \frac{2709}{5000}, \quad (275)$$

so trace error $1/10$ leaves signed expectation at least

$$\frac{2709}{5000} - \frac{1}{5} = \frac{1709}{5000} > \frac{1}{3}.$$

9.2 The robustness frontier

The preceding ℓ_1 theorem cannot be promoted to a constant ℓ_2 theorem on one unit-height path. In gauged variables set

$$p_i d_i = 1 - \frac{i}{N}, \quad d_j = 0 \quad (j \text{ in the copy tree}), \quad (276)$$

and take $q_j = 1, t_j = 1/2$. All cap and copy equations are exact, the N chain residuals equal $1/N$, and the total residual is $N^{-1/2}$, yet the amplified coordinates have no parity signal. Using $1 - 2i/N$ instead gives an entirely wrong copy plateau at residual $2/\sqrt{N}$. This is the basic drift obstruction.

A bounded-height repair, for power-of-two N , uses $R = N$ length- N signed paths in parallel. A binary fanout tree feeds the paths, and a binary tree with $L = 16N$ leaves is attached to every endpoint. The resulting tree has

$$P = (2N - 1) + N^2 + N(2L - 2) = 33N^2 - 1, \quad |S| = 16N^2 \quad (277)$$

vertices and designated output leaves S . At every vertex use $(u, v, h, t) \geq 0$, impose pinned labelled-copy equations on $d = u - v$, pinned unsigned-copy equations on h , and $q + t - 2h = 0$. Minimize $\sum_j (2u_j + 2v_j + h_j + t_j)$.

Theorem 9.2 (Parallel-path residual tube). *This LP has $4P$ variables, $3P$ full-rank equalities, coefficients of magnitude at most two, and row and column sparsity at most four. It has the same regularity properties as the capped family and a scalar reduced central Hessian for every $\mu > 0$. If $x \geq 0$, $x \neq 0$, and*

$$\frac{\|Ax - b\|_2}{\|b\|_2} \leq \frac{1}{200}, \quad \|b\|_2 = \sqrt{2}, \quad (278)$$

then the output-coordinate decoder has ideal bias greater than $1/80$. Consequently, preparing an amplitude state within trace distance $1/100$ of any such point needs $\Omega(N) = \Omega(\sqrt{P})$ raw queries. Any extra two-sided objective or centrality promise is immaterial.

Proof. Let λ_j be the exact propagated tree label. Gauging removes all signs. Descendant counting in each binary tree and Cauchy–Schwarz on the N parallel paths give

$$\frac{\|\text{Diag}(\lambda)z - \mathbf{1}\|_2}{\sqrt{P}} \leq 12\|(z_r - 1, Dz)\|_2, \quad \frac{\|\text{Diag}(\lambda_S)z_S - \mathbf{1}\|_2}{\sqrt{|S|}} \leq 12\|(z_r - 1, Dz)\|_2. \quad (279)$$

The binary-level operator norms sum to $\sum_{\ell \geq 1} 2^{-\ell/2} = 1 + \sqrt{2} < 5/2$, while the path term is $\sqrt{N/R} = 1$. With $E = \|Ax - b\|_2 \leq \sqrt{2}/200$, (279) for d and h , plus the cap and nonnegativity, gives

$$\|x\|_2^2 < 11P, \quad \sum_{j \in S} \lambda_j d_j q_j > \frac{3}{5}|S|. \quad (280)$$

The two loss terms in the second inequality, divided by $|S|$, are at most

$$12E \left(1 + 2.18\sqrt{P/|S|}\right) < \frac{12\sqrt{2}}{200} \left(1 + 2.18\sqrt{33/16}\right) < \frac{2}{5}.$$

Hence the measurement bias is larger than

$$\frac{(3/5)16}{2 \cdot 11 \cdot 33} = \frac{8}{605} > \frac{1}{80}.$$

Trace error $1/100$ leaves positive constant bias, and Lemma 9.1 applies. Full rank and regularity follow by pivoting successively on the pinned d , pinned h , and private t blocks. The public null vectors are the local vectors in (286); identical feasible intervals make the central curvature scalar. $\square$

The quadratic volume is necessary in the signed-edge gadget class, not in all bounded-coefficient real-linear gadgets.

Proposition 9.3 (Effective-resistance volume bound). *Let a connected graph have maximum degree Δ , a pinned root, and outputs S . Suppose every consistent edge constraint $d_v - a_{uv}(\sigma)d_u = 0$ has $a_{uv} \in \{\pm 1\}$ depending on at most k input bits, and every output label is parity. Then there is a vector with root value one and wrong-parity value at every output whose residual is at most*

$$\frac{k\sqrt{2\Delta|V|}}{N}. \quad (281)$$

Thus excluding all wrong-output vectors below an absolute residual η requires $|V| > \eta^2 N^2 / (2\Delta k^2)$.

Proof. The product of edge labels on a root-to-output path is parity. Fourier degree is subadditive under products, so every such path has length at least N/k . Gauge the consistent labels away and consider a unit root-to- S flow. Removing circulations and decomposing into paths gives $\sum_e |f_e| \geq N/k$. Cauchy–Schwarz and Thomson’s principle yield

$$R_{\text{eff}}(r, S) \geq \frac{N^2}{k^2|E|}, \quad C_{\text{eff}}(r, S) \leq \frac{\Delta k^2|V|}{2N^2}.$$

The Dirichlet energy for voltage difference two is $4C_{\text{eff}}$; its harmonic minimizer lies in $[-1, 1]$. Taking the square root and undoing the gauge proves (281). The next section develops the general mixture and cut frontiers; only this signed-edge specialization is needed here. $\square$

There is a load-bearing escape witness. On one path replace the signed recurrence by $d_i - 2\sigma_i d_{i-1} = 0$, add a public reference $h_i - 2h_{i-1} = 0$, and retain $q_i + t_i - 2h_i = 0$. The exact scale is $h_i = |d_i| = 2^i$. With absolute residual $E = \|Ax - b\|_2$, rescaling by 2^{-i} gives

$$|2^{-i}p_i d_i - 1| \leq \left(1 + \frac{1}{\sqrt{3}}\right) E. \quad (282)$$

Put $K_0 = 1 + 1/\sqrt{3}$, and assume $E \leq 10^{-2}$. Applying the same estimate to the reference rows gives

$$p_i d_i \geq (1 - K_0 E) 2^i, \quad 0 \leq h_i \leq (1 + K_0 E) 2^i.$$

If $c_i = q_i + t_i - 2h_i$ is the cap residual, nonnegativity gives $q_i, t_i \leq 2h_i + |c_i|$. Therefore

$$\|x\|_2^2 \leq \sum_i [(q_i + t_i)^2 + h_i^2] \leq 12(1 + K_0 E)^2 4^N + 2E^2, \quad x_{\text{correct}, N} \geq (1 - K_0 E) 2^N. \quad (283)$$

The endpoint consequently has constant normalized mass (and ideal decoder bias greater than $3/100$ at $E = 10^{-2}$), giving a linear-size robust family. It has exponential range and poor conditioning, showing that bounded local coefficients alone do not imply the quadratic volume law. The next construction replaces the long gain chain by a short gain prefix and a polynomial plateau.

9.3 The linear-size gain-plateau family

Set

$$T = \left\lceil \frac{1}{2} \log_2 N \right\rceil, \quad K = 16N, \quad P = 17N + 1, \quad H_i = 2^{\min\{i, T\}}, \quad H = 2^T \in [\sqrt{N}, 2\sqrt{N}). \quad (284)$$

There are nodes $i = 0, \dots, N + K$, gains $g_i = H_i/H_{i-1} \in \{1, 2\}$, and labels $a_i = \sigma_i$ for $i \leq N$ and $a_i = 1$ afterward. Define

$$\tau_0 = 1, \quad \tau_i = \prod_{j=1}^i a_j, \quad R_\tau = \text{diag}(\tau_0, \dots, \tau_{P-1}).$$

At every node introduce $(u_i, v_i, h_i, t_i) \geq 0$, put $d_i = u_i - v_i$, $q_i = u_i + v_i$, and impose

$$\begin{aligned} d_0 &= 1, & d_i - g_i a_i d_{i-1} &= 0, \\ h_0 &= 1, & h_i - g_i h_{i-1} &= 0, \\ q_i + t_i - 2h_i &= 0. \end{aligned} \quad (285)$$

The base objective is $\sum_i (2u_i + 2v_i + h_i + t_i)$. Its public orthonormal null basis is

$$W_i = \sqrt{\frac{2}{3}} \left(\frac{1}{2} e_{u_i} + \frac{1}{2} e_{v_i} - e_{t_i} \right). \quad (286)$$

Theorem 9.4 (Linear robust primal-state lower bound). *The LP (285) has $4P$ variables, $3P$ full-rank equalities, coefficients of magnitude at most two, and row and column sparsity at most four and three. It is bounded and strictly primal-dual feasible, with a unique nondegenerate strictly complementary optimum. The optimum is $\Theta(N^{3/2})$, its largest coordinate is $\Theta(\sqrt{N})$, and its Euclidean norm is $\Theta(N)$.*

For every nonzero $x \geq 0$ with

$$\text{res}_p(x) \leq \frac{1}{100}, \quad (287)$$

the fixed plateau decoder has ideal bias greater than $1/30$. Preparing its original-primal state to trace error $1/100$ therefore needs $\Omega(N) = \Omega(P)$ raw queries. Moreover,

$$\kappa_{\text{red}}(\mu) < \frac{7}{6} \quad (0 < \mu \leq 1/16). \quad (288)$$

Proof. The pinned signed and unsigned bidiagonal blocks fix $d_i = H_i \tau_i$ and $h_i = H_i$; private t_i columns finish the rank proof. Exact feasibility is

$$H_i \leq q_i \leq 2H_i, \quad t_i = 2H_i - q_i. \quad (289)$$

The local objective is $q_i + 3H_i$, so the unique optimum has $q_i = t_i = H_i$. The sign-selected pair member together with h_i, t_i forms a triangular positive basis, and the opposite coordinate has slack two. This proves all regularity claims. Since there are $\Theta(N)$ nodes of height $H = \Theta(\sqrt{N})$, the three scale estimates follow.

Write a central coordinate as $q_i = H_i + z_i$. Stationarity and curvature are

$$\frac{1}{\mu} = \frac{1}{2H_i + z_i} + \frac{1}{z_i} - \frac{1}{H_i - z_i}, \quad f_i'' = \mu [(2H_i + z_i)^{-2} + z_i^{-2} + (H_i - z_i)^{-2}]. \quad (290)$$

For $\mu \leq 1/16$, $15\mu/16 \leq z_i < \mu$, whence $1/\mu < f_i'' < 7/(6\mu)$. Changing to the orthonormal coordinates (286) multiplies every eigenvalue by $2/3$, proving (288).

For robustness put $E = \|Ax - b\|_2 \leq \sqrt{2}/100$. The weighted resistance is

$$\sum_{i \geq 1} H_i^{-2} < \frac{1}{3} + \frac{17N}{H^2} \leq \frac{52}{3}.$$

Weighted telescoping gives, simultaneously at every node,

$$\left| \frac{\tau_i d_i}{H_i} - 1 \right| < \frac{3}{40}, \quad \left| \frac{h_i}{H_i} - 1 \right| < \frac{3}{40}. \quad (291)$$

The cap then gives $q_i, t_i < 2.165H_i$, and

$$\|x\|_2^2 < 11 \sum_i H_i^2 < \frac{583}{3} NH^2. \quad (292)$$

On the $K = 16N$ output nodes, $p_N d_i q_i > (37H/40)^2$. The computational-basis decoder has bias

$$\frac{16N(37/40)^2 H^2}{2(583/3)NH^2} > \frac{1}{30}. \quad (293)$$

Trace error $1/100$ leaves bias greater than $7/300$, and Lemma 9.1 completes the proof. $\square$

The early-path conditioning claim cannot be strengthened: for $\mu/H \rightarrow \infty$, local curvature scales as μ/H_i^2 , so the unrigidified condition ratio can reach $\Theta(H^2) = \Theta(N)$. The gain is also the precise escape from the quadratic path-bundle law. If R internally disjoint length- N paths have reference height at most H , a harmonic wrong-endpoint interpolation has squared residual at most $4RH^2/N$. Therefore exclusion below absolute residual η requires

$$RH^2 > \frac{\eta^2 N}{4}, \quad P = \Theta(RN) \implies PH^2 = \Omega(N^2). \quad (294)$$

Both the bounded-height parallel construction and the linear-volume gain-plateau construction attain this frontier. The general gadget-class mixture and cut theorems are given in section 10; (294) is the only path-bundle statement used here. Naive balanced-XOR/PCP encodings do not improve it: boundary truth-table points obstruct strict feasibility, and homogenized shallow circuits still admit fractional drift across a small cut.

For one static input, Theorem 9.4 gives the tight total ledger

$$Q_{\text{prep}} + \sum_{t=1}^R (Q_{\text{form},t} + Q_{\text{solve},t} + Q_{\text{recover},t}) + Q_{\text{out}} = \Omega(P). \quad (295)$$

The bound is total, not per iteration. Querying all N signs and caching their prefixes costs $O(P)$ and makes every later raw query free, so no $\Omega(RP)$ fixed-instance theorem is possible.

9.4 Reduced geometry versus the affine offset

All reduced geometry in the gain-plateau family is public. Every exact feasible point has the form

$$x_\sigma(q)_i = \left(\frac{q_i + H_i \tau_i}{2}, \frac{q_i - H_i \tau_i}{2}, H_i, 2H_i - q_i \right), \quad (296)$$

and

$$\{x : A_\sigma x = b\} = x_\sigma^{\text{off}} + \text{range}(W), \quad (x_\sigma^{\text{off}})_i = H_i \left(\frac{1 + \tau_i}{2}, \frac{1 - \tau_i}{2}, 1, 1 \right). \quad (297)$$

Even the minimum-norm offset is hidden: minimizing the local squared norm in (296) gives $A_\sigma^\dagger b = x_\sigma(4H_i/3)$. By contrast, the full reduced barrier, all derivatives, central scalars, W , both projectors, and the diagonal reduced Hessian are independent of σ . For $\mu \leq 1/16$, its eigenvalues lie in $(2/(3\mu^2), 7/(9\mu^2))$ in the unscaled log-barrier convention of definition 2.9; the μ -weighted Hessian $W^\top \nabla^2 \Phi_\mu W$ has eigenvalues in $(2/(3\mu), 7/(9\mu))$.

Corollary 9.5 (Affine-offset oracle dichotomy). *Grant arbitrary exact input-independent reduced-geometry oracles for free. For the gain–plateau family, loading any original-primal state satisfying (287) costs $\Omega(P)$, while the free reduced Hessian has condition below $7/6$ on the late tail. For the parallel-path family, the analogous cost is $\Omega(\sqrt{P})$, while the reduced Hessian is an exact scalar identity for the entire central path. In both cases the hidden object is the affine offset, not the reduced solution.*

Proof. Every granted channel is the same for all σ and can be hardwired in the parity reduction. Theorem 9.4 and Theorem 9.2 then apply unchanged. A free feasible offset, prefix table, or loading isometry would not be input-independent: one known output pair reveals p_N , and a normalized offset state carries a constant amount of output-pair mass. $\square$

9.5 Exact Newton secants

The base objective in the gain–plateau family gives a uniformly conditioned late central tail, but its exact public-start construction initially used different complementarity parameters on the $O(\log N)$ gain nodes. A public reweighting removes that qualification. At height H_i , replace the local objective by

$$(1 + \omega_i)(u_i + v_i) + h_i + t_i, \quad \omega_i = \frac{7}{36H_i}. \quad (298)$$

On the feasible line its nonconstant part is $\omega_i q_i$, so the unique optimum and all regularity properties survive; the nonbasic reduced cost is $2\omega_i > 0$.

Set

$$\mu_1 = \frac{1}{16}, \quad x_i^1 = H_i \left(\frac{9}{8}, \frac{1}{8}, 1, \frac{3}{4} \right) \quad (299)$$

up to the hidden pair swap, and put $s_j^1 = \mu_1/x_j^1$. The reduced stationarity equation at $z_i = H_i/4$ is exact because

$$\frac{\omega_i}{\mu_1} = \frac{28}{9H_i} = \frac{1}{2H_i + H_i/4} + \frac{1}{H_i/4} - \frac{1}{H_i - H_i/4}. \quad (300)$$

Thus (x^1, y^1, s^1) is the genuine global μ_1 -central point.

Use the public start

$$\begin{aligned} x_i^0 &= H_i(5/6, 5/6, 20/27, 5/9), \\ s_i^0 &= H_i^{-1}(10/27, 10/27, 5/12, 5/9), \quad y^0 = 0. \end{aligned} \quad (301)$$

Every coordinate product is

$$\mu_0 = \frac{25}{81}, \quad \hat{\mu} = \frac{25}{162} = \frac{\mu_0}{2}. \quad (302)$$

Every Newton right-hand side is public, because the pair differences at the start vanish. For either hidden pair orientation,

$$\frac{10}{27H_i} \frac{9H_i}{8} + \frac{5H_i}{6} \frac{1}{18H_i} = \frac{10}{27H_i} \frac{H_i}{8} + \frac{5H_i}{6} \frac{1}{2H_i} = \frac{25}{54}. \quad (303)$$

The h, t coordinates give the same $25/54$. Hence

$$S^0(x^1 - x^0) + X^0(s^1 - s^0) = \left(\frac{25}{54} - 2\frac{25}{81} \right) e = (\hat{\mu} - \mu_0)e. \quad (304)$$

Primal and dual endpoint feasibility provide the other two rows of the standard system (14); full row rank makes the correction unique. One full algebraic Newton step therefore reaches the genuine global $\mu_1 = 1/16$ center. The endpoint parameter differs from $\hat{\mu}$ because the Newton equation omits $\Delta X \Delta s$.

Up to the hidden swap, the primal direction block is

$$\Delta x_i = H_i(7/24, -17/24, 7/27, 7/36), \quad \|\Delta x_i\|_2^2 = DH_i^2, \quad D = \frac{16139}{23328}. \quad (305)$$

The unselected-minus-selected squared pair gap is $5H_i^2/12$ (the unselected coordinate carries the larger direction mass, $17H_i/24$ against $7H_i/24$). Since $\sum_i H_i^2 < (17N + 4/3)H^2$, the fixed output-copy decoder has bias

$$\frac{16N(5/12)H^2}{2D(17N + 4/3)H^2} > \frac{1}{4}. \quad (306)$$

Thus the exact original-primal direction state at trace error $1/100$ needs $\Omega(P)$ raw queries.

In the unscaled log-barrier convention of [definition 2.9](#), the μ_1 -scaled endpoint reduced Hessian and the start reduced Newton matrix are

$$\mu_1 W^T \nabla^2 F(x^1) W = \frac{182}{243} \text{diag}(H_i^{-2}), \quad W^T (X^0)^{-1} S^0 W = \frac{22}{27} \text{diag}(H_i^{-2}). \quad (307)$$

Thus all but the $T = O(\log N)$ public gain modes form an exact scalar block. The public congruence $D_H = \text{diag}(H_i)$ makes both matrices scalar; as a standard-form diagonal change of variables it raises the coefficient scale to $\Theta(\sqrt{N})$. More generally, scaling by H_i^η gives

$$B_\eta = \Theta(H^\eta), \quad \kappa_\eta = H^{2(1-\eta)}, \quad B_\eta^2 \kappa_\eta = \Theta(N). \quad (308)$$

This is a property of this diagonal family, not a universal normalization-condition theorem.

The remaining outliers are forced inside this local ansatz.

Proposition 9.6 (Four-variable-node incompatibility). *Consider hidden-swappable endpoint pairs with values $L_i > S_i > 0$ at one common central parameter μ_1 , and suppose their differences are the prescribed heights $L_i - S_i = H_i$. Within the separable four-variable node gadget, the following three requirements are incompatible: a public start centered coordinatewise at one μ_0 ; one full standard Newton correction with a common target that works for both hidden orientations; and an $O(1)$ -conditioned endpoint Hessian in the raw orthonormal null basis. This statement does not cover added rows, weighted barriers, or other local gadgets.*

Proof. Write the two public start values as a_i, b_i , with slacks $\mu_0/a_i, \mu_0/b_i$. Equality of the first-coordinate secant cross terms for the two hidden orientations requires

$$\frac{\mu_0 L_i}{a_i} + \frac{a_i \mu_1}{L_i} = \frac{\mu_0 S_i}{a_i} + \frac{a_i \mu_1}{S_i}.$$

Since $L_i \neq S_i$, this gives $a_i^2 = \mu_0 L_i S_i / \mu_1$; the second coordinate similarly gives $b_i = a_i$. The common cross term is therefore

$$C_i = \sqrt{\mu_0 \mu_1} \frac{L_i + S_i}{\sqrt{L_i S_i}}.$$

One global Newton target makes C_i independent of i , hence the aspect ratio L_i/S_i is constant. The cap row $t_i = 2H_i - (L_i + S_i) > 0$ makes $L_i + S_i$ proportional to H_i and forces that constant aspect ratio to exceed three. Together with $L_i - S_i = H_i$, this makes both endpoint values proportional to H_i . The log-barrier curvature in the local orthonormal null direction consequently scales as H_i^{-2} , giving condition ratio $H^2 = \Theta(N)$. $\square$

The reweighted construction also exposes an important negative result. A fixed relative residual for the full Newton system alone does not force the primal direction ray. On the K output nodes let $w = \sum_{i \in S} W_i$, and perturb the exact solution by

$$e_x = Mw, \quad e_y = 0, \quad e_s = -(X^0)^{-1} S^0 e_x. \quad (309)$$

The primal and complementarity residual perturbations vanish. On the plateau, direct substitution in (301) gives the sharp bound

$$\|(X^0)^{-1}S^0w\|_2 \leq \frac{\|w\|_2}{H^2}$$

(the exact coefficient is below $0.856/H^2$), whereas the public cap RHS has norm $\Omega(H\sqrt{K})$. With $M = H^2$, the relative full-KKT residual is $O(H^{-1})$ and the normalized primal direction converges to the public zero-query state $|w/\|w\|\rangle$. The perturbed update has negative t -coordinates. Therefore the robust direction theorem below must retain the geometric condition $x^0 + \widetilde{\Delta}x \geq 0$; a bare fixed full-KKT residual promise is insufficient.

9.5.1 Prefix rigidification: the condition-one capstone

Return to the constraints (285). At each of the T small-height nodes $i < T$, add the public row

$$q_i - \frac{5}{4}h_i = 0, \quad (310)$$

and use the uniform objective

$$\sum_i [(1 + \omega)(u_i + v_i) + h_i + t_i], \quad \omega = \frac{7}{36H}. \quad (311)$$

Write the augmented matrix and RHS as A'_σ, b' .

Theorem 9.7 (Condition-one original-primal Newton hardness). *The prefix-rigidified family has $4P$ variables and $3P + T$ full-rank equalities. Coefficients have magnitude at most two, and row and column sparsity are at most four. It is bounded and strictly primal-dual feasible, with a unique nondegenerate strictly complementary optimum. In the raw orthonormal reduced coordinates,*

$$\kappa_{\text{red}}(\mu) = 1 \quad \text{for every } \mu > 0, \quad (312)$$

and the reduced Newton matrix at the public start (301) is also exactly condition one.

The unique standard Newton correction from that start with target $\hat{\mu} = 25/162$ lands after one full step at the genuine global $\mu_1 = 1/16$ center (299). Its original-primal direction state has decoder bias greater than $1/4$, so trace error $1/100$ requires $\Omega(N) = \Omega(P)$ raw queries.

The same lower bound holds for a state of every nonzero $x \geq 0$ satisfying $\text{res}_p(x) \leq 1/100$, and for every nonzero direction $\widetilde{\Delta}x$ whose update is nonnegative and obeys that residual, at state trace error $1/100$. In the latter case a fixed interference decoder has bias greater than $1/54$.

Proof. For the row r_i in (310) and the old null vectors,

$$r_i^T W_j = \sqrt{2/3} \delta_{ij}. \quad (313)$$

Hence the new rows are independent modulo the old row space and

$$\ker A'_\sigma = \text{span}\{W_i : i \geq T\}. \quad (314)$$

They fix $q_i = 5H_i/4$ on the prefix, leaving only equal-height H coordinates free. At the optimum all four prefix variables and three variables at every free node are positive, exactly $3P + T$ in total. A null vector supported on them must vanish because every nonzero free W_i touches that node's zero pair coordinate. The positive columns form a basis, and each nonbasic reduced cost is $2\omega > 0$. This proves rank and regularity.

All surviving nodes have the same height, objective slope, central scalar, and local curvature. Equation (314) therefore gives

$$W^T \nabla^2 F(x(\mu)) W = \lambda(\mu) I_{P-T}.$$

At $\mu_1 = 1/16$, the free coordinate is $q = 5H/4$, while the fixed prefix coordinates already have the same local template. Positive slacks $s^1 = \mu_1(X^1)^{-1}e$ are orthogonal to the surviving kernel after subtracting the objective, so full row rank supplies the central multiplier. The secant identities (303)–(304) are unchanged, and the endpoint satisfies the new rows. This proves the exact Newton statement. At the start,

$$W^T(X^0)^{-1}S^0W = \frac{22}{27H^2}I_{P-T}. \quad (315)$$

The fraction-to-boundary maximum is exactly $\alpha_{\max} = 20/17 > 1$, so the full step preserves strict positivity; any safety factor at least $17/20$ permits it. The direction formula (305) survives and (306) proves the exact-state lower bound.

For a robust feasible state, deleting the public new residual coordinates gives the base residual bound with the same RHS norm $\|b'\|_2 = \sqrt{2}$; Theorem 9.4 applies. For a robust direction, set $\tilde{x} = x^0 + \tilde{\Delta}x \geq 0$. On every plateau node the same telescoping estimates give

$$\tau_i \tilde{d}_i > \frac{37}{40}H, \quad \frac{37}{40}H < \tilde{h}_i < \frac{43}{40}H, \quad \tilde{q}_i, \tilde{t}_i < \frac{433}{200}H. \quad (316)$$

Together with the public start, these imply

$$\|\tilde{\Delta}x\|_2^2 < 100NH^2, \quad \tilde{\Delta}h_i > \frac{199}{1080}H \quad (i \in S). \quad (317)$$

Let $|g_i\rangle = (|u_i\rangle - |v_i\rangle)/\sqrt{2}$, and use the public norm-one observable

$$O = \bigoplus_{i \in S} (|h_i\rangle\langle g_i| + |g_i\rangle\langle h_i|). \quad (318)$$

Equations (316)–(317) give signed expectation greater than

$$\frac{7363\sqrt{2}}{270000} > \frac{1}{27}, \quad (319)$$

so its associated binary POVM has bias greater than $1/54$. Trace error $1/100$ leaves positive constant bias. Lemma 9.1 finishes the proof. $\square$

The condition-one claim is intentionally reduced-primal. At the same start, the raw OSS matrix $\mathcal{M}_0 = [-X^0(A')^T \ S^0W]$ and the eliminated augmented KKT matrix satisfy

$$\kappa_2(\mathcal{M}_0) \geq \sqrt{681/44}H^2, \quad \kappa_2(\mathcal{K}_0) \geq \sqrt{1701/178}H^2, \quad (320)$$

so both are $\Omega(P)$. Exact block preconditioning can reduce the second condition to φ^2 , but its difference-row Schur block has the gauge form

$$G_\sigma^{(d)} = R_\tau G_+^{(d)} R_\tau, \quad (G_\sigma^{(d)})^{-1/2} = R_\tau (G_+^{(d)})^{-1/2} R_\tau. \quad (321)$$

The all-positive block is an irreducible tridiagonal Stieltjes matrix, whose inverse square root is entrywise strictly positive by its resolvent integral. Thus the sign of a sign-resolving preconditioner entry reveals a prefix product. A factor block encoding without entry access requires a separate implementation analysis; section 12 treats that factor-access frontier.

The access localization can be exact. In the Moore–Penrose gauge write

$$\Delta x = p_\sigma + Wz, \quad p_\sigma = (A'_\sigma)^\dagger (b' - A'_\sigma x^0). \quad (322)$$

On every free node,

$$(p_d, p_q, p_h, p_t) = H(\tau_i, -4/27, 7/27, -2/27), \quad (323)$$

with pair coordinates recovered from p_d, p_q . The complete reduced system is public:

$$M = \frac{22}{27H^2}I, \quad z_i = -\frac{29}{108}\sqrt{\frac{3}{2}}H, \quad Mz = g. \quad (324)$$

Normalized at $\alpha_M = 22/(27H^2)$, it has $M/\alpha_M = I$, $\kappa = 1$, and

$$\alpha_M \frac{\|M^{-1}g\|_2}{\|g\|_2} = 1. \quad (325)$$

The states of g, z, Wz are public and free. For the fixed diagonal-pair observable

$$O_S = \sum_{i \in S} (|u_i\rangle\langle u_i| - |v_i\rangle\langle v_i|),$$

each output node contributes $u_i^2 - v_i^2 = p_d p_q = -(4/27)p_N H^2$. A free-node block has squared norm $(851/1458)H^2$, a prefix block has squared norm DH_i^2 with $D = 16139/23328 < 7/10$, and $\sum_i H_i^2 < (53/3)NH^2$. Hence

$$\left| \left\langle \frac{p_\sigma}{\|p_\sigma\|_2}, O_S \frac{p_\sigma}{\|p_\sigma\|_2} \right\rangle \right| > \frac{(64/27)NH^2}{(7/10)(53/3)NH^2} = \frac{1920}{10017} > \frac{1}{6}. \quad (326)$$

Its sign is $-p_N$, so preparing $|p_\sigma/\|p_\sigma\|\rangle$ to trace error $1/100$ costs $\Theta(P)$ raw queries. Because $p_\sigma \perp Wz$, their coherent combination succeeds with probability at least $1/2$, provided one has phase-referenced coherent preparation access to both normalized summands. A density-state contract alone does not supply that access, as [section 2.3](#) records. The localization is gauge-specific: an arbitrary particular solution can trade a null component with z ; the invariant claim is the end-to-end original-coordinate lower bound.

The capstone has the same tight static ledger [\(295\)](#). Its matching upper bound caches all signs in N queries. A genuine iteration multiplier is valid only for a streaming service in which round t releases a fresh independent sign block after output $t-1$ is committed. Before the next block is released, round t must return an unconditional state within trace distance $1/100$ of that round's hard original-coordinate Newton direction (or of a point under the robust contract in [theorem 9.7](#)). Let Q_t count every call to the current block's coefficient oracle or inverse, all setup after that block is released, and all queries used to prepare classical or quantum advice consumed in that round. Conditioned on all past memory, the new block is independent, so [theorem 9.7](#) applies in every round and

$$\sum_{t=1}^R Q_t = \Omega(RN) = \Omega(RP). \quad (327)$$

This is a freshness direct sum, not a per-iteration theorem for one fixed LP.

9.6 Full-KKT and preconditioned formulations

The preceding results request only the original-primal block. A related construction makes the complete primal-dual direction hard while keeping its unpreconditioned right-hand side public. Use the unrigidified gain-plateau constraints [\(285\)](#), the base objective $c_i = (2, 2, 1, 1)$, and the public start

$$x^0 = s^0 = e, \quad y^0 = 0, \quad \mu^0 = 1. \quad (328)$$

For a public centering target $\theta \in [0, 1]$, the Newton equations are

$$A_\sigma \Delta x = b - A_\sigma e, \quad A_\sigma^T \Delta y + \Delta s = c - e, \quad \Delta x + \Delta s = (\theta - 1)e. \quad (329)$$

The RHS is input-independent: every hidden coefficient multiplies the zero pair difference at the start.

Let $R_{r,j} = \sum_{i=j}^{N+K} H_i^r$. Direct elimination and backward substitution give

$$\begin{aligned} \Delta x_{u_i} &= (4H_i - 8 + 3H_i\tau_i)/6, & \Delta x_{v_i} &= (4H_i - 8 - 3H_i\tau_i)/6, \\ \Delta x_{h_i} &= H_i - 1, & \Delta x_{t_i} &= (2H_i - 1)/3, \\ \alpha_j &= \tau_j R_{2,j}/(2H_j), & \beta_j &= [7R_{2,j} + (4 - 9\theta)R_{1,j}]/(3H_j), \\ \gamma_j &= (2H_j + 2)/3 - \theta. \end{aligned} \tag{330}$$

Here $(\alpha, \beta, \gamma) = \Delta y$ correspond to the signed, reference, and cap rows.

Theorem 9.8 (Public-RHS full-KKT direction). *Let $w_\sigma = (\Delta x, \Delta y, \Delta s)$ solve (329). Preparing its normalized state to trace distance $1/100$ takes at least $N/2$ raw coefficient queries. The same holds if an approximate vector obeys $\|\tilde{w} - w_\sigma\|_2 \leq \|w_\sigma\|_2/300$ and its state has trace error at most $1/300$.*

For $\theta = 1$, the two-block elimination has an explicit canonical sparse block encoding of normalization six and a public RHS state. The same $N/2$ lower bound holds for calls to that fully specified block encoding, not merely to its principal matrix block.

Proof. On the K output rows define $A_S = H^2 \sum_{\ell=1}^K \ell^2/4$. The height sums and (330) imply

$$\|\alpha\|_2^2 < 3A_S, \quad \|\beta\|_2^2 < \frac{484}{3}A_S, \quad \|w_\sigma\|_2^2 < 165A_S. \tag{331}$$

Moreover α_j has sign p_N and $\beta_j > 0$ on the output suffix, with $\beta_j/|\alpha_j| \geq 3$. The public norm-one observable that swaps the corresponding α_j, β_j coordinates therefore has parity-signed expectation

$$p_N \langle O_{\text{KKT}} \rangle > \frac{6}{165} = \frac{2}{55} > \frac{1}{30}. \tag{332}$$

Trace error $1/100$ changes this by at most $1/50$. The resulting expectation is a real polynomial of degree at most twice the query count; its nonzero parity Fourier coefficient forces degree at least N . Thus the query count is at least $N/2$. The triangle inequality for normalized rays gives the approximate-vector statement.

At $\theta = 1$, eliminate $\Delta s = -\Delta x$, put $y' = -\Delta y$, and write

$$\mathcal{K}_\sigma = \begin{bmatrix} I & A_\sigma^T \\ A_\sigma & 0 \end{bmatrix}, \quad r_\sigma = \begin{pmatrix} e - c \\ b - A_\sigma e \end{pmatrix}. \tag{333}$$

The maximum absolute row and column sums are six. The standard magnitude-weighted row/column state-pair construction gives an exact normalization-six block encoding. Its support, magnitudes, failure arms, and left preparation are public; the right preparation uses one local sign phase, hence exactly one sign query per call, adjoint, or controlled call. Also $\|r_\sigma\|_2^2 = 3P + T + 1$, and its preparation is public. Removing Δs can only strengthen (332), so the same degree argument applies to this complete oracle implementation. $\square$

Ideal Schur preconditioning makes the matrix well conditioned but does not make the end-to-end raw-input task easy. Put

$$\mathcal{S}_\sigma = A_\sigma A_\sigma^T, \quad Q_\sigma = \mathcal{S}_\sigma^{-1/2} A_\sigma, \quad P_\sigma = \text{diag}(I, \mathcal{S}_\sigma^{-1/2}).$$

Define

$$\bar{w}_\sigma = P_\sigma^{-1} \begin{pmatrix} \Delta x \\ y' \end{pmatrix}, \quad \bar{r}_\sigma = P_\sigma r_\sigma, \quad \bar{K}_\sigma = P_\sigma \mathcal{K}_\sigma P_\sigma.$$

Then

$$\bar{K}_\sigma = \begin{bmatrix} I & Q_\sigma^T \\ Q_\sigma & 0 \end{bmatrix}, \quad \bar{K}_\sigma \bar{w}_\sigma = \bar{r}_\sigma, \tag{334}$$

and the primal block of $\bar{w}_\sigma$ is exactly Δx . Since $Q_\sigma Q_\sigma^T = I$, the spectrum is 1 on the nullspace and

$$\frac{1 + \sqrt{5}}{2}, \quad \frac{1 - \sqrt{5}}{2} \quad (335)$$

on every coupled two-plane. Therefore

$$\|\bar{K}_\sigma\|_2 = \|\bar{K}_\sigma^{-1}\|_2 = \varphi, \quad \kappa_{\text{abs}}(\bar{K}_\sigma) = \varphi^2 = \frac{3 + \sqrt{5}}{2}. \quad (336)$$

This is the classical ideal-Schur spectrum of Murphy–Golub–Wathen [110].

Theorem 9.9 (Residual-robust Schur-preconditioned KKT state). *Suppose $\tilde{w} \neq 0$ obeys*

$$\|\bar{K}_\sigma \tilde{w} - \bar{r}_\sigma\|_2 \leq 10^{-3} \|\bar{r}_\sigma\|_2. \quad (337)$$

Preparing its normalized state to trace error $1/200$ takes at least $N/2$ total raw coefficient queries, counting preconditioner, transformed RHS, solve, and recovery access. The result transfers to the prefix-rigidified family.

Proof. Let

$$O_x = \sum_{i \in S} (|D_i\rangle\langle h_i| + |h_i\rangle\langle D_i|), \quad |D_i\rangle = \frac{|u_i\rangle - |v_i\rangle}{\sqrt{2}}.$$

This norm-one, primal-supported observable has exact parity-signed expectation greater than $2/87$ on the preconditioned solution. Indeed,

$$\|\Delta x\|_2^2 \leq \frac{11}{3} PH^2, \quad p_N \frac{\langle \Delta x | O_x | \Delta x \rangle}{\|\Delta x\|_2^2} > \frac{1}{6}, \quad \|\bar{w}\|_2^2 < \frac{29}{4} \|\Delta x\|_2^2. \quad (338)$$

Equations (336)–(337) give $\|\tilde{w} - \bar{w}\|_2 \leq 10^{-3} \varphi^2 \|\bar{w}\|_2$. The total target-state distance is at most

$$\frac{1}{200} + \frac{2\varphi^2}{1000} < \frac{1}{87}, \quad (339)$$

which preserves the sign of the observable. The polynomial argument gives $N/2$.

For the prefix-rigidified transfer, use the same all-ones start. Every added row $q_i - \frac{5}{4}h_i = 0$ has the public primal residual $-3/4$, and the uniform objective is

$$c'_i = (1 + \omega, 1 + \omega, 1, 1), \quad \omega = \frac{7}{36H}.$$

The hidden difference coefficients still multiply $u_i - v_i = 0$, so the complete Newton RHS remains public. On every free node, null projection and exact feasibility give

$$\Delta q_i = \frac{4H - 6 - 2\omega}{3}, \quad \Delta h_i = H - 1, \quad \Delta u_i - \Delta v_i = H\tau_i.$$

The prefix variables are fixed by feasibility. These formulas yield

$$\|\Delta x\|_2^2 < 5 \sum_i H_i^2, \quad p_N \frac{\langle \Delta x | O_x | \Delta x \rangle}{\|\Delta x\|_2^2} > \frac{1}{8},$$

because the numerator is $K\sqrt{2}H(H-1)$, each prefix node contributes less than $5H_i^2$, and each free node less than $4H^2$. For the exact block preconditioner the transformed primal RHS is $e - c' = (-\omega, -\omega, 0, 0)$ per node, and

$$\frac{\|e - c'\|_2}{\|\Delta x\|_2} < \frac{21}{100}, \quad \|\bar{w}_\sigma\|_2^2 < \frac{5}{2} \|\Delta x\|_2^2.$$

Thus the parity-signed expectation on the complete preconditioned state exceeds $1/20$. Its condition remains φ^2 , while the solve and output errors in (339) change the expectation by less than $2/87 < 1/20$. The same $N/2$ conclusion follows. $\square$

There is no matrix-only normalization/condition tradeoff behind this theorem. The row-space projector is public because $\ker A_\sigma$ is public. If Q_0 is a public row isometry with $Q_0^T Q_0 = I - WW^T$, then $O_\sigma = Q_\sigma Q_0^T$ is orthogonal and

$$\text{diag}(I, O_\sigma)^T \bar{K}_\sigma \text{diag}(I, O_\sigma) = \begin{bmatrix} I & Q_0^T \\ Q_0 & 0 \end{bmatrix}, \quad (340)$$

which is wholly public. Input dependence has moved to the transformed RHS and recovery map. Accordingly the invariant accounting statement is

$$Q_{\text{pre}} + Q_{\text{on}} \geq N/2. \quad (341)$$

It is tight: after N cached sign queries, later raw-query cost can vanish. The separate factor-access frontier is given in [section 12](#); it is not proved here.

9.7 Bounded data and the sharp residual scale

Remove the gain prefix by setting every height and gain to one, retain a length- N signed chain followed by $K = 16N$ public copy edges, and put $P = 17N + 1$. Use the same four nonnegative variables at every node and the equations

$$\begin{aligned} d_0 &= 1, & d_i - a_i d_{i-1} &= 0, \\ h_0 &= 1, & h_i - h_{i-1} &= 0, \\ q_i + t_i - 2h_i &= 0, \end{aligned} \quad a_i = \begin{cases} \sigma_i & i \leq N, \\ 1 & i > N. \end{cases} \quad (342)$$

Use objective

$$c^T x = \sum_i \left[\frac{43}{36} (u_i + v_i) + h_i + t_i \right]. \quad (343)$$

All matrix, RHS, objective, feasible-primal, and relevant slack coordinates are bounded by absolute constants; $\|b\|_2 = \sqrt{2}$. Exact feasibility is $d_i = \tau_i, h_i = 1, t_i = 2 - q_i, 1 \leq q_i \leq 2$, so the same basis (286) spans the nullspace.

Theorem 9.10 (Bounded exact-state separation). *The all-unit family is bounded and strictly primal-dual feasible, with a unique nondegenerate strictly complementary optimum. Its raw orthonormal reduced central Hessian has condition one for every $\mu > 0$, and its reduced Newton matrix at the public start below is $(22/27)I$.*

At $\mu_1 = 1/16$, one exact Newton correction from

$$x_i^0 = (5/6, 5/6, 20/27, 5/9), \quad s_i^0 = (10/27, 10/27, 5/12, 5/9), \quad y^0 = 0 \quad (344)$$

with $\mu_0 = 25/81$ and target $25/162$ lands at the exact center

$$x_i^1 = (9/8, 1/8, 1, 3/4), \quad s_i^1 = (1/18, 1/2, 1/16, 1/12), \quad (345)$$

up to the hidden pair swap. Preparing either the exact primal direction state or exact central-primal state to trace error $1/100$ requires $\Omega(N) = \Omega(P)$ raw queries.

Proof. On the feasible line the local objective is $3 + (7/36)q_i$, so the unique optimum has $q_i = 1$. The positive columns form a basis and the zero pair coordinate has reduced cost $7/18$. All central coordinates solve the same scalar equation

$$\frac{7}{36\mu} = \frac{1}{q+1} + \frac{1}{q-1} - \frac{1}{2-q},$$

so the reduced Hessian is scalar. At $q = 5/4, \mu = 1/16$, both sides are $28/9$, and direct evaluation gives the μ_1 -scaled reduced curvature $182/243$. Equations (303)–(304) with $H_i = 1$ prove the same exact Newton secant and $(22/27)I$ start matrix.

The primal direction is the constant block in (305). Its plateau decoder has bias

$$\frac{K(5/12)}{2DP} > \frac{1}{4}.$$

Each central block has squared norm $91/32$ and oriented pair gap $5/4$, so the central-state decoder has bias

$$\frac{K(5/4)}{2(91/32)P} = \frac{20K}{91P} > \frac{1}{5}.$$

Lemma 9.1 completes both lower bounds. $\square$

The all-unit family is robust, but only at the sharp vanishing scale.

Theorem 9.11 (Sharp all-unit residual threshold). *For every nonzero $x \geq 0$ satisfying the absolute residual bound*

$$\|A_\sigma x - b\|_2 \leq \frac{1}{16\sqrt{N}}, \quad (346)$$

the normalized primal state has constant correct-parity bias. Preparing it to trace error $1/100$ needs $\Omega(N) = \Omega(P)$ raw queries. In relative form, (346) is $1/(16\sqrt{2N})$; in particular it contains the $1/(100\sqrt{N})$ absolute residual tube.

The order is sharp: a nonnegative zero-signal plateau exists at absolute residual $1/\sqrt{N}$, and a nonnegative entirely wrong-parity plateau exists at absolute residual $2/\sqrt{N}$, even with exact optimal objective. After public homogeneous rigidification, the wrong witness can also have exact dual feasibility and exact coordinatewise complementarity.

Proof. Let $E = \|Ax - b\|_2$, $z_i = \tau_i d_i$, and $\delta = \sqrt{PE}$. Telescoping the signed and reference recurrences gives

$$|z_i - 1| \leq \sqrt{PE}, \quad |h_i - 1| \leq \sqrt{PE}. \quad (347)$$

The factor $\sqrt{P}$ is exact by equality in Cauchy–Schwarz. Under (346), $P/N \leq 35/2$ gives $\delta < 4/15$, so on every output node $p_N d_i > 11/15$. The cap and nonnegativity give $q_i, t_i < 31/12$, $h_i < 19/15$, and $\|x\|_2^2 < 15P$. Thus the output decoder has bias

$$\frac{K(11/15)^2}{30P} > \frac{1}{70}. \quad (348)$$

Trace error $1/100$ leaves bias at least $1/70 - 1/100 = 3/700 > 0$.

For the zero-signal witness take

$$z_i = 1 - i/N \ (i \leq N), \quad z_i = 0 \ (i > N), \quad (349)$$

and for the wrong-parity witness replace this by

$$z_i = 1 - 2i/N \ (i \leq N), \quad z_i = -1 \ (i > N). \quad (350)$$

Set $d_i = \tau_i z_i$ and $h_i = q_i = t_i = 1$. All reference and cap rows are exact. The N signed residuals are respectively $1/N$ and $2/N$, proving the claimed norms. Both points are nonnegative and have the exact optimal objective.

For the stronger failure, add $q_i - 5h_i/4 = 0$ on all non-output nodes and average the exact centers of the N inputs obtained by flipping one sign. The average has the wrong common output template and the same $2/\sqrt{N}$ primal residual. Every coordinate is positive. Setting $\bar{s} = \mu_1 \bar{X}^{-1}e$ gives exact complementarity. The remaining nullspace is supported on output nodes, where $\bar{x}$ is itself an exact central template; hence $c - \bar{s} \perp \ker A$, which supplies an exact dual multiplier. $\square$

The recurrence-only cutoff is exactly $E < 1/\sqrt{P}$: taking $z_i = 1 - (i + 1)/P$ makes all gauged residuals $-1/P$, reaches zero at the last node, and has norm $1/\sqrt{P}$. Inverting the robust scale over all hidden lengths L with $17L + 1 \leq P$ gives the worst-case accuracy law

$$Q(P, \epsilon) = \Omega(\min\{P, \epsilon^{-2}\}) \quad (351)$$

for absolute or RHS-relative residual, up to constants. This is a reparameterized family bound, not accumulating hardness at several accuracies on one instance.

The public copy plateau is necessary for constant one-copy distinguishability at unit height. Flipping only σ_N swaps exactly $K + 1$ target blocks. For the central and direction states the normalized overlaps are

$$1 - \frac{32(K + 1)}{91P}, \quad 1 - \frac{K + 1}{DP}, \quad (352)$$

respectively. If $K = o(N)$, their trace distance is $o(1)$ even for opposite parities. Hence the correct bounded-data resource is $K = \Theta(N)$ copies, not $\Theta(\sqrt{N})$ dynamic range.

9.8 Dual homogeneity and the bounded full direction

There is one exact way to suppress the multiplier accumulation of a unit-height chain. Positive dual scaling is an elementary symmetry of the central and Newton equations, consistent with the general scale-invariance viewpoint in interior-point theory [129].

Lemma 9.12 (Dual homogeneity). *In the standard Newton system, leave (A, b, x^0) fixed and replace*

$$(c, y^0, s^0, \hat{\mu}) \mapsto (\lambda c, \lambda y^0, \lambda s^0, \lambda \hat{\mu}), \quad \lambda > 0. \quad (353)$$

Then $(\Delta x, \Delta y, \Delta s)$ is replaced by $(\Delta x, \lambda \Delta y, \lambda \Delta s)$. The primal central trajectory is unchanged, with μ reparameterized as $\mu' = \lambda \mu$.

Proof. The primal equation is unchanged, while substitution multiplies the dual and complementarity equations by λ . Full row rank and positivity make the Newton solution unique. The same substitution in $Ax = b, A^T y + s = c, Xs = \mu e$ proves the central-path statement. $\square$

For the full-state theorem use the all-unit signed chain and copy plateau, but pin each reference coordinate independently by $h_i = 1$. This changes the row representation, not the primal feasible set. Starting from (344), scale the objective, initial slacks, and complementarity data by

$$\lambda = \frac{1}{P}. \quad (354)$$

The exact primal correction remains the block (305). If the first coordinate is paired with the larger endpoint coordinate, the unscaled slack correction is

$$\Delta \bar{s}_i = (-17/54, 7/54, -17/48, -17/36), \quad (355)$$

and, with difference, height-pin, and cap multipliers,

$$\Delta y = \frac{1}{P}(\bar{\alpha}, \bar{\beta}, \bar{\gamma}), \quad \bar{\alpha}_j = \frac{2}{9}\tau_j(P - j), \quad \bar{\beta}_j = \frac{133}{48}, \quad \bar{\gamma}_j = \frac{11}{12}. \quad (356)$$

Theorem 9.13 (Bounded full-KKT direction). *Every coordinate of the complete $11P$ -dimensional direction $w = (\Delta x, \Delta y, \Delta s)$ in (354)–(356) is bounded by an absolute constant, and $\|w\|_2 = \Theta(\sqrt{P})$. Preparing its normalized state to trace error $1/100$ requires $\Omega(N) = \Omega(P)$ raw queries.*

The explicit canonical sparse block encoding of either the three-block Newton matrix or its two-block elimination has normalization five, a public RHS, and the same linear call lower bound. The reduced Newton condition remains one, but the central endpoint is at $\mu'_1 = 1/(16P)$, and both raw full-KKT matrices have condition $\Theta(P)$.

Proof. Equations (305), (355), and (356) give

$$\|\Delta x\|_\infty \leq 17/24, \quad \|\Delta y\|_\infty \leq 2/9, \quad \|\Delta s\|_\infty \leq 17/(36P).$$

Writing $D = 16139/23328$, direct summation gives

$$DP \leq \|w\|_2^2 < \left(D + \frac{1}{40}\right)P < \frac{3}{4}P. \quad (357)$$

Use the diagonal decoder convention $+1$ on every v coordinate and -1 on every u coordinate. The underlying u -minus- v squared-mass difference on each output node is $-p_N(5/12)$, so the decoder expectation contributes $+p_N(5/12)$ per node and has magnitude

$$\frac{(5/12)K}{(3/4)P} = \frac{5K}{9P} \geq \frac{32}{63} > \frac{1}{2}. \quad (358)$$

Trace error preserves constant bias. The polynomial method gives at least $N/2$ sign queries.

The maximum absolute row and column sums of both canonical matrices are five. The same magnitude-weighted state-pair construction used in Theorem 9.8 therefore gives normalization five and one sign query per full oracle call. The RHS remains public because the start has zero pair difference. This proves the canonical-oracle claim.

Dual homogeneity gives $\mu'_1 = 1/(16P)$ and the reduced start matrix $(22/(27P))I$. If $z \in \ker A$ is unit, applying either raw KKT matrix to the corresponding nullspace test vector has norm at most $1/P$, while the largest singular value is bounded below by a constant. Conversely, the pinned signed incidence has smallest singular value $\Theta(P^{-1})$; bounded local changes reduce the remaining blocks to identities. The paired saddle eigenvalues then show that both conditions are $\Theta(P)$. $\square$

The price is real: objective coefficients, positive slacks, reduced costs, and complementarity are $\Theta(P^{-1})$. The theorem is not a simultaneous constant-scale and constant-full-condition result. It also does not inherit the sharp relative-residual theorem, because independent height pins make $\|b\|_2 = \Theta(\sqrt{P})$.

9.9 Why the inverse-scale price remains open

The obstruction can be stated without signed pairs. At any positive start, eliminate Δs from the Newton equations and put

$$H = (X^0)^{-1}S^0, \quad g = (X^0)^{-1}r_c - r_d.$$

Then

$$H\Delta x - A^T\Delta y = g, \quad \boxed{r_p^T\Delta y = \Delta x^T H\Delta x - g^T\Delta x}. \quad (359)$$

This exact work identity is representation-independent. If $g = 0$, $H \succeq \rho I$, and $\|r_p\|_1 \leq R$, it gives

$$\|\Delta y\|_\infty \geq \frac{\rho\|\Delta x\|_2^2}{R}. \quad (360)$$

A constant-scale correction with $\Theta(P)$ squared norm behind a sparse public source therefore forces an $\Omega(P)$ multiplier, independently of redundant rows or branching.

There is also a two-input version for constant-dimensional copy blocks. Let a flat output region S carry $z_v \in \mathbb{R}^k$ and orthogonal copy equations $z_v - U_e z_u = 0$. After public transports G_v , suppose stationarity is

$$\operatorname{div}_U(f)_v = D_v\Delta z_v + g_v, \quad D_v \succeq \rho I, \quad (361)$$

with the same numerical D_v, g_v for two opposite-parity inputs, and their transported codewords differ by a common w , $\|w\|_2 \geq 2a$. If a boundary row scale is at most B , summing the difference of (361) over S cancels every public source and internal force. Taking the inner product with w yields

$$\max_{p \in \{+, -\}} \|\Delta y_{\partial S}^p\|_{2,\infty} \geq \frac{\rho a |S|}{B |\partial S|}. \quad (362)$$

Thus swaps, simplex permutations, and public flat rotations do not remove dual accumulation behind a bounded interface.

The most tempting counterexample is a real quarter-turn $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, with $z_i = \sigma_i J z_{i-1}$ and closure $z_0 - J z_N = e_1$. For $4 \mid N$,

$$z_0(p) = \frac{1}{2}(1, p)^T, \quad (363)$$

so all blocks are bounded and locally reveal parity after a public inverse rotation. Nevertheless the work identity gives the closure multiplier

$$e_1^T \Delta y_{\text{close}} = \sum_{i=0}^N \|z_i\|_2^2 = \frac{N+1}{2}. \quad (364)$$

After gauging, the constraint is a twisted cycle whose singular values are

$$2 \left| \sin \frac{2\pi\ell \pm \pi/2}{2(N+1)} \right|, \quad \ell = 0, \dots, N, \quad (365)$$

so its least singular value is $2 \sin(\pi/[4(N+1)]) = \Theta(N^{-1})$. A positive/negative standard-form LP realization preserves bounded primal and slack directions and even allows a positive half-centering step, but (364) survives exactly. Circulations, public antisymmetric sources, and dense public residuals fail for the same work or two-parity cancellation reason unless they introduce linearly many genuinely independent hard interfaces.

Open Problem 9.14 (Bounded-scale full-KKT parity amplifier). Does a sparse standard-form LP family exist that simultaneously has:

- (i) linear dimension and bounded row and column degree;
- (ii) bounded coefficients with one-bit-local input dependence;
- (iii) public, simply preparable Newton right-hand sides;
- (iv) primal and slack starts bounded above and below by positive constants;
- (v) a coordinatewise bounded complete Newton direction; and
- (vi) a parity-hard primal component with constant mass in that complete direction state?

The work identity and orthogonal-copy cut law rule out the natural signed, permutation, rotation, and flat-copy constructions. They do not cover arbitrary nonorthogonal mixed-mode constraints, input-dependent sources whose loading is separately charged, or output interfaces that quotient the equality multipliers.

The distinction closing the section is therefore exact. The prefix-rigidified construction proves condition-one *reduced Newton geometry* together with a linear original-coordinate loading/recovery cost. The preconditioned full-KKT result proves a constant-condition residual theorem only after charging construction of its data-dependent access. The bounded full direction obtained by dual homogeneity instead has canonical full-system condition $\Theta(P)$. None is a constant- κ QLS lower bound with free matrix and RHS preparation oracles; each is an end-to-end LP access and output theorem.

10 Gadget-Class Frontiers

Section 9 exhibited three ways to make a local parity signal robust: replicate constraints, scale rows, or increase the height of the propagated signal. This section asks whether a different sparse LP gadget can avoid all three costs. The answer is negative for several broad, precisely stated classes. The strongest default result is a graph-free convex-mixture theorem. Electrical resistance gives a geometric refinement for signed-copy graphs and bounded-arity Walsh rows, while multiplier conservation explains why complete central and Newton states must pay through row scale, dual scale, multiplier size, or input incidence.

Throughout this section the hidden input is $\varsigma \in \{\pm 1\}^N$, with $N \geq 2$, and $p(\varsigma) = \prod_{i=1}^N \varsigma_i$. Input locality always refers to the raw sparse coefficient access of Definition 2.11. Residuals are absolute when displayed as $\|Ax - b\|_2$; when a relative threshold is named, it is the global RHS-relative residual of Definition 2.26. These are structural obstructions to output contracts. They become query lower bounds only after a separate reduction verifies local oracle simulation and decodes parity.

10.1 The convex-mixture frontier

Consider a fixed-support family

$$\mathcal{F}_\varsigma = \{x \in \mathbb{R}_{\geq 0}^n : A_\varsigma x = b\}, \quad (366)$$

where b is public. Every row has at most s nonzeros, every coefficient has magnitude at most B , and each input-dependent position (r, j) is a function of one designated bit $\varsigma_{\ell(r, j)}$. Let M_{dep} be the number of those positions. A bit may label many positions, including several in one row. For a fixed base input, let $\varsigma^{(i)}$ flip bit i , and suppose there is an exact witness

$$x^{(i)} \in \mathcal{F}_{\varsigma^{(i)}}, \quad \|x^{(i)}\|_\infty \leq H \quad (1 \leq i \leq N). \quad (367)$$

The magnitude bounds B, H are nonnegative. The residual estimates allow $s = 0$; the divided resource inequalities below assume $s > 0$.

Theorem 10.1 (Single-flip convex mixture). *The average $\bar{x} = N^{-1} \sum_i x^{(i)}$ is nonnegative and satisfies*

$$\boxed{\|A_\varsigma \bar{x} - b\|_2 \leq \frac{2BH\sqrt{sM_{\text{dep}}}}{N}}. \quad (368)$$

If a public objective c has the same optimum v_ on every instance and the witnesses in (367) are optimal, then $c^\top \bar{x} = v_*$: the mixture is objective-exact, although generally infeasible for the base instance.*

Proof. Feasibility for the neighboring instances gives

$$N(A_\varsigma \bar{x} - b) = \sum_{i=1}^N (A_\varsigma - A_{\varsigma^{(i)}})x^{(i)}.$$

Let D_r be the dependent positions in row r , and set $m_r = |D_r|$. Expanding the r -th coordinate by positions, only the summand $i = \ell(r, j)$ can change position (r, j) . The numerator is therefore a sum of at most m_r terms, each of magnitude at most $2BH$. Since $m_r \leq s$,

$$|N(A_\varsigma \bar{x} - b)_r|^2 \leq 4B^2H^2m_r^2 \leq 4sB^2H^2m_r.$$

Summing over rows and using $\sum_r m_r = M_{\text{dep}}$ proves (368). Convexity proves nonnegativity, and linearity proves the objective statement. $\square$

The theorem also handles a one-bit-local right-hand side by applying it to $[A_\varsigma, -b_\varsigma](x, 1) = 0$. One then uses row sparsity $s + 1$, height $\max\{H, 1\}$, replaces B by a bound on every entry of the augmented matrix $[A_\varsigma, -b_\varsigma]$, and counts the dependent RHS positions. Inequalities are covered after standard slack conversion only when the selected slack coordinates satisfy the same height bound.

The output hypothesis cannot be omitted. For an affine public score $\phi(x) = a^\top x - a_0$, suppose every selected witness for input ξ obeys $p(\xi)\phi(x^\xi) \geq \gamma > 0$. All single-flip witnesses have parity $-p(\varsigma)$, so $p(\varsigma)\phi(\bar{x}) \leq -\gamma$. Thus a soundness contract excluding every nonnegative, objective-exact, wrong-output point with relative residual at most $\eta \geq 0$, with $b \neq 0$ as in Definition 2.26, requires

$$M_{\text{dep}} B^2 H^2 > \frac{\eta^2 \|b\|_2^2}{4s} N^2. \quad (369)$$

In particular, with fixed $s, B, \eta, \|b\|_2$, linear incidence and bounded height are incompatible with a fixed residual tube. Here and in the fixed-tube asymptotic consequences below, $\eta > 0$.

The same conclusion applies to the normalized primal-state contract under a concrete mixture-stability condition. Let $\mathcal{S}$ be $L \geq 1$ pairwise disjoint, public output pairs (u_ℓ, v_ℓ) , and suppose

$$p(\xi)(u_\ell^\xi - v_\ell^\xi) \geq a > 0 \quad (\ell \in \mathcal{S}) \quad (370)$$

for every selected witness. Averaging gives $-p(\varsigma)(\bar{u}_\ell - \bar{v}_\ell) \geq a$. Nonnegativity gives $\bar{u}_\ell + \bar{v}_\ell \geq a$, hence

$$-p(\varsigma)(\bar{u}_\ell^2 - \bar{v}_\ell^2) \geq a^2.$$

For $Q_{\mathcal{S}} = \sum_{\ell \in \mathcal{S}} (|u_\ell\rangle\langle u_\ell| - |v_\ell\rangle\langle v_\ell|)$,

$$-p(\varsigma) \left\langle \frac{\bar{x}}{\|\bar{x}\|_2} \middle| Q_{\mathcal{S}} \middle| \frac{\bar{x}}{\|\bar{x}\|_2} \right\rangle \geq \frac{La^2}{nH^2}. \quad (371)$$

Measure a coordinate, output its pair sign on the selected pairs, and use an independent fair sign on all other coordinates. This binary measurement has half the displayed signed expectation as its wrong-parity bias. Thus $La^2 = \Omega(nH^2)$ gives a constant wrong-parity bias. Without an affine score, a common output template, the pair margin (370), or another explicit mixture-stability property, Theorem 10.1 makes no claim about a normalized quadratic decoder.

Two useful refinements expose the role of sensitivity and uneven input incidence. For a Boolean function f , fix a base input with sensitive set S_f of size $k > 0$, and average only its sensitive neighbors. If M_S counts positions labelled by bits in S_f , the same proof gives

$$\|A_\varsigma \bar{x}_S - b\|_2 \leq \frac{2BH\sqrt{sM_S}}{k}. \quad (372)$$

Under the common-objective and mixture-stable output hypotheses, relative soundness at radius η therefore requires

$$M_S B^2 H^2 > \frac{\eta^2 \|b\|_2^2 k^2}{4s}.$$

More sharply, let M_i be the number of positions labelled by sensitive bit i . For weights $w_i \geq 0$, $\sum_i w_i = 1$, rowwise Cauchy–Schwarz, with $x_w = \sum_i w_i x^{(i)}$, gives

$$\|A_\varsigma x_w - b\|_2^2 \leq 4sB^2H^2 \sum_i M_i w_i^2. \quad (373)$$

When every $M_i > 0$, its right side is minimized by $w_i = M_i^{-1} / \sum_h M_h^{-1}$, yielding

$$\|A_\varsigma x_w - b\|_2 \leq 2BH \sqrt{\frac{s}{\sum_i M_i^{-1}}}.$$

Table 3: Sharp endpoints of the one-bit-local mixture frontier.

construction	M_{dep}	B	H
bounded-height parallel replication	$\Theta(N^2)$	$\Theta(1)$	$\Theta(1)$
direct row scaling	$\Theta(N)$	$\Theta(\sqrt{N})$	$\Theta(1)$
gain-plateau scaling	$\Theta(N)$	$\Theta(1)$	$\Theta(\sqrt{N})$

The corresponding relative soundness condition is

$$B^2 H^2 > \frac{\eta^2 \|b\|_2^2}{4s} \sum_i M_i^{-1}.$$

If some $M_i = 0$, that neighbor is exactly feasible for the base matrix and already defeats the contract. These mixtures preserve any uniform affine or pair margin and the common objective. Their governing resource is therefore the harmonic incidence profile, not merely its arithmetic sum.

For a fixed relative tube and fixed $s, \eta, \|b\|_2$, the three pure amplification currencies are as follows. Every row has $M_{\text{dep}} B^2 H^2 = \Theta(N^2)$. The parallel and gain-plateau families of Theorems 9.2 and 9.4 attain the first and third endpoints. Scaling the propagation rows by $\sqrt{N}$ attains the second without changing the feasible set, while exposing the cost in matrix normalization. This sharpness is about the frontier's scales, not an assertion that the formulations have identical conditioning or QIPM cost.

This is not an extension-complexity lower bound. Carr–Konjevod give an $O(N)$ -size dynamic-programming extended formulation for the parity polytope, and Ermel–Walter develop further flow formulations [36, 60]. Those formulations do not promise a robust parity output from every point in a fixed equality-residual tube. Sparse LP decoding likewise has classical fractional pseudocodewords and graph-cover descriptions [62, 87, 132]; this is a conceptual collision, not the quantitative bound (368). In particular, expander LP decoding may use an input-dependent decoding objective, whereas exact objective preservation above requires one common objective and optimum.

10.2 Supplements: broader rows, KKT stacks, cones, and centrality

If coefficient position (r, j) may depend on a set $I_{rj} \subseteq [N]$, define

$$d = \max_r \left| \bigcup_j I_{rj} \right|, \quad d_{\text{coeff}} = \max_{r,j} |I_{rj}| \leq d, \quad M = \sum_{(r,j)} |I_{rj}|,$$

while retaining row sparsity s , coefficient bound B , and witness height H .

Proposition 10.2 (Multi-bit coefficient supplement). *The single-flip average obeys*

$$\|A_\zeta \bar{x} - b\|_2 \leq \frac{2BH\sqrt{d_{\text{coeff}} s M}}{N} \leq \frac{2BH\sqrt{dsM}}{N}. \quad (374)$$

For fixed d_{coeff}, s , the product frontier remains $MB^2 H^2 = \Omega(N^2)$ under the objective and output assumptions above. In particular, the stronger bound recovers the one-bit constant when every coefficient depends on at most one bit, regardless of how many distinct bits appear in a row.

Proof. Write $f_{rji} = (A_{\zeta, rj} - A_{\zeta^{(i)}, rj}) x_j^{(i)}$. It vanishes outside $i \in I_{rj}$, and $|f_{rji}| \leq 2BH$. Cauchy–Schwarz first over the at most s dependent positions in a row, and then over each coefficient's input set, gives

$$\left(\sum_j \sum_{i \in I_{rj}} f_{rji} \right)^2 \leq s \sum_j |I_{rj}| \sum_{i \in I_{rj}} f_{rji}^2 \leq 4s d_{\text{coeff}} B^2 H^2 \sum_j |I_{rj}|.$$

Sum over rows and divide by N^2 . The row-union estimate follows from $d_{\text{coeff}} \leq d$. $\square$

There is also a simultaneous primal–dual version. Suppose A_ς has maximum row and column sparsities s_r, s_c , its one-bit-dependent positions number M , and selected exact optimal triples (x^ξ, y^ξ, s^ξ) have all coordinates bounded in magnitude by H and common optimum v . Split $y = y^+ - y^-$, stack primal feasibility and stationarity, and put

$$s_{\text{KKT}} = \max\{s_r, 2s_c + 1\}, \quad B_{\text{KKT}} = \max\{B, 1\}.$$

Every dependent position occurs once in the primal block and twice in the split-dual block. Theorem 10.1 therefore gives

$$\left\| \begin{bmatrix} A_\varsigma \bar{x} - b \\ A_\varsigma^\top \bar{y} + \bar{s} - c \end{bmatrix} \right\|_2 \leq \frac{2B_{\text{KKT}}H\sqrt{3s_{\text{KKT}}M}}{N}. \quad (375)$$

Linearity preserves $c^\top \bar{x} = b^\top \bar{y} = v$, so the algebraic gap is exactly zero. This covers combined feasibility, stationarity, and gap contracts; it does not preserve coordinatewise complementarity.

The mixture proof uses only convexity of the domain. Let $\mathcal{K}$ be a fixed convex set or cone, and suppose that for every selected input $\xi \in \{\varsigma, \varsigma^{(1)}, \dots, \varsigma^{(N)}\}$ there is a base witness $x^\xi \in \mathcal{K}$ with $A_\xi x^\xi = b$, $\|x^\xi\|_\infty \leq H$ in one fixed basis, and $c^\top x^\xi = v$ for one public objective and common value. If, in addition, a fixed parity-neutral linear map satisfies $Lx^\xi = p(\xi)y$ for a fixed nonzero y , then the half-base mixture

$$\hat{x} = \frac{1}{2}x^\varsigma + \frac{1}{2N} \sum_i x^{\varsigma^{(i)}}$$

belongs to $\mathcal{K}$, has the common objective, satisfies $L\hat{x} = 0$, and has half the residual in (374). For a sparse SDP this statement applies after one fixed svec identification; all row sparsity, locality, coefficient, and coordinate bounds must be measured in that basis. It gives no normalized-state, rank, determinant-neighborhood, or complementarity conclusion by itself.

Centrality can be retained only with an explicit stability premise. Retain the preceding row, column, one-bit-locality, and height bounds, and suppose the selected neighboring triples are exact central points at a common $\mu > 0$, so $x_j^{(i)} s_j^{(i)} = \mu$. Define their arithmetic averages. For $t_i = x_j^{(i)}$, direct symmetrization gives the exact multiplicative-variance identity

$$\boxed{\frac{\bar{x}_j \bar{s}_j}{\mu} - 1 = \frac{1}{2N^2} \sum_{i,k=1}^N \frac{(t_i - t_k)^2}{t_i t_k}}. \quad (376)$$

In particular, the excess is nonnegative and vanishes exactly when the neighboring primal coordinates agree. If, for every coordinate j , $\max_i x_j^{(i)} / \min_i x_j^{(i)} \leq R$, where $R \geq 1$, then

$$\mu \leq \bar{x}_j \bar{s}_j \leq K(R)\mu, \quad K(R) = \frac{(R+1)^2}{4R} = 1 + \frac{(R-1)^2}{4R}. \quad (377)$$

Indeed, for $\alpha \leq t_i \leq \beta$, $t_i + \alpha\beta/t_i \leq \alpha + \beta$. Averaging and maximizing the product under this linear constraint gives $(N^{-1} \sum_i t_i)(N^{-1} \sum_i t_i^{-1}) \leq (\alpha + \beta)^2/(4\alpha\beta)$, the scalar Kantorovich inequality [81].

Proposition 10.3 (Central-mixture soundness dichotomy). *Assume independently a uniform mixture-stable wrong-output premise for the base input. Concretely, write $Z^{(i)} = (x^{(i)}, y^{(i)}, s^{(i)})$, and suppose either that a fixed affine decoder Φ satisfies $p(\varsigma)\Phi(Z^{(i)}) \leq -\gamma < 0$ for every i , or that the primal components have the public pair decoder of (371), with $-p(\varsigma)(u_\ell^{(i)} - v_\ell^{(i)}) \geq a$ for every i, ℓ and $La^2 = \Omega(nH^2)$. The decoded object is respectively the stacked triple Z_w or the*

normalized primal state $|x_w\rangle = x_w/\|x_w\|_2$; under either premise every probability mixture has a uniform wrong output for the base input.

Let w be any probability vector and set $Z_w = (x_w, y_w, s_w) = \sum_i w_i(x^{(i)}, y^{(i)}, s^{(i)})$. With M_i the one-bit incidence of bit i , and with $t_{ij} = x_j^{(i)}$, the stacked residual and centrality defect obey

$$\left\| \begin{bmatrix} A_\zeta x_w - b \\ A_\zeta^\top y_w + s_w - c \end{bmatrix} \right\|_2^2 \leq 12s_{\text{KKT}} B_{\text{KKT}}^2 H^2 \sum_i M_i w_i^2, \quad (378)$$

$$\delta_j(w) := \frac{(x_w)_j (s_w)_j}{\mu} - 1 = \frac{1}{2} \sum_{i,k} w_i w_k \frac{(t_{ij} - t_{kj})^2}{t_{ij} t_{kj}}. \quad (379)$$

The weighted residual estimate admits the sharper constant

$$\left\| \begin{bmatrix} A_\zeta x_w - b \\ A_\zeta^\top y_w + s_w - c \end{bmatrix} \right\|_2^2 \leq 4(s_r + s_c) B^2 H^2 \sum_i M_i w_i^2. \quad (380)$$

Apply the weighted matrix-change estimate separately to the primal block and to the transposed coefficient changes in stationarity. The latter permits signed multipliers and has the same incidence profile, with column sparsity s_c . Adding the squared bounds proves (380); the earlier bound follows by increasing the constant. Thus the displayed soundness inequalities also hold with $12s_{\text{KKT}} B_{\text{KKT}}^2$ replaced by $4(s_r + s_c) B^2$. The algebraic objective difference remains exactly $n\mu$. This follows from feasibility, stationarity, and complementarity of each neighboring triple; common individual primal or dual objective values are unnecessary. A sound contract imposing absolute combined residual at most ε and an ℓ_∞ neighborhood of width θ around the common μ therefore requires, for every w ,

$$12s_{\text{KKT}} B_{\text{KKT}}^2 H^2 \sum_i M_i w_i^2 > \varepsilon^2 \quad \text{or} \quad \max_j \delta_j(w) > \theta. \quad (381)$$

Here $\varepsilon, \theta \geq 0$. For the point-centered convention, with $n > 0$, $\mu_w = x_w^\top s_w / n$, replace the second term by the exact width

$$\frac{\max_j |\delta_j - \bar{\delta}|}{1 + \bar{\delta}}, \quad \bar{\delta} = \frac{1}{n} \sum_j \delta_j.$$

For uniform weights, suppose for every coordinate j that

$$\frac{\max_i t_{ij}}{\min_i t_{ij}} \leq R.$$

Either convention then has width at most $K(R) - 1$, so the simpler necessary dichotomy is

$$\frac{12s_{\text{KKT}} B_{\text{KKT}}^2 H^2 M}{N^2} > \varepsilon^2 \quad \text{or} \quad \frac{(R-1)^2}{4R} > \theta. \quad (382)$$

Proof. The stacked mixture estimate is the single-flip argument applied to primal feasibility and stationarity. Pairwise symmetrization gives the displayed formula for $\delta_j(w)$, and linearity preserves the algebraic objective difference. The stated affine or pair-margin premise makes the named decoded object wrong for the base input. If neither strict inequality in (381) held, that wrong-output mixture would meet every certificate in the contract. The coordinatewise Kantorovich bound gives (382) for uniform weights. $\square$

The stability premise is necessary. Fix $0 < \mu < 1$, let $b = \mu e, c = 0$, and let bit j choose $a_j \in \{\mu, 1\}$ in the diagonal LP $\text{Diag}(a)x = b$. Its exact μ -central point is

$$x_j = \mu/a_j, \quad s_j = a_j, \quad y_j = -1.$$

At the all- μ base input, each single-flip neighbor changes one coordinate from $(x, s) = (1, \mu)$ to $(\mu, 1)$. Thus every coordinate of the average has product

$$\bar{x}_j \bar{s}_j = \frac{N-1+\mu}{N} \frac{1+(N-1)\mu}{N},$$

which is $\Theta(1/N)$, rather than $\Theta(\mu)$, when $\mu \ll 1/N$, although primal and stationarity residuals vanish as $O(N^{-1/2})$. Bounded coefficients and bounded central coordinates alone therefore do not imply central-mixture stability around the common μ . This particular uniform mixture has the same complementarity product in every coordinate, so its point-centered width is exactly zero. Its algebraic objective difference is still $N\mu$, whereas its complementarity sum is $N\bar{x}_j \bar{s}_j$; these need not agree at an infeasible mixture.

There is also a bounded-coordinate counterexample for the point-centered convention. Append N public diagonal coordinates with coefficient μ and central coordinates $(x, s, y) = (1, \mu, -1)$. The moving coordinates have the common product $q = (N-1+\mu)(1+(N-1)\mu)/N^2$, while the public coordinates have product μ . The point-centered mean is $(q+\mu)/2$, and every coordinate has relative deviation $(q-\mu)/(q+\mu)$. For $N \geq 8$ and $\mu = N^{-2}$, the exact estimates are

$$\left\| \begin{bmatrix} A\bar{x} - b \\ A^\top \bar{y} + \bar{s} - c \end{bmatrix} \right\|_2^2 = \frac{(1+\mu^2)(1-\mu)^2}{N} \leq \frac{2}{N}, \quad \frac{q-\mu}{q+\mu} \geq \frac{1}{3}.$$

The original unpadded example has the same residual and common-parameter defect at least $N/8$ for $\mu = N^{-2}$, $N \geq 2$. Both constructions have row and column sparsity one, coefficient and coordinate magnitudes at most one, and exactly N dependent positions. The padded construction therefore rules out an inference from these bounds and vanishing residual to point-centered width below any fixed $\theta < 1/3$.

Formal verification. The Lean development in `formal/QipmFormal/Mixture/` verifies the residual and centrality results in these two subsections from the actual finite matrices and neighboring witnesses. It includes weighted and sensitive-subset mixtures, the stronger coefficientwise multibit bound, RHS and slack augmentation, the split KKT construction, the direct KKT bound, harmonic incidence weights, affine and normalized pair decoders, the scalar variance and Kantorovich identities, both neighborhood conventions, and the explicit counterexamples. The soundness hypotheses are stated as contracts on candidate outputs; the proofs derive the mixture's residual, objective, and decoder properties. The development uses Euclidean sums of squares and permits only Lean's standard axioms `propext`, `Classical.choice`, and `Quot.sound`. The claim map and reproduction command are in `formal/MIXTURE.md`. The separate sharpness constructions in Table 3, the electrical frontiers below, and quantum query reductions are outside this verification scope. The noncommutative extension has a separate Lean development and claim map in `formal/SDP_MIXTURE.md`; see [section 14.2](#).

10.3 Electrical frontiers

The graph route constructs an obstruction within one fixed instance and exposes the geometry hidden by the general mixture argument. Let $G = (V, E)$ be a connected oriented multigraph with root r , nonempty output set S , public heights $h_v > 0$ with $h_r = 1$, and propagation rows

$$\lambda_e(d_v - g_e a_e(\varsigma) d_u) = 0, \quad g_e = h_v/h_u, \quad (383)$$

where $e = (u, v)$, $\lambda_e \neq 0$ is a public scale, a_e is a fixed sign or a fixed sign times one input bit, and the rows are cycle-consistent. Thus signs τ_v exist with $\tau_r = 1$ and $\tau_v = a_e \tau_u$. Assume independently that the designated outputs carry parity:

$$\tau_s(\varsigma) = \epsilon_s p(\varsigma) \quad (s \in S), \quad \epsilon_s \in \{\pm 1\} \text{ public.} \quad (384)$$

Embed each d_v as the capped pair of Section 9.1: write $d_v = u_v - v_v$, $q_v = u_v + v_v$, pin a reference variable $h_v^{\text{var}} = h_v$, impose $q_v + t_v - 2h_v^{\text{var}} = 0$, and use local objective $2u_v + 2v_v + h_v^{\text{var}} + t_v$. Put

$$L = \max_{e=(u,v)} |\lambda_e| h_v \leq BH, \quad M = |E|,$$

when $0 < |\lambda_e| \leq B$ and $h_v \leq H$.

Theorem 10.4 (Cut-resistance frontier). *There is a nonnegative, objective-exact point satisfying every root, reference, and cap row exactly, erasing all output differences, and obeying*

$$\|Ax - b\|_2^2 \leq \frac{ML^2}{N^2} \leq \frac{MB^2H^2}{N^2}. \quad (385)$$

There is also a fully sign-flipped point with four times this squared-residual bound. If the two unit anchors are the only nonzero RHS entries, so $\|b\|_2 = \sqrt{2}$, any relative-residual contract at threshold η that rules out the erased point requires

$$MB^2H^2 > 2\eta^2N^2. \quad (386)$$

If each edge label contains at most k raw signs, the erased residual in (385) acquires a factor k^2 , and soundness requires $Mk^2B^2H^2 = \Omega(N^2)$.

Proof. For bit i , let E_i be the edges whose label changes under its flip. Cycle consistency makes every cycle meet E_i evenly, so signed-graph switching identifies E_i with a cut separating the root from every output [75, 144]. One-bit locality makes the N cuts edge-disjoint.

Give edge $e = (u, v)$ conductance $c_e = (\lambda_e h_v)^2 \leq L^2$, and wire the outputs together. For any unit root-to-output flow f , one unit crosses every cut. Weighted Cauchy–Schwarz gives

$$\sum_{e \in E_i} \frac{f_e^2}{c_e} \geq \frac{1}{C_i}, \quad C_i = \sum_{e \in E_i} c_e.$$

Because the cuts are disjoint, Thomson’s principle and a second Cauchy–Schwarz inequality give the weighted Nash–Williams bound

$$\mathcal{R}_{r,S} \geq \sum_i \frac{1}{C_i} \geq \frac{N^2}{\sum_i C_i} \geq \frac{N^2}{ML^2}. \quad (387)$$

This proof is included to expose the oracle-specific role of disjoint input cuts; the resistance inequality itself is classical [111].

Let $w \in [0, 1]^V$ be the harmonic voltage with $w_r = 1$ and $w_s = 0$ on every output. Dirichlet’s principle and (387) give energy at most ML^2/N^2 . Set

$$d_v = \tau_v h_v w_v, \quad q_v = h_v^{\text{var}} = t_v = h_v, \quad u_v = (h_v + d_v)/2, \quad v_v = (h_v - d_v)/2.$$

The maximum principle proves nonnegativity. Since every λ_e is nonzero and the graph is connected, every feasible point has $|d_v| = h_v$, hence $q_v = u_v + v_v \geq |d_v| = h_v$; the local objective reduces to $q_v + 3h_v$, so this point, with $q_v = h_v$, is objective-exact. All nonpropagation rows are exact, while edge e has residual $\lambda_e \tau_v h_v (w_v - w_u)$, whose squared sum is the Dirichlet energy. Every output has $d_s = 0$. Replacing the output boundary value zero by -1 doubles every voltage difference and proves the sign-flipped statement.

For k -local labels, an edge lies in at most k bit cuts. Hence $\mathcal{R}_{r,S} \geq k^{-1} \sum_i C_i^{-1}$, while $\sum_i C_i \leq kML^2$. This gives $\mathcal{R}_{r,S} \geq N^2/(k^2ML^2)$ and the stated loss. $\square$

Theorem 10.4 makes the gain–plateau family optimal throughout its cycle-consistent signed-graph class: linear size and bounded rows force $H = \Omega(\sqrt{N})$. The two other rows of Table 3 attain the remaining endpoints. An adaptive trajectory does not repair the pointwise output contract, because the harmonic witness exists at every requested stage. Conversely, it gives no unconditional fresh per-iteration lower bound: after all N fixed signs are cached, later raw-query cost may be zero under Definition 2.27.

The same resistance mechanism extends to genuine multi-variable rows. Give wire v public height $h_v \in (0, H]$ and intended value $h_v \chi_{J_v}(\varsigma)$, where $\chi_J = \prod_{i \in J} \varsigma_i$. Let each of m homogeneous rows have arity at most q :

$$\sum_{v \in V_a} \alpha_{av} \chi_{I_{av}}(\varsigma) d_v = 0, \quad |\alpha_{av}| \leq A, \quad |I_{av}| \leq k. \quad (388)$$

Roots have $J_v = \emptyset$, outputs have $J_v = [N]$, and every row is valid on the intended wires for every input. The comparison graph in the proof may be assumed to connect the anchored roots to the outputs: otherwise the output character is unconstrained and can be erased with zero propagation residual. For the objective-exact conclusion, assume also that the intended full-height assignment is an optimum of the capped LP. A sufficient magnitude-floor hypothesis is that every exact feasible assignment obeys $|d_v| \geq h_v$ for every wire, since nonnegativity then forces $q_v \geq h_v$.

Theorem 10.5 (Bounded-arity Walsh-row frontier). *The capped-pair embedding has a nonnegative, objective-exact point that satisfies anchors, references, and caps, erases every output, and has*

$$\|r\|_2^2 \leq \frac{4k^2 q^3 m A^2 H^2}{N^2}. \quad (389)$$

Consequently fixed k, q and constant residual soundness require $m A^2 H^2 = \Omega(N^2)$.

Proof. Substitution of the intended wires into row a gives $\sum_v \beta_{av} \chi_{K_{av}} = 0$, where $\beta_{av} = \alpha_{av} h_v$ and $K_{av} = I_{av} \triangle J_v$. Independence of Walsh characters forces $\sum_{v \in G_{a,K}} \beta_{av} = 0$ within each equal-character group. Choose one pivot per group and join it to the other group members by comparison edges of conductance $q^2 \beta_{av}^2$. There are at most qm edges, each with conductance at most $q^2 A^2 H^2$. For arbitrary interpolation scalars w_v , a character group with pivot p has residual

$$\sum_{v \in G} \beta_{av} w_v = \sum_{v \in G \setminus \{p\}} \beta_{av} (w_v - w_p).$$

Cauchy–Schwarz first over a group’s at most q terms and then over a row’s at most q groups gives

$$\|r\|_2^2 \leq \sum_e c_e (w_{h(e)} - w_{t(e)})^2.$$

Endpoints of a comparison edge satisfy $J_u \triangle J_v = I_{au} \triangle I_{av}$, so that edge crosses at most $2k$ of the vertex cuts $\{v : i \in J_v\}$. The bounded-overlap form of the Nash–Williams calculation already proved above gives

$$\mathcal{R}_{\text{eff}} \geq \frac{N^2}{4k^2 q^3 m A^2 H^2}.$$

Take the harmonic interpolation equal to one at the roots and zero at the outputs. Its energy gives (389); the maximum principle and the capped-pair construction used in the preceding proof give nonnegativity and exact auxiliary rows. The obstructing point has $q_v = h_v$ for every wire and therefore has the same local objective as the intended full-height assignment; the explicit capped-optimum hypothesis gives the exact optimal objective. $\square$

Exact local XOR hulls have complementary, concrete failures. If a set U of unfixed variables meets every parity check in zero or at least two positions, then setting $x_j = 1/2$ on U and

retaining a satisfying Boolean codeword elsewhere lies in every local parity polytope. For a check meeting U in $r \geq 2$ variables, it is the uniform average of the 2^{r-1} assignments with the required residual parity, proving the claim. This stopping-set pseudocodeword can make an output exactly unbiased. If checks instead peel along a dependency chain of length D , linearly interpolating the gauged wire error from zero to one violates one opposing facet at each gate by $1/D$, for total Euclidean violation $1/\sqrt{D}$. Finally, every satisfying truth-table assignment is a vertex of its exact gate hull, so some standard-form inequality slacks vanish; a Boolean circuit forced by its inputs is not strictly primal feasible. Thickness restores positive slacks only by admitting fractional drift. These facts explain the LP-decoding analogy but are not a universal lower bound for all inequality extended formulations.

10.4 Multiplier conservation

Residual robustness constrains primal witnesses. Complete central and Newton outputs face a second conservation law. In a signed-copy graph write $G = (V \cup T, E)$, vertices in V carry variables with $d_v = x_v^+ - x_v^-$, $q_v = x_v^+ + x_v^-$, and vertices in T are fixed terminals used for anchor rows. A variable–variable edge $e = (u, v)$ represents $d_v - a_e d_u = 0$. A terminal edge is oriented from its terminal to its variable endpoint, with the fixed terminal term moved to the RHS. Let B_a and B be respectively the signed and ordinary oriented incidence matrices with all terminal columns omitted. For $S \subseteq V$, its boundary includes edges to $V \setminus S$ and all terminal edges incident to S . Each normalized edge equality has multiplier y_e ; an actual LP row may be multiplied by a public nonzero scale $r_e \neq 0$, in which case its invariant row force is $f_e = r_e y_e$. Assume exact primal–dual centrality, $d_v \neq 0$, a pair-symmetric objective, and that all nongraph equality columns are pair-symmetric.

Theorem 10.6 (Gauge flow and isoperimetric law). *Let $\tau_v = \text{sgn}(d_v)$. After the edge gauge $z_e = \tau_{h(e)} y_e$, signed stationarity is the positive-divergence law*

$$B^\top z = g, \quad g_v = \frac{2\mu|d_v|}{q_v^2 - d_v^2} > 0. \quad (390)$$

For every vertex set S with nonempty boundary,

$$\sum_{e \in \partial S} |y_e| \geq G_S, \quad \|y_{\partial S}\|_\infty \geq \frac{G_S}{|\partial S|}, \quad \|y_{\partial S}\|_2 \geq \frac{G_S}{\sqrt{|\partial S|}}, \quad G_S = \sum_{v \in S} g_v. \quad (391)$$

If $S \neq \emptyset$ but $\partial S = \emptyset$, no such positive central point exists. If $|d_v| \geq a > 0$ and $q_v \leq Q$ on S , then

$$\|y_{\partial S}\|_\infty \geq \frac{2\mu a}{Q^2 - a^2} \frac{|S|}{|\partial S|}, \quad \|y_{\partial S}\|_2 \geq \frac{2\mu a}{Q^2 - a^2} \frac{|S|}{\sqrt{|\partial S|}}. \quad (392)$$

For a row written as $r_e(d_v - a_e d_u) = 0$, all statements hold with the row force $f_e = r_e y_e$; hence $|r_e| \leq B_{\text{row}}$ charges $B_{\text{row}} \|y\|_\infty$, not the basis-dependent multiplier alone.

Proof. Subtract pair stationarity equations:

$$2(B_a^\top y)_v + s_v^+ - s_v^- = 0.$$

Since $x_v^\pm = (q_v \pm d_v)/2$ and $s_v^\pm = \mu/x_v^\pm$, this becomes $(B_a^\top y)_v = 2\mu d_v/(q_v^2 - d_v^2)$. Feasibility implies $a_e = \tau_u \tau_v$ on variable–variable edges; terminal rows have the same single variable coefficient in B_a and B . Switching the signed incidence matrix therefore gives (390). Summing over S cancels internal edges; the triangle inequality and Cauchy–Schwarz give (391). Positivity gives $q_v > |d_v|$, and the margin hypotheses lower-bound every g_v by $2\mu a/(Q^2 - a^2)$. A diagonal row scaling enters stationarity only as $r_e y_e$, proving the final statement. $\square$

Pair symmetry is not needed for a two-input comparison if the pair-antisymmetric source is the same numerical vector in the original pair coordinates. Let two exact central points, denoted $+$ and $-$, have row forces $f_e^\pm = r_e y_e^\pm$, and suppose their subtracted stationarity equations are

$$(B_{a^\pm}^\top f^\pm)_v = h_v + \frac{2\mu_\pm d_v^\pm}{(q_v^\pm)^2 - (d_v^\pm)^2}.$$

If $d_v^+ > 0 > d_v^-$ on S , switching the two inputs, adding their equations, and summing over S cancels both the common source and every internal edge. Hence

$$\sum_{e \in \partial S} (|f_e^+| + |f_e^-|) \geq \sum_{v \in S} (g_v^+ + g_v^-), \quad g_v^\pm = \frac{2\mu_\pm |d_v^\pm|}{(q_v^\pm)^2 - (d_v^\pm)^2}. \quad (393)$$

If $\mu_\pm \geq \mu_0$, $|d_v^\pm| \geq a$, $q_v^\pm \leq Q$, and $|r_e| \leq B_{\text{row}}$, this gives

$$\max_{p \in \{+, -\}} \|y_{\partial S}^p\|_\infty \geq \frac{2\mu_0 a}{Q^2 - a^2} \frac{|S|}{B_{\text{row}} |\partial S|}. \quad (394)$$

Thus a public asymmetric objective can cancel one parity orientation, but not both behind a small interface. Coefficients of another public asymmetric constraint do not suffice unless their multiplier-mediated contribution is also the same vector h on both inputs.

The graph is not essential to the global work. Let $Cd = b_a$ be an arbitrary pair-antisymmetric equality block, with columns C , $-C$, and let all other local data be pair-symmetric.

Theorem 10.7 (Central and Newton work identities). *At an exact central point,*

$$\boxed{b_a^\top y_a = \sum_v \frac{2\mu d_v^2}{q_v^2 - d_v^2} \geq 0.} \quad (395)$$

The work is strictly positive exactly when $d \neq 0$. At a pair-symmetric Newton start with pair-symmetric dual and complementarity right-hand sides, pair-symmetric other equality columns, $x_v^{0,+} = x_v^{0,-} = \xi_v$, and $s_v^{0,+} = s_v^{0,-} = \zeta_v$, the antisymmetric block satisfies

$$C_\zeta \Delta d = r_a, \quad C_\zeta^\top \alpha = D \Delta d, \quad D = \text{Diag} \left(\frac{\zeta_v}{2\xi_v} \right), \quad \boxed{r_a^\top \alpha = \Delta d^\top D \Delta d.} \quad (396)$$

Both pairings are invariant under $(C, b_a, y_a) \mapsto (UC, Ub_a, U^{-\top} y_a)$, and likewise for (C_ζ, r_a, α) . If $D \succeq \rho I$, $|\Delta d_v| \geq a$ on L coordinates, and $\|r_a\|_1 \leq R$, then $\|\alpha\|_\infty \geq \rho a^2 L / R$. Likewise, if $|d_v| \geq a$ and $q_v \leq Q$ on L coordinates, then

$$\|y_a\|_\infty \geq \frac{2\mu a^2 L}{(Q^2 - a^2) \|b_a\|_1}, \quad \|y_a\|_2 \geq \frac{2\mu a^2 L}{(Q^2 - a^2) \|b_a\|_2}. \quad (397)$$

A zero denominator with nonzero work is an impossibility statement.

Proof. Pair subtraction gives $C^\top y_a = 2\mu d / (q^2 - d^2)$ coordinatewise. Its inner product with d , followed by $Cd = b_a$, proves (395). At the Newton start, subtracting the dual and complementarity equations and eliminating the slack difference gives $C_\zeta^\top \alpha = D \Delta d$; primal antisymmetric feasibility gives $C_\zeta \Delta d = r_a$. Taking inner products proves (396), and Hölder gives the norm bounds. The contragredient transformations preserve both scalar pairings. $\square$

For the bounded secant start of Section 9.8, $D = (2/9)I$, $\Delta d_v = \tau_v$, and the root antisymmetric RHS is a unit vector. Equation (396) therefore gives the root multiplier $2P/9$. Scaling all dual data by λ changes this to $2\lambda P/9$; bounded complete directions require $\lambda = O(1/P)$, exactly the inverse dual-scale price paid by Theorem 9.13. Section 9.9 states the corresponding unrestricted

Newton virtual-work identity. We use the paired forms here because they expose gadget resources; [section 14](#) will place them in the general involution/symmetry framework.

The cut laws also hold directly for Newton directions. Under a pair-symmetric start, pair-symmetric Newton right-hand sides, and homogeneous signed-copy primal equations, switching by $\text{sgn}(\Delta d_v)$ gives

$$B^\top z = g^N, \quad g_v^N = \frac{\zeta_v}{2\xi_v} |\Delta d_v|. \quad (398)$$

The proof is the Newton half of Theorem [10.7](#) followed by the cut summation in Theorem [10.6](#). In particular, if $\zeta_v/\xi_v \geq r$ and $|\Delta d_v| \geq a$ on S , then

$$\|\Delta y_{\partial S}\|_\infty \geq \frac{ra}{2} \frac{|S|}{|\partial S|}.$$

Now assume terminal values, labels, and terminal Newton residuals are public and fixed, and fix a base input for which it and all single-flip neighbors have exact central points with nonzero differences. Let O be a fixed region of K output vertices whose orientation flips under every one-bit change, and define

$$S_j = \{v : \text{sgn } d_v(\zeta^{(j)}) = -\text{sgn } d_v(\zeta)\}.$$

Let M_j count the signed rows changing under flip j , and put $M_{\text{dep}} = \sum_j M_j$. Feasibility on an internal edge gives $a_e = \text{sgn}(d_u) \text{sgn}(d_v)$, so its label changes exactly when one endpoint is in S_j . Public terminal signs act as fixed outside vertices, so the same boundary accounting includes terminal edges. Hence $|\partial S_j| \leq M_j$, with equality when rows rather than expanded coefficient positions are counted, and $O \subseteq S_j$. The base multiplier must satisfy all N cut inequalities simultaneously. Assume explicitly that the base central point has $|d_v| \geq a = \Omega(1)$ and $q_v \leq Q = O(1)$ on O . Applying this central margin only on O and summing over j proves

$$M_{\text{dep}} B_{\text{row}} \|y\|_\infty \geq \frac{2\mu a}{Q^2 - a^2} KN. \quad (399)$$

For Newton directions, suppose instead that the base and every single-flip system satisfy the hypotheses of [\(398\)](#), define S_j from the signs of Δd , and assume that every output direction flips with $|\Delta d_v| \geq a = \Omega(1)$ on O . The same argument gives

$$M_{\text{dep}} B_{\text{row}} \|\Delta y\|_\infty \geq \rho a KN, \quad \rho = \min_{v \in O} \frac{s_v^0}{2x_v^0}. \quad (400)$$

Thus $M_{\text{dep}} = \Theta(N)$, $K = \Theta(N)$, bounded row scale, and bounded multipliers, together with the explicit constant margin $a = \Omega(1)$ and central range $Q = O(1)$, force $\mu = O(1/N)$ in the central system and $\rho = O(1/N)$ in the Newton system. Equation [\(400\)](#) is the general conservation framework behind the beyond-pair design obstruction in [Section 9.9](#). It says that every signed-copy gadget pays somewhere: in incidences, row scale, multiplier height, or inverse dual scale. Branching and redundant copy rows merely move the payment into the relevant boundary count.

Pair-asymmetric local data do not provide a cheap universal cancellation. Suppose a prescribed direct source $h_v(\zeta)$ changes central stationarity to

$$(B_a^\top f)_v = \frac{2\mu d_v}{q_v^2 - d_v^2} + h_v(\zeta), \quad f_e = r_e y_e.$$

On a fixed output region O , assume $\text{sgn}(d_v) = \kappa_v p(\zeta)$, $|d_v| \geq a$, and $q_v \leq Q$ for every input. With the uniform Fourier convention $\hat{h}_v([N]) = 2^{-N} \sum_\zeta p(\zeta) h_v(\zeta)$, put $\varepsilon_O = K^{-1} \sum_{v \in O} |\hat{h}_v([N])|$.

Theorem 10.8 (Fourier obstruction to direct cancellation). *For some input ς^* ,*

$$\sum_{e \in \partial O} |r_e y_e(\varsigma^*)| \geq K \left(\frac{2\mu a}{Q^2 - a^2} - \varepsilon_O \right). \quad (401)$$

In particular, bounded-bit juntas that depend on strictly fewer than N input bits, and every source of Fourier degree less than N , have zero full-parity coefficient and cannot cancel the demand on all inputs. The bound is tight: a direct source $h_v = -\kappa_v p(\varsigma) g_0$ cancels a constant demand g_0 exactly.

Proof. After switching, sum the stationarity equation over O . The direct contribution is $H(\varsigma) = \sum_{v \in O} \kappa_v p(\varsigma) h_v(\varsigma)$, whose input average is at least $-K\varepsilon_O$. Some ς^* has $H(\varsigma^*)$ at least this average. The positive central demand is at least $2\mu a K / (Q^2 - a^2)$; internal edge forces cancel, and the triangle inequality proves (401). $\square$

There is an exact Newton analogue. At a pair-symmetric start put $x_v^{0,\pm} = \xi_v$, $s_v^{0,\pm} = \zeta_v$, and allow asymmetric dual and complementarity right-hand sides. Pair subtraction gives

$$(B_a^\top \Delta f)_v = \frac{\zeta_v}{2\xi_v} \Delta d_v + h_v^N(\varsigma), \quad h_v^N = \frac{r_{d,v}^+ - r_{d,v}^-}{2} - \frac{r_{c,v}^+ - r_{c,v}^-}{2\xi_v}.$$

If $\text{sgn}(\Delta d_v) = \kappa_v p(\varsigma)$, $|\Delta d_v| \geq a$, and $\zeta_v / (2\xi_v) \geq \rho$ on O , then the same proof gives some input ς^* with

$$\sum_{e \in \partial O} |r_e \Delta y_e(\varsigma^*)| \geq K(\rho a - \varepsilon_O^N), \quad \varepsilon_O^N = K^{-1} \sum_{v \in O} |\hat{h}_v^N([N])|. \quad (402)$$

This Fourier coefficient must be checked after division by the start coordinates: input dependence in $1/\xi_v$ can manufacture full parity from otherwise local raw data. The word *direct* also matters. An unknown multiplier of another antisymmetric constraint may depend on all input bits even when that row is local, so Theorem 10.8 does not silently assign it low Fourier degree.

Switching and incidence-flow conservation are classical [75, 144]; the work identities are standard KKT virtual-work and Schur-energy specializations. Electrical flows on interior-point central paths are also established algorithmic tools [98]. The contribution claimed here is only the design-constraint conjunction: the local input cuts, parity-sized output region, multiplier conservation, row scaling, and incidence accounting must all be met at once. Effective resistance has a separate established role in quantum graph algorithms [9, 23, 79]; those works do not imply these LP residual or multiplier frontiers.

10.5 Relative value and shortcut remnants

The path family also gives a clean value-output statement. Put $p_0 = 1$ and $p_i = \prod_{k=1}^i \varsigma_k$. On one unit signed path minimize

$$\frac{1}{N+1} \sum_{i=0}^N (u_i + v_i) + \frac{1}{4} (u_N - v_N) \quad (403)$$

subject to $d_0 = 1$ and $d_i - \varsigma_i d_{i-1} = 0$.

Theorem 10.9 (Relative-value gap). *In raw sparse coefficient access, estimating the optimum of (403) to target-relative error $\epsilon < 1/4$ requires $\Omega(N)$ queries. The LP has a finite attained optimum, is strictly primal-dual feasible, and has a unique nondegenerate strictly complementary optimum. Its objective norm is bounded by an absolute constant, and its reduced primal log-barrier Hessian has condition one along the entire central path.*

Proof. Exact feasibility fixes $d_i = p_i$ and $q_i = u_i + v_i \geq 1$. On this affine space the objective is $(N+1)^{-1} \sum_i q_i + p_N/4$, so the unique optimum has every $q_i = 1$ and

$$\text{OPT} = 1 + \frac{1}{4}p(\varsigma) \in \left\{ \frac{3}{4}, \frac{5}{4} \right\}. \quad (404)$$

The two target-relative error intervals are disjoint for every $\epsilon < 1/4$; hence such value estimation requires $\Omega(N)$ raw queries by the parity lower bound. Even an exactly feasible point with objective at most $3\text{OPT}/2$ distinguishes the cases by a threshold between $9/8$ and $5/4$. Every sparse coefficient query is simulated by one raw sign query. Also $\|c\|_2^2 = 2/(N+1) + 1/8 = O(1)$.

The equality matrix is full row rank, and its public orthonormal null basis consists of $(e_{u_i} + e_{v_i})/\sqrt{2}$. Taking every $q_i = 2$ proves strict primal feasibility. Taking $s_{u_i} = s_{v_i} = 1/(N+1)$ gives a positive slack for which $c - s$ is the endpoint difference term; it is orthogonal to the nullspace and hence lies in $\text{range}(A^\top)$, proving strict dual feasibility. At the optimum, one positive coordinate from each pair gives a triangular basis, and every opposite coordinate has positive reduced cost $2/(N+1)$. This proves nondegeneracy and strict complementarity.

Finally, on the feasible slice the barrier objective is

$$\sum_{i=0}^N \left[\frac{q_i}{N+1} - \mu \log \frac{q_i^2 - 1}{4} \right] + \frac{1}{4}p_N.$$

Every central coordinate solves the same scalar equation $(N+1)^{-1} = 2\mu q/(q^2 - 1)$, so the Hessian in the displayed orthonormal null basis is a positive scalar multiple of the identity. $\square$

This is a normalization-strengthened value reduction, not a generic LP query lower bound.

One tempting repair is a long two-coordinate shortcut. If $0 \leq i < j \leq N$ and a homogeneous row $a(\varsigma)d_i + b(\varsigma)d_j = 0$ is valid on every exact signed path, with both $a(\varsigma)$ and $b(\varsigma)$ nonzero for every input covered by the conclusion, then substitution gives

$$-\frac{a(\varsigma)}{b(\varsigma)} = \frac{p_j}{p_i} = \prod_{k=i+1}^j s_k. \quad (405)$$

Thus the row itself carries interval parity, invariant under common row rescaling, and generally costs $\Omega(j-i)$ raw queries to simulate. The older replicated-path and expander calculations are otherwise superseded by the mixture, resistance, and Walsh-row theorems above.

10.6 Boundary of the conclusions

The parallel-path and gain-plateau constructions of [section 9](#) are optimal in their stated bounded-height and linear-size gadget classes. The all-unit family's $\Theta(N^{-1/2})$ residual threshold in [Theorem 9.11](#) is the matching unamplified boundary, and dual homogeneity attains the $\rho = \Theta(1/N)$ escape demanded by [\(400\)](#). None of these conclusions says that parity needs a quadratic-size LP, that every bounded-range LP fails, or that an adaptive QIPM pays a fresh linear query cost at every iteration.

A bounded-range, linear-incidence construction must leave at least one premise used here. The recorded escape routes are:

- (i) use a nonlinear or genuinely non-mixture-stable output promise and prove its robustness directly;
- (ii) require only central or Newton outputs and permit the neighboring central coordinates to leave every fixed Kantorovich neighborhood;
- (iii) use nonlocal input coefficients and charge their raw-query implementation honestly;

- (iv) abandon the common objective or common optimum, so neighbor mixtures need not be objective-exact; or
- (v) impose integrality or another nonconvex promise, leaving linear programming.

The second route alone is not automatic: equations (381)–(382) show exactly what central instability it must supply.

Two problems remain open within the present scope. First, can an arbitrary sparse bounded-range equality LP with a nonlinear normalized-state decoder achieve a fixed global residual tube without nonlocal coefficient access? Second, can the bounded-scale complete-direction problem in Open Problem 9.14 be solved by a nonorthogonal mixed-mode gadget while charging all input-dependent sources and recovery maps? Together with the orthogonal-copy cut law in section 9.9, the theorems here rule out the natural convex-mixture-stable, signed-copy, constant-arity Walsh, permutation, rotation, and flat-copy routes; they do not settle either question.

11 SDP Hardness Families

This section gives SDP-native lower bounds rather than embedding the LP families of section 9. Three output levels must be kept separate. The first family has an intrinsic, constant gap in the optimal value. The second has a public Schur-complement solve but makes the requested original-coordinate state expensive to load. At the third level, even a constant-normalization encoding of a canonical KKT matrix, or the first synthesis of a tangent projector, may require linear raw-input work. All claims use the access and output contracts of sections 2.2 and 2.3.

The geometric boundary is equally important. The first trace-normalized family exposes the density slice of $\mathbb{S}_+^3$. In the Schur family the 3×3 blocks leave only $\mathbb{S}_+^2$ and hence are SOCP-representable; 4×4 blocks are the first members whose free Schur complement is $\mathbb{S}_+^3$ and has no finite second-order-cone lift.

11.1 An intrinsic-spectrum trace-normalized family

Let $n_{\text{bit}} \geq 1$, let

$$K = 16n_{\text{bit}}, \quad P = 2K + n_{\text{bit}} = 33n_{\text{bit}}, \quad h = \prod_{j=1}^{n_{\text{bit}}} \varsigma_j, \quad (406)$$

and put $D_s = \text{diag}(s, 1, 1)$. There are P blocks $X_0, \dots, X_{P-1} \in \mathbb{S}_+^3$. Consecutive blocks obey

$$X_i = G_i X_{i-1} G_i^T. \quad (407)$$

The first $K - 1$ and last K edges have $G_i = I$; the intervening n_{bit} edges have $G_i = D_{\varsigma_j}$ in input order. Add $\text{tr } X_0 = 1$. If $R_0 = I$ and $R_i = G_i \cdots G_1$, elimination gives exactly

$$X_i = R_i Z R_i^T, \quad Z \succeq 0, \quad \text{tr } Z = 1. \quad (408)$$

In the Frobenius-isometric coordinates

$$\text{svec}(X) = (X_{11}, \sqrt{2}X_{12}, \sqrt{2}X_{13}, X_{22}, \sqrt{2}X_{23}, X_{33}), \quad (409)$$

congruence by D_s is $\text{diag}(1, s, s, 1, 1, 1)$. Each copy row has two nonzeros, the trace row has three, and each scalar column occurs in at most two rows. One input sign changes only the two predecessor coefficients in the 12 and 13 rows. Ordering the copy rows along the path proves that they are independent; their kernel is the six-dimensional root family in (408), and the trace row removes one further dimension.

For $a < b$, write $H_{ab} = e_a e_b^T + e_b e_a^T$, and set

$$u = \frac{1}{10}, \quad q = \frac{K}{P} = \frac{16}{33}, \quad \gamma = \frac{u}{q} = \frac{33}{160}. \quad (410)$$

Use the public costs

$$C_i = \begin{cases} 3I + uH_{23} + \gamma H_{12}, & 0 \leq i < K, \\ 3I + uH_{23}, & K \leq i < K + n_{\text{bit}}, \\ 3I + uH_{23} + \gamma H_{13}, & K + n_{\text{bit}} \leq i < P, \end{cases} \quad (411)$$

and consider

$$\min \left\{ \frac{1}{P} \sum_{i=0}^{P-1} \langle C_i, X_i \rangle : (407), \text{ tr } X_0 = 1, X_i \succeq 0 \right\}. \quad (412)$$

Because $\sqrt{u^2 + \gamma^2} < 1/4$, all local costs have spectrum in $(11/4, 13/4)$. The point $X_i = I/3$ is strictly primal feasible, and zero equality multipliers with slacks C_i/P prove strict dual feasibility.

Theorem 11.1 (Intrinsic qutrit parity). *The program (412) has optimal values*

$$v_+ = \frac{29}{10}, \quad v_- = \frac{14}{5}. \quad (413)$$

Consequently additive-1/25 or target-relative-1/100 value estimation requires $\Omega(P)$ raw coefficient queries. More exactly,

$$Q_{\text{add}}(\epsilon) = \begin{cases} \lceil n_{\text{bit}}/2 \rceil, & \epsilon < 1/20, \\ 0, & \epsilon \geq 1/20, \end{cases} \quad Q_{\text{rel}}(\eta) = \begin{cases} \lceil n_{\text{bit}}/2 \rceil, & \eta < 1/57, \\ 0, & \eta \geq 1/57. \end{cases} \quad (414)$$

The feasible spectrahedron has no lift over any finite product of second-order cones.

Proof. The H_{23} term is invariant under every R_i . The first anchor remains in the root frame and the last acquires the final sign h . Since $q\gamma = u$, pulling the objective through (408) gives

$$\overline{C}_h = 3I + uA_h, \quad A_h = H_{12} + H_{23} + hH_{13}. \quad (415)$$

For $h = +1$, A_h has eigenvalues $2, -1, -1$; for $h = -1$, its signed cycle product is negative and its eigenvalues are $1, 1, -2$. Hence

$$\text{spec}(\overline{C}_+) = \left\{ \frac{16}{5}, \frac{29}{10}, \frac{29}{10} \right\}, \quad \text{spec}(\overline{C}_-) = \left\{ \frac{31}{10}, \frac{31}{10}, \frac{14}{5} \right\}. \quad (416)$$

Rayleigh–Ritz on $Z \succeq 0$, $\text{tr } Z = 1$, proves (413). The common trace nine rules out an orthogonal congruence followed by nontrivial positive rescaling. Even after adding a scalar multiple of I , the simple-largest/repeated-smallest and repeated-largest/simple-smallest multiplicities cannot be exchanged. The gap is therefore intrinsic to the optimization problem.

The additive intervals are disjoint exactly when $\epsilon < 1/20$ and meet at $57/20$. The relative intervals meet exactly at

$$\eta_* = \frac{v_+ - v_-}{v_+ + v_-} = \frac{1}{57}, \quad v_+(1 - \eta_*) = v_-(1 + \eta_*) = \frac{812}{285}. \quad (417)$$

Below either threshold, comparison with a public separating point computes parity with bounded error. One coefficient query uses at most one sign query, and the approximate-degree bound recalled in section 2.4 gives the bounded-error lower bound $\lceil n_{\text{bit}}/2 \rceil$. Conversely, apply Deutsch’s one-query circuit to each pair and query an unpaired last sign; this computes h in exactly that

many queries. At and above the thresholds the displayed intersection points are public zero-query answers.

Projection onto X_0 identifies the feasible set with $\mathcal{D}_3 = \{Z \succeq 0 : \text{tr } Z = 1\}$. If it had a finite SOC lift, replace each affine lift equation $Ay = b$ by $Ay = tb$ and impose $t \geq 0$. For $t > 0$ the projection is $t\mathcal{D}_3$. At $t = 0$, a homogeneous lift direction could be added at arbitrary scale to a fixed feasible lift; compactness of $\mathcal{D}_3$ forces its matrix projection to be zero. The homogenized projection is exactly $\mathbb{S}_+^3$, contradicting Fawzi's theorem [61]. $\square$

With $r_+ = (1, 1, 1)^T/\sqrt{3}$ and $r_- = (1, -1, 1)^T/\sqrt{3}$, strictly complementary reduced pairs are

$$Z_+^* = \frac{1}{2}(I - r_+ r_+^T), \quad Z_-^* = r_- r_-^T, \quad \bar{S}_+^* = \frac{3}{10}r_+ r_+^T, \quad \bar{S}_-^* = \frac{3}{10}(I - r_- r_-^T). \quad (418)$$

Lifting by $X_i^* = R_i Z_h^* R_i^T$ and $S_i^* = P^{-1} R_i \bar{S}_h^* R_i^T$ gives a strictly complementary product-cone optimum. Stationarity follows because the remaining coefficient tuple is orthogonal to every trace-zero feasible tangent and hence lies in the equality adjoint's range.

11.1.1 A public, condition-one Newton step

Use

$$\Phi_\mu(X) = \frac{1}{P} \sum_i \langle C_i, X_i \rangle - \mu \sum_i \log \det X_i, \quad X_i^0 = I/3, \quad \mu = 1/P. \quad (419)$$

At this point, in (409),

$$g_i = (C_i - 3I)/P, \quad \nabla^2 \Phi_\mu(X^0) = (9/P)I. \quad (420)$$

Thus both right-hand sides of

$$\tilde{A}_\zeta \Delta x = 0, \quad (9/P) \Delta x - \tilde{A}_\zeta^T \Delta y = -g \quad (421)$$

are public. For trace-zero Y , define the isometry $\mathcal{W}_\zeta(Y) = P^{-1/2}(R_0 Y R_0^T, \dots, R_{P-1} Y R_{P-1}^T)$. Then

$$\mathcal{W}_\zeta^* \nabla^2 \Phi_\mu(X^0) \mathcal{W}_\zeta = (9/P)I_5. \quad (422)$$

In the physical root coordinate the Hessian is $9I_5$ and

$$\Delta Z_h = -\frac{u}{9} A_h. \quad (423)$$

Its decrement and the standard full-step bound are

$$\lambda = \frac{1}{\sqrt{150}}, \quad \lambda_{\text{next}} \leq \frac{1}{(\sqrt{150} - 1)^2} < \frac{1}{126}. \quad (424)$$

Indeed $\lambda^2 = \langle u A_h, (9I)^{-1} u A_h \rangle = 1/150$. The post-step root eigenvalues are $(14/45, 31/90, 31/90)$ for $h = +1$ and $(29/90, 29/90, 16/45)$ for $h = -1$, so the undamped step remains positive in every block. The next-decrement estimate is the universal self-concordant bound, not a claim about a paper-specific primal–dual neighborhood.

Theorem 11.2 (Newton and complete-KKT state hardness). *At (419), preparing the normalized root or global primal Newton-direction state to trace distance $1/100$ requires $\Omega(P)$ raw queries. This remains true for every exact feasible lift of a reduced solve Y satisfying*

$$\frac{\|9Y + u A_h\|_F}{\|u A_h\|_F} \leq \frac{1}{10}. \quad (425)$$

It also holds for the normalized complete primal-plus-multiplier solution of (421) in the displayed unit row scaling and multiplier convention. The latter has a public parity observable of magnitude at least $1489/2097 > 7/10$.

Proof. Up to global sign,

$$|\delta z_h\rangle = \frac{|12\rangle + |23\rangle + h|13\rangle}{\sqrt{3}}. \quad (426)$$

The public contraction $T = |12\rangle\langle 13| + |13\rangle\langle 12|$ has expectation $2h/3$. At block i , both swapped coordinates acquire the same prefix sign, so the same expectation holds for $T_{\text{glob}} = \bigoplus_i T$. The POVM $(I \pm T)/2$ succeeds with probability $5/6$, and trace error $1/100$ leaves constant bias.

For the residual claim, (425) is exactly $\|Y - \Delta Z_h\|_F \leq \eta \|\Delta Z_h\|_F$ with $\eta \leq 1/10$. The projective sine angle from the exact ray is at most η . Direct minimization gives the sharp bound

$$h\langle T \rangle \geq m(\eta) := \frac{2}{3} - \frac{\eta^2}{3} - \frac{2\sqrt{2}}{3}\eta\sqrt{1-\eta^2}, \quad 0 \leq \eta \leq \frac{1}{3}. \quad (427)$$

To verify it, at sine angle s write the only useful tangent coefficient as $x \in [-1, 1]$. The expectation is $\frac{2}{3}(1-s^2) - s^2 + \frac{2\sqrt{2}}{3}s\sqrt{1-s^2}x + \frac{4}{3}s^2x^2$. For $s \leq 1/3$ its unconstrained minimizer is below -1 , so $x = -1$, and the result decreases with s . At $\eta = 1/10$ it is $199/300 - \sqrt{198}/150 > 17/30$. Also $\lambda_{\min}(I/3 + Y) > 3/10$.

For the complete vector, put

$$\alpha = 16/33, \quad c = \sqrt{2}\gamma/P, \quad d = \sqrt{2}u/9, \quad v_e = \begin{cases} c(1-\alpha)e, & e \leq K, \\ c\alpha(P-e), & K < e < P. \end{cases} \quad (428)$$

If τ_i is the prefix sign, the three primal chains are $-d\tau_i, -dh\tau_i, -d$ in coordinates 12, 13, 23. Switching signed path incidence to ordinary incidence shows that the only nonzero copy multipliers are $-\tau_e v_e$ on 12 and $h\tau_e v_{P-e}$ on 13. The public within-sector swap then has expectation $2h(D-C)/(3D+2V)$, where

$$\begin{aligned} D &= \frac{11n_{\text{bit}}}{1350}, \\ V &= \frac{289n_{\text{bit}}}{4950} + \frac{17}{158400n_{\text{bit}}}, \\ C &= \frac{577n_{\text{bit}}}{9900} + \frac{1}{9900n_{\text{bit}}}. \end{aligned} \quad (429)$$

Its sign is $-h$. The leading ratio is $1489/2097$; the ratio of positive $1/n_{\text{bit}}$ corrections is $16/17 > 1489/2097$, so finite size only increases the magnitude. Reversing outcome labels decodes parity. Each preparer would therefore compute parity and costs $\Omega(n_{\text{bit}}) = \Omega(P)$ queries. $\square$

The complete-state claim is coordinate-specific: rescaling equality rows changes multiplier amplitudes. Also $DA_-D = -A_+$ for $D = \text{diag}(1, -1, 1)$, so direction rays are related by an input-dependent orthogonal gauge. The value gap is invariant; the state theorem requests fixed public coordinates, consistent with [section 6](#).

11.1.2 Ambient robustness and access objects

For arbitrary $X_i \succeq 0$, define

$$e_0 = \text{tr } X_0 - 1, \quad E_i = X_i - G_i X_{i-1} G_i^T, \quad e_0^2 + \sum_{i=1}^{P-1} \|E_i\|_F^2 \leq \epsilon^2. \quad (430)$$

Gauging to $Z_i = R_i^T X_i R_i$ and telescoping gives

$$\left| P^{-1} \sum_i \text{tr } X_i - 1 \right| < \sqrt{P}\epsilon, \quad \left| P^{-1} \sum_i \langle C_i, X_i \rangle - \langle \bar{C}_h, Z_0 \rangle \right| < \frac{13}{4} \sqrt{P}\epsilon. \quad (431)$$

The second bound writes the difference as suffix costs against $Z_i - Z_{i-1}$ and uses $\|C_i\|_{\text{op}} < 13/4$. If the objective is at most $v_h + 1/100$ and $\epsilon \leq 1/(400\sqrt{P})$, define

$$\rho(X) := \frac{\bigoplus_i X_i}{\sum_i \text{tr } X_i}. \quad (432)$$

The trace bounds in (431) make this density operator well-defined. The public contraction

$$M = \frac{1}{2/5} \left(\bigoplus_i C_i - \frac{57}{20} I \right) \quad (433)$$

satisfies $h \text{tr}(M\rho(X)) > 1/16$. Specifically the normalized cost is $> 23/8$ for $h = +1$ and $< 113/40$ for $h = -1$. Thus any procedure which, for every input, returns a density operator $\tilde{\rho}$ satisfying $D_{\text{tr}}(\tilde{\rho}, \rho(X)) \leq 1/100$ for some PSD tuple X meeting the displayed residual and objective conditions requires $\Omega(P)$ raw queries. The tuple may be selected adversarially by the procedure. The residual order is necessary: averaging the n_{bit} neighboring negative-parity optimizers against a fixed positive-parity input gives wrong value and residual

$$\frac{4}{3\sqrt{n_{\text{bit}}}} = \Theta(P^{-1/2}). \quad (434)$$

For access objects, use the following state-pair construction. If a real matrix B has public support and magnitudes and maximum absolute row and column sums at most a , give each supported pair (r, c) its own token. Row and column states with amplitudes $\sqrt{|B_{rc}|/a}$ and $\text{sgn}(B_{rc})\sqrt{|B_{rc}|/a}$, completed on disjoint failure tokens, are orthonormal and have overlap B_{rc}/a . Unitary completions therefore give an exact normalization- a block encoding.

Put $y' = -\Delta y$ in (421) and apply this to

$$M_{\zeta} = \begin{pmatrix} (9/P)I & \tilde{A}_{\zeta}^T \\ \tilde{A}_{\zeta} & 0 \end{pmatrix}. \quad (435)$$

Its maximum absolute row and column sums equal three; the trace row attains three and a primal row has sum at most $2 + 9/P < 3$. One query phases the four symmetric hidden KKT tokens. This gives a fully specified exact normalization-three encoding, and substitution into the decoders proves

$$q_{\text{BE}} \geq \left\lceil \frac{n_{\text{bit}}}{2} \right\rceil \quad (436)$$

for root, global, or complete-KKT state preparation. The claim is relative to this completion; input-dependent junk may contain parity and must be charged.

Let Π_{ζ} project onto $\ker \tilde{A}_{\zeta}$. Projecting (421) and calculating norms gives

$$\|\Pi_{\zeta} g\|_2^2 = \frac{3}{50P}, \quad \|g\|_2^2 = \frac{41}{400P}, \quad \|\Pi_{\zeta} |\hat{g}\rangle\|_2^2 = \frac{24}{41}. \quad (437)$$

One use of an operator-error- $1/32$ normalization-one projector encoding then leaves signed expectation at least $16/41 - \sqrt{24/41}/16 - 1/1024 > 0.341$. Hence first synthesis costs at least $\lceil n_{\text{bit}}/2 \rceil$ raw queries. Conversely, applying all prefix signs to the 12, 13 chains constructs the tangent isometry; conjugating the public trace-zero root projector gives an exact encoding with $2n_{\text{bit}}$ queries and no QRAM. Projector synthesis is therefore $\Theta(n_{\text{bit}})$; a supplied projector already performs the aggregation.

Condition one in (422) does not describe the unscaled saddle matrix. Switching removes all path signs without changing singular values. On each individual off-diagonal coordinate, the path-difference operator has singular values $2\sin(j\pi/(2P))$, $1 \leq j < P$; in particular, this gives $s_{\min}^+(\tilde{A}_{\zeta}) \leq \pi/P$.

The trace-coupled diagonal subsystem supplies the missing lower bound. Let the three-coordinate vector $x = (x^{(1)}, x^{(2)}, x^{(3)})$ be orthogonal to its kernel, and write $m_j = P^{-1}\mathbf{1}^T x^{(j)}$. Orthogonality to every constant triple whose coefficients sum to zero implies $m_1 = m_2 = m_3 =: m$. Put $x^{(j)} = m\mathbf{1} + z^{(j)}$, with each $z^{(j)}$ mean zero, and define

$$E^2 = \sum_{j=1}^3 \|Bz^{(j)}\|_2^2, \quad t = \sum_{j=1}^3 x_0^{(j)}, \quad (438)$$

where B is ordinary path incidence. The path Poincaré inequality and telescoping give

$$\sum_j \|z^{(j)}\|_2^2 \leq \frac{P^2}{4} E^2, \quad |z_0^{(j)}| \leq \sqrt{P} \|Bz^{(j)}\|_2. \quad (439)$$

Since $3m = t - \sum_j z_0^{(j)}$, it follows that $\|x\|_2^2 \leq (9/4)P^2(t^2 + E^2)$. The last parenthesis is exactly the squared norm of the diagonal subsystem applied to x . Combining diagonal and off-diagonal coordinates, and using the maximum absolute row and column sums three and two, yields

$$\frac{2}{3P} \leq s_{\min}^+(\tilde{A}_\zeta) \leq \frac{\pi}{P}, \quad s_{\max}(\tilde{A}_\zeta) \leq \sqrt{6}. \quad (440)$$

The kernel gives the saddle eigenvalue $9/P$, while every positive equality singular value s gives eigenvalues $\{9/P \pm \sqrt{(9/P)^2 + 4s^2}\}/2$. The two-sided bounds in (440) therefore prove

$$\kappa_2(M_\zeta) = \Theta(P). \quad (441)$$

Its scalar off-diagonal graph is nevertheless a forest: each coordinate forms a subdivided path and the trace multiplier joins three distinct diagonal paths at one new vertex. Thus its treewidth is one.

11.2 Non-SOCP Schur-block families

Fix $k \geq 2$, let $K = 16n_{\text{bit}}$ and $L = 17n_{\text{bit}} + 1$, and set $a_i = \zeta_i$ for $i \leq n_{\text{bit}}$ and $a_i = 1$ thereafter, with $\tau_i = \prod_{j=1}^i a_j$. For $X_i \in \mathbb{S}_+^{k+1}$, put $F_j = (e_j e_{k+1}^T + e_{k+1} e_j^T)/\sqrt{2}$ and $E = e_{k+1} e_{k+1}^T$. Impose

$$\begin{aligned} \langle F_1, X_0 \rangle &= \sqrt{2}, & \langle F_1, X_i \rangle - a_i \langle F_1, X_{i-1} \rangle &= 0, \\ \langle F_j, X_0 \rangle &= 0, & \langle F_j, X_i \rangle - \langle F_j, X_{i-1} \rangle &= 0 \quad (2 \leq j \leq k), \\ \langle E, X_0 \rangle &= 1, & \langle E, X_i \rangle - \langle E, X_{i-1} \rangle &= 0 \end{aligned} \quad (442)$$

for $1 \leq i < L$. Rows and scalar columns have incidence at most two. Every feasible block is

$$X_i = \begin{pmatrix} A_i & \tau_i e_1 \\ \tau_i e_1^T & 1 \end{pmatrix}, \quad Z_i = A_i - e_1 e_1^T \succeq 0. \quad (443)$$

Put $\mu_0 = 1/L$ and use $C_0 = \mu_0 \text{diag}(I_k, 2)$ on every block.

Theorem 11.3 (Exact Schur path and SOCP boundary). *The unique, strictly complementary optimum has $Z_i = 0$. For $t = \mu/\mu_0$, the complete central path is*

$$X_{\tau_i}^{(k)}(t) = \begin{pmatrix} tI_k + e_1 e_1^T & \tau_i e_1 \\ \tau_i e_1^T & 1 \end{pmatrix}, \quad S_i(\mu) = \begin{pmatrix} \mu_0 I_k & -\mu_0 \tau_i e_1 \\ -\mu_0 \tau_i e_1^T & \mu + \mu_0 \end{pmatrix}. \quad (444)$$

In every public Frobenius-orthonormal tangent basis, $H_{\text{red}}(\mu) = (\mu_0/\mu)^2 I = t^{-2} I$. All primal, slack, and multiplier coordinates are bounded for $0 < \mu \leq \mu_0$. The feasible block slice has a finite SOCP lift exactly when $k \leq 2$.

Proof. Schur complementation gives (443), $\det X_i = \det Z_i$, and $\langle C_0, X_i \rangle = \mu_0(\text{tr } Z_i + 3)$. Thus $Z_i = 0$ is the unique optimum, and $\mu_0 \text{tr } Z_i - \mu \log \det Z_i$ is uniquely minimized at tI_k , proving the primal formula. Direct inversion gives the slack. With $S = C - \mathcal{A}^*y$, the nonzero chain multipliers are

$$\alpha_i = \sqrt{2}\mu_0(L-i)\tau_i, \quad \beta_i = (\mu_0 - \mu)(L-i). \quad (445)$$

Their divergences reproduce $C - S$. On $0 < t \leq 1$, primal entries are at most two, slack entries at most $2/L$, and $\|\alpha\|_\infty \leq \sqrt{2}$, $\|\beta\|_\infty \leq 1$. At $t = 0$ the rank-one primal and rank- k displayed slack are strictly complementary. For dual uniqueness, put $w_i = (\tau_i e_1, 1)^T$, so $X_i^{\text{opt}} = w_i w_i^T$. Zero gap for any dual optimum gives $S_i w_i = 0$. Stationarity in the unconstrained upper-left coordinates fixes $(S_i)_{1:k, 1:k} = \mu_0 I_k$; annihilating w_i then forces the last column and row to be $(-\mu_0 \tau_i e_1, \mu_0)^T$, exactly the $t = 0$ limit in (444). Finally, full row rank of $\mathcal{A}$ makes $\mathcal{A}^*$ injective, so the equality multipliers are unique as well.

Every feasible tangent is $\Delta X_i = \begin{pmatrix} H_i & 0 \\ 0 & 0 \end{pmatrix}$. Using the inverse of (444), whose upper-left block is $t^{-1}I_k$ because $\tau_i^2 = 1$, the reduced log-det Hessian is

$$\sum_i \text{tr}(X_i^{-1} \Delta X_i X_i^{-1} \Delta Y_i) = \frac{\mu_0^2}{\mu^2} \sum_i \langle H_i, K_i \rangle,$$

proving scalarity. The two local centers do not commute: $[X_+^{(k)}(t), X_-^{(k)}(t)] = 2t(e_{k+1}e_1^T - e_1e_{k+1}^T)$. Finally, (443) is an affine bijection from $\mathbb{S}_+^k$. $\mathbb{S}_+^2$ is a Lorentz cone, while $\mathbb{S}_+^3$ has no finite SOC lift; larger k contain it as a face. This proves the exact boundary. $\square$

The noncommutativity is removable by an input-dependent congruence; in the Z_i coordinates the optimization is public. The product-cone representation is essential: replacing it by one unrestricted $(k+1)L$ -dimensional PSD block and killing off-block entries takes quadratically many scalar equalities.

11.2.1 Tight loading and the accuracy frontier

For $k = 3$ and $0 < t \leq 1$, define

$$\rho_\varsigma(t) = \frac{\bigoplus_i X_{\tau_i}^{(3)}(t)}{L(3t+2)}. \quad (446)$$

On the suffix $T = \{n_{\text{bit}} + 1, \dots, n_{\text{bit}} + K\}$, every $\tau_i = h$. Let O equal $e_1 e_4^T + e_4 e_1^T$ on these blocks and zero elsewhere. Then

$$h \text{tr}(O \rho_\varsigma(t)) = \frac{2K}{L(3t+2)} \geq \frac{32n_{\text{bit}}}{5(17n_{\text{bit}}+1)} > \frac{1}{3}. \quad (447)$$

Thus trace-distance-1/100 density preparation costs $\Omega(L)$ queries.

For the complete central triple $g_\varsigma(t) = (\text{svec } X, y, \text{svec } S)$,

$$\|g_\varsigma(t)\|_2^2 = L(3t^2 + 2t + 4) + [2 + (1-t)^2] \frac{(L+1)(2L+1)}{6L} + \frac{5 + (1+t)^2}{L} < 11L. \quad (448)$$

Swapping the suffix primal coordinates 1 and $\sqrt{2}\tau_i$ gives a public contraction W with $h\langle W \rangle = 2\sqrt{2}K/\|g_\varsigma(t)\|_2^2 > 1/5$. Hence complete-triple amplitude preparation is also $\Omega(L)$ throughout the late tail.

Here is the promised canonical normalization-two access object. The equality matrix has public unit-magnitude entries and maximum absolute row and column sums two. Give each row and column two public slots. A degree-two row or column prepares the equal superposition of its two entry tokens; a degree-one row or column sends the unused slot to its own failure label, and a degree-zero free column goes entirely to its failure label. Distinct rows, columns, and failure arms

use disjoint labels. Public two-slot Hadamards, reversible token maps, and a fixed lexicographic bijection on invalid addresses give fixed unitary completions L_A and $R_A^{(0)}$, with every hidden sign set to $+1$ in $R_A^{(0)}$ while retaining the public predecessor minus signs. A phase unitary S_ς recognizes the predecessor token in signed transition i and phases it by ς_i , while acting as the identity on all other entry, failure, and invalid labels. Extending the sign oracle by the public dummy value $\varsigma_0 = 1$ makes this one coherent raw query. Therefore

$$U_A = L_A^\dagger S_\varsigma R_A^{(0)}, \quad \langle 0, r | U_A | 0, c \rangle = \frac{(\mathcal{A}_\varsigma)_{rc}}{2}. \quad (449)$$

After padding the row and column address spaces to a common public dimension, this is an exact, fully specified normalization-two encoding, and every ordinary, adjoint, or controlled call uses one raw query. Thus $q_{\text{BE}} \geq n_{\text{bit}}/2$ for either output. An alternative input-dependent completion, junk action, prefix table, or preprocessing stage is admissible only when all queries used to construct it are charged.

For arbitrary $X_i \succeq 0$, let $d_i = \sqrt{2}(X_i)_{14}$ and suppose

$$\|\mathcal{A}_\varsigma(X) - b\|_2 \leq \epsilon, \quad \sum_i \langle C_0, X_i \rangle \leq 6, \quad \epsilon \leq \frac{1}{4\sqrt{L}}. \quad (450)$$

Define the trace-normalized density output of such a point by

$$\rho(X) := \frac{\bigoplus_i X_i}{\sum_i \text{tr } X_i}. \quad (451)$$

It is well-defined because the root residual forces a nonzero off-diagonal entry and a PSD matrix with such an entry has positive trace. If $r_0 = d_0 - \sqrt{2}$ and $r_i = d_i - a_i d_{i-1}$, switching the recurrence gives $\tau_i d_i = \sqrt{2} + \sum_{j \leq i} \tau_j r_j$. Summing over T and applying Cauchy–Schwarz gives $h \sum_{i \in T} d_i > K$. PSD and the objective cap give $\sum_i \text{tr } X_i \leq 6L$. Since $\text{tr}(O_i X_i) = \sqrt{2}d_i$ on the suffix,

$$h \text{tr}(O\rho(X)) = \frac{\sqrt{2}h \sum_{i \in T} d_i}{\sum_i \text{tr } X_i} > \frac{\sqrt{2}K}{6L} > \frac{1}{5}. \quad (452)$$

Therefore preparing any $\tilde{\rho}$ with $D_{\text{tr}}(\tilde{\rho}, \rho(X)) \leq 1/100$ for some blockwise PSD point meeting (450) costs $\Omega(L)$ raw queries. The point may depend adversarially on the input; no centrality or matrix-norm proximity is needed. Conversely, averaging the exact centers obtained by flipping each sign once gives the wrong suffix parity and

$$\|\mathcal{A}_\varsigma(\bar{X}) - b\|_2 = 2\sqrt{2/n_{\text{bit}}} = \Theta(L^{-1/2}). \quad (453)$$

For precision parameters $\epsilon, \eta > 0$, let $Q_{\text{abs}}(P, \epsilon)$ be the worst-case first-preparation raw-query cost, including setup, of producing an unconditional density output $\tilde{\rho}$ within trace distance $1/100$ of $\rho(X)$ for some blockwise PSD X with at most P blocks, absolute equality residual at most ϵ , and the displayed objective cap. Define $Q_{\text{rel}}(P, \eta)$ identically with $\|\mathcal{A}_\varsigma(X) - b\|_2 / \|b\|_2 \leq \eta$. For a heralded procedure, its success probability and all repetition or amplification costs are charged as in [definition 2.25](#); reusable preprocessing is charged as in [definition 2.27](#).

Theorem 11.4 (Accuracy–query frontier). *For at most P blocks and the relational output contract above,*

$$Q_{\text{abs}}(P, \epsilon) = \Omega(\min\{P, \epsilon^{-2}\}), \quad Q_{\text{rel}}(P, \eta) = \Omega(\min\{P, \eta^{-2}\}) \quad (454)$$

for $0 < \epsilon, \eta \leq 1/8$. The relative residual is the global RHS-relative residual of [definition 2.26](#). Exact- P public padding preserves the theorem, and the bounds are tight for the tuned family.

Proof. For absolute error choose $n_{\text{bit}} = \lfloor \min\{(P-1)/17, 1/(2048\epsilon^2)\} \rfloor$. Apart from a constant-size range, this is $\Theta(\min\{P, \epsilon^{-2}\})$ and ensures the last inequality in (450). The decoder reduces parity to every valid output. Since $\|b\|_2 = \sqrt{3}$, replacing 2048 by 6144 proves the relative claim.

For exact P , add $P-L$ public 4×4 dummy blocks fixed by ten one-coordinate rows to $P^{-1}I_4$, and set the common cost scale and parameter to $1/P$. The cap becomes $6L/P + 5(P-L)/P^2$, so total trace is at most $6L + 5$ and active bias remains above $1/5$. Dummy blocks have no tangent coordinates, have bounded multipliers and a strictly complementary optimum, and do not change the active non-SOCP slice. Their RHS makes $\|b_{\text{pad}}\|_2^2 < 4$, so 8192 replaces 6144 for exact- P relative padding. Reading all active signs gives an exact valid output in $O(n_{\text{bit}})$ queries, while (453) matches the residual exponent. $\square$

A hidden coordinate in block i is the prefix τ_i and has complexity $\Theta(\min\{i, n_{\text{bit}}\})$. A classical scan streams all blocks with exactly n_{bit} queries and constant extra workspace. Quantumly, use

$$X_{\tau}^{(k)}(t) = t \sum_{j=1}^k e_j e_j^T + w_{\tau} w_{\tau}^T, \quad w_{\tau} = \tau e_1 + e_{k+1}, \quad (455)$$

to prepare a public local purification. A coherent prefix-phase loop queries and unqueries each sign, using at most $2n_{\text{bit}}$ queries, logarithmic workspace, and no QRAM. Public coordinate predicates extend it to the complete-triple state. Both first-preparation complexities are therefore $\Theta(L)$.

If $q_k = (k+1)(k+2)/2$, a path bag containing two adjacent block cliques and their $k+1$ transition multipliers proves

$$\text{tw}(K_{\text{KKT}}) \leq 2q_k + k. \quad (456)$$

Thus the free central solve uses zero input queries while original-coordinate loading uses $\Theta(L)$. This complements bounded-treewidth chordal conversion [146]. The bound is for first preparation: charged cached prefixes can make later centers cheap. It gives no lower bound for a reduced Z output or the public value.

11.3 Abstract tree-holonomy loading

Theorem 11.5 (Tree-holonomy loading). *Let $\mathcal{T} = (V, E)$ be a public rooted tree with edge signs ς_e and transported labels $\tau_v = \prod_{e \in \text{path}(r, v)} \varsigma_e$. For public constant-dimensional $B_+, B_- \succeq 0$ of common trace T_0 , put*

$$\rho_{\varsigma} = \frac{1}{|V|T_0} \bigoplus_{v \in V} B_{\tau_v}, \quad d = \frac{1}{2} \|B_+/T_0 - B_-/T_0\|_1. \quad (457)$$

If public constant-size circuits prepare local purifications, a purification of (457) needs at most $4|E|$ sign queries and logarithmic workspace, without QRAM. If an unknown path of length n_{bit} is followed by $K = \Theta(n_{\text{bit}})$ public descendants, $|V| = \Theta(n_{\text{bit}})$, and $d > 0$, preparation to trace distance below a constant depending only on d and $K/|V|$ needs at least $n_{\text{bit}}/2$ queries on the first preparation.

Proof. In a public depth-first order, an edge is on the root-to- v path exactly when v lies in the child subtree's preorder interval. Prepare a uniform entangled vertex address, query every edge, toggle a label on that interval when the sign is negative, and unquery. Select B_{τ_v}/T_0 , then reverse the loop to erase the label. The two compute-uncompute loops use four queries per edge and logarithmic workspace.

For the lower bound, project onto the K descendants. This has public probability $\alpha = K/|V|$. Conditioned on it, the local state is B_h/T_0 , whose Helstrom success is $(1+d)/2$. Guessing randomly off the tail gives $1/2 + \alpha d/2$. Preparation error below $\alpha d/4$ leaves constant advantage, so parity costs $n_{\text{bit}}/2$ queries. A classical tree scan matches the linear order and may restrict itself to the union of requested root paths. $\square$

11.4 Matrix-valued group holonomy

Let $U : G \rightarrow O(d)$, put $k = 3d$, and define $W = U_{g_{n_{\text{bit}}}} \cdots U_{g_1}$. Use the copied construction with $P = 33n_{\text{bit}}$ and hidden transports $\mathcal{D}_g = \text{diag}(U_g, I_d, I_d)$. Replace $3I$ in (411) by kI_k and replace scalar H_{ab} by $H_{ab}(I_d)$. Elimination gives

$$\bar{C}_W = kI_k + uA_W, \quad A_W = \begin{pmatrix} 0 & I_d & W^T \\ I_d & 0 & I_d \\ W & I_d & 0 \end{pmatrix}. \quad (458)$$

Conjugating W by Q conjugates A_W by $\text{diag}(Q, Q, Q)$.

Theorem 11.6 (Holonomy spectrum).

$$\text{spec}(A_W) = \bigcup_{e^{i\theta} \in \text{spec}(W)} \left\{ 2 \cos \frac{\theta + 2\pi\ell}{3} : \ell = 0, 1, 2 \right\}. \quad (459)$$

Thus $v(W) = k + u\lambda_{\min}(A_W)$ is a conjugacy-class function. For the natural permutation representation of S_3 , the identity, transposition, and three-cycle values are

$$\frac{89}{10}, \quad \frac{44}{5}, \quad 9 + \frac{1}{5} \cos \frac{8\pi}{9}. \quad (460)$$

Proof. On a W -eigenvector with eigenvalue $e^{i\theta}$, the complexified operator is $\begin{pmatrix} 0 & 1 & e^{-i\theta} \\ 1 & 0 & 1 \\ e^{i\theta} & 1 & 0 \end{pmatrix}$, whose characteristic polynomial is $\lambda^3 - 3\lambda - 2 \cos \theta$. Substituting $\lambda = 2 \cos \phi$ proves the spectrum. The natural S_3 eigenvalue lists are $(1, 1, 1)$, $(1, 1, -1)$, and $(1, e^{2\pi i/3}, e^{-2\pi i/3})$, with least connection eigenvalues $-1, -2, 2 \cos(8\pi/9)$. Identity and three-cycles agree in the abelianization but have different values, so the unrestricted observable is genuinely nonabelian. $\square$

Signed-permutation representations keep each svec copy row at two unit-magnitude nonzeros, though the hidden symbol may move the predecessor position. The natural S_3 example uses 9×9 blocks, public costs in $(9 - 1/4, 9 + 1/4)$, and is strictly primal-dual feasible.

Theorem 11.7 (S_3 value and Newton hardness). *Restrict every symbol to $\{e, t\}$ for one fixed transposition. Additive $1/25$ or target-relative $1/200$ value estimation takes $\Omega(P)$ raw queries. At $X_i = I_9/9$, $\mu = 1/P$, the ambient and reduced primal Hessians have condition number one, the KKT right-hand side is public, and*

$$\Delta Z_W = -\frac{u}{81}A_W, \quad \lambda = \frac{\sqrt{2}}{30}, \quad \lambda_{\min}(I_9/9 + \Delta Z_W) \geq \frac{44}{405}. \quad (461)$$

Preparing the normalized root or global primal direction, or the normalized complete primal-plus-multiplier KKT state in the stated row scaling, to trace distance $1/100$ requires $\Omega(P)$ raw queries. The complete-state observable has magnitude greater than $33/100$.

Proof. The restricted word is identity or t according to parity. The first two values in (460) differ by $1/10$ and are separated at $177/20$; their relative intervals are disjoint below $1/177$. A sparse position query is simulated by a constant number of bit queries, so parity proves value hardness.

At the public start the gradient is $(C_i - 9I)/P$ and the ambient Hessian is $(81/P)I$. Orthogonal transport makes the normalized tangent map isometric, so the physical root Hessian is $81I$. Since $\text{tr } A_W = 0$, the direction is as displayed. Also $\|A_W\|_F^2 = 18$ and $\|A_W\|_{\text{op}} \leq 2$, so $\lambda^2 = 2u^2/9 = 1/450$ and $1/9 - 2u/81 = 44/405$. The next decrement is $2/(30 - \sqrt{2})^2 < 1/400$.

For identity and a transposition, the normalized A_W overlap is $7/9$. The root-direction states have trace distance $4\sqrt{2}/9$, so their fixed Helstrom measurement succeeds with probability $\frac{1}{2}(1 + 4\sqrt{2}/9) > 0.81$. Let J swap corresponding Frobenius-isometric svec coordinates in the

(1, 2) and (1, 3) connection blocks and act as zero on every other coordinate. It has expectation $2/3$ for identity and $2/9$ for a transposition; centering as $T_G = (9J - 4I)/13$ gives $2h/13$ on every physical block and hence on the global direction.

For the complete state, a public basis makes $U_t = \text{diag}(-1, 1, 1)$. Two modes are public and one is the signed qutrit mode. With $d_0 = \sqrt{2}u/81$, its primal contribution is $D = Pd_0^2 = 11n_{\text{bit}}/109350$ and its multiplier tent has the V, C in (429). Swapping 12, 13 only in the signed mode gives

$$|\langle T_{\text{sign}} \rangle| = \frac{2239504n_{\text{bit}}^2 + 3888}{6759216n_{\text{bit}}^2 + 12393} > \frac{33}{100}, \quad (462)$$

by cross multiplication. These fixed decoders retain positive bias at trace error $1/100$, and parity proves all state lower bounds. $\square$

For canonical KKT access, 45 scalar coordinates lie in each block. A copy-row state uses amplitude $1/3$ on two tokens and a trace-row state uses amplitude $1/3$ on nine. The column preparation routes predecessor tokens through the svec involution induced by t , which one symbol query implements by toggling its public orbit bit. This exactly encodes the equality dilation with normalization nine. Adding $(81/P)$ times the public primal-sector projector P_x by a two-term LCU gives

$$\alpha_M = 9 + \frac{81}{P} < 12, \quad (463)$$

and $q_{\text{BE}} \geq \lceil n_{\text{bit}}/2 \rceil$ for the three outputs above. The completion is fixed; other input-dependent junk must be charged.

The tangent projector again has signal probability $24/41$. Combining it with the global decoder gives $48/533$. At operator error $1/64$, the signed expectation remains $48/533 - 2\sqrt{24/41}/64 - 1/4096 > 0.065$. First synthesis is therefore $\Omega(n_{\text{bit}})$. Accumulating each prefix svec involution and conjugating the public trace-zero projector gives an exact $2n_{\text{bit}}$ query construction, so the bound is tight. The scalar KKT graph is a forest: a layer permutation makes disjoint paths, and the trace multiplier joins distinct diagonal paths without a cycle.

If setup uses S queries and later iterations use q_t , the canonical XOR position convention yields

$$S + \sum_t q_t \geq n_{\text{bit}}/4, \quad (464)$$

improved to $n_{\text{bit}}/2$ for the clean one-query convention. Linear setup may store the word or prefixes and make later iterations cheap.

The unrestricted objective detects nonabelian information, but every proved $\Omega(P)$ query theorem above restricts inputs to the abelian subgroup $\{e, t\} \cong \mathbb{Z}_2$. This is parity hardness, not a bound for a genuinely noncommuting ordered-word promise.

The scalar value is invariant under simultaneous orthogonal gauge by construction. The direction decoders instead use the fixed natural permutation basis and root frame. A common public gauge merely conjugates both public measurements, whereas an input-dependent output gauge must be learned or supplied and is outside the stated output contract.

Open Problem 11.8 (Intrinsically nonabelian word hardness). Prove a local-symbol query lower bound for distinguishing conjugacy classes of a genuinely noncommuting word promise, and transfer it through (458), without restricting the alphabet to one cyclic subgroup.

11.5 Historical boundary and synthesis

Signed switching and cycle products are classical graph invariants [80]; group synchronization organizes the mechanism as gauge transport and holonomy [64], and orthogonal synchronization is routinely relaxed by SDPs [96]. General group oracle classification and symmetric-oracle

problems use different promises [45, 145]. Fawzi’s theorem gives the exact lift boundary, not a coefficient-query lower bound.

The early 2×2 sparse-block examples reduce to paired LPs in their exact-scalar form; their shared-trace variant is genuinely PSD. The first noncommuting 3×3 Schur example is still SOCP-representable, which is why 4×4 blocks are the honest headline. The two-anchor precursor used asymmetric anchors: its two effective costs, and hence its central matrices, have different spectra and are not related by any orthogonal similarity.

Separately, the free- $\mathbb{S}_+^3$ parity-central-Hessian precursor gives one small operator result. For

$$\bar{C}_h = I + a(H_{12} + hH_{13}), \quad a = 8/33, \quad \mathcal{L}_h(Y) = \bar{C}_h Y \bar{C}_h, \quad (465)$$

the reduced Hessian has condition number below five, but synthesizing a public-constant-normalization block encoding to unscaled error $1/8$ costs at least $n_{\text{bit}}/2$ raw queries. Indeed $\text{svec}(\mathcal{L}_h(E_{33})) = (a^2, 0, \sqrt{2}ha, 0, 0, 1)^T$; swapping its 13, 33 coordinates has signed expectation $2\sqrt{2}a/(1+a^2)^2 > 0.61$, and error $1/8$ leaves constant bias. This is a first-synthesis statement: a supplied Hessian oracle reveals parity in one call. In this free- $\mathbb{S}_+^3$ family—unlike the two-anchor family—the two Hessians are related by an input-dependent orthogonal gauge.

Linear sparse optimal-value lower bounds already follow from LP constructions, including Theorem 8.4 of Apers–Gribling [10], and general SDP value bounds predate these families [130]. Mori et al.’s sparse QLS bound also uses parity propagation [109]. The contribution here is the optimization-native conjunction, not parity, propagation, or the linear exponent.

The qutrit component imported by section 7 is exactly (406) with the costs (411); under that import $N = n_{\text{bit}}$. Its spectra are (416) and its value gap is $1/10$. Nothing here makes that section’s public-start Newton state winner-revealing; trace allocation is governed by section 6. As in the LP families of section 9, reduced conditioning, KKT sparsity, data access, and requested coordinates are separate resources. On these parity promises the only quantum improvement is Deutsch’s factor of two; classical reading takes n_{bit} signs, so there is no asymptotic quantum speedup.

12 Preconditioning and Interface Obstructions

“Just precondition” and “just keep the state” are incomplete complexity arguments. Preconditioning may exchange spectral condition for factor normalization, right-hand-side preparation, or recovery sensitivity. A normalized Newton state may in turn be too weak to supply the coordinatewise diagonal needed by the next Newton system. This section prices those exchanges in the access models of section 2.2. The conclusions are products and complete ledgers, never lower bounds on one spectral factor in isolation.

The analytic condition/normalization statements, including the fixed-factor minimax result, are model free. Only their compilation/accounting corollaries, the parity reductions, and the lazy-oracle results use the raw sparse coefficient model of definition 2.11. The state-interface results instead use the distinct state-preparation access of definition 2.14; they are not raw-coefficient query lower bounds. The lazy-oracle results finally explain why the caching ceiling in section 9.3 prevents an unconditional per-iteration multiplier on a fixed instance.

12.1 Factor normalization and whitening

We first isolate the signed gain-propagation block used by theorem 9.4. Let $N \geq 2$, $M = 17N$, and

$$T = \lceil \tfrac{1}{2} \log_2 N \rceil, \quad H = 2^T, \quad H_i = 2^{\min\{i, T\}}, \quad g_i = H_i/H_{i-1},$$

and take $a_i = \varsigma_i$ for $i \leq N$ and $a_i = 1$ afterward. Thus ς_i here is the hidden sign denoted σ_i in theorem 9.4. Here H is the plateau height; the normal matrix H_μ used later is unrelated. On

$d = (d_0, \dots, d_M)^T$, define the pinned lower-bidiagonal factor

$$(B_\varsigma d)_0 = d_0, \quad (B_\varsigma d)_i = d_i - g_i a_i d_{i-1}. \quad (466)$$

Put $\tau_0 = 1$, $\tau_i = \prod_{j=1}^i a_j$, and $D_\tau = \text{diag}(\tau_0, \dots, \tau_M)$. Let B_+ be (466) with every sign positive and set $S_+ = B_+ B_+^T$. Then

$$B_\varsigma = D_\tau B_+ D_\tau, \quad S_\varsigma := B_\varsigma B_\varsigma^T = D_\tau S_+ D_\tau. \quad (467)$$

Thus the singular values are independent of the signs, although implementing the gauge requires their prefix products. This switching identity and the corresponding inverse formulas are standard signed-graph facts [3].

Lemma 12.1 (Plateau singular scale). *If $L = M - T + 1$ is the number of plateau nodes, then*

$$\sigma_{\min}(B_\varsigma) \leq 2 \sin \frac{\pi}{2(L+1)} \leq \frac{\pi}{L+1} < \frac{\pi}{16N}, \quad \|B_\varsigma^{-1}\|_2 = \Theta(N). \quad (468)$$

Proof. Take a Dirichlet sine vector supported on nodes $T, \dots, M$, with virtual zeros at $T-1$ and $M+1$. On this support B_+ is the first-difference operator. Adding the omitted final difference can only increase its energy, and the full Dirichlet path quotient is $4 \sin^2(\pi/(2(L+1)))$. Equation (467) gives the signed case, while $\sin x \leq x$ and $L+1 > 16N$ give the two remaining inequalities.

For $i \geq j$, direct forward substitution gives

$$(B_+^{-1})_{ij} = H_i / H_j. \quad (469)$$

The column $B_+^{-1} e_0 = (H_i)_i$ has norm $\Theta(N)$, whereas summing the squares in (469) gives $\|B_+^{-1}\|_F = O(N)$. Orthogonal equivalence in (467) proves the final claim. $\square$

Theorem 12.2 (Sharp plateau whitening frontier). *Let P_ς be any square factor such that, for $K \geq 1$,*

$$\frac{1}{K} I \preceq P_\varsigma S_\varsigma P_\varsigma^* \preceq I. \quad (470)$$

Then

$$\|P_\varsigma\|_2 \sqrt{K} \geq \frac{1}{\sigma_{\min}(B_\varsigma)} > \frac{16N}{\pi}. \quad (471)$$

Consequently every exact $(\alpha_P, a_{\text{anc}}, 0)$ block encoding of P_ς obeys

$$\boxed{\alpha_P \sqrt{K} > \frac{16N}{\pi}}. \quad (472)$$

For $1 \leq a \leq \Theta(N)$, the regularized family

$$P_\varsigma^{(a)} = D_\tau (S_+ + a^{-2} I)^{-1/2} D_\tau \quad (473)$$

has $\|P_\varsigma^{(a)}\|_2 = \Theta(a)$ and, after a constant input-independent normalization of the largest eigenvalue,

$$K = \Theta((N/a)^2), \quad \|P_\varsigma^{(a)}\|_2 \sqrt{K} = \Theta(N). \quad (474)$$

Thus the order of the frontier is attained throughout, not only at its endpoints.

Proof. Equation (470) implies $\sigma_{\min}(P_\varsigma B_\varsigma) \geq K^{-1/2}$. On a right singular vector of B_ς for its least singular value,

$$\sigma_{\min}(P_\varsigma B_\varsigma) \leq \|P_\varsigma\|_2 \sigma_{\min}(B_\varsigma).$$

This and [lemma 12.1](#) prove (471); the normalization of an exact block encoding is at least the encoded norm.

For tightness, the eigenvalues of the congruence in (473) are

$$\frac{\lambda}{\lambda + a^{-2}}, \quad \lambda \in \sigma(S_+).$$

The bidiagonal factor has bounded norm, a diagonal entry of one, and inverse norm $\Theta(N)$ by [lemma 12.1](#). Hence $\lambda_{\max}(S_+) = \Theta(1)$ and $\lambda_{\min}(S_+) = \Theta(N^{-2})$. It follows that the factor norm is $\Theta(a)$ and that the ratio of the displayed largest and smallest eigenvalues is $\Theta((N/a)^2)$ in the stated range. This proves (474). $\square$

The theorem is a specialization, not a new general quantum-preconditioning obstruction. Theorem 3 and Corollary 4 of Nie, Lai, and An already prove the separate-block-encoding limitation for arbitrary preconditioners [114]. What is specific here is the constant $16/\pi$, the tight interpolation (473)–(474), and the preprocessing-aware raw-query theorem below. Directly encoding a classically formed or algebraically regrouped product is a different access model, as both that work and [proposition 5.8](#) emphasize.

Two useful approximate operator consequences are immediate. If the factor is supplied through an exact $(\alpha_P, a_{\text{anc}}, 0)$ block encoding, then, for fixed $\delta < 1$,

$$\|P_\zeta S_\zeta P_\zeta^* - I\|_2 \leq \delta \implies \alpha_P \geq \frac{\sqrt{1-\delta}}{\sigma_{\min}(B_\zeta)}, \quad (475)$$

$$\|P_\zeta B_\zeta - I\|_2 \leq \delta \implies \alpha_P \geq \frac{1-\delta}{\sigma_{\min}(B_\zeta)}. \quad (476)$$

Indeed, the first inequality gives $\sigma_{\min}(P_\zeta B_\zeta) \geq \sqrt{1-\delta}$, while the second gives $\sigma_{\min}(P_\zeta B_\zeta) \geq 1-\delta$; the same singular-vector argument then applies. Constant-error approximate whitening or inversion supplied through such an exact factor encoding therefore has $\alpha_P = \Omega(N)$. More generally, if $\beta = \|P_\zeta S_\zeta P_\zeta^*\|_2$ and the congruence has condition at most K , the scale-invariant statement is

$$\|P_\zeta\|_2 \sqrt{K/\beta} \geq \sigma_{\min}(B_\zeta)^{-1}. \quad (477)$$

This normalization is essential: scalar rescaling defeats an unnormalized statement. For a left factor normalized by $\sigma(P_\zeta B_\zeta) \subseteq [1/K, 1]$ and supplied through an exact $(\alpha_P, a_{\text{anc}}, 0)$ block encoding, the same argument gives

$$\alpha_P K \geq \|P_\zeta\|_2 K > \frac{16N}{\pi}. \quad (478)$$

The factor norm still does not price compilation from raw signs. Define

$$F_\zeta = \text{diag}(B_\zeta^{-1}, B_+^{-1}), \quad G_\zeta = \text{diag}(B_\zeta, B_+), \quad |r\rangle = \frac{|0, e_0\rangle + |1, e_0\rangle}{\sqrt{2}}. \quad (479)$$

By (469),

$$F_\zeta |r\rangle = \frac{1}{\sqrt{2}}((H_i \tau_i)_i \oplus (H_i)_i), \quad \|F_\zeta r\|_2 = \left(\sum_i H_i^2 \right)^{1/2} = \Theta(N). \quad (480)$$

Pair the signed and unsigned coordinates and measure their side qubit in the X basis on the $16N$ plateau-copy nodes. Its parity-signed expectation is

$$\frac{16NH^2}{\sum_i H_i^2} > \frac{16N}{17N + 4/3} > \frac{9}{10}. \quad (481)$$

Theorem 12.3 (Reusable inverse access–normalization tradeoff). *Suppose an offline stage makes s raw sign queries and produces a reusable exact $(\alpha, a, 0)$ block encoding of $\tilde{F}_\zeta$. One call, its inverse, or a controlled call makes at most q further raw queries. If, for fixed $\delta \leq 1/20$,*

$$\|\tilde{F}_\zeta G_\zeta - I\|_2 \leq \delta, \quad (482)$$

then

$$\boxed{s + O(q\alpha/N) = \Omega(N)}, \quad \boxed{Ns + q\alpha = \Omega(N^2)}. \quad (483)$$

The constants hidden in O and Ω are absolute.

Proof. Since $G_\zeta F_\zeta = I$,

$$\|(\tilde{F}_\zeta - F_\zeta)r\|_2 \leq \delta \|F_\zeta r\|_2.$$

Thus the normalized approximate-inverse state retains a constant parity bias under the decoder in (481), and $\|\tilde{F}_\zeta r\|_2 \geq (1 - \delta)\Theta(N)$. Fixed-point amplitude amplification prepares it with constant success using $O(\alpha/N)$ block-encoding calls. Including setup, the resulting parity algorithm uses $s + O(q\alpha/N)$ raw queries. Quantum parity needs $\Omega(N)$ queries [20], proving the first inequality; multiplication by N gives the second. Equation (476) also gives $\alpha = \Omega(N)$, so the amplification count is never subconstant. $\square$

Three no-go qualifications delimit these statements. First, a preconditioned product supplied directly can be the identity: $P_\zeta = B_\zeta^{-1}$ gives $P_\zeta B_\zeta = I$, and exact Schur whitening gives $P_\zeta S_\zeta P_\zeta^* = I$. No lower bound on the product’s normalization and condition alone is possible. Second, applying D_τ is not free. Any operator-norm- $< 1/3$ implementation needs $\Omega(N)$ raw queries: apply it to $(|0\rangle + |M\rangle)/\sqrt{2}$ and measure in the $(|0\rangle \pm |M\rangle)/\sqrt{2}$ basis to recover endpoint parity. Third, the factor-independent conservation law needs the hard input/output contract. Consider specifically the plateau solve $G_\zeta x = r$ from (479), and require an output state within trace distance $1/20$ of $F_\zeta|r\rangle/\|F_\zeta r\|_2$. For any such end-to-end preconditioned implementation, compose its output with the fixed X -basis decoder of (481). The reduction gives

$$q_{\text{set}} + q_{\text{rhs}} + q_{\text{rec}} + mq_{\text{BE}} = \Omega(N), \quad (484)$$

where m is the number of solver calls, q_{BE} is the raw cost per call, the solve phase makes no other raw queries, and the other terms charge setup, transformed-RHS preparation, and recovery. Indeed, composing all phases with that decoder gives a bounded-error parity algorithm, so their total raw-query cost is $\Omega(N)$. Without this particular RHS/output guarantee, no such law is claimed; a public solution may carry no hidden information. An orthogonal factor rotation can move parity among the charged phases. The statement is for the pinned plateau block, not for every unrestricted full-KKT preconditioner.

12.2 Congruence condition versus physical-state recovery

A second frontier arises even when the preconditioner is an elementary active–inactive scaling. Let the fixed Euclidean space split orthogonally as $\mathbb{R}^r = U \oplus K$, with projectors Q and R , and suppose

$$H_\mu = \mu^{-1}C_\mu + \mu D_\mu, \quad 0 < \mu \leq \mu_0. \quad (485)$$

Assume $C_\mu = QC_\mu Q$ and $C_\mu|_U \rightarrow C \succ 0$; $D_\mu \succeq 0$ is uniformly bounded; $D_{KK,\mu} \rightarrow D_{KK} \succ 0$; and all other blocks converge. For a fixed nonzero $f \in U$, put

$$p = C^{-1}f, \quad g = D_{KK}^{-1}D_{KU}p, \quad (486)$$

and assume the nontrivial coupling $g \neq 0$. These are fixed-instance strict-complementarity asymptotics, not uniform claims in a growing dimension.

For $0 \leq \gamma \leq 1$, define

$$T_{\mu,\gamma} = \mu^{\gamma/2}Q + \mu^{-\gamma/2}R, \quad G_{\mu,\gamma} = T_{\mu,\gamma}H_\mu T_{\mu,\gamma}, \quad \eta_\mu = H_\mu^{-1}f, \quad z_{\mu,\gamma} = T_{\mu,\gamma}^{-1}\eta_\mu. \quad (487)$$

Because $f \in U$, $T_{\mu,\gamma}f = \mu^{\gamma/2}f$; the normalized transformed RHS is exactly the original RHS state. The tradeoff below is therefore a condition–recovery effect, not a loading effect. The normalized contraction that recovers physical coordinates is

$$R_{\mu,\gamma} = T_{\mu,\gamma}/\|T_{\mu,\gamma}\|_2 = \mu^\gamma Q + R, \quad \Phi_{\mu,\gamma}(w) = \frac{R_{\mu,\gamma}w}{\|R_{\mu,\gamma}w\|_2}. \quad (488)$$

For a unit w , measure its local normalized-state sensitivity by

$$\chi_{\mu,\gamma}(w) = \|D\Phi_{\mu,\gamma}(w)|_{w^\perp}\|_{2 \rightarrow 2}. \quad (489)$$

Theorem 12.4 (Power congruence frontier). *Under the preceding assumptions, for each fixed $\gamma \in [0, 1]$ as $\mu \downarrow 0$,*

$$\kappa_2(G_{\mu,\gamma}) = \Theta(\mu^{-2(1-\gamma)}), \quad (490)$$

$$\eta_\mu = \mu \begin{pmatrix} p \\ -g \end{pmatrix} + o(\mu), \quad (491)$$

$$z_{\mu,\gamma} = \begin{pmatrix} \mu^{1-\gamma/2}(p + o(1)) \\ -\mu^{1+\gamma/2}(g + o(1)) \end{pmatrix}. \quad (492)$$

Writing $\hat{z} = z/\|z\|_2$, exact normalized recovery holds and direct postselection has

$$\Phi_{\mu,\gamma}(\hat{z}) = \frac{\eta_\mu}{\|\eta_\mu\|_2}, \quad P_{\text{rec}} = \|R_{\mu,\gamma}\hat{z}\|_2^2 = \Theta(\mu^{2\gamma}), \quad (493)$$

while

$$\chi_{\mu,\gamma}(\hat{z}) = \Theta(\mu^{-\gamma}). \quad (494)$$

Consequently

$$\boxed{\sqrt{\kappa_2(G_{\mu,\gamma})} \chi_{\mu,\gamma}(\hat{z}) = \Theta(1/\mu)}, \quad \sqrt{\kappa_2(G_{\mu,\gamma})} P_{\text{rec}}^{-1/2} = \Theta(1/\mu). \quad (495)$$

Proof. Set $a_\mu = \mu^{-(1-\gamma)}$ and $b_\mu = \mu^{1-\gamma} = a_\mu^{-1}$. In $U \oplus K$ blocks,

$$G_{\mu,\gamma} = \begin{pmatrix} a_\mu C_\mu + \mu^{1+\gamma} D_{UU,\mu} & \mu D_{UK,\mu} \\ \mu D_{KU,\mu} & b_\mu D_{KK,\mu} \end{pmatrix}. \quad (496)$$

The upper-left block A_μ is $\Theta(a_\mu)I_U$, and its Schur complement is

$$b_\mu D_{KK,\mu} - \mu^2 D_{KU,\mu} A_\mu^{-1} D_{UK,\mu} = \Theta(b_\mu)I_K.$$

The elimination factor has off-diagonal norm $O(\mu b_\mu) = o(1)$. A uniformly conditioned near-identity congruence therefore reduces (496) to blocks of sizes $\Theta(a_\mu)$ and $\Theta(b_\mu)$, proving (490).

The K block of $H_\mu \eta_\mu = f$ gives exactly

$$\eta_{K,\mu} = -D_{KK,\mu}^{-1} D_{KU,\mu} \eta_{U,\mu}.$$

Substitution in the U block gives

$$[\mu^{-1}C_\mu + \mu E_\mu] \eta_{U,\mu} = f, \quad E_\mu = D_{UU,\mu} - D_{UK,\mu} D_{KK,\mu}^{-1} D_{KU,\mu},$$

where E_μ is bounded. Hence $\eta_{U,\mu} = \mu(C_\mu + \mu^2 E_\mu)^{-1} f = \mu(p + o(1))$, which proves (491); applying T^{-1} proves (492).

Because $Rz = \mu^{\gamma/2} \eta$, normalized recovery is exact. The two expansions give

$$\|R\hat{z}\|_2 = \mu^\gamma \frac{(\|p\|_2^2 + \|g\|_2^2 + o(1))^{1/2}}{(\|p\|_2^2 + \mu^{2\gamma} \|g\|_2^2 + o(1))^{1/2}} = \Theta(\mu^\gamma),$$

proving (493). Differentiation gives, with $\hat{\eta} = \Phi(w)$,

$$\chi_{\mu,\gamma}(w) = \frac{\|(I - \hat{\eta}\hat{\eta}^T)R_{\mu,\gamma}(I - ww^T)\|_2}{\|R_{\mu,\gamma}w\|_2}. \quad (497)$$

This is $O(\mu^{-\gamma})$ at $w = \hat{z}$. For $\gamma > 0$, project $g/\|g\|$ onto $\hat{z}^\perp$. The resulting unit tangent is $g/\|g\| + o(1)$ in K , while the recovered state tends to $(p, -g)/(\|p\|^2 + \|g\|^2)^{1/2}$. Since $p \neq 0$, the numerator in (497) is bounded below by $\|p\|/(\|p\|^2 + \|g\|^2)^{1/2} > 0$. This proves the matching lower bound. For $\gamma = 0$, $R = I$ and $\chi = 1$. Multiplying the estimates proves (495). $\square$

The same two-sparse LP witnesses every exponent. Consider

$$\min_{x \geq 0} x_2 + x_3, \quad x_1 + x_2 = 1, \quad x_2 - x_3 = 0, \quad (498)$$

so $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$, $b = e_1$, and $c = (0, 1, 1)^T$. Its unique optimum is $(1, 0, 0)$. Along the central path $x_2 = x_3 = t_\mu$, $x_1 = 1 - t_\mu$, with $t_\mu = \mu + O(\mu^2)$, and

$$H_\mu = A(XS^{-1})A^T = \frac{(1 - t_\mu)^2}{\mu} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{t_\mu^2}{\mu} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}. \quad (499)$$

Indeed, dual feasibility and complementarity give equal inactive slacks $q_\mu = 1 + \mu/[2(1 - t_\mu)]$ and $t_\mu q_\mu = \mu$, which yields the expansion. For $U = \text{span}\{e_1\}$ and $K = \text{span}\{e_2\}$, the limiting blocks in (485) are $C = e_1 e_1^T$ and $D = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$. Thus $p = 1$, $g = 1/2$, and $H_\mu^{-1}b = \mu(1, -1/2)^T + o(\mu)$. This fixed two-row, three-variable LP therefore realizes all parts of theorem 12.4; at $\gamma = 1$ its scaled condition has a finite positive limit and its recovery amplitude is asymptotic to $\sqrt{5/4}\mu$.

The power family is also the complete order-optimal interpolation within all scalar block-diagonal SPD congruences. Let

$$T_\mu = t_U(\mu)Q + t_K(\mu)R, \quad r_\mu = t_U/t_K, \quad M_\mu = \max\{r_\mu/\mu, \mu/r_\mu\}, \quad K_\mu = \max\{r_\mu, r_\mu^{-1}\}. \quad (500)$$

Put

$$\tilde{G}_\mu = T_\mu H_\mu T_\mu, \quad \tilde{z}_\mu = T_\mu^{-1} \eta_\mu, \quad \tilde{R}_\mu = T_\mu / \|T_\mu\|_2, \quad \tilde{\Phi}_\mu(w) = \frac{\tilde{R}_\mu w}{\|\tilde{R}_\mu w\|_2}. \quad (501)$$

Define $\tilde{P}_{\text{rec}}$ by applying the squared norm in (493) to $\tilde{R}_\mu$ and $\tilde{z}_\mu$, and define $\tilde{\chi}_\mu$ by (489) with $\tilde{\Phi}_\mu$.

Theorem 12.5 (General scalar-block frontier). *Uniformly over all positive scalar functions t_U, t_K ,*

$$\sqrt{\kappa_2(\tilde{G}_\mu)} = \Theta(M_\mu), \quad \tilde{P}_{\text{rec}}^{-1/2} = \Theta(K_\mu), \quad \tilde{\chi}_\mu = \Theta(K_\mu). \quad (502)$$

Consequently

$$\sqrt{\kappa_2(\tilde{G}_\mu)} \tilde{\chi}_\mu = \Theta(M_\mu K_\mu) \geq c/\mu. \quad (503)$$

The same holds with $\tilde{P}_{\text{rec}}^{-1/2}$ in place of $\tilde{\chi}_\mu$. Equality in order, $M_\mu K_\mu = \Theta(1/\mu)$, holds exactly for the maximal class

$$r_\mu = \Omega(\mu), \quad r_\mu = O(1). \quad (504)$$

More explicitly,

$$M_\mu K_\mu = \begin{cases} \mu/r_\mu^2, & r_\mu < \mu, \\ 1/\mu, & \mu \leq r_\mu \leq 1, \\ r_\mu^2/\mu, & r_\mu > 1. \end{cases} \quad (505)$$

Proof. The assumptions imply uniform spectral equivalence

$$c_0(\mu^{-1}Q + \mu R) \preceq H_\mu \preceq C_0(\mu^{-1}Q + \mu R). \quad (506)$$

For the lower bound, write a vector as $u + k$. Fixed $c, d, M > 0$ give

$$(u + k)^T H_\mu (u + k) \geq (c/\mu - 2M^2\mu/d)\|u\|^2 + (d\mu/2)\|k\|^2,$$

after Young's inequality; for small μ this proves the lower half of (506), and boundedness proves the upper half. Congruence by T_μ shows that the two spectral scales are t_U^2/μ and μt_K^2 , proving the condition estimate in (502).

Equation (491) gives $\|\eta_{U,\mu}\| = \Theta(\mu)$ and $\|\eta_{K,\mu}\| = \Theta(\mu)$. Therefore

$$\|\tilde{R}_\mu \hat{z}_\mu\| = \frac{\|\eta_\mu\|}{\max\{t_U, t_K\} \sqrt{\|\eta_{U,\mu}\|^2/t_U^2 + \|\eta_{K,\mu}\|^2/t_K^2}} = \Theta(K_\mu^{-1}).$$

This proves the recovery-probability claim. In the plane spanned by the normalized U and K components, differentiating the output angle of $\text{diag}(t_U, t_K)$ gives tangent gain

$$\frac{r_\mu^{-1}\|\eta_{U,\mu}\|^2 + r_\mu\|\eta_{K,\mu}\|^2}{\|\eta_{U,\mu}\|^2 + \|\eta_{K,\mu}\|^2} = \Theta(K_\mu).$$

This is a lower bound on $\tilde{\chi}_\mu$; differentiation as in (497) gives the matching upper bound. The elementary three-case calculation (505) proves the product and the equality class. $\square$

This is not a universal preconditioning lower bound. It excludes arbitrary left preconditioners, input-dependent changes mixing U and K , direct preparation of the physical state, variable-time recovery, and algorithms that request only an observable. It also charges neither construction of Q nor block-encoding normalization. If $g = 0$, the recovery conclusion can fail. In particular, the state-preserving left equilibration in [theorem 5.7](#) lies outside this congruence-recovery contract and is a genuine positive escape. The theorem is therefore a sharp structural lemma for a specified family, realized by (498), rather than a claim against all preconditioners.

12.3 The parity preconditioner dichotomy

The parity LP gives a complementary dichotomy. Its reduced primal Hessian is condition one, but its equality normal matrix is a signed grounded-tree Laplacian. Fixed preconditioners cannot serve all signings, whereas a perfect data-dependent sparse factor is locally trivial. The global information then appears in factor application, transformed-RHS preparation, or recovery.

The minimax claim already holds on a path. Let

$$B = \begin{pmatrix} 1 & & & \\ -1 & 1 & & \\ & -1 & 1 & \\ & & \ddots & \ddots \end{pmatrix}, \quad C = BB^T, \quad R = \text{diag}(1, -1, 1, -1, \dots) \quad (507)$$

for $m \geq 1$. Both C and RCR are legal sign inputs.

Theorem 12.6 (Minimax fixed-preconditioner bound). *For SPD matrices write $\kappa = \lambda_{\max}/\lambda_{\min}$. For every input-independent SPD matrix M ,*

$$\max \left\{ \kappa(M^{-1/2}CM^{-1/2}), \kappa(M^{-1/2}RCRM^{-1/2}) \right\} \geq \frac{4m^2 - 1}{3} \geq m^2. \quad (508)$$

The identity gives condition number at most $4m^2$ for either signing, so it is minimax optimal in order. The statement allows dense M and holds more generally for every fixed invertible factor P , with the two matrices $P^T C P$ and $P^T R C R P$. The bound $(4m^2 - 1)/3$ is a witness lower bound, not a claim of the exact minimax value. It strengthens the former $m^2/4$ bound for $m \geq 4$.

Proof. For SPD A_0, A_1, M , generalized Rayleigh quotients give

$$\kappa(A_0^{-1/2}A_1A_0^{-1/2}) \leq \kappa(M^{-1/2}A_0M^{-1/2})\kappa(M^{-1/2}A_1M^{-1/2}). \quad (509)$$

Indeed, bound the numerator and denominator of $x^T A_1 x / (x^T A_0 x)$ by the extremal eigenvalues in M coordinates; the independent scales cancel in the quotient of extrema.

For $r = (m, m-1, \dots, 1)^T$,

$$x^T C x = \sum_{j=0}^{m-2} (x_j - x_{j+1})^2 + x_{m-1}^2, \quad r^T C r = m.$$

Summing the alternating witness exactly gives

$$r^T R C R r = \sum_{k=1}^m (2k-1)^2 = \frac{m(4m^2-1)}{3}.$$

Put $t = (4m^2-1)/3$. The generalized quotient is t at r and $1/t$ at Rr , so $\kappa(C^{-1/2}R C R C^{-1/2}) \geq t^2$. Apply (509) with $A_0 = C$ and $A_1 = R C R$; one of its two factors must be at least t . The same argument in coordinates $P^{-1}r$ and $P^{-1}Rr$ proves the assertion for any invertible P , without a polar-decomposition assumption.

For the upper bound, write $d_i = x_i - x_{i+1}$ for $i < m-1$ and $d_{m-1} = x_{m-1}$. Then $x_i = \sum_{j=i}^{m-1} d_j$ and $x^T C x = \sum_j d_j^2$. Cauchy–Schwarz gives $\|x\|_2^2 \leq m^2 x^T C x$, while $(a-b)^2 \leq 2a^2 + 2b^2$ gives $x^T C x \leq 4\|x\|_2^2$. Thus $\kappa(C) \leq 4m^2$. Since R is orthogonal, the same bound holds for $R C R$; more generally every signed path is an orthogonal diagonal conjugate of C . $\square$

The fixed-preconditioner result is formalized in Lean, including the actual incidence matrices, their positive definiteness, extremal-eigenvalue condition numbers, the simultaneous-preconditioning inequality, the exact witness ratio, and the identity upper bound for every edge signing. The development also proves the arbitrary-factor statement directly and identifies the SPD specialization with the positive inverse square root. It does not formalize the data-dependent factor or quantum-query results that follow. The repository’s `formal/PRECONDITIONER.md` maps these claims to the proofs and records the verification commands.

For data-dependent preconditioning, let D_ζ be the square grounded signed incidence matrix of the chain-plus-copy tree: its root row is e_0^T and the row of a nonroot node j is $e_j^T - \tau_j e_{\pi(j)}^T$, where τ_j is its edge sign. With pair variables the equality matrix is

$$A_\zeta = D_\zeta[I, -I], \quad C_\zeta = D_\zeta D_\zeta^T. \quad (510)$$

Every row of D_ζ has at most two nonzeros, each determined by one local sign, yet

$$D_\zeta^{-1} C_\zeta D_\zeta^{-T} = I, \quad D_\zeta^{-1} e_0 = (p_j)_j, \quad (511)$$

where p_j is the root-to- j sign product. Set $R_\zeta = \text{diag}(p_j)$, and let D_+ be the unsigned grounded incidence matrix on the same rooted tree. The switching identity is $D_\zeta = R_\zeta D_+ R_\zeta$. Thus construction of a sparse perfect preconditioner is not hard; applying its inverse exposes all prefixes. This distinction is the classical requirement that a useful preconditioner be cheap to apply, expressed here in support-theory language [27].

The right-hand side e_0 in (511) occurs in an actual Newton equation. At the infeasible-start point $x = s = \mathbf{1}$, $y = 0$, $\mu = 1$ for the pair LP with $c = \mathbf{1}$ and $b = e_0$, the point is centered and dual feasible, and $A_\zeta \mathbf{1} = 0$. With the residual convention of (13), any centering target $0 < \sigma_c \leq 1$ gives

$$A_\zeta \Delta x = e_0, \quad A_\zeta^T \Delta y + \Delta s = 0, \quad \Delta x + \Delta s = (\sigma_c - 1)\mathbf{1}, \quad A_\zeta A_\zeta^T \Delta y = e_0. \quad (512)$$

Since $A_\varsigma A_\varsigma^T = 2D_\varsigma D_\varsigma^T$, the signed incidence congruence transforms this exact Newton RHS into $D_\varsigma^{-1}e_0$ up to a public scalar.

For the state lower bound, take $N \geq 2$ and attach two isomorphic complete copy trees, one to the root and one to the endpoint of the N -edge hidden spine. Each has M leaves and $J = 2M - 2$ new nodes, where

$$M = 2^{\lceil \log_2(16(N+1)) \rceil}. \quad (513)$$

Here M and $m = N + 1$ below are local to this amplified-tree construction and are unrelated to the plateau length M and path dimension m used earlier. All new edges have positive sign; write $\widehat{D}_\varsigma$ for this grounded incidence factor and define

$$\widehat{A}_\varsigma = \widehat{D}_\varsigma[I, -I], \quad \widehat{C}_\varsigma = \widehat{D}_\varsigma \widehat{D}_\varsigma^T. \quad (514)$$

At the same infeasible-start point $x = s = \mathbf{1}$, $y = 0$, $\mu = 1$, now for the amplified pair LP with $c = \mathbf{1}$ and $b = e_0$, the Newton equations are

$$\widehat{A}_\varsigma \Delta x = e_0, \quad \widehat{A}_\varsigma^T \Delta y + \Delta s = 0, \quad \Delta x + \Delta s = (\sigma_c - 1)\mathbf{1}, \quad \widehat{A}_\varsigma \widehat{A}_\varsigma^T \Delta y = e_0. \quad (515)$$

Thus the signed-incidence congruence maps this actual amplified Newton RHS to $\widehat{D}_\varsigma^{-1}e_0$ up to a public scalar.

The weighted factor below is also the exact local factor away from the start. At any positive interior point let $\Theta = XS^{-1} = \text{diag}(\Theta_+, \Theta_-)$, where $\Theta_\pm = \text{diag}(\theta_j^\pm)_j$, and set

$$\Omega = \text{diag}(\omega_j)_j, \quad \omega_j = \theta_j^+ + \theta_j^- > 0. \quad (516)$$

Then

$$\widehat{A}_\varsigma \Theta \widehat{A}_\varsigma^T = \widehat{D}_\varsigma \Omega \widehat{D}_\varsigma^T = L_\varsigma L_\varsigma^T, \quad L_\varsigma := \widehat{D}_\varsigma \Omega^{1/2}. \quad (517)$$

Here g below is the corresponding eliminated Newton RHS. In particular, $4/5 \leq x_i, s_i \leq 6/5$ implies $4/3 \leq \omega_j \leq 3$, hence $\Gamma := \max_j \omega_j / \min_j \omega_j \leq 9/4 < 4$.

Theorem 12.7 (Factor-specific Newton-RHS hardness). *Every unconditional algorithm that, on each sign input, outputs a state within trace distance $1/20$ of*

$$|z_\varsigma\rangle = \frac{\widehat{D}_\varsigma^{-1}e_0}{\|\widehat{D}_\varsigma^{-1}e_0\|_2} \quad (518)$$

makes $\Omega(N)$ raw coefficient queries.

The conclusion remains true for the weighted nearby factor L_ς in (517) and eliminated RHS g whenever

$$\frac{\max_j \omega_j}{\min_j \omega_j} \leq 4, \quad \|g - e_0\|_1 \leq \frac{1}{100}; \quad (519)$$

the target is the normalized state proportional to $\Omega^{-1/2} \widehat{D}_\varsigma^{-1}g$. If g or Ω is input-dependent, all raw queries used to construct and access it are included. Thus the theorem applies at any nearby central iterate satisfying these two promises. Freely supplied side oracles whose numerical values themselves encode parity are outside the theorem.

Proof. In the exact case, the prefix vector is $+1$ on every node of the root copy tree and endpoint parity p_N on every node of the endpoint copy tree. Pair corresponding nodes and measure their side qubit in the X basis. If $m = N + 1$, then $J \geq 31m$ and the probability of a paired node is

$$\frac{2J}{m + 2J} > \frac{62}{63}. \quad (520)$$

Conditional on that event the measurement returns p_N exactly; outputting a fair bit otherwise succeeds with probability greater than $1/2 + 31/63 = 125/126$. Trace error $1/20$ leaves success above $2/3$, so the fixed decoder computes parity and requires $\Omega(N)$ queries.

For (519), first take $g = e_0$ and put $\Gamma = \max \omega / \min \omega \leq 4$. Paired nodes have squared mass at least

$$\alpha \geq \frac{2J}{2J + \Gamma m} \geq \frac{31}{33}.$$

The amplitudes of a corresponding pair are a and $p_N b$ with $a/b \in [1/2, 2]$. The X measurement is correct with probability $1/2 + ab/(a^2 + b^2) \geq 9/10$. The full ideal success is therefore at least $1/2 + (2/5)(31/33) = 289/330$.

For $\delta = g - e_0$, every coordinate of $\hat{D}_\zeta^{-1}\delta$ is a signed ancestor sum, so, if P is the number of tree nodes,

$$\|\hat{D}_\zeta^{-1}\delta\|_2 \leq \sqrt{P}\|\delta\|_1, \quad \frac{\|\Omega^{-1/2}\hat{D}_\zeta^{-1}\delta\|_2}{\|\Omega^{-1/2}\hat{D}_\zeta^{-1}e_0\|_2} \leq \sqrt{\Gamma}\|\delta\|_1 \leq \frac{1}{50}.$$

The corresponding normalized pure states are at Euclidean distance at most $1/25$. Including trace error $1/20$ changes outcome probabilities by at most $9/100$; subtracting this from $289/330$ leaves success above $2/3$. Parity again proves the claim. $\square$

The ℓ_1 hypothesis is essential to this proof. On the signed spine set $\tilde{d}_j = p_j(1 - j/N)$, keep the root copy tree at value one, and set the endpoint copy tree to zero. Then

$$\|\hat{D}_\zeta \tilde{d} - e_0\|_2 = N^{-1/2}, \quad (521)$$

although the amplified endpoint has vanished and $\tilde{d}$ stays a constant relative distance from the prefix vector. Thus even a vanishing ℓ_2 feasibility residual does not imply the state closeness used above; (521) is a failure of this proof route, not a no-go for every possible ℓ_2 -robust reduction.

Nor is transformed-RHS hardness invariant under the factor. On the amplified tree, let $\hat{R}_\zeta = \text{diag}(p_j)$ be the prefix gauge and let $\hat{D}_+$ be the unsigned grounded incidence matrix. Then $\hat{D}_\zeta = \hat{R}_\zeta \hat{D}_+ \hat{R}_\zeta$, and the equally perfect factor

$$\hat{P}_\zeta = \hat{D}_+^{-1} \hat{R}_\zeta = \hat{R}_\zeta \hat{D}_+^{-1} \quad (522)$$

satisfies

$$\hat{P}_\zeta \hat{C}_\zeta \hat{P}_\zeta^T = I, \quad \hat{P}_\zeta e_0 = \hat{D}_+^{-1} e_0 = \mathbf{1}. \quad (523)$$

The RHS is now input independent. Recovery uses $\hat{P}_\zeta^T = \hat{R}_\zeta \hat{D}_+^{-T}$, so the parity cost has moved to factor application or original-coordinate recovery. This orthogonal-factor loophole is why [theorem 12.7](#) is deliberately factor-specific.

The invariant statement is end to end. Specialize (515) to $\sigma_c = 1$ and take the full step. With $p = \hat{D}_\zeta^{-1}e_0$, the unique minimum-norm direction and its update are

$$\Delta x = \frac{1}{2}(p, -p), \quad \Delta s = -\Delta x, \quad x^+ = \mathbf{1} + \Delta x. \quad (524)$$

Every pair of x^+ is $(3/2, 1/2)$ or its swap, and $X^+ s^+ = (3/4)\mathbf{1}$; this is the exact feasible center at $\mu = 3/4$. The endpoint copy tree occupies more than $15/32$ of all pairs. Conditional on measuring such a pair, the computational-basis coordinate identifies p_N with probability $9/10$. Returning a fair bit otherwise gives ideal success greater than $1/2 + (2/5)(15/32) = 11/16$, and trace error $1/100$ leaves success above $2/3$.

Theorem 12.8 (Preconditioner-oracle accounting). *Let P_ζ be any possibly data-dependent invertible factor for the normal equation in (515), with no sparsity or locality assumption, and let an end-to-end Newton procedure output a state within trace distance $1/100$ of the normalized x^+ in (524). Partition all raw coefficient queries into setup, transformed-RHS preparation, iterative solve, and original-coordinate recovery/update. Then*

$$\boxed{Q_{\text{set}} + Q_{\text{rhs}} + Q_{\text{solve}} + Q_{\text{rec}} = \Omega(N)}. \quad (525)$$

If the preconditioned system has condition at most K and the solve phase's only raw queries occur inside at most $T(K, \epsilon)$ calls to its preconditioned oracles, each costing at most q_{on} raw queries, then

$$Q_{\text{set}} + Q_{\text{rhs}} + T(K, \epsilon)q_{\text{on}} + Q_{\text{rec}} = \Omega(N). \quad (526)$$

In particular, for $T(K, \epsilon) = \tilde{O}(K \log(1/\epsilon))$ and hidden polylogarithmic factor $L(N, \epsilon)$,

$$Q + O(Kq_{\text{on}}L(N, \epsilon)) = \Omega(N), \quad Q := Q_{\text{set}} + Q_{\text{rhs}} + Q_{\text{rec}}. \quad (527)$$

Proof. All four phases together form one coefficient-query algorithm. Compose its output with the fixed decoder following (524); this computes parity of the N hidden spine signs with bounded error. Each LP coefficient query is simulated by $O(1)$ sign queries, proving (525). Under the oracle-only solve hypothesis, $Q_{\text{solve}} \leq T(K, \epsilon)q_{\text{on}}$, which gives (526), and substitution of the stated solver call bound gives (527). $\square$

A hybrid solver that also queries raw data directly remains covered by (525), where such queries stay in Q_{solve} .

The ledger is tight and is not a quantum–classical separation. A classical algorithm reads the N signs, computes every prefix, and stores the gauge in $\Theta(N)$ queries and work. Ladner–Fischer prefix scan reduces parallel depth to $O(\log N)$ with $O(N)$ bounded-fan-in work [89]. Caching then removes further hidden-sign queries, but it does not make the unsigned triangular operations free: applying $\hat{D}_+^{-1}$ or $\hat{D}_+^{-T}$ still takes $\Theta(N)$ work and $O(\log N)$ parallel depth by prefix or subtree aggregation. Multiplication by the cached diagonal gauge is local. Quantum parity changes only the constant in the query count. The right referee-level interpretation is therefore a lemma/counterexample suite: [theorem 12.6](#) is a finite-dimensional minimax lower bound, [theorem 12.7](#) locates cost for a canonical factor, and [theorem 12.8](#) records how another factor can relocate it. It is not a standalone universal preconditioning lower bound. Contemporary preconditioned QIPM analyses that assume prepared transformed data are consistent with this conclusion; reducing those oracles to raw LP access is a separate charged step [139].

12.4 Newton-state output is not an iterate oracle

We now change oracle models. Let $v \in \mathbb{R}_{>0}^n$, let its known norm be $r_v = \|v\|_2$, and grant controlled state-preparation access

$$U_v|0\rangle = |\hat{v}\rangle = \frac{1}{r_v} \sum_{j=1}^n v_j |j\rangle, \quad (528)$$

together with $U_v^\dagger$ and a fixed real phase. This is precisely [definition 2.14](#); it is stronger than receiving copies but distinct from coordinate-value, raw-coefficient, QRAM, or already compiled diagonal access. Those interfaces are not generally ordered by strength. The next results count calls to $U_v, U_v^\dagger$, not raw LP queries.

Theorem 12.9 (State-to-diagonal normalization–query product). *Suppose a controlled circuit W_v , using q calls to $U_v, U_v^\dagger$, is an $(\alpha, a, 1/8)$ block encoding of $\text{diag}(v)$. If the reported normalization satisfies the uniform known bound $\alpha \leq A$, where $A \geq 1$, then in the worst case*

$$\boxed{qA = \Omega(\sqrt{n})}. \quad (529)$$

The product is tight in order on the hard near-uniform family.

Proof. For $n \geq 10$, take the positive equal-norm pair

$$v^{(0)} = (1, \dots, 1), \quad v^{(1)} = (2, \beta, \dots, \beta), \quad \beta = \sqrt{\frac{n-4}{n-1}}. \quad (530)$$

Both norms are $\sqrt{n}$, and

$$\frac{1}{\sqrt{n}} \leq \|v^{(0)}/\sqrt{n} - v^{(1)}/\sqrt{n}\|_2 \leq \sqrt{2/n}. \quad (531)$$

Choose unitary extensions U_0, U_1 related by the minimum planar rotation between the prepared states. Then $\|U_0 - U_1\|_2 = \Theta(n^{-1/2})$, so the hybrid argument requires $\Omega(\sqrt{n})$ oracle calls for constant-bias distinction.

On the other hand, $O(A)$ controlled calls to W_v estimate $\text{Re}\langle 0^a, 1|W_v|0^a, 1\rangle$ to additive error $1/(16A)$. Multiplication by the reported α contributes error at most $1/16$, and block-encoding error contributes at most $1/8$. The first diagonal entry is therefore known within $1/4$, distinguishing one from two. The composed distinguisher uses $O(qA)$ original oracle calls, proving (529).

Rattew and Rebentrost give an exact constant-query block encoding of $\text{diag}(v/r_v)$ from U_v [121]. As an encoding of $\text{diag}(v)$ it has normalization $r_v = \sqrt{n}$ on (530), attaining the product in order. This is worst-case tightness, not a per-vector lower bound for every structured preparation circuit. $\square$

With copy-only access, the pair in (530) requires $\Omega(n)$ copies: one-copy overlap is $1 - \Theta(1/n)$, so only $k = \Omega(n)$ copies have constant trace distance. Thus inverse access in definition 2.14 changes the exponent but does not erase the interface cost.

The same obstruction sharpens near a strictly complementary endpoint. Let $n = 2m$, $m \geq 5$, $0 < \mu \leq 1$, and define

$$\beta = \sqrt{\frac{m-4}{m-1}}, \quad v = (2, \beta \mathbf{1}_{m-1}), \quad r = (m-1)(1-\beta) = \frac{3}{1+\beta}, \quad z = 2(r, \mathbf{1}_{m-1}). \quad (532)$$

For the one-sparse LP

$$A = [I_m \ 0], \quad b = \mathbf{1}_m, \quad c = (z, \mathbf{1}_m), \quad (533)$$

take the common $x = (\mathbf{1}_m, \mu \mathbf{1}_m)$ and

$$\begin{aligned} s^{(0)} &= (\mu \mathbf{1}_m, \mathbf{1}_m), & y^{(0)} &= z - \mu \mathbf{1}_m, \\ s^{(1)} &= (\mu v, \mathbf{1}_m), & y^{(1)} &= z - \mu v. \end{aligned} \quad (534)$$

Both triples are feasible. Since $\|v\| = \|\mathbf{1}_m\|$ and $z^T(v - \mathbf{1}_m) = 0$, the corresponding slack norms and dual norms agree.

Theorem 12.10 (Strict-complementarity state-access barrier). *In controlled state-preparation access to the slack and dual states in (534), distinguishing*

$$\|Xs - \bar{\mu} \mathbf{1}\|_2 \leq \bar{\mu}/4 \quad \text{from} \quad \|Xs - \bar{\mu} \mathbf{1}\|_2 \geq 3\bar{\mu}/4 \quad (535)$$

requires

$$\boxed{\Omega(\sqrt{n}/\mu)} \quad (536)$$

total controlled preparation/inverse calls. With copy-only access the bound is $\Omega(n/\mu^2)$. The result holds although both triples approach the same strictly complementary optimum. The fixed LP data, the common x , all vector norms, positivity, and the real phase conventions are free.

Proof. The zero triple is exactly centered with $\bar{\mu}_0 = \mu$. For the other, $\bar{\mu}_1 < \mu$ while the first complementarity product is 2μ , so

$$\|Xs^{(1)} - \bar{\mu}_1 \mathbf{1}\|_2 > \mu > \bar{\mu}_1. \quad (537)$$

The normalized slack-state distance is $\Theta(\mu/\sqrt{n})$. The duals differ by the same unnormalized vector $\mu(v - \mathbf{1}_m)$ and have common norm $\Omega(\sqrt{m})$, so their normalized-state distance is also $O(\mu/\sqrt{n})$.

Choose close minimum-rotation extensions for both oracles. A hybrid over queries to either oracle gives (536). For independent copies, fidelity multiplicativity gives the square of this dependence. Finally both triples converge to $((\mathbf{1}_m, 0), z, (0, \mathbf{1}_m))$, proving strict complementarity. $\square$

The triples can also be generated by a sparse LP's objective-state oracle. Put $d = v - \mathbf{1}_m$, $z_0 = 2(r, \mathbf{1}_{m-1})$, and $z_\mu = z_0 - (\mu/2)d$. Then

$$z_0^T d = 0, \quad \mathbf{1}^T d = 1 - r, \quad 1 \leq \|d\|_2^2 \leq 2. \quad (538)$$

Theorem 12.11 (Input-generated state-only barrier). *Use the common one-sparse $A = [I_m \ 0]$, $b = \mathbf{1}_m$, and objectives*

$$c^{(0)} = (z_\mu, \mathbf{1}_m), \quad c^{(1)} = (z_\mu + \mu d, \mathbf{1}_m). \quad (539)$$

Let the only input-dependent access be a controlled clean preparation of the normalized objective and its inverse; the common norm, A , and b are free. Distinguishing the inputs, certifying which one is centered at the common pair $x = (\mathbf{1}_m, \mu \mathbf{1}_m)$, $y = z_\mu - \mu \mathbf{1}_m$, or estimating the optimum to additive error $\mu/8$, needs $\Omega(\sqrt{n}/\mu)$ controlled oracle calls. Copy-only access needs $\Omega(n/\mu^2)$ copies.

Moreover, multiply both objectives symmetrically by n , take $b = \mathbf{1}_m/6$, and use

$$\tilde{c}^{(0)} = n(z_0 - \frac{\mu}{2}d, \mathbf{1}_m), \quad \tilde{c}^{(1)} = n(z_0 + \frac{\mu}{2}d, \mathbf{1}_m).$$

The normalized objective oracles do not change, the value gap obeys

$$|\text{OPT}_1 - \text{OPT}_0| = \frac{n\mu}{6}(r - 1) \geq \frac{n\mu}{12}, \quad (540)$$

and additive value accuracy $n\mu/48$ has the same query lower bound. This is a fixed fraction of the centered total duality gap $n\mu$; here $n = \nu$ is the canonical LP barrier parameter of [section 2.1](#).

Proof. The objective norms agree because $2\mu z_\mu^T d + \mu^2 \|d\|^2 = 0$, and their normalized-state distance is $\Theta(\mu/\sqrt{n})$. Minimum-rotation objective-state oracles therefore require $\Omega(\sqrt{n}/\mu)$ calls to distinguish; fidelity multiplicativity gives $\Omega(n/\mu^2)$ copies. The displayed common pair is centered for input zero and violates (535) for input one. Every feasible point has first block $\mathbf{1}_m$ and its positive-cost second block vanishes at optimum, so the original value gap is $\mu|\mathbf{1}^T d| \geq \mu/2$. Additive error $\mu/8$ distinguishes the inputs.

For the scaled claim, the common point

$$\tilde{x} = (\mathbf{1}_m/6, (\mu/n)\mathbf{1}_m), \quad \tilde{y} = n(z_0 - \frac{\mu}{2}d) - 6\mu\mathbf{1}_m$$

has every complementarity product equal to μ for input zero; input one remains positive because $n(1 - \beta) \leq 5 < 6$ and lies outside every fixed N_2 neighborhood of radius below one. Its centered total duality gap is $n\mu$. Feasibility fixes the first primal block to $\mathbf{1}_m/6$, which gives (540); accuracy $n\mu/48$ distinguishes the inputs. The inverse- μ factor is therefore not caused by demanding accuracy finer than the centered duality-gap scale. $\square$

This input-generated theorem has a sharp boundary. A binary coefficient-value oracle exposes the known exceptional coordinate, whose values differ by $\Theta(\mu)$, in one query at sufficient requested precision. Distinct value-register strings are not close unitaries merely because the encoded reals are close. Likewise, the naturally normalized Newton diagonals for the two inputs differ by a constant in their first entry, so a freely supplied Newton block encoding distinguishes them in $O(1)$ calls. The $\Omega(\sqrt{n}/\mu)$ theorem is therefore an end-to-end result only for the state-only objective/update interface; it does not lift to raw exact-value or precompiled-Newton access.

State access also obstructs fraction-to-boundary line search.

Corollary 12.12 (Fraction-to-boundary). *For $n \geq 4$, let $x = \mathbf{1}_n$ and compare the equal-norm directions*

$$\delta^{(0)} = -\frac{1}{2}\mathbf{1}_n, \quad \delta^{(1)} = (-1, -\gamma, \dots, -\gamma), \quad \gamma = \frac{1}{2}\sqrt{\frac{n-4}{n-1}}.$$

Their maximal positive steps are two and one, while their normalized states are $\Theta(n^{-1/2})$ apart. A state-only rule that is never larger than one on the second input but is at least $3/2$ on the first needs $\Omega(\sqrt{n})$ controlled state-preparation calls.

Proof. Choose the two preparation unitaries as minimum rotations. Any valid output distinguishes them with constant bias, while a Q -query hybrid has distance $O(Q/\sqrt{n})$. Thus $Q = \Omega(\sqrt{n})$. $\square$

The next theorem rules out bypassing diagonal compilation by directly solving the following easy system. For $n \geq 4$, fix $0 < \eta \leq 1/4$ and define

$$v^{(0)} = \mathbf{1}_n, \quad v^{(1)} = (1 + \eta, \beta_\eta, \dots, \beta_\eta), \quad \beta_\eta = \sqrt{1 - \frac{2\eta + \eta^2}{n-1}}. \quad (541)$$

Both norms equal $\sqrt{n}$, and their normalized distance d_η satisfies

$$\frac{\eta}{\sqrt{n}} \leq d_\eta \leq \frac{2\eta}{\sqrt{n}}. \quad (542)$$

Let $D_b = \text{diag}(v^{(b)})$, $r = (e_1 + e_2)/\sqrt{2}$, and $|z_b\rangle = D_b^{-1}r/\|D_b^{-1}r\|_2$. These systems are one-sparse and $\kappa(D_b) < 2$.

Theorem 12.13 (State to next solve). *Give an algorithm controlled access to the minimum-rotation preparations $U_b, U_b^\dagger$ of the vectors in (541). If its output ρ_b satisfies*

$$D_{\text{tr}}(\rho_b, |z_b\rangle\langle z_b|) \leq \eta/12 \quad (543)$$

for both inputs, then it uses $\Omega(\sqrt{n})$ oracle calls. With copy-only iterate access it needs $\Omega(n)$ copies. In particular, fixed $\eta = 1/4$ is a constant-accuracy lower bound for a condition-below-two system.

Proof. For $b = 0$, measuring coordinate one in $|z_b\rangle$ has probability $1/2$. For $b = 1$, put $q = \beta_\eta/(1 + \eta)$. The probability is $q^2/(1 + q^2)$, and

$$\frac{1}{2} - \frac{q^2}{1 + q^2} \geq \frac{1}{2} - \frac{1}{1 + (1 + \eta)^2} \geq \frac{\eta}{3}. \quad (544)$$

Contractivity and (543) therefore make the two algorithm outputs at least $\eta/6$ apart in trace distance. A Q -query hybrid is at most $Qd_\eta \leq 2Q\eta/\sqrt{n}$ apart, whence $Q \geq \sqrt{n}/12$. For k copies the input trace distance is at most $\sqrt{k}d_\eta = O(\sqrt{k}\eta/\sqrt{n})$, forcing $k = \Omega(n)$. $\square$

The Hermitian dilation $\begin{pmatrix} 0 & D_b \\ D_b & 0 \end{pmatrix}$ has the same sparsity and condition number, so this is also a standard Hermitian QLS instance. It does not contradict a polylogarithmic-dimensional QLS algorithm: such an algorithm assumes the matrix oracle that this theorem requires one to construct or bypass from the iterate state.

Per-round costs add only under an explicit fresh-oracle contract. At round $t = 1, \dots, T$, after earlier outputs are committed, an adversary chooses a fresh bit b_t independent of the current workspace and supplies only a new minimum-rotation pair $U_{t,b_t}, U_{t,b_t}^\dagger$ for (541); future oracles are unavailable. Before the next round, the algorithm must either satisfy (543), or, after S_t setup queries, expose a reusable controlled $(\alpha_t, a, \eta/8)$ block encoding of D_{b_t} with $\alpha_t \leq A_t$. One invocation may use another q_t state-oracle calls.

Theorem 12.14 (Online fresh-oracle direct sum). *In the preceding interface, every fixed worst-case per-round budget satisfies*

$$\sum_{t=1}^T Q_t = \Omega(T\sqrt{n}) \quad (545)$$

for next-solution-state outputs, and

$$\boxed{\sum_{t=1}^T (\eta S_t + q_t A_t) = \Omega(T\sqrt{n})} \quad (546)$$

for reusable diagonal compilers. For fixed short-step η , the latter is equivalently $\sum_t (S_t + q_t A_t) = \Omega(T\sqrt{n})$.

Proof. Condition on the complete workspace and transcript at the start of round t . Freshness makes that side information independent of b_t , so it is a fixed ancilla in [theorem 12.13](#), which gives $Q_t = \Omega(\sqrt{n})$. For a compiler, run setup once and estimate the first encoded entry to $O(\eta/A_t)$ using $O(A_t/\eta)$ invocations. This distinguisher makes $S_t + O(q_t A_t/\eta)$ state-oracle calls, but the close input oracles require $\Omega(\sqrt{n}/\eta)$ calls. Hence $\eta S_t + q_t A_t = \Omega(\sqrt{n})$ in each conditioned round. Summation proves both claims. $\square$

The full oracle ledger in [theorem 12.14](#) is load-bearing:

- The atomic charged operations are controlled calls to the current U_{t,b_t} and its inverse. Gate complexity is not lower-bounded.
- The common norm $\sqrt{n}$, the two vector formulas, r , and the round number are free; only b_t is hidden.
- No coordinate-value, QRAM, objective-coefficient, or compiled matrix oracle is available. Each could reveal the exceptional coordinate in one query.
- The complete off-zero action is promised: the adversary may choose the close minimum-rotation completions. Specifying only $U|0\rangle$ would not define a query lower bound.
- One-time setup and reusable advice are charged through S_t , while per-invocation use is charged through q_t . Advice consumed by calls must be prepared in sufficient quantity and cannot be cloned for free.
- The required output is committed before the next fresh oracle arrives; this causality makes the conditional bounds additive.

A heralded compiler with success probability bounded below obeys the same ledger after charging repetitions. An unheralded constant failure probability is not covered when $\eta \rightarrow 0$, because it can exceed the target states' $\Theta(\eta)$ separation.

The hard stream remains positive and obeys a coordinatewise short-update promise: $|v_i^{(b_{t+1})} - v_i^{(b_t)}| \leq \eta v_i^{(b_t)}$. Thus positivity and short relative motion do not defeat the online theorem. They also do not turn it into a fixed-instance QIPM iteration bound. The exact obstructions are:

1. *Freshness is stronger than a fixed optimization instance.* Every QIPM iterate is determined by one static input and prior computation; no independent oracle normally arrives after each output.
2. *One-vector worst-case hardness is not automatically a direct sum.* If every iterate depends on one bit, learning it once makes every later diagonal easy.

3. *Circuit generation must be expanded to atomic queries.* A circuit for $U_{x_{t+1}}$ may call U_{x_t} , which contains the prior history. Unit-cost treatment hides inlining, while charging both as independent primitives double-counts it.
4. *Off-zero leakage is model dependent.* The hybrid chooses close extensions, whereas a concrete solver circuit may reveal coordinate or trajectory data away from $|0\rangle$. Its entire oracle action must be fixed.
5. *All-round coherent access changes the direct-sum problem.* If a single multiplexed query reaches every time-indexed oracle, the causal proof fails and a separate coherent direct-sum theorem is needed.
6. *The QIPM equations may permit a bypass.* The diagonal systems above have not been embedded as consecutive Newton systems of one fixed sparse LP; cancellation, scale-free formulations, or history-state methods might avoid separate compilation.
7. *The output contract matters.* An architecture that exposes a reusable diagonal or a Newton state every round meets the theorem's contract. An algorithm that proves its step analytically and emits only one final observable need not.

Therefore a fixed-LP $\Omega(T\sqrt{n})$ result would require T independent hard pieces in one sparse instance, a trajectory that reveals them one by one, and a direct-sum theorem against the static input oracle. None is proved here.

Finally, coherent update errors accumulate. If

$$\|\tilde{v}\rangle - |v/r_v\rangle\|_2 \leq \xi, \quad |\tilde{r}_v - r_v| \leq \delta_r, \quad (547)$$

amplitude-to-diagonal lifting has absolute operator error at most $\delta_r + \tilde{r}_v\xi$. For coherent LCU updates $X_{t+1} = X_t + h_t\Delta_t$, let $E_t := \|\tilde{X}_t - X_t\|_2$ be the accumulated absolute operator error before update t . If the step lengths and LCU arithmetic are exact and the only new error is the lifted direction error, the operator-norm triangle inequality gives

$$E_{t+1} \leq E_t + |h_t|(\delta_{r,t} + \tilde{r}_t\xi_t), \quad E_T \leq E_0 + \sum_{t < T} |h_t|(\delta_{r,t} + \tilde{r}_t\xi_t). \quad (548)$$

Rounding, imperfect LCU coefficients, or postselection errors must be added as separate terms. Late-path centrality therefore constrains the cumulative weighted error, not only each local preparation. Literal circuit reuse nests as well. Let C_t be the cost of the current iterate-preparation circuit, let Q_t be the number of its calls made by the next direction circuit, and let G_t be all nonrecursive work in that update. With literal inlining, additive uniform gate/query costs, and no sharing or memoization, the exact accounting is

$$C_{t+1} = G_t + (1 + Q_t)C_t. \quad (549)$$

This identity is for literal composition, not a universal lower bound against direct structured or history-state oracles.

12.5 Lazy central oracles and the caching ceiling

Changing all diagonal weights does not imply that they must all be rewritten. The following fixed sparse LP makes this failure explicit. For K disjoint blocks of length L , let $d_{a,i} = u_{a,i} - v_{a,i}$ and impose

$$d_{a,0} = 1, \quad d_{a,i} - s_{a,i}d_{a,i-1} = 0 \quad (1 \leq i \leq L), \quad (550)$$

with all $u, v \geq 0$. Minimize

$$\sum_{a=1}^K \left[\sum_{i=0}^L (u_{a,i} + v_{a,i}) + \alpha(u_{a,L} - v_{a,L}) \right], \quad 0 < \alpha < 1. \quad (551)$$

Rows and columns have constant sparsity and the instance size is $\Theta(KL)$. With

$$p_{a,0} = 1, \quad p_{a,i} = \prod_{j=1}^i \varsigma_{a,j}, \quad q(\mu) = \mu + \sqrt{\mu^2 + 1}, \quad (552)$$

the primal central path is exactly

$$(u_{a,i}(\mu), v_{a,i}(\mu)) = \left(\frac{q(\mu) + p_{a,i}}{2}, \frac{q(\mu) - p_{a,i}}{2} \right). \quad (553)$$

Indeed feasibility fixes the difference $p_{a,i}$; differentiating the one-dimensional pair barrier gives the common sum $q(\mu)$. In the fixed orthonormal null basis $(e_{u_{a,i}} + e_{v_{a,i}})/\sqrt{2}$, the reduced log-barrier Hessian $W^\top X^{-2}W$ is the scalar

$$2[(q(\mu) + 1)^{-2} + (q(\mu) - 1)^{-2}]I,$$

so it has condition one throughout the trajectory. All changing numerical information is the public scalar $q(\mu)$; all hidden structural information is the static prefix table.

Give reversible raw access to bits $\varsigma_{a,i} = (-1)^{b_{a,i}}$. An endpoint central lookup at block a returns (553) for $i = L$ and therefore reveals $P_a = p_{a,L}$, even with coordinate error strictly below $1/2$.

Theorem 12.15 (Lazy endpoint lookup). *One exact coherent endpoint lookup, or one whose decoded sign is correct with probability at least $2/3$, needs $\Omega(L)$ raw queries in the worst case, even when the block index is in superposition. The bound is tight.*

Proof. Fix the block register to any basis value a and fix all other blocks. Decoding the returned pair computes parity of the remaining L bits, which has bounded-error quantum query complexity $\Omega(L)$. Conversely, the exact quantum parity algorithm computes the selected block's parity in $O(L)$ queries while carrying a coherently as a spectator; reversible arithmetic then combines it with $q(\mu)$. The same circuit works in superposition over a . $\square$

Call a resource an exact reusable endpoint oracle if, once constructed, a fixed processor can obtain every one of the K endpoint values without further raw queries, and the resource survives or otherwise supports those K invocations.

Theorem 12.16 (Reusable endpoint preprocessing). *Constructing an exact reusable endpoint oracle needs at least*

$$\left\lceil \frac{KL}{2} \right\rceil \quad (554)$$

raw queries. A bounded-error resource also needs $\Omega(KL)$ queries when all K successive answers are jointly correct with probability at least $2/3$ on every input.

Proof. After preprocessing, invoke the resource on every block and multiply the decoded signs. The result is

$$\prod_{a=1}^K P_a = (-1)^{\bigoplus_{a=1}^K \bigoplus_{i=1}^L b_{a,i}},$$

the parity of all KL raw bits. Exact quantum parity requires $\lceil KL/2 \rceil$ queries, and bounded-error parity requires $\Omega(KL)$. The proof permits entangled program states and implicit formulas; reusability, not classical table materialization, is the decisive promise. $\square$

Theorem 12.17 (Explicit-coordinate trajectory service). *Fix arbitrary parameters $\mu_1, \dots, \mu_K > 0$. A service that at stage a returns the classical endpoint pair of block a at μ_a , with both coordinate errors below $1/2$ and all answers jointly correct with probability at least $2/3$ on every input, uses*

$$\boxed{\Omega(KL)} \tag{555}$$

total raw queries across preprocessing and all stages. Thus its amortized cost is $\Omega(L)$.

Proof. Each returned pair reveals the independent P_a . Multiplying the K answers computes parity of all KL input bits, so bounded-error quantum parity proves (555). $\square$

Theorem 12.17 is a lower bound for an explicit-cell service contract, not for every QIPM. Quantum cell-probe lower bounds likewise require a specified storage-and-probe model and do not transfer automatically to an analytic central path [124]. There is no unconditional per-IPM-iteration refresh theorem here, for five separate reasons:

1. A fixed LP has no fresh per-iteration information. Once all KL bits have been processed, unrestricted persistent memory can amortize that setup over arbitrarily many iterations.
2. The path parameter may be low dimensional. In (553) every changing coordinate is generated by one scalar $q(\mu)$, so a stored structural oracle needs only $O(1)$ extra arithmetic when μ changes.
3. Cell-probe lower bounds need a query contract. An IPM follows an analytic, algorithm-chosen trajectory rather than an adversarial sequence of cell updates and requests; theorem 12.17 applies only after the distinct endpoint outputs are required.
4. Coherent lookup defeats a naive block direct sum. One $O(L)$ circuit evaluates the selected block in superposition. A factor K requires a classical-output contract or a suitable direct-product theorem.
5. A normalized state is weaker than a data structure. It need not support reusable random access to every coordinate; assuming that it does silently adds tomography or QRAM construction.

This is the same caching ceiling as (32) and the static ledger following theorem 9.4: caching all signs can make later raw-query cost vanish. A genuine refresh lower bound would require a memory cap, mandatory coordinate monitoring, nonshareable independent invocations, or a proven online-hard fixed-LP trajectory. Sparsity and changing diagonal weights alone provide none of these.

12.6 Synthesis

Across all five interfaces, the invariant is a product or a ledger. Separate exact factor access gives $\alpha_P \sqrt{K} = \Omega(N)$ and, with reusable compilation, $Ns + q\alpha = \Omega(N^2)$. Scalar congruence obeys $\sqrt{\kappa} \chi \geq c/\mu$; equality in order holds exactly for the maximal class $r_\mu = \Omega(\mu)$ and $r_\mu = O(1)$. The parity instance preserves the total setup–RHS–solve–recovery cost even when a perfect factor makes one phase trivial. State preparation obeys $qA = \Omega(\sqrt{n})$, and its iterative version becomes additive only under a fresh-oracle contract. Lazy evaluation then shows why static preprocessing can defeat any unqualified per-iteration claim.

These are access-model obstructions, not a verdict against preconditioning or state reuse. The positive modules in section 13 are designed to respect the same ledgers: they directly assemble favorable operators, identify which projectors or cached structures are charged, and state what output interface is actually produced.

13 Positive Algorithmic Modules

The lower bounds in [sections 9](#) and [12](#) do not rule out useful quantum subroutines. They say what a positive result must count. This section gives such results at their natural interfaces: a pathwise reuse theorem, a partition-free tail solver and face repair, an implicit-coordinate dual method with compressed output, and an explicit border solver for block-angular systems. Some compose to conditional trajectory theorems; none is an unconditional end-to-end speedup for arbitrary sparse LPs.

We use the following rules throughout. We refer to these as the referee rules; they are also a checklist for assessing any positive QIPM claim.

1. A small spectral condition number does not make loading, normalization, RHS preparation, or recovery free. Every block encoding and output compiler must be expanded into the access model of [sections 2.2](#) and [2.3](#).
2. A strict-complementarity construction applies only after its stated, data-dependent tail threshold; the separation margin and active singular gaps are parameters, not consequences of sparsity.
3. Compressed dual output is dimensionally useful only when the retained dimension satisfies $m = o(n)$, and only if the classical comparator is charged in the same model.
4. A coherent primal-coordinate oracle is substantive output. When its values are well spread, constant-overhead value-to-amplitude conversion makes it at least as informative as a primal-state preparer.
5. A genuinely dual-only certificate can fall outside a primal-output lower bound. A claimed contradiction requires an explicit low-query reduction to the hard output contract.

These rules instantiate rather than repeat the ledgers of [section 12](#). In particular, all costs below retain operator normalization, RHS amplification, residual conversion, tomography, and static input construction.

13.1 Projective lazy refresh and its exact boundary

Consider a realized sequence of nonsingular real systems, indexed by $t = 0, \dots, T$, with $f_t \neq 0$:

$$H_t z_t = f_t, \quad q_t = \|z_t\|, \quad \psi_t = z_t/q_t, \quad M_t = \frac{q_t}{\|f_t\|} H_t, \quad (556)$$

and define

$$\delta_t = \|M_{t+1}(\psi_{t+1} - \psi_t)\|, \quad V_{\text{proj}} = \sum_{t < T} \delta_t. \quad (557)$$

Here z_t is the coordinate vector of the solution in whatever fixed basis the module stores it in (for the OSS systems below, the fixed V spanning $\ker A$), so δ_t and V_{proj} depend on that basis. Fix $\eta > 0$ and $\epsilon \geq 0$. At the initial checkpoint $s = 0$ and every subsequent refresh, store a classical $\hat{z}_s$ satisfying $\|H_s \hat{z}_s - f_s\|/\|f_s\| \leq \epsilon$. At each later time t , let s be the most recent checkpoint and choose the scalar minimizing $\|H_t(a\hat{z}_s) - f_t\|$. For $v = H_t \hat{z}_s \neq 0$ this scalar is $\langle v, f_t \rangle / \|v\|^2$; for $v = 0$ choose zero. Reuse the vector if this minimum relative residual is at most η ; otherwise refresh at t . The minimization and threshold comparison here are exact. Write R for the total number of checkpoint solves, including the initial solve.

Theorem 13.1 (Pathwise residual-certified refresh). *Suppose $G > 0$ and, for every test at time t with most recent checkpoint s , including a test that triggers a refresh,*

$$\|M_t M_s^{-1}\| \leq G \quad (s \leq j \leq t). \quad (558)$$

If $\epsilon \leq \eta/(2G)$, then the number R of checkpoint solves satisfies

$$\boxed{R \leq 1 + \frac{2GV_{\text{proj}}}{\eta}}. \quad (559)$$

The statement is pathwise: the systems may depend adaptively on earlier inexact steps.

Proof. Put $u_s = \widehat{z}_s/q_s$ and $e_s = M_s(u_s - \psi_s) = (H_s \widehat{z}_s - f_s)/\|f_s\|$, so $\|e_s\| \leq \epsilon$. The nonzero right-hand sides imply $q_s > 0$. For a test at t , the least-squares scalar is no worse than $a = q_t/q_s$. The candidate's normalized residual vector is

$$M_t(u_s - \psi_t) = M_t M_s^{-1} e_s - \sum_{j=s}^{t-1} M_t M_{j+1}^{-1} M_{j+1}(\psi_{j+1} - \psi_j).$$

The triangle inequality and (558) give

$$\frac{\|H_t(a\widehat{z}_s - z_t)\|}{\|f_t\|} \leq G \left(\epsilon + \sum_{j=s}^{t-1} \delta_j \right).$$

Consequently every failed test after a checkpoint consumes more than $\eta/(2G)$ of projective variation. The consumed intervals are disjoint, and are contained in $\{0, \dots, T-1\}$, which proves (559), including when no test fails. $\square$

Formal verification and scope. The Lean development in `formal/QipmFormal/Refresh/` verifies the normalization identities, scalar least-squares minimization, residual transport, and the bound for the sequential reuse/refresh procedure. It checks the initial solve, equality at the acceptance threshold, and the final interval with no further refresh. The claim map and reproduction command are in `formal/REFRESH.md`. The audit permits only `propext`, `Classical.choice`, and `Quot.sound`. This verifies the pathwise theorem under its stated gain and checkpoint accuracy hypotheses; it does not establish those hypotheses for an IPM trajectory, nor verify the tail, OSS, cycling, or quantum-cost results below. A test with numerical error needs an explicit rejection guarantee as well as safe acceptance: rejecting a truly acceptable vector can add checkpoint solves without consuming the variation used by this proof.

For $H_t = AD_t A^T$, the test uses $A^T \widehat{z}_s$, the current diagonal, and A ; it costs $O(\text{nnz}(A))$ classical arithmetic without forming H_t . This per-iteration term remains in the ledger. The theorem amortizes QLS and readout, not all outer-loop work. It is a residual-screened classical cache, distinct from coherent eigenpath traversal [26] and from classical Krylov recycling or inverse maintenance [93, 120]. Classical warm starts and reoptimization instead transfer information between perturbed problem instances [68, 140]; they do not certify reuse of a direction along one fixed central tail.

The projective parameters are finite on the exact central tail even when the unscaled normal matrix becomes ill conditioned.

Theorem 13.2 (Directed strict-complementarity tail). *Let Q project onto U , let $Q^\perp = I - Q$, and suppose*

$$H_\mu = \mu^{-1} C_\mu + \mu D_\mu, \quad C_\mu = Q C_\mu Q, \quad (560)$$

where, uniformly on $0 \leq \mu \leq \mu_0$,

$$c_- Q \preceq C_\mu \preceq c_+ Q, \quad D_\mu \succeq 0, \quad \|D_\mu\| \leq d_+, \quad Q^\perp D_\mu Q^\perp \succeq d_- Q^\perp.$$

Assume C_μ, D_μ are C^1 with bounded derivatives. For nonzero $f \in U$, set $z_\mu = H_\mu^{-1} f$, $\psi_\mu = z_\mu/\|z_\mu\|$, and $N_\mu = (\|z_\mu\|/\|f\|)H_\mu$. There are finite constants B, L, G depending only on the displayed bounds and μ_0 such that

$$\sup_\mu \|N_\mu\| \leq B, \quad \sup_{0 < \mu \leq \nu \leq \mu_0} \|N_\mu N_\nu^{-1}\| \leq G, \quad \sup_\mu \|\psi'_\mu\| \leq L. \quad (561)$$

Hence every decreasing discretization, using the refresh policy above with $\epsilon \leq \eta/(2G)$, satisfies

$$V_{\text{proj}} \leq BL\mu_0, \quad R \leq 1 + \frac{2GBL\mu_0}{\eta}, \quad (562)$$

independently of its endpoint and number of steps.

Proof. Put $\mathcal{A}_\mu = C_\mu + \mu^2 D_\mu$ and $w_\mu = \mathcal{A}_\mu^{-1} f$. Then $z_\mu = \mu w_\mu$ and $N_\mu = a_\mu \mathcal{A}_\mu$, where $a_\mu = \|w_\mu\|/\|f\|$. In $U \oplus U^\perp$ blocks, Schur elimination gives

$$E_\mu = D_{UU} - D_{U\perp} D_{\perp\perp}^{-1} D_{\perp U} \succeq 0, \quad S_\mu = C_{UU} + \mu^2 E_\mu, \\ w_U = S_\mu^{-1} f, \quad w_\perp = -D_{\perp\perp}^{-1} D_{\perp U} w_U.$$

The hypotheses bound these factors and their derivatives and bound $\|w_\mu\|$ above and away from zero. Thus a_μ and the normalized ray are C^1 at zero, proving the first and third bounds.

The only unbounded block of $\mathcal{A}_\nu^{-1}$ is $\nu^{-2} D_{\perp\perp, \nu}^{-1}$ in its lower-right block. Every block of $\mathcal{A}_\mu$ that can multiply it contains μ^2 . For $\mu \leq \nu$ this ratio is at most one, so $\|\mathcal{A}_\mu \mathcal{A}_\nu^{-1}\| = O(1)$. The bounded ratio a_μ/a_ν proves the directed cross-gain. Finally the fundamental theorem of calculus gives

$$\sum_t \|N_{\mu_{t+1}}(\psi_{\mu_{t+1}} - \psi_{\mu_t})\| \leq B \int_0^{\mu_0} \|\psi'_\mu\| d\mu \leq BL\mu_0,$$

and [theorem 13.1](#) gives the refresh count. $\square$

The order in (561) is essential. For $C = \text{diag}(1, 0)$, $D = \text{diag}(0, 1)$, and $f = e_1$, the ray and forward gain are constant, whereas the reverse gain is $(\nu/\mu)^2$. On an exact LP central path, $C_\mu = A_B \text{diag}(x_B^2) A_B^T$, $D_\mu = A_N \text{diag}(s_N^{-2}) A_N^T$, and $f = b \in \text{range}(A_B)$. Boundary analyticity supplies the stated regularity even for degenerate optimal faces [73, 108]. Its constants are instance dependent and need not be bounded by sparsity or dimension alone.

At checkpoint s , let $\Gamma_{f,s}$ be the RHS-state amplification, $\mathcal{K}_s = \alpha_s \|H_s^{-1}\|$ the effective QLS parameter, and $\rho_s = \alpha_s \|z_s\|/\|f_s\|$ the normalized-state-to-residual conversion. Suppose these are bounded by $\Gamma_f, \mathcal{K}, \rho$ on the tail. A refresh must have residual accuracy $O(\eta/G)$, as required by [theorem 13.1](#); hence its QLS precision contributes $\log(G/\eta)$, and tight coherent tomography uses $\tilde{O}(mG\rho/\eta)$ controlled state preparations. The resulting ledger is

$$\tilde{O} \left[\left(1 + \frac{GBL\mu_0}{\eta} \right) \Gamma_f \mathcal{K} \frac{mG\rho}{\eta} \log \frac{G}{\eta} \right], \quad (563)$$

where RHS preparation and amplification are explicit rather than hidden in the operator cost. Copy tomography instead has quadratic inverse dependence on the required $O(\eta/(G\rho))$ normalized-state accuracy [131]. The ordinary $T = O(\sqrt{n} \log(\mu_0/\epsilon_{\text{opt}}))$ sparse updates and residual tests remain.

Corollary 13.3 (Exact-central feasible OSS). *With a fixed full-rank V spanning $\ker A$, let*

$$\mathcal{M}_\mu = [-X_\mu A^T \quad S_\mu V], \quad f_\mu = -\tau \mu e. \quad (564)$$

On the exact central path, its normalized solution ray has finite tail variation and bounded directed cross-gain. Every coefficient vector reconstructs $\Delta x = V\lambda$, $\Delta s = -A^T \Delta y$ and hence preserves both feasibility equations exactly. Consequently residual-certified reuse with checkpoint accuracy $\epsilon \leq \eta/(2G)$ has a final-precision-independent refresh count on the exact-central tail.

Proof. At the center,

$$\mathcal{M}_\mu^T \mathcal{M}_\mu = \text{diag}(A X_\mu^2 A^T, V^T S_\mu^2 V),$$

so $\|\mathcal{M}_\mu^{-1}\| = O(1/\mu)$. Analyticity gives $\|\mathcal{M}_\mu - \mathcal{M}_\nu\| = O(|\mu - \nu|)$, and therefore $\|\mathcal{M}_\mu \mathcal{M}_\nu^{-1}\| = O(1)$ for $\mu \leq \nu$. Differentiating centrality shows that the OSS solution divided by μ is an analytic nonzero vector. Thus the normalized exact-central OSS ray has finite variation. The reconstruction identities prove exact feasibility, and [theorem 13.1](#) gives the refresh conclusion. $\square$

This does not extend to every standard inexact trajectory.

Proposition 13.4 (Cycling and proximal no-go). *There is a row-3-sparse LP and a feasible three-phase trajectory in a fixed $N_2(\theta)$ neighborhood, satisfying the standard fixed relative OSS residual contract, for which*

$$\sum_{t < T} \delta_t = \Omega\left(\log \frac{\mu_0}{\epsilon_{\text{opt}}}\right). \quad (565)$$

Moreover, no Tikhonov rule inferred only from that neighborhood and residual contract can simultaneously guarantee finite projective variation, admissible OSS residual, and bounded reconstructed steps. A residual-safe ridge scale does not improve worst-case conditioning.

Proof. Use

$$\min \sum_{j=4}^n x_j \quad \text{subject to} \quad x_1 + x_2 + x_3 = 1, \quad x \geq 0.$$

Set $\tau = a_0/\sqrt{n}$, $\mu_t = (1 - \tau)^t \mu_0$, and use

$$u_0 = \frac{1}{\sqrt{2}}(1, -1, 0), \quad u_1 = \frac{1}{\sqrt{2}}(0, 1, -1), \quad u_2 = \frac{1}{\sqrt{2}}(-1, 0, 1).$$

Cycle

$$x_{1:3}^{(t)} = e/3 + d u_{t \bmod 3}, \quad x_{4:n}^{(t)} = \mu_t e, \quad y^{(t)} = -3\mu_t, \quad s_{1:3}^{(t)} = 3\mu_t e, \quad s_{4:n}^{(t)} = e.$$

Then $\|X_t s_t - \mu_t e\| = 3d\mu_t$, and the feasible step to the next phase has OSS residual $3d\mu_t(u_{j+1} - \tau u_j)$. Taking $d = \Theta(1/\sqrt{n})$ meets the usual relative contract $\eta_n = \Theta(1/\sqrt{n})$. With the fixed sparse basis $V = [e_1 - e_3, e_2 - e_3, e_4, \dots, e_n]$, the exact OSS coefficient rays converge cyclically to the three distinct lines $(1, -1)$, $(0, 1)$, and $(-1, 0)$. The normalized destination operator acts on their differences at constant scale, so $\delta_t = \Omega(d)$ eventually. Since $T = \Theta(\sqrt{n} \log(\mu_0/\epsilon_{\text{opt}}))$, this proves (565).

The preceding cycling half uses both $d = \Theta(1/\sqrt{n})$ and $\eta_n = \Theta(1/\sqrt{n})$. The proximal no-go switches both regimes: fix constants $d > 0$ and $0 < \eta < \theta < 1$ such that

$$3d \leq \theta, \quad \frac{c_1 d}{\sqrt{a_0^2 + 9d^2}} > \eta, \quad (566)$$

where $c_1 > 0$ is the constant separation of the three projective lines. For the ridge selector

$$z_\lambda = \arg \min_z \{\|Mz - f\|^2 + \lambda \|z - c\|^2\},$$

let $z_* = M^{-1}f$, v be a right singular vector of singular value σ , and $e = c - z_*$. Direct diagonalization gives the exact frontier

$$\begin{aligned} \langle v, z_\lambda - c \rangle &= -\frac{\sigma^2}{\sigma^2 + \lambda} \langle v, e \rangle, \\ \left| \left\langle \frac{Mv}{\sigma}, Mz_\lambda - f \right\rangle \right| &= \frac{\lambda \sigma}{\sigma^2 + \lambda} |\langle v, e \rangle|. \end{aligned}$$

On the cycling face $\sigma = \Theta(\mu)$. A ray-locked sequence of bounded steps and finite projective variation is Cauchy, so on at least one of every three phases it stays a fixed distance from that phase's exact face target. If the ridge moves a nonvanishing fraction toward those targets infinitely often, its projective variation diverges. Otherwise its residual is at least $c_1 d \mu_t$ on an infinite separated subsequence. Since $\|f_t\| = \mu_t \sqrt{a_0^2 + 9d^2}$, (566) makes this exceed the admissible $\eta \|f_t\|$. This is the fixed-constant regime, not the $d = \Theta(1/\sqrt{n})$ regime used above.

Conversely the universal bound

$$\|Mz_\lambda - f\| \leq \frac{\sqrt{\lambda}}{2} \|e\| \quad (567)$$

is sharp, since $\sigma\lambda/(\sigma^2 + \lambda)$ is maximized at $\sigma = \sqrt{\lambda}$. With $\|f\| = \Theta(\mu)$ and $\|e\| = \Theta(1)$, the uniform worst-case residual-safe scale is $\lambda = O(\eta^2\mu^2)$. The augmented system $[M^T \sqrt{\lambda}I]^T$ then retains condition $\Theta(1/\mu)$; its normal matrix retains $\Theta(1/\mu^2)$. Thus regularization cannot establish the missing trajectory property or improve the safe worst-case scale. $\square$

Residual-certified reuse nevertheless gives an honest conditional QIPM. If the feasible short-step theorem accepts every coefficient vector satisfying $\|\mathcal{M}_t \hat{z}_t - f_t\| \leq \eta \|f_t\|$, then exact OSS reconstruction preserves feasibility and the published neighborhood, gap, and iteration guarantees apply unchanged [107]. Under finite realized V_{proj} , the directed cross-gain bound, and checkpoint accuracy $\epsilon \leq \eta/(2G)$, [theorem 13.1](#) also applies. Assume block encodings of A^T/α_A and V/α_V , coherent value access to the current x_t, s_t , and known bounds $R_x \geq \|x_t\|_\infty$ and $R_s \geq \|s_t\|_\infty$. If $F_{\infty,t} \geq \|f_t\|_\infty$, define

$$\alpha_{\mathcal{M},t} = R_x \alpha_A + R_s \alpha_V, \quad \mathcal{K}_t = \alpha_{\mathcal{M},t} \|\mathcal{M}_t^{-1}\|, \quad \Gamma_{f,t} = \frac{\sqrt{n} F_{\infty,t}}{\|f_t\|},$$

and the conversion factor $\rho_t = \alpha_{\mathcal{M},t} \|z_t\| / \|f_t\|$, its dense-output ledger is

$$C_{\text{iter}} + O(T \text{nnz } \mathcal{M}) + \sum_{s \in \mathcal{R}} \tilde{O}\left(\Gamma_{f,s} \mathcal{K}_s \frac{n \rho_s}{\epsilon}\right), \quad (568)$$

plus $O(nR)$ checkpoint writes. Here C_{iter} is the full cost of maintaining the promised coherent iterate oracles. In particular, rebuilding dense x_t, s_t memories each round gives $C_{\text{iter}} = \Theta(nTB_{\text{bit}})$; the ledger omits that term only when the outer implementation already supplies coherent reads without such rebuilds. A quantum residual test costs $\tilde{O}((\Lambda_{t,s} + \Gamma_{f,t})/\eta)$, where $\Lambda_{t,s} = \alpha_{\mathcal{M},t} \|\hat{z}_s\| / \|\mathcal{M}_t \hat{z}_s\|$; on the cycling witness $\Lambda_{t,s} = \Omega(1/\mu_t)$. Generic verification is therefore not free [126]. Proximal QLS warm starts remain useful for a certified instance, but do not change this sequence-level conclusion [85]. Iterative-refinement tomography can change how one solves a single high-accuracy system, but it does not remove the per-refresh/output factors in (568) [105].

13.2 Partition-free equilibration and exact face repair

The stable solver in [section 5.4](#) assumes a supplied active range. On a strict-complementarity tail it can instead be constructed from current coordinates. Suppose

$$\underline{x} \leq x_i \leq \bar{x} \quad (i \in B), \quad \underline{s} \leq s_i \leq \bar{s} \quad (i \in N), \quad (569)$$

and $(1 - \theta)\mu \leq x_i s_i \leq (1 + \theta)\mu$.

Lemma 13.5 (Constant-margin tail selector). *If*

$$\mu \leq \frac{1}{2(1 + \theta)} \min\{\underline{x}^2, \underline{s}^2\}, \quad (570)$$

then $E = \text{diag}(\mathbf{1}\{x_i \geq s_i\})$ equals P_B , and every comparison has absolute margin at least $\frac{1}{2} \min\{\underline{x}, \underline{s}\}$.

Proof. For $i \in B$, $s_i \leq (1 + \theta)\mu/\underline{x} \leq \underline{x}/2 \leq x_i/2$. The symmetric calculation gives $x_i \leq s_i/2$ for $i \in N$. $\square$

This is the classical tail rule of Mehrotra–Ye, now used coherently [37, 102]. It requires current value oracles and enough bits to resolve the promised margin; it does not convert amplitude states into coordinate values.

Theorem 13.6 (Constructive stable active-range equilibration). *Under lemma 13.5, define $F = I - E$,*

$$\begin{aligned} C_\mu &= \mu A E X S^{-1} E A^T, & D_\mu &= \mu^{-1} A F X S^{-1} F A^T, \\ Q_U &= \Pi_{\text{range}(AE)}, & \widehat{M}_\mu &= C_\mu + D_\mu - (1 - \mu^2) Q_U D_\mu. \end{aligned} \quad (571)$$

Then $H_\mu = A X S^{-1} A^T = \mu^{-1} C_\mu + \mu D_\mu$, and direct encodings have

$$\alpha_C \leq \alpha_A^2 \frac{\bar{x}^2}{1 - \theta}, \quad \alpha_D \leq \alpha_A^2 \frac{1 + \theta}{\underline{s}^2}. \quad (572)$$

If $\sigma_B = \sigma_{\min}^+(A_B) > 0$, QSVT constructs Q_U to error δ in

$$O\left(\frac{\alpha_A}{\sigma_B} \log \frac{1}{\delta}\right) \quad (573)$$

queries. If the fixed tail block bounds of theorem 5.7 hold, $\widehat{M}_\mu$ has μ -independent normalization and inverse norm. For $f \in \text{range}(AE)$,

$$\widehat{M}_\mu^{-1} f = \mu^{-1} H_\mu^{-1} f \quad (574)$$

for the exact projector, and constant projector error gives constant solution-state error. With RHS factor Γ_f , state preparation costs

$$\tilde{O}\left(\Gamma_f \Gamma_M \|\widehat{M}_\mu^{-1}\| \frac{\alpha_A}{\sigma_B}\right), \quad \Gamma_M = \alpha_C + (2 - \mu^2) \alpha_D, \quad (575)$$

with no polynomial $1/\mu$ factor.

Proof. The selector splits the columns exactly. On B , $\mu x_i/s_i = \mu x_i^2/(x_i s_i) \leq \bar{x}^2/(1 - \theta)$; on N , $\mu^{-1} x_i/s_i = (x_i s_i)/(\mu s_i^2) \leq (1 + \theta)/\underline{s}^2$. This proves (572). QSVT thresholding of AE/α_A at its smallest nonzero singular value gives (573) [65]. With $P_\mu = \mu Q_U + \mu^{-1}(I - Q_U)$, multiplication gives $\widehat{M}_\mu = P_\mu H_\mu$. Its $U \oplus U^\perp$ diagonal blocks are uniformly invertible and the upper off-diagonal block is $O(\mu^2)$; the Schur complement bounds its inverse. Since $P_\mu f = \mu f$, the solution identity follows. Direct LCU gives Γ_M , and a Neumann bound makes constant projector error sufficient. Multiplying projector, QLS, RHS, and direct encoding factors yields (575). $\square$

Thus theorem 13.6 removes exactly the supplied-partition assumption left open in section 5.4. It is narrower than the published partition-estimated infeasible-QIPM preconditioner [139]: it gives constant tail conditioning for active-range RHSs by direct assembly, not a full infeasible-QIPM theorem. The input-dependent gaps and output remain charged, consistent with classical active-partition preconditioning [39].

The dangerous face components of an OSS residual can also be isolated exactly. Let Z span $\ker A_B$ and Y span $\ker A_B^T$ orthonormally, and define

$$\begin{aligned} G_p &= Z^T X_B^{-1} S_B Z, & g_p(r) &= Z^T X_B^{-1} r_B, \\ G_d &= Y^T A_N S_N^{-1} X_N A_N^T Y, & g_d(r) &= Y^T A_N S_N^{-1} r_N. \end{aligned} \quad (576)$$

Write $\sigma_N = \sigma_{\min}(A_N^T|_{\ker A_B^T})$ when $\ker A_B^T \neq \{0\}$.

Theorem 13.7 (Face leakage, repair, and restricted stability). *For any feasible OSS direction error with residual r , the primal and dual face errors are*

$$p_F = ZG_p^{-1}g_p(r), \quad q_F = -YG_d^{-1}g_d(r). \quad (577)$$

Under (569), the primal map has sharp two-sided $\Theta(1/\mu)$ amplification when $\ker A_B \neq \{0\}$, and the dual map has the same amplification when $\ker A_B^T \neq \{0\}$; otherwise the corresponding map is zero. The correction

$$\widehat{\Delta x}_B^+ = \widehat{\Delta x}_B - p_F, \quad \widehat{\Delta y}^+ = \widehat{\Delta y} + YG_d^{-1}g_d(r), \quad \widehat{\Delta s}^+ = \widehat{\Delta s} - A^TYG_d^{-1}g_d(r) \quad (578)$$

preserves primal and dual feasibility and makes $g_p(r^+) = g_d(r^+) = 0$ exactly. It also obeys $\|r^+\| \leq C_{\text{rep}}\|r\|$, and the inverse of the OSS map restricted by these two equations is uniformly bounded as $\mu \downarrow 0$. The sharp worst-case amortized condition is

$$\sum_t \left(\frac{\|g_p(r_t)\|}{\mu_t} + \frac{\|g_d(r_t)\|}{\mu_t} \right) < \infty. \quad (579)$$

Proof. Writing the direction error as (e_x, e_y, e_s) gives $Ae_x = 0$, $e_s = -A^Te_y$, and $Se_x - XA^Te_y = r$. Multiply the B rows by $Z^TX_B^{-1}$; the dual term vanishes and $Z^TX_B^{-1}S_B(e_x)_B = g_p(r)$. This is the normal equation for the $X_B^{-1}S_B$ -weighted projection of $(e_x)_B$ onto $\ker A_B$. Eliminating e_x gives $AXS^{-1}A^Te_y = -AS^{-1}r$; left-multiplying by Y^T and using $Y^TA_B = 0$ gives $Y^T(A_NX_NS_N^{-1}A_N^T)e_y = -g_d(r)$, the normal equation for the H_μ -weighted projection of e_y onto $\ker A_B^T$, whose coordinate is $-G_d^{-1}g_d(r)$ because $Y^TH_\mu Y = G_d$. These are (577).

Complementarity and the tail margins imply

$$c_p\mu I \preceq G_p \preceq C_p\mu I, \quad c_d\mu I \preceq G_d \preceq C_d\mu I,$$

where, when $\ker A_B^T \neq \{0\}$, the dual constants also contain $\sigma_N^2 > 0$. This proves both sides of the $1/\mu$ amplification on each nonzero face. The primal correction lies in $\ker A$ and is supported on B ; the dual correction lies in $\ker A_B^T$. They therefore preserve feasibility and affect disjoint projected residual components, giving $g_p(r^+) = g_d(r^+) = 0$. The displayed spectral bounds and the bounded tail coordinates give $\|r^+\| \leq C_{\text{rep}}\|r\|$.

At $\mu = 0$, the kernel of the limiting OSS map consists precisely of primal motions in $\ker A_B$ and dual motions in $\ker A_B^T$. The two corrected equations remove those components. Compactness of the unit sphere on the remaining complement and continuity give a uniform smallest restricted singular value. Finally each step can realize face displacement comparable to $\|g_p\|/\mu + \|g_d\|/\mu$; aligned successive residuals attain that order. Summability is therefore sufficient and worst-case sharp. $\square$

The object controlled by (579) is the residual-induced total variation inside the primal and dual optimal faces. It is not by itself a bound on the realized projective variation. A local strict-complementarity chart also needs the transverse-forcing ledger

$$W_r = \sum_t \left\| \frac{r_{t+1}}{\mu_{t+1}} - \frac{r_t}{\mu_t} \right\| < \infty. \quad (580)$$

Define the total face leakage

$$\mathcal{L}_\infty = \sum_t (\|p_F(r_t)\| + \|q_F(r_t)\|). \quad (581)$$

These ledgers bound the residual-induced face motion and the transverse forcing. They do not bound realized V_{proj} , even together with the tail bound $\sum_{k \geq t} (\|p_F(r_k)\| + \|q_F(r_k)\|) = O(\mu_t)$: the row-3-sparse family of proposition 13.4 with face displacement $d_t = \varrho_t \bmod 3\mu_t$ and $\varrho = (1, 3, 1)$

has convergent $\mathcal{L}_\infty$ and W_r , satisfies that tail bound, and stays inside the neighborhood and residual contract, yet its realized V_{proj} grows linearly in the number of steps. An oscillating relative face displacement d_t/μ_t moves the normalized ray by $\Theta(1)$ per step, while both ledgers are absolute sums that do not see it; with the constant ratio $\varrho = (1, 1, 1)$ the same family has bounded V_{proj} when δ_t is measured in an orthonormal basis of $\ker A$, but not in the skew stored basis $V = [e_1 - e_3, e_2 - e_3, e_4, \dots, e_n]$ of [proposition 13.4](#): there the three cycling directions have coefficient norms $1, 1/\sqrt{2}, 1/\sqrt{2}$, so $\|z_t\|$ is phase dependent, $\delta_t = \Theta(1)$ on two of every three steps, and V_{proj} still grows linearly. The constant-ratio family therefore does not certify finite V_{proj} for [theorem 13.1](#) in the basis the module stores. Neither $o(\mu_t)$ leakage nor convergence of d_t/μ_t suffices either: $d_t = \varrho_t \bmod 3\mu_t/\sqrt{t}$ gives convergent ledgers and an $o(\mu_t)$ tail while V_{proj} diverges logarithmically. Finite realized V_{proj} therefore remains a separate trajectory hypothesis, as in [theorem 13.1](#); it is the hypothesis under which that theorem bounds the refresh count, and its failure is not shown here to preclude reuse. If repaired directions are used, r_t here is the corrected and retested residual.

The repair is basis free. With $T = E\sqrt{XS^{-1}}E$, $B_r = E(XS)^{-1/2}E$, $K_F = AT$, $Q = \Pi_{\ker(AE)^T}$, $D_N = FXS^{-1}F$, and $H_N = AD_NA^T$,

$$p_F(r) = T\Pi_{\ker K_F}B_r r, \quad q_F(r) = -(QH_NQ)^+QAFS^{-1}Fr. \quad (582)$$

Indeed the weighted kernel projector equals $W^{1/2}Z(Z^TWZ)^{-1}Z^TW^{1/2}$ on B , with $W = X_B^{-1}S_B$; sandwiching it yields the first expression. The second is the inverse of the H_N form on $\ker A_B^T$. Given known positive lower bounds on the relevant gaps σ_B and σ_N , QSVT implements the projectors with

$$\chi_p = O\left(\frac{\alpha_A \bar{x}}{\sigma_B \underline{x}} \sqrt{\frac{1+\theta}{1-\theta}}\right), \quad \chi_d = O\left(\frac{\alpha_A^2}{\sigma_N^2} \frac{1+\theta \bar{s}^2}{1-\theta \underline{s}^2}\right), \quad (583)$$

both independent of μ . With

$$R_T = \frac{\bar{x}}{\sqrt{(1-\theta)\mu}}, \quad R_B = \frac{1}{\sqrt{(1-\theta)\mu}}, \quad \alpha_p = R_T R_B = \Theta(1/\mu),$$

the primal correction map has RHS factor $\Gamma_p = \alpha_p \|Er\|/\|p_F(r)\|$. If $c_N(r) = QAFS^{-1}Fr$, the dual factor is

$$\Gamma_d(r) = \frac{\alpha_A \bar{s}^{-1} \|Fr\|}{\|c_N(r)\|}.$$

Writing $\Gamma_{r,B}, \Gamma_{r,N}$ for selected-residual preparation, the conservative normalized correction-state ledgers are

$$Q_p = \tilde{O}(\Gamma_{r,B} \Gamma_p \chi_p), \quad Q_d = \tilde{O}(\Gamma_{r,N} \Gamma_d \chi_p \chi_d), \quad (584)$$

with norm recovery and the RHS-specific inverse success amplitude charged separately. These are operator modules, not free corrections.

Replacing the fixed face radius in the row-3-sparse example by $d_t > 0$ gives exactly

$$r_t = 3\mu_t(d_{t+1}u_{j+1} - \tau d_t u_j),$$

and, for adjacent radii within a fixed factor, its leakage is $\Theta(d_t + d_{t+1})$. This verified identity establishes the leakage ledger as a necessary worst-case condition. It does not establish the projective threshold claimed in a preliminary version of this analysis: exact solves show that when $d_t \gg \mu_t$, the phase-dependent transverse coefficient changes by $\Theta(\mu_t/d_t)$, while the normalized OSS operator amplifies that component by $\Theta(d_t/\mu_t)$, so $\delta_t = \Theta(1)$ rather than $\Theta(d_t + d_{t+1})$.

For example, on a geometric schedule with $d_t = \Theta(\mu_t^\gamma)$ and $0 < \gamma < 1$, both $\mathcal{L}_\infty$ and W_r converge, but realized V_{proj} grows linearly with the number of steps. This family also violates the tail requirement $\sum_{k \geq t} (\|p_F(r_k)\| + \|q_F(r_k)\|) = O(\mu_t)$; the oscillating-displacement family

above shows that even that tail bound together with both convergent ledgers does not bound projective variation. Thus the sharpness-family claim from that preliminary version is corrected here. A blanket $O(\mu^2)$ residual rule is sufficient but is not the sharp leakage boundary.

There is a sharp internal dichotomy. If the selector is omitted and one uses $T_{\text{all}} = (X^{-1}S)^{-1/2}$ and $K_{\text{all}} = AT_{\text{all}}$, the weighted kernel formula reconstructs the entire Newton error, not merely face leakage. For

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}, \quad x = (1/2, 1/2, \mu, \mu), \quad s = (2\mu, 2\mu, 1, 1),$$

$\kappa(K_{\text{all}}) = 1/(2\mu)$ and $\kappa(AXS^{-1}A^T) = 1/(4\mu^2)$. A Tikhonov kernel filter $\lambda(K_{\text{all}}^T K_{\text{all}} + \lambda I)^{-1}$ accurate to $\delta < 1/2$ forces $\lambda \leq 2\delta\mu/(1-\delta)$ and hence condition $\Omega(1/(\delta\mu^2))$. Cluster resolution, by the coordinate selector or an equivalent spectral cutoff, is necessary.

Finally, if a materialized repair must meet the pointwise budget $O(\tau\mu_t)$, an arbitrary constant-norm correction in dimension d needs absolute tomography error $O(\tau\mu_t)$. Generic coherent tomography costs

$$\tilde{\Theta}\left(\frac{d}{\tau\mu_t}\right) \quad (585)$$

state-unitary calls. This is a boundary for that generic extraction route, not an LP-specific output lower bound. A concise formula, sparse output, or the compressed representation below can evade it.

13.3 Implicit dual coordinates and compressed output

The state-to-diagonal obstruction of [section 12.4](#) is avoided when the changing diagonal is computed from a smaller explicit iterate instead of from an amplitude state. Let the columns of A have at most d_c nonzeros and store only $y_t \in \mathbb{R}^m$. A static block encoding of A/α_A costs C_A per use, a coherent read of one retained-iterate entry costs $C_{y,\text{read}}$, and B_{bit} is the common arithmetic word width. The one-time cost $C_{\text{build}}(A)$ is separate. Coherent RAM may make $C_{y,\text{read}}$ polylogarithmic; otherwise the actual multiplexed-read cost must be substituted below.

Theorem 13.8 (Implicit slack oracle and affine closure). *Given static coherent sparse-column access, coherent value access to c_i , and coherent reads of y_t , a reversible circuit computes*

$$s_i(y_t) = c_i - a_i^T y_t \quad (586)$$

to absolute error ϵ_s using d_c column accesses and

$$\tilde{O}(d_c B_{\text{bit}} + d_c C_{y,\text{read}} + \log(1/\epsilon_s))$$

gates. If $s_i \geq \underline{s}_t > 0$, it gives a unit-normalized encoding of $\underline{s}_t S_t^{-1}$ with operator error

$$O\left(\frac{\epsilon_s}{\underline{s}_t} + \epsilon_{\text{arith}}\right), \quad (587)$$

and therefore an encoding of $AS_t^{-2}A^T$ with normalization $\alpha_A^2/\underline{s}_t^2$. It suffices to make the error in [\(587\)](#) at most

$$O\left(\frac{\eta}{J_t \rho_t \mathcal{K}_{M,t}}\right),$$

with ρ_t as in [\(594\)](#); the factor $1/\mathcal{K}_{M,t}$ converts operator error into relative vector error, and $1/(J_t \rho_t)$ is the vector accuracy target of [theorem 13.9](#). On active coordinates this follows, for example, from absolute slack error $O(\eta\mu_t/(J_t \rho_t \mathcal{K}_{M,t}))$. Thus finite precision is charged against the transformed residual budget. Updating $y_{t+1} = y_t + \lambda_t \Delta y_t$ updates every implicit slack exactly with no history factor and no n -coordinate rewrite.

Proof. Query c_i and the support and values of a_i , then read the matching entries of y_t and reversibly accumulate (586). Approximate $\underline{s}_t/z$ on the promised interval and use it as a controlled-rotation amplitude; uncomputation yields the diagonal encoding. Product encodings give the normal matrix. The affine identity $c_i - a_i^T y_{t+1} = s_i(y_t) - \lambda_t a_i^T \Delta y_t$ proves exact closure. No state-preparation oracle for s is queried, so the state-to-diagonal premise is absent. $\square$

The dual method does not store a primal iterate. Its selector instead uses the implicit exact-barrier coordinate

$$x_i^b = \frac{\mu_t}{s_i(y_t)}, \quad E_t = \text{diag}(\mathbf{1}\{x_i^b \geq s_i(y_t)\}) = \text{diag}(\mathbf{1}\{s_i(y_t)^2 \leq \mu_t\}). \quad (588)$$

This is lemma 13.5 with exact complementarity ($\theta = 0$), applied to $(x^b, s(y_t))$; the tail-margin hypotheses in this application are explicitly assumed for this implicit pair. Below $\mu_t \leq \frac{1}{2} \min\{\underline{x}^2, \underline{s}^2\}$, an active coordinate has $x_i^b \geq 2s_i$ and an inactive coordinate has $s_i \geq 2x_i^b$. Consequently the x^b - s comparison has the constant margin of that lemma; equivalently, the directly queryable squared test is at least $\mu_t/2$ from its threshold. The squared test is resolved with $O(\log(1/\mu_t))$ bits and does not require a stored x_t .

The partition-free stable dual equation is

$$G_t \Delta y_t = g_t, \quad G_t = \mu_t A S_t^{-2} A^T, \quad g_t = b - \mu_t A S_t^{-1} e. \quad (589)$$

Using (588), define $C_t = \mu_t^2 A E S_t^{-2} E A^T$, $D_t = A F S_t^{-2} F A^T$, $Q_t = \Pi_{\text{range}(AE)}$, and

$$M_t = C_t + D_t - (1 - \mu_t^2) Q_t D_t = P_t G_t, \quad P_t = \mu_t Q_t + \mu_t^{-1} (I - Q_t). \quad (590)$$

With direct component normalizations $\alpha_{C,t}$ and $\alpha_{D,t}$,

$$\alpha_{M,t} \leq \alpha_{C,t} + (2 - \mu_t^2) \alpha_{D,t}, \quad \mathcal{K}_{M,t} = \alpha_{M,t} \|M_t^{-1}\|. \quad (591)$$

The assembled block encoding of M_t is an $(\alpha_{M,t}, \cdot, \epsilon_{M,t})$ encoding in the sense of definition 2.13; its precision is fixed so that $\epsilon_{M,t} \leq \alpha_{M,t}$, a convention used below. Both component normalizations and $\mathcal{K}_{M,t}$ are independent of μ_t under the explicit tail gaps. Since $b \in \text{range}(A_B)$, the transformed RHS can be assembled without encoding the badly normalized P_t :

$$\begin{aligned} \tilde{g}_t &= P_t g_t = \mu_t b - \mu_t^2 A E S_t^{-1} E e - \mu_t^2 Q_t A F S_t^{-1} F e \\ &\quad - (I - Q_t) A F S_t^{-1} F e. \end{aligned} \quad (592)$$

Let $\Gamma_{g,t}$ be its actual LCU numerator divided by $\|\tilde{g}_t\|$, so cancellation is charged. Thus the transformed system actually solved is

$$M_t \Delta y_t = \tilde{g}_t. \quad (593)$$

Theorem 13.9 (Conditional compressed dual QIPM). *Assume the implicit-access model of theorem 13.8, the strict-complementarity margins and known gaps used above, and the inexact dual-barrier theorem of [138] (their Theorem 3.3), which with $B_t = S_t^{-1} A^\top$ and $\delta_t^N = \|B_t \Delta y_t\| \leq 1/2$ accepts any step $\hat{z}_t$ satisfying $\|B_t(\hat{z}_t - \Delta y_t)\| \leq \eta \delta_t^N$ for a fixed $\eta \leq 0.1$, in $T = O(\sqrt{n} \log(n\mu_0/\epsilon_{\text{opt}}))$ iterations. Assume throughout the tail that $0 < \mu_t \leq 1$ and $\text{range } Q_t \neq \{0\}$. For $g_t \neq 0$ set*

$$J_t = \frac{\|P_t^{-1}\| \|P_t g_t\|}{\|g_t\|}, \quad \rho_t = \frac{\alpha_{M,t} \|\Delta y_t\|}{\|\tilde{g}_t\|}. \quad (594)$$

Solving (593) and coherently reconstructing its m -vector direction $\hat{z}_t$ with

$$\|\hat{z}_t - \Delta y_t\| \leq \frac{\eta}{2J_t \rho_t} \|\Delta y_t\|$$

produces

$$(y_T, \mu_T, O_{s,T}, O_{x,T}), \quad x_i = \frac{\mu_T}{s_i} + \frac{\mu_T}{s_i^2} a_i^T \Delta y_T,$$

with the positivity, neighborhood, and convergence guarantees of the outer method. The T outer updates use solves $t < T$; one additional terminal solve at (y_T, μ_T) supplies the displayed Δy_T and final coordinate oracle. Its query count is

$$Q_{\text{path}} = \tilde{O} \left(\sum_{t \leq T} \frac{m J_t \rho_t}{\eta} \Gamma_{g,t} \mathcal{K}_{M,t} \chi_{Q,t} \right), \quad (595)$$

where $\chi_{Q,t} = O((\alpha_A/\sigma_B) \log(1/\delta_Q))$. Its data and gate ledger is

$$C_{\text{build}}(A) + \tilde{O} \sum_{t \leq T} \left[\frac{m J_t \rho_t}{\eta} \Gamma_{g,t} \mathcal{K}_{M,t} \chi_{Q,t} (C_A + d_c(B_{\text{bit}} + C_{y,\text{read}})) + m B_{\text{bit}} \right]. \quad (596)$$

There is no nT iterate-write term. If every displayed factor and η^{-1} is $O(1)$, the favorable tail regime is $Q_{\text{path}} = \tilde{O}(Tm)$.

Proof. Let $e_t = \hat{z}_t - \Delta y_t$ with $\|e_t\| \leq (\eta/(2J_t \rho_t)) \|\Delta y_t\|$. Since $G_t = P_t^{-1} M_t$ and $\|M_t\| \leq \alpha_{M,t} + \epsilon_{M,t} \leq 2\alpha_{M,t}$ by the block-encoding contract and the precision convention $\epsilon_{M,t} \leq \alpha_{M,t}$ fixed after (591), $\|G_t\| \leq 2\|P_t^{-1}\| \alpha_{M,t}$, while (594) gives $J_t \rho_t = \alpha_{M,t} \|P_t^{-1}\| \|\Delta y_t\| / \|g_t\|$. Using $g_t = G_t \Delta y_t$ and $G_t^2 \preceq \|G_t\| G_t$,

$$J_t \rho_t \geq \frac{\|G_t\| \|\Delta y_t\|}{2\|G_t \Delta y_t\|} \geq \frac{\|G_t\|^{1/2} \|\Delta y_t\|}{2\|\Delta y_t\|_{G_t}}.$$

With $G_t = \mu_t B_t^\top B_t$ we have $\|B_t e_t\| = \mu_t^{-1/2} \|e_t\|_{G_t} \leq \mu_t^{-1/2} \|G_t\|^{1/2} \|e_t\|$ and $\delta_t^N = \|B_t \Delta y_t\| = \mu_t^{-1/2} \|\Delta y_t\|_{G_t}$, so $\|B_t e_t\| \leq \eta \delta_t^N$. Thus the reconstructed step meets the weighted contract of the outer theorem.

The operator-error budget of [theorem 13.8](#) supplies this vector accuracy; its factor $1/\rho_t$ enters only logarithmic precision terms. Normalized-state and norm errors $O(\eta/(J_t \rho_t))$ suffice; tight coherent tomography uses $\tilde{O}(m J_t \rho_t / \eta)$ controlled preparations. Multiplication by RHS, QLS, and projector costs gives (595). The slack and selector queries depend only on the current explicit y_t , so each accepted direction costs $\tilde{O}(m B_{\text{bit}})$ to store and never materializes an n -vector. Expanding one state preparation yields (596). The transferred step accuracy invokes the outer theorem, completing the composition. $\square$

Stable conditioning alone is insufficient. On the normalized tail $0 < \mu_t \leq 1$, assuming range $Q_t \neq \{0\}$ and $g_t \neq 0$, if $u_t = \|Q_t g_t\|$ and $v_t = \|(I - Q_t)g_t\|$, direct substitution gives

$$J_t = \frac{\sqrt{u_t^2 + \mu_t^{-4} v_t^2}}{\sqrt{u_t^2 + v_t^2}}. \quad (597)$$

Thus $v_t \leq C \mu_t^2 u_t$ implies $J_t \leq \sqrt{1 + C^2}$, while purely inactive forcing gives $J_t = \mu_t^{-2}$. The favorable theorem explicitly requires stable forcing, bounded RHS cancellation, and stable output conversion, not only a stable matrix. Full face-repair tomography still obeys (585); classically, the same uniformly conditioned sparse repair can be computed by Krylov iteration with only logarithmic precision dependence.

The matched classical baseline applies the same stable operator matrix-free. It costs $\Theta(nd_c)$ arithmetic per multiplication and

$$\tilde{O} \left(nd_c \log \frac{J_t}{\eta} \right) \quad (598)$$

per direction under the uniform condition promise. Ignoring the different query and gate units, a necessary regime for the compressed quantum route to improve this comparator is

$$\frac{mJ_t\rho_t\Gamma_{g,t}\mathcal{K}_{M,t}\chi_{Q,t}}{\eta} = o(nd_c). \quad (599)$$

Randomized column sketches may improve the classical side further. Moreover, $C_{\text{build}}(A) = \Omega(nd_c B_{\text{bit}})$ for an explicit sparse input file, so no sublinear end-to-end wall-clock claim follows unless a reusable oracle or concise input circuit is supplied.

The condition-one prefix family of [section 9](#) neither contradicts nor validates [theorem 13.9](#). Its hard public start is primal–dual infeasible, its selector tail begins only at $\mu = O(1/P)$, and its retained dimension is $m = \Theta(P)$, already giving a $\tilde{\Omega}(P)$ generic tomography budget. The compatibility is exact:

Proposition 13.10 (Compressed-coordinate compiler bound). *On the prefix family, suppose a compiler uses S raw coefficient queries and returns data for a coherent coordinate oracle of either the exact central primal point or a nonzero $x \geq 0$ satisfying the robust contract $\text{res}_p(x) \leq 1/100$ of [theorem 9.7](#). If one oracle use costs q further raw queries, then*

$$S + q = \Omega(P). \quad (600)$$

The same conclusion holds for a direction oracle whose update is nonnegative and satisfies that residual contract. In particular, the affine compressed output above, including its Newton correction, has $q = O(1)$; whenever it meets this robust approximately feasible contract, $S = \Omega(P)$.

Proof. Here the plateau height is $H = \Theta(\sqrt{P})$. Both the central vector and every output admitted by the robust contract have $\Theta(P)$ plateau coordinates with $\|x\| = \Theta(H\sqrt{P})$ and $\|x\|_\infty = \Theta(H)$. Query the value oracle in a uniform superposition and rotate by $x_j/(CH)$. The success probability is constant, so constant-round amplification prepares the hard primal state with $S + O(q)$ raw queries. The central and robust primal-state clauses of [theorem 9.7](#) give $\Omega(P)$, proving the claim. For a direction oracle, add the public start before applying the same argument. $\square$

This is where hardness belongs: loading or retained-dimension tomography, not the equilibrated condition number. A dual-only certificate without a primal coordinate oracle remains outside that reduction.

13.4 Checkpoint spans and the classical sketching boundary

Lazy refresh strengthens the implicit representation. If the checkpoint directions are $\hat{z}_{s_1}, \dots, \hat{z}_{s_R}$, then scalar reuse gives the exact affine representation

$$y_t = y_0 + \sum_{r \leq R_t} \theta_{r,t} \hat{z}_{s_r}, \quad R_t \leq R. \quad (601)$$

Theorem 13.11 (Checkpoint-span trajectory oracle). *Under the hypotheses of [theorem 13.1](#), if the stored vectors have B_{bit} bits per entry and columns of A have sparsity d_c , a y_t coordinate uses $R_t + 1$ coherent reads and a slack coordinate uses $O(d_c R_t)$ reads. Including reversible arithmetic, a y_t query costs $\tilde{O}(R_t B_{\text{bit}})$ gates after the public y_0 term is supplied, and a slack query costs $\tilde{O}(d_c(R_t + 1)B_{\text{bit}})$ gates. All vector loads and scalar updates over the path cost*

$$\tilde{O}\left(mB_{\text{bit}}\left(1 + \frac{GV_{\text{proj}}}{\eta}\right) + TB_{\text{bit}}\right). \quad (602)$$

Proof. Collect consecutive scalar steps using one checkpoint to obtain (601). A coordinate query reads that coordinate from each stored vector and evaluates the affine sum. A slack query repeats this for the at most d_c entries of its column. Each checkpoint loads one m -vector and each ordinary iteration changes one scalar. Sum these costs and apply theorem 13.1. $\square$

This representation assumes accepted steps meet the outer residual contract. A scheduled exact-tail version can avoid per-iteration residual tests; an online version must charge them.

For a checkable tall example, let B be the oriented incidence matrix of a fixed-degree balanced bipartite expander on m vertices. For each vertex i and copy $k \in [K]$, introduce u_{ik}, v_{ik} with respective columns $K^{-1/2}e_i$ and $-K^{-1/2}e_i$ and costs $a_{ik} \in [1, 2]$:

$$\begin{aligned} \min \quad & \sum_{i,k} a_{ik}(u_{ik} + v_{ik}) \\ \text{subject to} \quad & Bx + K^{-1/2} \sum_{k=1}^K (u_{\cdot k} - v_{\cdot k}) = B\mathbf{1}, \quad x, u, v \geq 0. \end{aligned} \tag{603}$$

Here $n = |E(B)| + 2mK = \Theta(mK)$; K is the copy multiplicity, unrelated to the effective QLS symbols $\mathcal{K}$ above. On a sufficiently small tail, $|y_i|/\sqrt{K} \leq 1/2$. The quantum algorithm and its comparator receive the same random/coherent-query access to (a_{ik}) and the formula-defined sparse columns. If the costs arrive as an explicit file, loading already costs $\Omega(nB_{\text{bit}})$. This family has constant direct normalization and condition and projective constants independent of K , but it is not a quantum-advantage example.

Proposition 13.12 (Bounded-average sketch obstruction). *For the scaled-copy family, the inactive contribution at vertex i is*

$$D_{ii}(y) = \frac{1}{K} \sum_{k=1}^K \left[\frac{1}{(a_{ik} - y_i/\sqrt{K})^2} + \frac{1}{(a_{ik} + y_i/\sqrt{K})^2} \right], \tag{604}$$

where all summands lie in a fixed bounded interval on the promised tail. Sampling $O(\varepsilon^{-2} \log(m/\delta))$ terms independently for every vertex produces $\|\hat{D} - D\| \leq \varepsilon$ with probability at least $1 - \delta$, in $O(m\varepsilon^{-2} \log(m/\delta))$ classical queries, independently of K .

Proof. Scalar Hoeffding at each vertex and a union bound give the diagonal operator bound. Uniform invertibility of the equilibrated matrix and the resolvent identity convert it to $O(\varepsilon)$ relative direction error. $\square$

The column scale is essential. The inactive block has row sparsity $2K$, column sparsity one, and entries $K^{-1/2}$, so its standard sparse-access normalization is $O(1)$. Making one hidden column contribute $\Theta(1)$ requires increasing its magnitude to $\Theta(1)$, which raises the standard sparse-access factor normalization and its two-factor product Gram normalization to

$$\alpha_{A_N}^{\text{sparse}} = \Theta(\sqrt{K}), \quad \alpha_{H_N}^{\text{sparse}} = \Theta(K). \tag{605}$$

Since H_N remains diagonal with $\|H_N\| = \Theta(1)$, a direct diagonal encoding can keep constant normalization, but assembling its value oracle from raw column access requires locating the exceptional columns, which is the Grover preprocessing charged below.

At constant IPM accuracy this $\tilde{O}(m)$ sketch matches the unavoidable m -coordinate checkpoint readout. More generally, well-spread bounded contributions admit classical leverage or matrix-concentration sampling. If a rare column is made influential, a standard sparse block encoding acquires large normalization; locating the exceptions separately yields only Grover preprocessing. The open target is therefore a structured access model that supplies a constant-normalized quantum operator without also supplying a bounded-leverage classical sampling distribution. No explicit sparse LP family with that conjunction is known here.

13.5 A block-angular border module

Consider the arrowhead Newton system

$$\begin{pmatrix} D_1 & & & B_1 \\ & \ddots & & \vdots \\ & & D_N & B_N \\ B_1^T & \cdots & B_N^T & -D_0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} f \\ g \end{pmatrix}, \quad (606)$$

where $D_i \succ 0$, $D_0 \succeq 0$, local blocks have size at most p , and the border has size k . Classical block-angular IPMs use the same Schur decomposition [25, 71, 82].

Theorem 13.13 (Explicit border and coherent local directions). *Let*

$$S = D_0 + \sum_{i=1}^N B_i^T D_i^{-1} B_i \succ 0, \quad h = \sum_{i=1}^N B_i^T D_i^{-1} f_i - g, \quad v = S^{-1}h.$$

Suppose one coherent block query returns (D_i, B_i, f_i) and local factorization costs $\text{poly}(p, k)$. Put $M = Np + k$, $\alpha_z = \max_i \|D_i^{-1/2} f_i\|_\infty$, and $\chi_h = \alpha_z / \|v\|_S$ for $v \neq 0$, and suppose a known upper bound $\bar{\chi}_h \geq \chi_h$ is supplied. There is a bounded-error algorithm using

$$\tilde{O}\left(\sqrt{Mk} \left(\frac{1}{\epsilon} + \frac{\bar{\chi}_h}{\eta}\right)\right) \quad (607)$$

block queries that returns an explicit $\tilde{v}$ satisfying

$$\frac{\|\tilde{v} - v\|_S}{\|v\|_S} \leq \frac{\epsilon + \eta}{1 - \epsilon}. \quad (608)$$

It also implements, with one additional block query, the coherent local oracle $|i, 0\rangle \mapsto |i, D_i^{-1}(f_i - B_i \tilde{v})\rangle$. No tomography of the Np local coordinates and no condition-number factor is required. Without a known relative bound, one must instead request absolute inverse-energy error δ_h , replacing the second term by α_z / δ_h , or charge adaptive estimation of $\|v\|_S$.

Proof. Stack $C_i = D_i^{-1/2} B_i$ and $D_0^{1/2}$ into C , so $S = C^T C$, and stack $z_i = D_i^{-1/2} f_i$. The tall-matrix spectral approximation and inverse-local-norm mean-estimation primitives used by Apers–Gribbling [10] return explicit $\tilde{S}, \tilde{h}$ with $(1 - \epsilon)S \preceq \tilde{S} \preceq (1 + \epsilon)S$ and $\|\tilde{h} - h\|_{S^{-1}} \leq \eta \|v\|_S$ at the cost in (607). Cancellation with g is exactly the $\bar{\chi}_h$ factor. With $\tilde{v} = \tilde{S}^{-1} \tilde{h}$,

$$\tilde{v} - v = \tilde{S}^{-1}[\tilde{h} - h + (S - \tilde{S})v].$$

The spectral sandwich bounds the first operator by $(1 - \epsilon)^{-1}$ in S norm and the bracket by $(\epsilon + \eta)\|v\|_S$, proving (608). Block elimination gives each local formula. $\square$

The $\sqrt{N}$ exponent is optimal in the block-oracle model. Set the border data $D_0 = 0$ and $g = 0$. For scalar blocks take a known base scenario with $D_{\text{base}} = B_{\text{base}} = f_{\text{base}} = 1$ and hidden $z_i \in \{0, 1\}$ of Hamming weight zero or one with $D_i = 1$, $B_i = z_i$, and $f_i = 0$. Then $S = 1 + \sum_i z_i$, $h = 1$, and the border value is 1 or 1/2, while the augmented system remains uniformly conditioned. Constant additive accuracy decides OR, proving $\Omega(\sqrt{N})$ quantum and $\Omega(N)$ randomized block-query lower bounds.

An exact LP realization exists at the public center $x = s = e$, $y = 0$ using one dense equality row: hidden coefficient pairs are (δ, δ) or $(1, 2\delta - 1)$ with $\delta = 1/(2N)$. Its scalar normal matrix has condition one and its centering multiplier has a constant marked/unmarked gap. This is a column/coefficient-query lower bound, not a constant-row-sparsity lower bound. Indeed, for normalized uniformly conditioned bounded-degree matrices, polynomial approximation of $1/x$

implies exponential inverse decay with graph distance; a bounded local perturbation far from a fixed border cannot have constant effect. Growing condition, RHS norm, output support, or graph degree can evade that qualified obstruction.

The module is not yet an end-to-end QIPM. First, the product log barrier still has parameter $\Theta(Np)$: partial minimization preserves that certificate and can give $-N \log x$ even when the projected set admits $-\log x$. Second, static blocks can be cached classically. Third, later iterations need a nonrecursive representation of updated local variables. The concrete open target is a projected barrier of parameter $\tilde{O}(k + \text{poly}(p))$ whose local center and derivatives are recomputable from one scenario block and the explicit border. Only with such a barrier would [theorem 13.13](#) yield a genuine $\tilde{O}(\sqrt{N} \text{poly}(p, k))$ border-output QIPM against the $\Omega(N)$ classical block-query bound.

13.6 Supporting exact crossover

For a tall integer LP $Az \geq b$ in dimension d , with coefficient magnitude at most 2^ℓ , row sparsity r , and a unique nondegenerate strictly complementary optimum, let

$$D = (\sqrt{r} 2^\ell)^d, \quad M = \max\{1, \sqrt{r} 2^\ell \sqrt{d} (\sqrt{r} 2^\ell)^{d-1}\}.$$

A feasible point of objective error

$$E = \frac{1}{8D^2M} \tag{609}$$

gives true row slacks at most $1/(8D)$ on active rows and at least $7/(8D)$ on inactive rows. Evaluate each slack with error at most $1/(8D)$ and mark it when the computed value is at most $1/(2D)$; these inequalities make the predicate exact. Its reversible arithmetic uses $O(d(\ell + \log d))$ -bit precision (and the corresponding polynomial gate cost). Thus $\log(1/E) = O(d(\ell + \log d))$, independent of the number n of rows. Coherent enumeration identifies all binding rows in $\tilde{O}(\sqrt{nd})$ row queries; exact rational solves for the primal and dual variables and a Grover feasibility check then return an exact KKT certificate. The enumeration cost is tight for this sparse-certificate wrapper by d independent search blocks.

This is only a supporting crossover lemma for the tall-row-oracle QIPM of Apers–Gribling. It assumes the structural promise and coherent row access, and explicit input loading can erase sublinear wall time. More importantly, Objois–Vladu give an exact quantum Clarkson algorithm with $\tilde{O}(\sqrt{n} d^3)$ row queries under a comparable nondegeneracy promise (a unique optimal basis of d constraints) but without the integrality, bit-size, or strict-complementarity structure, and Dadush–Végh–Zambelli give an exact primal–dual oracle framework [\[50, 117\]](#). The wrapper is therefore not a best-known exact-LP result.

13.7 Synthesis

The positive modules identify three real escape routes from the obstructions of [sections 5, 9](#) and [12](#): amortize classical readout by projective variation, generate changing diagonals from a smaller affine iterate, or request an explicit low-dimensional border rather than the full Newton vector. Each route has a matching condition that cannot be dropped. Lazy reuse, as bounded here, needs finite realized V_{proj} and a fixed residual tolerance; partition-free equilibration needs a certified tail, active gaps, stable forcing, and bounded RHS/output factors; compressed output needs $m = o(n)$ and charges its coordinate compiler; block aggregation needs a small-parameter projected barrier and a nonrecursive local-state representation.

The first-class negative results explain why simply combining the modules is not enough. A standard feasible inexact neighborhood permits logarithmically divergent face cycling. Proximal locking trades motion for residual and gives no residual-safe condition improvement. Unfiltered global repair recreates the ill-conditioned Newton inverse. Generic full-vector tomography

recreates inverse-precision cost. Benign tall averages admit equally benign classical sketches, while rare influential columns move the cost into normalization or Grover search.

Accordingly, this section proves reusable operators and conditional end-to-end compressed-output theorems, not an unconditional quantum advantage. A genuine advantage still requires one explicit problem family and access model for which all of the following hold simultaneously: stable direct normalization and inverse, stable forcing and RHS preparation, finite projective variation or a low-parameter barrier, cheap static construction, a useful compressed output, and a classical lower bound surviving randomized sketching and preprocessing. None of the modules above silently supplies a missing item from that list.

14 Structural Identities and Obstruction Tools

The preceding lower bounds use several small pieces of structure that are useful beyond any one hard family. This section collects them in one place. The first two tools turn symmetry and cross-instance averaging into invariant certificates. The third says how much matrix rank an SDP Newton step must carry to make complementarity progress. The remaining tools delimit what sparse support, bounded treewidth, and geometric locality can and cannot buy. They are supporting results: most of their ingredients are classical, and we separate the new obstruction formulations from those ingredients throughout.

14.1 Symmetry as virtual work

Work in a finite-dimensional Euclidean space. Let R be an orthogonal involution, let Ω be an open convex set with $R\Omega = \Omega$, and let $F : \Omega \rightarrow \mathbb{R}$ be differentiable, convex, and R -invariant. Assume also that $R^T c = c$ and use the stationarity convention

$$A^T y = c + \mu \nabla F(x), \quad \mu > 0. \quad (610)$$

Thus the equality term in the Lagrangian is $-y^T(Ax - b)$. Put

$$x_- := \frac{I - R}{2}x, \quad d := x - Rx = 2x_-, \quad r_- := Ax_-. \quad (611)$$

The definition $r_- = Ax_-$, rather than a selected block of b , is what makes the following statement insensitive to mixed equality rows.

Theorem 14.1 (Involution work identity). *Under the preceding assumptions,*

$$\begin{aligned} r_-^T y &= \mu \langle x_-, \nabla F(x) \rangle \\ &= \frac{\mu}{4} \langle x - Rx, \nabla F(x) - \nabla F(Rx) \rangle \\ &= \frac{\mu}{2} D_F(Rx, x) = \frac{\mu}{2} D_F(x, Rx) \geq 0, \end{aligned} \quad (612)$$

where $D_F(u, v) = F(u) - F(v) - \langle \nabla F(v), u - v \rangle$. If $F \in C^2(\Omega)$ and

$$\overline{H}_x := \int_0^1 \nabla^2 F(Rx + td) dt, \quad (613)$$

then, exactly,

$$r_-^T y = \frac{\mu}{4} d^T \overline{H}_x d = \mu x_-^T \overline{H}_x x_-. \quad (614)$$

For a standard self-concordant barrier, set $x_+ = (x + Rx)/2$ and $\delta = \|x_-\|_{x_+} < 1$. Then the integrated chord comparison gives sharper local constants than a worst-point comparison, and the universal lower bound can be strengthened further:

$$\boxed{\mu \left(1 - \delta + \frac{\delta^2}{3}\right) \delta^2 \leq \frac{\mu \delta^2}{1 + \delta} \leq r_-^T y \leq \frac{\mu \delta^2}{1 - \delta}.} \quad (615)$$

Proof. The invariant objective does no antisymmetric work:

$$\langle x_-, c \rangle = \frac{1}{2} \langle x, c - R^T c \rangle = 0.$$

Pairing (610) with x_- gives the first equality in (612). Differentiating $F(Rx) = F(x)$ gives $\nabla F(Rx) = R\nabla F(x)$. Since $Rd = -d$,

$$\langle d, \nabla F(Rx) \rangle = \langle Rd, \nabla F(x) \rangle = -\langle d, \nabla F(x) \rangle,$$

which proves the second equality. Gradient monotonicity proves nonnegativity. Invariance also gives

$$D_F(Rx, x) = D_F(x, Rx) = \langle \nabla F(x), x - Rx \rangle,$$

so the Bregman equalities follow. The fundamental theorem of calculus on the chord gives (614).

For the last claim, parameterize the chord as $x_+ + ux_-$, $-1 \leq u \leq 1$. Self-concordant Hessian comparison bounds its directional energy by

$$(1 - |u|\delta)^2 \delta^2 \leq x_-^T \nabla^2 F(x_+ + ux_-) x_- \leq (1 + |u|\delta)^2 \delta^2.$$

The relevant chord averages are

$$\int_0^1 (1 - u\delta)^2 du = 1 - \delta + \delta^2/3, \quad \int_0^1 (1 - u\delta)^{-2} du = (1 - \delta)^{-1},$$

so these constants are the exact integrals of the comparison envelopes, not generally attained bounds. For the stronger lower bound, put

$$\phi(u) = x_-^T \nabla^2 F(x_+ + ux_-) x_-.$$

Self-concordance gives $|(\phi^{-1/2})'| \leq 1$ and $\phi(0) = \delta^2$, hence $\phi(u) \geq \delta^2/(1 + u\delta)^2$ for $0 \leq u \leq 1$. Invariance makes ϕ even and $r_-^T y = \mu \int_0^1 \phi(u) du$, which proves the middle lower bound. It strictly dominates the chord-comparison lower constant for $0 < \delta < 1$, since

$$\frac{1}{1 + \delta} - \left(1 - \delta + \frac{\delta^2}{3}\right) = \frac{\delta^2(2 - \delta)}{3(1 + \delta)} > 0.$$

□

No rank assumption on A is hidden here. If $\tilde{y}$ is another multiplier, then $A^T(\tilde{y} - y) = 0$ and hence $(Ax_-)^T(\tilde{y} - y) = 0$. Moreover, under any invertible row change

$$A' = UA, \quad y' = U^{-T}y, \quad r'_- = Ur_-, \quad (r'_-)^T y' = r_-^T y. \quad (616)$$

Only the source-multiplier pairing is invariant; an ill-conditioned U can shrink a multiplier norm while enlarging the source or the input normalization. If rows genuinely split as $A_+R = A_+$ and $A_-R = -A_-$ and $Ax = b = (b_+, b_-)$, then $r_- = (0, b_-)$ and the pairing is the familiar $b_-^T y_-$. Arbitrary row mixing preserves (616), but not that coordinate shortcut.

Two specializations fix the constants used earlier. For paired positive variables $x = (u, v)$, $R(u, v) = (v, u)$, and $F = -\sum_j (\log u_j + \log v_j)$, put $d = u - v$ and $q = u + v$. Then

$$r_-^T y = \frac{\mu}{2} \sum_j \frac{d_j^2}{u_j v_j} = \sum_j \frac{2\mu d_j^2}{q_j^2 - d_j^2}. \quad (617)$$

This is exactly (395); the proof in theorem 10.7 is the paired-coordinate specialization, not a second general identity.

For the log-determinant barrier on $\mathbb{S}_{++}^r$, let $\mathcal{R}(X) = QXQ$ for a symmetric orthogonal Q , and set $X_- = (X - QXQ)/2$. With the Frobenius pairing,

$$(\mathcal{A}X_-)^T y = \frac{\mu}{2} [\text{tr}(QXQX^{-1}) - r] \quad (618)$$

$$= \mu \int_0^1 \|Z_t^{-1/2} X_- Z_t^{-1/2}\|_F^2 dt, \quad Z_t = QXQ + t(X - QXQ). \quad (619)$$

The bracketed trace quantity in the first line is half the symmetrized Stein loss, so the work is $\mu/4$ times that symmetrization. This is not the distinct Jensen–Bregman LogDet, or S-divergence. The second line is the exact noncommutative integrated-Hessian formula; no commuting assumption is used. If $\ell I \preceq X \preceq UI$, it lies between $\mu\|X_-\|_F^2/U^2$ and $\mu\|X_-\|_F^2/\ell^2$. Here U is a spectral endpoint, unrelated to the row-change matrix in (616).

At a fixed start $x^0 = Rx^0$, the Newton version is equally direct. If $G_0 \succ 0$ commutes with R and

$$A\Delta x = r_p, \quad G_0\Delta x - A^T\Delta y = h_+, \quad Rh_+ = h_+, \quad (620)$$

then, for $\Delta x_- = (I - R)\Delta x/2$,

$$\boxed{(A\Delta x_-)^T \Delta y = \Delta x_-^T G_0 \Delta x_- \geq 0.} \quad (621)$$

This recovers (396). A nonzero antisymmetric forcing h_- adds $-\langle \Delta x_-, h_- \rangle$ and destroys the sign.

The involution is only the two-element case. Let a finite or compact group $\mathcal{G}$ act orthogonally, and assume $g\Omega = \Omega$ for every $g \in \mathcal{G}$. Let

$$P_0 = \int_{\mathcal{G}} g d\nu(g), \quad x_0 = P_0 x, \quad x_{\perp} = (I - P_0)x \quad (622)$$

be the Reynolds projection. The orbit is compact, and Carathéodory's theorem shows that its convex hull is a compact subset of the convex domain; thus $x_0 \in \Omega$ and the segment from x_0 to x stays in Ω .

Theorem 14.2 (Compact-group work). *If F and c are $\mathcal{G}$ -invariant and (610) holds, then*

$$(Ax_{\perp})^T y = \mu \langle x_{\perp}, \nabla F(x) - \nabla F(x_0) \rangle \geq 0. \quad (623)$$

When the Hessian exists, the complete chain is

$$\begin{aligned} (Ax_{\perp})^T y &= \mu \langle x_{\perp}, \nabla F(x) - \nabla F(x_0) \rangle \\ &= \mu \int_0^1 \langle x_{\perp}, \nabla^2 F(x_0 + tx_{\perp}) x_{\perp} \rangle dt \\ &\geq 0. \end{aligned} \quad (624)$$

At a group-fixed Newton start, suppose $G_0 \succeq 0$ commutes with the action and

$$A\Delta x = r_p, \quad G_0\Delta x - A^T\Delta y = h_0, \quad gh_0 = h_0 \quad (g \in \mathcal{G}). \quad (625)$$

Define $\Delta x_{\perp} = (I - P_0)\Delta x$. Then

$$(A\Delta x_{\perp})^T \Delta y = \Delta x_{\perp}^T G_0 \Delta x_{\perp} \geq 0. \quad (626)$$

Proof. Both c and $\nabla F(x_0)$ are fixed by the group and hence orthogonal to $x_{\perp}$. Pair stationarity with $x_{\perp}$, then apply gradient monotonicity between x_0 and x . Integration gives (624). The Newton identity follows by pairing the stationarity equation with the nontrivial-representation component; G_0 preserves the orthogonal representation decomposition. $\square$

For coordinate permutations, P_0 replaces each orbit by its average; for the logarithmic barrier the work on an orbit O is $\mu(\bar{x}_O \sum_{i \in O} x_i^{-1} - |O|)$. The simplex gives the same formula with $\bar{x} = 1/n$. For the standard $SO(k)$ action and the unit-ball barrier, it is $2\mu\|z\|^2/(1 - \|z\|^2)$. These examples explain the permutation and rotation gadgets excluded in [section 10.6](#).

Remark 14.3 (Scope and provenance). The sign reverses with the opposite multiplier convention. An asymmetric objective or Newton forcing contributes an uncontrolled linear term. Mixed rows require the canonical source Ax_- ; convexity without strict curvature gives only nonnegativity. The scalar identity extends to a nonorthogonal linear involution by using $\nabla F(Rx) = R^{-T} \nabla F(x)$, but the Euclidean orthogonal decomposition does not. Finally, arbitrary edge transports need not arise from one simultaneous public symmetry action and may have no public Reynolds projector. Even when they do come from a common representation, the Reynolds identity supplies no cut-local conservation law; the signed/gain incidence and holonomy analysis of [section 11](#) is separate.

The divergence in (612) is the classical symmetrized Bregman divergence [32, 115]; Reynolds reduction and symmetric criticality are likewise classical [119]. The contribution used in this paper is narrower: the row-mixing-invariant source (611) is an obstruction template for sparse QIPM gadgets.

14.2 Central mixtures in semidefinite programming

The scalar Kantorovich argument in [section 10.2](#) has a fully noncommutative analogue. Consider SDPs with common C, b and input-dependent measurement maps $\mathcal{A}_\xi$, with matrix order $r \geq 1$ and $N \geq 1$ neighboring inputs. Fix one Frobenius-isometric $\text{svec} : \mathbb{S}^r \rightarrow \mathbb{R}^{r(r+1)/2}$. All sparsity and coefficient bounds below refer to the resulting matrix A_ξ , not to the unvectorized data. At a base input ς , suppose every one-bit neighbor $\varsigma^{(i)}$ has an exact center at the same $\mu > 0$:

$$\mathcal{A}_{\varsigma^{(i)}}(X_i) = b, \quad \mathcal{A}_{\varsigma^{(i)}}^*(y_i) + S_i = C, \quad S_i = \mu X_i^{-1}, \quad X_i \succ 0. \quad (627)$$

For weights $w_i \geq 0$, $\sum_i w_i = 1$, write

$$\begin{aligned} X_w &= \sum_i w_i X_i, & y_w &= \sum_i w_i y_i, & S_w &= \sum_i w_i S_i, \\ D_i &= X_i - X_w, & Z_w &= X_w^{1/2} (S_w / \mu) X_w^{1/2}. \end{aligned} \quad (628)$$

Lemma 14.4 (Matrix multiplicative variance). *The following identity is exact:*

$$\boxed{Z_w - I = X_w^{-1/2} \left(\sum_i w_i D_i X_i^{-1} D_i \right) X_w^{-1/2}.} \quad (629)$$

Consequently $Z_w \succeq I$, with equality exactly when all positively weighted X_i equal X_w .

Proof. Expanding the middle term and using $\sum_i w_i X_i = X_w$ gives

$$\sum_i w_i D_i X_i^{-1} D_i = X_w \left(\sum_i w_i X_i^{-1} \right) X_w - X_w.$$

Congruence by $X_w^{-1/2}$ proves the identity. Every un-congrued summand is positive semidefinite, and a positive weighted sum of them vanishes only when each corresponding D_i vanishes. $\square$

Identity (629) is an elementary exact expansion of the arithmetic–harmonic defect; it is not claimed as a new matrix inequality.

Theorem 14.5 (Noncommutative central sandwich). *If $0 < m \leq M$, $mI \preceq X_i \preceq MI$ for every i , and $R = M/m$, then*

$$\boxed{I \preceq Z_w \preceq K(R)I, \quad K(R) = \frac{(R+1)^2}{4R}.} \quad (630)$$

The constant is sharp, already for an equal scalar mixture of m and M . No two matrices in the mixture need commute.

Here and below $R = M/m$ is a spectral ratio, unrelated to the involution R in [section 14.1](#).

Proof. The lower bound is [lemma 14.4](#). Functional calculus, applied to one X_i at a time, gives

$$X_i + mMX_i^{-1} \preceq (m+M)I. \quad (631)$$

Averaging and congruencing by $X_w^{1/2}$ yields

$$Z_w \preceq \frac{(m+M)X_w - X_w^2}{mM}.$$

Every eigenvalue λ of X_w is in $[m, M]$, and

$$\frac{\lambda(m+M-\lambda)}{mM} \leq \frac{(m+M)^2}{4mM} = K(R).$$

Only X_i and X_w individually entered functional calculus, so no commutation was assumed. $\square$

The variance identity supplies a dimension-free refinement of the spectral ratio bound. If only $X_i \succeq mI$ with $m > 0$ is assumed, then

$$\|Z_w - I\|_F \leq \frac{1}{m^2} \sum_i w_i \|D_i\|_F^2, \quad (632)$$

$$\|Z_w - I\|_{\text{op}} \leq \frac{1}{m^2} \sum_i w_i \|D_i\|_{\text{op}}^2. \quad (633)$$

Indeed, the two inverse factors come separately from the outer congruence and X_i^{-1} in [\(629\)](#). Multiplying by μ gives the corresponding complementarity-residual bounds. They survive standard recentering: if

$$\mu_w = \frac{\langle X_w, S_w \rangle}{r} = \mu\alpha, \quad \alpha = \frac{\text{tr } Z_w}{r} \geq 1, \quad (634)$$

then

$$\mu_w^{-1} X_w^{1/2} S_w X_w^{1/2} - I = \frac{(Z_w - I) - [\text{tr}(Z_w - I)/r]I}{\alpha}.$$

Removing the trace component is an orthogonal projection in Frobenius norm, and it does not enlarge the operator range because $Z_w - I \succeq 0$.

The objective identity uses no feasibility of the average. Every point in [\(627\)](#) has algebraic gap $r\mu$, so linearity gives

$$\boxed{\langle C, X_w \rangle - b^T y_w = r\mu.} \quad (635)$$

By contrast, $\langle X_w, S_w \rangle - [\langle C, X_w \rangle - b^T y_w] = \mu \text{tr}(Z_w - I)$ is nonnegative. Under the common sandwich $0 < m \leq M$ and $mI \preceq X_i \preceq MI$, it is at most $r[K(M/m) - 1]\mu$. This is an arithmetic–harmonic Jensen gap at a generally base-infeasible point, not an application of weak duality.

We next combine the operator statement with the sparse single-flip residual. Let s_r, s_c bound the row and column sizes of one fixed union support of the matrices A_ξ , let M_{dep} count input-dependent positions, each of which is a function of one designated bit $\varsigma_{\ell(r,c)}$ (a bit may designate several positions), let their magnitudes be at most $B \geq 0$, and assume that every coordinate of $\text{svec}(X_i), y_i, \text{svec}(S_i)$ has magnitude at most $H \geq 0$. Set

$$s_{\text{KKT}} = \max\{s_r, s_c + 1\}, \quad B_{\text{KKT}} = \max\{B, 1\}. \quad (636)$$

For the uniform average of all N one-bit neighbors, the same incidence count as [theorem 10.1](#), now once in the primal block and once in the dual block, gives

$$\left\| \begin{bmatrix} \mathcal{A}_\varsigma(X_w) - b \\ \text{svec}(\mathcal{A}_\varsigma^*(y_w) + S_w - C) \end{bmatrix} \right\|_2 \leq \frac{2B_{\text{KKT}}H\sqrt{2s_{\text{KKT}}M_{\text{dep}}}}{N}. \quad (637)$$

Keeping the dual multipliers signed gives the stronger weighted estimate

$$\left\| \begin{bmatrix} \mathcal{A}_\varsigma(X_w) - b \\ \text{svec}(\mathcal{A}_\varsigma^*(y_w) + S_w - C) \end{bmatrix} \right\|_2^2 \leq 4(s_r + s_c)B^2H^2 \sum_i M_i w_i^2, \quad (638)$$

where M_i counts dependent coefficient positions assigned to bit i . For uniform weights this is $4(s_r + s_c)B^2H^2M_{\text{dep}}/N^2$, which implies (637). The PSD variables need no coordinatewise sign split; convexity preserves the cone, and the identity block on S is not input dependent. For multibit coefficients the degree factor of [proposition 10.2](#) enters, as in [section 10.2](#).

Proposition 14.6 (SDP mixture phase boundary). *Suppose $0 < m \leq M$, every neighboring center obeys the common sandwich $mI \preceq X_i \preceq MI$, and $R = M/m$. Assume the one-bit-local dependence stated before (637), that all N bits are sensitive, and that the selected neighboring centers have a common mixture-stable wrong output. This may be either a fixed affine score with a uniform signed margin, or a fixed symmetric contraction O such that*

$$f(\xi)\langle O, X^\xi \rangle \geq \gamma \text{tr } X^\xi \quad (639)$$

for an output $f \in \{-1, +1\}$ and a positive margin $0 < \gamma \leq 1$. The contraction condition means $-I \preceq O \preceq I$; the binary measurement has effects $(I + O)/2$ and $(I - O)/2$ on $X/\text{tr } X$. If a base-instance contract demands the correct output from every positive-definite triple with combined residual at most $\varepsilon \geq 0$, algebraic gap $r\mu$, and either common- μ or point-centered operator width at most θ , then soundness requires

$$\boxed{\frac{8s_{\text{KKT}}B_{\text{KKT}}^2H^2M_{\text{dep}}}{N^2} > \varepsilon^2 \quad \text{or} \quad \frac{(R-1)^2}{4R} > \theta.} \quad (640)$$

For a Frobenius neighborhood of width η_F , the second alternative is

$$\sqrt{r} \frac{(R-1)^2}{4R} > \eta_F, \quad (641)$$

unless (632) is smaller. The residual alternative can be sharpened to $4(s_r + s_c)B^2H^2M_{\text{dep}}/N^2 > \varepsilon^2$. Under point centering the algebraic gap remains $r\mu$, not $r\mu_w$.

Proof. Equation (637) gives the first certificate, (635) gives the second, and [theorem 14.5](#) gives operator width at most $K(R) - 1 = (R - 1)^2/(4R)$ under either centering convention. The affine score is preserved by averaging. Under (639), trace weighting gives

$$-f(\varsigma) \text{tr} \left(O \frac{X_w}{\text{tr } X_w} \right) \geq \gamma,$$

so the normalized density matrix has constant bias for the wrong sign. If neither strict inequality in (640) held, this wrong output would satisfy the contract. Finally, $\|Z_w - I\|_F \leq \sqrt{r}\|Z_w - I\|_{\text{op}}$ gives (641); the same projection argument used after (633) covers point centering. $\square$

Since

$$K(R) - 1 = \sinh^2\left(\frac{\log R}{2}\right), \quad (642)$$

under a common sandwich $0 < m \leq M$ and $mI \preceq X_i \preceq MI$, with $R = M/m \geq 1$, the worst-case common- μ Frobenius guarantee of width $\eta_F \geq 0$ based only on this global ratio is equivalent to

$$\log R \leq 2 \operatorname{arsinh}(\sqrt{\eta_F} r^{-1/4}) = O(r^{-1/4}). \quad (643)$$

Indeed, the equal mixture $X_1 = mI$, $X_2 = MI$ attains $Z_w = K(R)I$ in every dimension. The same ratio criterion is sufficient for point centering, but this example has zero point-centered width, so it does not establish necessity for that convention. A finite consequence of the criterion is $\log R \leq 2\sqrt{\eta_F} r^{-1/4}$. Actual Frobenius variance can be small even when the global spectral ratio is not.

The common sandwich is load-bearing. Individual condition numbers cannot replace its ratio: the scalars $X_1 = 1$, $X_2 = 9$ each have condition number one, but their equal mixture has $\|Z_w - I\|_{\text{op}} = 16/9$.

Remark 14.7 (Seven scope limits). The common-basis bounds $mI \preceq X_i \preceq MI$ are strong. Operator control incurs a generic $\sqrt{r}$ loss in Frobenius norm. The factors m^{-2} in the additive-variance bounds are real. Every sparsity and height parameter depends on the chosen Frobenius-isometric svec. Output stability is a separate hypothesis, and (639) says nothing about normalized svec amplitudes. The mixture is existential, so the theorem is not a query lower bound without a reduction. Finally, (635) is an algebraic identity at an infeasible point. These qualifications are part of the theorem’s usable boundary.

The arithmetic–harmonic operator inequality and its Kantorovich reverse are classical [8, 88, 95, 100]. The nearest slogan-level collision is conic-IPM warm starting [42], which moves one old solution to a centered start for a perturbed problem. The conjunction here is different: cross-instance averaging, a sparse incidence budget, simultaneous primal and dual residuals, and a wrong-output boundary. The scalar LP corollary was already proved in [proposition 10.3](#); the present result is the SDP operator theory behind it.

Formal verification. The Lean development in `formal/QipmFormal/SDPMixture/` verifies the finite matrix-mixture argument in this subsection. It includes the variance identity and equality case, the sharp noncommutative sandwich, operator and Frobenius variance bounds, trace recentering, and the algebraic-gap discrepancy. A constructed Frobenius-isometric symmetric vectorization connects actual SDP measurement maps and their adjoints to weighted, uniform, and multibit residual bounds. The soundness results combine these properties with explicit affine or trace-normalized observable output contracts under both centering conventions. The scalar ratio criterion, endpoint sharpness, and the distinction between the two centering conventions are also checked. The claim map, qualifications, and reproduction command are in `formal/SDP_MIXTURE.md`. The development permits only `propext`, `Classical.choice`, and `Quot.sound`. It does not assert existence of the neighboring centers for arbitrary inputs or formalize an IPM trajectory or quantum query reduction.

14.3 Rank required for SDP complementarity progress

Low rank at an SDP optimum does not imply low rank of an accurate Newton direction. The clean statement is in Nesterov–Todd (NT) coordinates. At an exact center $XS = \mu I$, let W be the NT scaling, so $WSW = X$, put $P = W^{-1/2}$, and define

$$U = P\Delta XP^T, \quad V = P^{-T}\Delta SP^{-1}. \quad (644)$$

Here U denotes the NT-scaled primal direction and is unrelated to the row-change matrix and spectral endpoint denoted by U above. Write the symmetric scaled complementarity equation operationally as

$$\sqrt{\mu}(U + V) = -(1 - \sigma)\mu I + \tilde{R}. \quad (645)$$

This definition avoids dependence on competing residual conventions.

Theorem 14.8 (Sharp rank-progress barrier). *Let $0 \leq \sigma < 1$. For $1 \leq p < \infty$, if*

$$\|\tilde{R}\|_{S_p} \leq \theta(1 - \sigma)\mu n^{1/p}, \quad 0 \leq \theta < 1, \quad (646)$$

then

$$\boxed{\text{rank}(\Delta X) + \text{rank}(\Delta S) \geq \lceil (1 - \theta^p)n \rceil}. \quad (647)$$

For spectral error,

$$\|\tilde{R}\|_{\text{op}} < (1 - \sigma)\mu \implies \text{rank}(\Delta X) + \text{rank}(\Delta S) \geq n. \quad (648)$$

The result is invariant under primal congruence and holds blockwise on a product PSD cone.

Proof. Congruence preserves rank, so, with $k = \text{rank}(\Delta X) + \text{rank}(\Delta S)$,

$$\text{rank}(U + V) \leq \text{rank}(U) + \text{rank}(V) = k.$$

Equation (645) makes $\tilde{R}$ the error of approximating $(1 - \sigma)\mu I$ by a rank-at-most- k matrix. Eckart–Young–Mirsky gives

$$\|\tilde{R}\|_{S_p} \geq (1 - \sigma)\mu(n - k)_+^{1/p}.$$

Combining this with (646) proves (647). If $k < n$, a kernel vector of $U + V$ sees residual exactly $(1 - \sigma)\mu$, proving (648). $\square$

The bound is sharp as a matrix statement: take $U = -(1 - \sigma)\sqrt{\mu} \text{diag}(I_k, 0)$ and $V = 0$. The residual is precisely $(1 - \sigma)\mu \text{diag}(0, I_{n-k})$. The approximation theorem used in the proof is classical [58, 104]; the rank-progress interpretation is the structural content here.

14.3.1 A path witness and the norm audit

Let L be the Laplacian of the n -vertex path and consider

$$\min \langle L, X \rangle \quad \text{subject to} \quad \text{tr } X = 1, \quad X \succeq 0. \quad (649)$$

For $u = \mathbf{1}/\sqrt{n}$, the unique optimum is $X^* = uu^T$, while the dual optimum has slack $S^* = L$; hence strict complementarity holds. Its exact central path is

$$y(t) = -t, \quad S(t) = L + tI, \quad Z(t) = \text{tr}(S(t)^{-1}), \quad \mu(t) = Z(t)^{-1}, \quad X(t) = \mu(t)S(t)^{-1}. \quad (650)$$

With $T(t) = \text{tr}(S(t)^{-2})$, differentiation gives

$$\dot{S} = I, \quad \dot{\mu} = \mu^2 T, \quad \dot{X} = \mu S^{-1} - \mu S^{-2}. \quad (651)$$

If $0 = \lambda_0 < \lambda_1 \leq \dots$ are the eigenvalues of L , the eigenvalues of $\dot{X}$ are

$$\frac{\mu}{(t + \lambda_i)^2} \left[\frac{T}{Z}(t + \lambda_i) - 1 \right]. \quad (652)$$

The zero-mode entry is negative. Every other entry is positive whenever $t \leq \lambda_1/(2\sqrt{n})$, because $T/Z > 1/(t + \lambda_1)$ follows from $\lambda_1^2 + \lambda_1 t - (n - 1)t^2 > 0$. Thus both tangent matrices are full

rank arbitrarily close to a rank-one optimum. The exact Newton direction targeting $\sigma\mu$ is the scalar multiple

$$(\Delta X, \Delta S, \Delta y) = -\frac{(1-\sigma)\mu}{\dot{\mu}}(\dot{X}, \dot{S}, \dot{y}). \quad (653)$$

Full algebraic rank alone is weak, so the scaled spectrum matters. Put $d_i = t + \lambda_i$ and $\beta = T/Z$. Since $W = \sqrt{\mu}S^{-1}$ on (650), the NT-scaled tangent eigenvalues are

$$u_i = \sqrt{\mu}(\beta - d_i^{-1}), \quad v_i = \sqrt{\mu}d_i^{-1}, \quad u_i + v_i = \sqrt{\mu}\beta. \quad (654)$$

If $t \leq \lambda_1/(4\sqrt{n})$, then $\beta d_i \geq 3$ for $i \geq 1$:

$$\beta\lambda_1 \geq \frac{\lambda_1/t}{1 + (n-1)t/\lambda_1} > 3.$$

Hence

$$\frac{2}{3}\sqrt{\mu}\beta < u_i < \sqrt{\mu}\beta \quad (i \geq 1), \quad (655)$$

and every rank- $k < n-1$ approximation obeys

$$\|U - U_k\|_F \geq \frac{2}{3}\sqrt{\mu}\beta\sqrt{n-1-k}. \quad (656)$$

Using the path trace identities below to control the exceptional zero mode, any fixed relative Frobenius target error below $2/3$ therefore requires rank $\Omega(n)$ in the NT metric. More generally, for any fixed target error below one, shrinking the constant in $t \leq \lambda_1/(4\sqrt{n})$ makes all $n-1$ nonexceptional singular values an arbitrarily large fixed fraction of $\sqrt{\mu}\beta$ and again forces rank $\Omega(n)$.

The coordinate system is essential. For the path,

$$\lambda_k = 4\sin^2 \frac{\pi k}{2n}, \quad \text{tr } L^+ = \frac{n^2-1}{6}, \quad \text{tr}(L^+)^2 = \frac{2n^4+5n^2-7}{180}. \quad (657)$$

As $t \downarrow 0$, the raw tangent spectrum tends to $\{-\text{tr } L^+, \lambda_1^{-1}, \dots, \lambda_{n-1}^{-1}\}$, whose stable rank tends to $7/5$ and whose best rank-one relative Frobenius error tends to $\sqrt{2/7}$. At $t = \lambda_1/(2\sqrt{n})$, by contrast, the $n-1$ nonzero-mode NT singular values are asymptotically flat. This regime is late and ill conditioned:

$$t = \frac{\pi^2}{2n^{5/2}}(1 + o(1)), \quad \text{cond}(L + tI) = \frac{8}{\pi^2}n^{5/2}(1 + o(1)). \quad (658)$$

For smaller t only the lower bound $\Omega(n^{5/2})$ is uniform.

14.3.2 Why the barrier is self-limiting

The rank theorem obstructs low-rank *factor tomography* of the one-block NT direction. It does not prove an $\Omega(n^2)$ storage bound, and the path witness makes the escapes explicit.

First, chordal conversion replaces the single block by edge blocks

$$Y_i = \begin{pmatrix} x_{ii} & x_{i,i+1} \\ x_{i,i+1} & x_{i+1,i+1} \end{pmatrix} \succeq 0 \quad (1 \leq i < n), \quad (659)$$

with overlap equations $(Y_i)_{22} = (Y_{i+1})_{11}$ and trace equation

$$(Y_1)_{11} + \sum_{i=1}^{n-1} (Y_i)_{22} = 1. \quad (660)$$

The objective is the sum of inner products with $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$. Restriction of a global PSD matrix is feasible in this formulation. Conversely, consistent PSD edge matrices have a PSD completion because the path is chordal: construct Gram vectors successively, adding one orthogonal coordinate per new vertex. Objective and trace use only specified entries, so the reformulation is exact. Its barrier Hessian has constant-size blocks; elimination gives a tridiagonal normal matrix with one dense border. A tridiagonal factorization and one scalar Schur complement use $O(n)$ memory and arithmetic per Newton solve. The conservative short-step total is $O(n^{3/2} \log(n/\epsilon))$ arithmetic and $O(n)$ memory. This is a concrete instance of the formulation dependence emphasized in the chordal conversion literature [146].

Second, the tangent itself has the implicit resolvent representation (651). More generally, accurate low-rank SDP methods may use low-rank endpoint structure to precondition or reformulate a dense direction rather than factorizing that direction [21, 44]. These are compatible with theorem 14.8.

Third, high approximate matrix rank does not imply a hard amplitude state. Writing $a = \sqrt{\mu}\beta$, the path estimates give

$$\|U/a - (I - uu^T)\|_{\text{op}} = O(n^{-1/2}), \quad \|U/a - (I - uu^T)\|_F = O(1). \quad (661)$$

After normalization, $\text{vec}(U)$ is $O(n^{-1/2})$ -close to

$$\frac{|I\rangle - |uu^T\rangle}{\sqrt{n-1}}. \quad (662)$$

For $u = \mathbf{1}/\sqrt{n}$, $|I\rangle/\sqrt{n}$ is a maximally entangled index state and $|uu^T\rangle$ is a product of uniform states. A two-term LCU prepares (662) with constant success probability and $\tilde{O}(\log n)$ gates. The theorem therefore makes no state-preparation lower-bound claim.

Boundary analyticity under strict complementarity is classical [74]. The rank law is a sharp local obstruction, but cone reformulation, implicit resolvents, and compact state-preparation circuits are first-class escapes rather than footnotes.

14.4 Sparse support and sparse-system embeddings

The next statements concern standard-form complementarity, not arbitrary optimization algorithms. At an exact LP center $XSe = \mu e$, let $0 \leq \sigma < 1$. An inexact centering direction satisfies

$$S\Delta x + X\Delta s = -(1 - \sigma)\mu e + r, \quad \|r\|_2 \leq \eta\mu. \quad (663)$$

Let $\text{psupp}(\Delta x, \Delta s)$ be the indices where at least one member of the coordinate pair is nonzero.

Theorem 14.9 (Complementarity support barrier). *For $0 \leq \sigma < 1$, every direction satisfying (663) obeys*

$$|\text{psupp}(\Delta x, \Delta s)| \geq n - \frac{\eta^2}{(1 - \sigma)^2}. \quad (664)$$

For a short step $1 - \sigma = a/\sqrt{n}$ with $\eta < a$, this is $[1 - (\eta/a)^2]n$. If a k -sparse latent vector z is reconstructed as $(\Delta x, \Delta s) = Bz$, and each column of B touches at most g complementarity pairs, then

$$gk \geq n - \frac{\eta^2}{(1 - \sigma)^2}. \quad (665)$$

Proof. Let Z contain the indices at which both update coordinates vanish. For each $i \in Z$, (663) forces $r_i = (1 - \sigma)\mu$. Hence

$$|Z|(1 - \sigma)^2\mu^2 \leq \|r\|_2^2 \leq \eta^2\mu^2,$$

which proves (664). The support of Bz is contained in the union of at most k column pair-supports, of total size at most gk , proving (665). $\square$

The count is realized by a sparse, incompressible direction. For $n \geq 3$, consider

$$\min x_2 \quad \text{subject to} \quad x_1 + x_2 = 1, \quad x_{i-1} - x_i = 0 \quad (3 \leq i \leq n), \quad x \geq 0. \quad (666)$$

Its equality matrix has maximum row and column sparsity two (the endpoint columns have only one nonzero), and its unique optimum is e_1 . The feasible segment is $x(t) = (1 - t, t, \dots, t)$, and, for $0 < t < (n - 1)/n$, the central parameter is

$$\mu(t) = \frac{t(1 - t)}{n - 1 - nt}. \quad (667)$$

Indeed, eliminating the equalities gives the one-variable barrier $t - \mu[\log(1 - t) + (n - 1) \log t]$; its stationarity equation is the displayed formula. At $t = 1/2$, every nonzero central tangent is proportional to $(-1, 1, \dots, 1)$. Its normalized best k -sparse approximation error is exactly $\sqrt{1 - k/n}$; a sparse optimum therefore does not make the central direction compressible.

There is also no hidden geometric restriction on a positive direction profile. Given any $u \in \mathbb{R}_{++}^n$ with $\|u\|_2 = 1$, choose $\mu_0 > 0$ so that $u_i^{-2} > 4\mu_0$ and set

$$c_1 = -\sqrt{u_1^{-2} - 4\mu_0}, \quad c_i = \sqrt{u_i^{-2} - 4\mu_0} \quad (i \geq 2). \quad (668)$$

For the bound-constrained QP

$$\min_{x \geq 0} \quad \frac{1}{2} \|x\|_2^2 + c^T x, \quad (669)$$

the unique optimum is the one-sparse vector $(-c_1)e_1$, with strict complementarity. Central stationarity gives $s = x + c$ and

$$x_i(\mu) = \frac{-c_i + \sqrt{c_i^2 + 4\mu}}{2}, \quad x'_i(\mu) = \frac{1}{\sqrt{c_i^2 + 4\mu}}, \quad (670)$$

so $x'(\mu_0) = u$. The row-scaled complementarity Jacobian is $\text{diag}(2x(\mu_0) + c) = \text{diag}(1/u_i)$, which is one-sparse and constant conditioned for near-uniform u . Combining this realization with coherent tomography gives the established $\tilde{\Omega}(n/\epsilon)$ solution-unitary readout bound on the near-uniform hard family [131]. It is a modular readout bound: an algorithm allowed to inspect the explicit coefficients (668) can learn u there.

The condition-one box-LP family behind the older version of this observation is superseded in strength by the original-primal and bounded-data families of sections 9.5.1 and 9.7. Its remaining lesson is only that perfect reduced conditioning does not remove coefficient-query or classical-output cost. Likewise, the older coordinate-reset ledger is superseded by the projective path theorem of section 13.1; the QRAM-write limitation is stated once in section 14.5.

14.4.1 A query-preserving OSS progress step

The support theorem says what a physical update must touch. A distinct question is whether standard sparse-QLS lower bounds occur inside a valid QIPM Newton solve. They do. Let $H \in \mathbb{R}^{n \times n}$ be real symmetric, invertible, with $\|H\|_2 = 1$ and bidirectional row/column-slot sparse access. Let d be the declared slot bound; declared candidate slots may contain zeros. Let $\|q\|_2 = 1$. Fix $\theta \in (0, 1)$, choose

$$0 < \delta \leq \min\{\theta, 1/2\}, \quad 0 \leq \alpha \leq \delta/4, \quad a = \delta q/\sqrt{2}, \quad D = \text{diag}(a), \quad \sigma = 1 - \alpha/\sqrt{n}. \quad (671)$$

Consider

$$\min c^T x \quad \text{subject to} \quad Ax = 0, \quad x \geq 0, \quad A = [H \quad -H], \quad c = (e + a, e - a). \quad (672)$$

At $x^0 = e_{2n}$, $y^0 = 0$, $s^0 = c$, the complementarity average is one and the N_2 residual norm is δ , while the N_∞ residual is at most $\delta/\sqrt{2}$. Thus the point belongs to both radius- θ neighborhoods.

Theorem 14.10 (Feasible OSS embedding). *With null-space basis $V = (I, I)^T$ and $r = 1 - \sigma$, the feasible OSS step is*

$$\underbrace{\begin{bmatrix} -H & I + D \\ H & I - D \end{bmatrix}}_{M_{\text{OSS}}} \begin{pmatrix} \Delta y \\ \lambda \end{pmatrix} = \begin{pmatrix} -re - a \\ -re + a \end{pmatrix}. \quad (673)$$

It has row sparsity at most $d + 1$, column sparsity at most $2d$, and $\text{cond}(M_{\text{OSS}}) = \Theta_\delta(\text{cond}(H))$. Its exact solution is

$$\lambda = -re, \quad \Delta y = \sigma H^{-1}a. \quad (674)$$

The normalized solution has probability at least $2/3$ in the Δy block. The physical step lands at

$$x^1 = \sigma e, \quad s^1 = (e + ra, e - ra), \quad \mu^1 = \sigma, \quad (675)$$

with centrality-residual norm $\sigma r \delta$. Every sparse query to the LP or OSS data is simulated with $O(1)$ queries to H and evaluations of the known q .

Proof. Primal and dual feasibility at the start are immediate, and $\|X^0 s^0 - e\|_2 = \|(a, -a)\|_2 = \delta$. Substitution of $\Delta x = V\lambda$ and $\Delta s = -A^T \Delta y$ into the centering equation gives (673). The orthogonal row transform

$$Q = 2^{-1/2} \begin{bmatrix} -I & I \\ I & I \end{bmatrix}$$

gives

$$QM_{\text{OSS}} = \sqrt{2} \begin{bmatrix} H & -D \\ 0 & I \end{bmatrix}. \quad (676)$$

The triangular factor has constant norm and inverse $\begin{bmatrix} H^{-1} & H^{-1}D \\ 0 & I \end{bmatrix}$; since $\|D\| \leq \delta/\sqrt{2}$, its inverse norm is $\Theta_\delta(\|H^{-1}\|)$.

Applying Q to the right-hand side gives $H\Delta y - D\lambda = a$ and $\lambda = -re$. Since $De = a$, this proves (674). Also $\|\Delta y\|_2^2 \geq \sigma^2 \delta^2 / 2$, whereas $\|\lambda\|_2^2 = \alpha^2$, so the block probability is at least

$$\frac{\sigma^2 \delta^2 / 2}{\sigma^2 \delta^2 / 2 + \alpha^2} > \frac{2}{3}.$$

Direct substitution proves (675) and gives $X^1 s^1 - \mu^1 e = \sigma r(a, -a)$. All remaining entries and sparse positions are fixed, proving the oracle simulation. $\square$

After symmetric dilation, the theorem transfers the $\Omega(\kappa\sqrt{d})$ fixed-error and $\Omega(\kappa \log(1/\epsilon))$ constant-sparsity QLS lower bounds of Mori et al. to a strictly feasible short progress step [109]. It is an embedding of known QLS restrictions, not a new QLS technique or an end-to-end LP lower bound. Indeed, the physical $(\Delta x, \Delta s)$ above is explicit; the hard block is Δy , and algorithms that bypass accurate OSS-state preparation are outside the statement. Here d is the declared sparse-oracle slot bound, which may count padded zero slots. A symmetric dilation of a nonsymmetric source matrix requires bidirectional row- and column-slot access. Mori et al.'s clock family has known successor/predecessor slot locations and reversible hidden updates, so both directions are simulated with constant query overhead.

14.4.2 What bounded treewidth preserves

The same precision restriction survives at treewidth two. Mori et al.'s hard matrix has the form

$$A = I - \rho B, \quad (677)$$

where B is a clock permutation and every row and column has one identity and one permutation position. Form its symmetric dilation

$$K_A = \begin{bmatrix} 0 & A^T \\ A & 0 \end{bmatrix}, \quad \hat{b} = \begin{pmatrix} 0 \\ b \end{pmatrix}. \quad (678)$$

Theorem 14.11 (Treewidth-two KKT precision). *There are real symmetric, indefinite, two-sparse equality-KKT matrices K_A whose support graphs have treewidth at most two and for which preparing an ϵ -approximation to the normalized solution requires*

$$\Omega(\text{cond}(K_A) \log(1/\epsilon)) \quad (679)$$

queries in the parameter range of the underlying QLS theorem.

Proof. One has $K_A^{-1}\hat{b} = (A^{-1}b, 0)^T$, and the singular values of K_A are those of A , each repeated. The identity positions of A give one perfect matching in the bipartite support graph of K_A , while the clock-permutation positions give a second. Their union is a disjoint union of paths and even cycles, so its treewidth is at most two. Successor and predecessor clock queries simulate every sparse query with constant overhead. The state and condition number are preserved, so the QLS precision lower bound transfers. Finally, K_A is the equality-KKT matrix of $\min 0$ subject to $Ax = b$. $\square$

The two-sparse clock dilation of Wang–Zhang has the same low-treewidth support phenomenon but proves a different query-depth statement and not the $\log(1/\epsilon)$ precision factor [133]. This does not say that bounded-treewidth optimization is generically hard. The stronger classical fact points the other way: supplied width- τ elimination gives $O(N\tau^2)$ scalar Cholesky work and $O(N\tau)$ storage for an N -dimensional SPD Newton system. A dense classical output already costs $\Omega(N)$, so the dimension-only full-output speedup ceiling is $O(\tau^2)$. Nearly linear bounded-treewidth IPMs sharpen this baseline [56, 72].

There is nevertheless scalar LP value hardness at an exact chosen width. For Boolean matrices $Z^i \in \{0, 1\}^{c \times b}$, start from the pre-dual program

$$\begin{aligned} & \max_{w, v^1, \dots, v^a} \sum_{k=1}^c w_k \\ \text{subject to} \quad & w - \sum_{i=1}^a Z^i v^i \leq 0, \quad e^T v^i \leq 1 \quad (1 \leq i \leq a), \\ & w, v^i \geq 0. \end{aligned} \quad (680)$$

Its value is

$$\sum_{i=1}^a \max_{j \in [b]} \sum_{k=1}^c Z_{ijk}. \quad (681)$$

Add a nonnegative, zero-objective z_i to every top inequality and bottom inequality i , always with coefficient $+1$.

Theorem 14.12 (LP value hardness at exact treewidth). *For integers $a, b, c \geq 1$, the augmented LP has $p = c + a(b + 1)$ variables, $q = c + a$ constraints, and constraint-intersection treewidth exactly $\tau = c$. When $c \leq a$, constant-additive approximation of its scalar value requires*

$$\Omega(a\sqrt{b}c) = \Omega(\tau\sqrt{pq}) \quad (682)$$

fixed-position quantum coefficient queries. The corresponding randomized bound is $\Omega(abc) = \Omega(p\tau)$.

Proof. For fixed v^i , each top constraint is tight at an optimum, so summing the coordinates of w separates the a simplex problems in v^i and gives (681). Setting $z = 0$ extends every original feasible point. Conversely, deleting positive z from an augmented point only relaxes its inequalities, so the value is unchanged.

Let C be the c top-constraint vertices. The column of z_i makes $C \cup \{i\}$ a clique, and no variable column contains two bottom vertices. Thus the intersection graph is exactly $K_c \vee I_a$,

with bags $C \cup \{i\}$, and has treewidth c . The dimensions follow by counting w, v, z . The published Boolean reduction for (681) gives $\Omega(a\sqrt{b}c)$ quantum and $\Omega(abc)$ randomized fixed-position queries. Since

$$a\sqrt{b}c = \tau\sqrt{(q-\tau)(p-q)}, \quad abc = \tau(p-q),$$

and $c \leq a$ makes $q - \tau = \Theta(q)$ and $p - q = \Theta(p)$, the displayed bounds follow. $\square$

This is a reparameterized structural corollary of an existing reduction [10], not a new adversary argument. It applies to individual coefficient queries; a whole column reveals $\Theta(\tau)$ coefficients, and a whole top row can reveal far more. The broad claim that treewidth was absent from QLS lower-bound work would also be false in view of Khanpour–Talkington [84]. The new restriction here is only the treewidth-two KKT precision embedding.

14.5 Geometric locality of central-point access

Sparse-query access suppresses where the input is stored. On bounded-range hardware, that geometry can itself be an obstruction. Let hidden signs $\varsigma_1, \dots, \varsigma_N \in \{\pm 1\}$ and introduce $d_i = u_i - v_i$ with $u_i, v_i \geq 0$. For fixed $\alpha \in (0, 1)$, consider

$$\begin{aligned} \min \quad & \sum_{i=0}^N (u_i + v_i) + \alpha(u_N - v_N), \\ \text{subject to} \quad & d_0 = 1, \\ & d_i - \varsigma_i d_{i-1} = 0 \quad (1 \leq i \leq N). \end{aligned} \tag{683}$$

Every row has at most four nonzeros, every column at most two, and the constraint-intersection graph is a path. Put $p_i = \prod_{j=1}^i \varsigma_j$ and $q_i = u_i + v_i$. Feasibility gives $d_i = p_i$. Taking every $q_i = 2$ is strictly primal feasible; with the convention $A^T y + s = c$, the choice $y = 0$ gives $s = c > 0$ and strict dual feasibility. The barrier restricted to the feasible affine space is

$$\sum_{i=0}^N q_i + \alpha p_N - \mu \sum_{i=0}^N \log \frac{q_i^2 - 1}{4}. \tag{684}$$

Thus every central coordinate equals

$$q(\mu) = \mu + \sqrt{\mu^2 + 1}. \tag{685}$$

At $\mu = 3/4$, $q = 2$ and $(u_i, v_i) = (3/2, 1/2)$ or $(1/2, 3/2)$ according to p_i . Reading the final cell with coordinate error below $1/2$ therefore reveals parity.

This information cost is invisible to reduced conditioning. The public vectors $V_i = (e_{u_i} + e_{v_i})/\sqrt{2}$ form an orthonormal null-space basis, and for the full logarithmic-barrier Hessian $H_x = X^{-2}$,

$$V^T H_x V = 2[(q+1)^{-2} + (q-1)^{-2}]I. \tag{686}$$

It has condition number one for every $\mu > 0$.

Place input register ς_i at a site v_i of a hardware graph Γ . A depth- T , range- r circuit has gates of hardware diameter at most r in each layer. Its fixed initial workspace may contain arbitrary input-independent entanglement. A measurement at site o is locally parity-readable if it returns p_N with probability at least $2/3$ for every input.

Theorem 14.13 (Central-oracle light cone). *Let $R_\Gamma(o; v) = \max_{1 \leq i \leq N} \text{dist}_\Gamma(o, v_i)$ and let $b = \max_{w \in \Gamma} |\{i : v_i = w\}|$ be the maximum site occupancy. Every such circuit constructing a locally readable final central cell obeys*

$$T \geq \left\lceil \frac{R_\Gamma(o; v)}{r} \right\rceil. \tag{687}$$

Consequently, when at most b input registers share a site, a path requires $T \geq (\lceil N/b \rceil - 1)/(2r)$ for every choice of output site, which is $\Omega(N/(br))$ once $N \geq 2b$, and a d -dimensional nearest-neighbor grid requires $(2rT + 1)^d \geq \lceil N/b \rceil$, hence $T \geq ((N/b)^{1/d} - 1)/(2r)$, which is $\Omega((N/b)^{1/d}/r)$ once $N \geq 2^d b$. Pairwise-distinct placement is the case $b = 1$.

Proof. Pull a local output observable backwards through the circuit. Its support can grow by hardware distance at most r per layer, so after T layers it is contained in $B_\Gamma(o, rT)$. If some v_j lies outside this ball, two inputs differing only at j have identical reduced input states on the backward light cone. Their workspace state is also identical, including all of its input-independent entanglement. The local output distributions are therefore identical, although flipping ς_j flips p_N ; one distribution cannot return both answers with probability $2/3$.

Thus the ball $B_\Gamma(o, rT)$ contains every occupied site, proving (687). Under occupancy at most b , at least $\lceil N/b \rceil$ distinct sites are occupied. On a path a radius- rT ball has at most $2rT + 1$ sites, and on the grid at most $(2rT + 1)^d$, which proves the two consequences. $\square$

This is the standard backward-light-cone argument [17]; the claimed contribution is only its conjunction with the strictly feasible, condition-one central-point witness. It is not a lower bound in the all-to-all or free-QRAM query model, for a homogeneous self-dual embedding with a universal start, for an answer left in a global observable whose measurement is declared free, or when unboundedly many input registers may share one site. The same lower bound holds classically, so it is an obstruction to a claimed quantum depth advantage, not a quantum–classical separation. Input-dependent pre-entanglement can already contain the answer and merely moves its construction cost into advice.

A stronger per-iteration QRAM-write claim does not follow. A lazy oracle may store the input once and evaluate a changed weight from the queried index and a scalar path parameter. An $\Omega(N)$ refresh theorem would need a materialized-array or cell-probe model, or a lower bound for the composed evaluation oracle. Numerical change in all coordinates is not enough.

14.6 Synthesis

These tools charge different resources and should not be collapsed into one informal claim that sparsity fails. Symmetry charges invariant virtual work even after equality rows are mixed. Central mixtures trade sparse one-bit-local input incidence against operator spread. Complementarity progress charges pair support or NT aggregate rank. Sparse-QLS hardness survives a legitimate OSS step and even a treewidth-two KKT representation, while exact treewidth also admits strong classical algorithms. Physical locality charges communication only when the input and output are locally materialized with bounded per-site occupancy.

Their escape routes are equally structural. A nonlinear output may not be mixture stable; central matrices may have a large common-basis spectral spread; a nonlocal reconstruction map may keep a latent state sparse; chordal conversion or an implicit resolvent may avoid matrix-factor tomography; an algorithm may bypass the OSS; and a global or lazy oracle may avoid local materialization. The results above identify where such an escape must enter. They do not supply a universal quantum query or runtime lower bound.

15 Corrections to Prior Work and Recorded Dead Ends

Corrections and negative results delimit what the positive theorems actually establish. The first result below corrects a norm estimate in a publicly posted QCPM complexity derivation. The next two results isolate representation and initialization costs on sparse Hamiltonian and central-path constructions. We then record false shortcuts that were tested and rejected. None of these statements is a general verdict against Hamiltonian optimization, preconditioning, or state reuse.

15.1 Correction to the QCPM simulation norm

We consider arXiv:2311.03977v2 (16 October 2024), the quantum central-path method (QCPM) of Augustino, Leng, Nannicini, Terlaky, and Wu [15]. All formulas in this subsection refer to that version. Let $d = m + n + 2$ be the dimension of its self-dual embedding, and use its notation

$$s(z) = Mz + q, \quad F(z) = z \odot s(z), \quad f_\mu(z) = \frac{1}{2} \|F(z) - \mu e\|_2^2. \quad (688)$$

The central-path Hamiltonian is $H(\mu) = -(h^2/2)\nabla^2 + f_\mu$, and Algorithm 1 sets

$$h(t) = a\mu(t)^2, \quad a = \frac{\gamma^2}{2R_1C_{d,\delta}}, \quad (689)$$

where $C_{d,\delta} = \sqrt{d}/2 + (3/4)\log(2/\delta)$ is the concentration factor displayed in Algorithm 1. We assume $d \geq 1$, $a, \eta > 0$, and $0 < \varepsilon < 1$. The spatial domain Ω_D is fixed and compact and contains e ; statements involving $z(\mu)$ also require that point to belong to Ω_D . The invoked real-space simulation theorem measures a potential V on a fixed domain Ω_D by

$$\|V\|_{\infty,1} = \int \sup_{z \in \Omega_D} |V(z, \tau)| d\tau. \quad (690)$$

The QCPM proof truncates to a fixed box containing the central-path neighborhood and replaces the supremum in (690) by an assumed bound $\sup_{\Omega_D} f_\mu \leq K\gamma^2\mu^2$. That bound is false.

Proposition 15.1 (Global-supremum counterexample). *If Ω_D contains the initial center $e = z(1)$, then, for every $0 < \mu \leq 1$,*

$$\sup_{z \in \Omega_D} f_\mu(z) \geq f_\mu(e) = \frac{d}{2}(1 - \mu)^2. \quad (691)$$

Consequently, at $\mu = \varepsilon \leq 1/2$, the asserted bound would require $K \geq d/(8\gamma^2\varepsilon^2)$.

Proof. The self-dual embedding is initialized so that $s(e) = e$. Thus $F(e) = e$, and (688) gives

$$f_\mu(e) = \frac{1}{2} \|(1 - \mu)e\|_2^2 = \frac{d}{2}(1 - \mu)^2.$$

Taking the supremum proves (691). When $\varepsilon \leq 1/2$, its right-hand side is at least $d/8$, which gives the claim about K . $\square$

Ground-state concentration controls state-weighted mass, not the operator supremum on a fixed domain. A scalar phase shift does not restore the claimed scaling. If $z(\mu) \in \Omega_D$, then $f_\mu(z(\mu)) = 0$, and for every real scalar c the triangle inequality gives

$$\sup_{z \in \Omega_D} |f_\mu(z) - c| \geq \frac{1}{2} |f_\mu(e) - f_\mu(z(\mu))| = \frac{d}{4}(1 - \mu)^2.$$

There is a second, independent issue in the time dilation. Let $\mu : [0, 1] \rightarrow [\varepsilon, 1]$ be continuous. After division by $h(t)\mu(t)$, the evolution equation is

$$i\eta \partial_t \Psi = \left[-\frac{\theta(t)}{2} \nabla^2 + \frac{f_{\mu(t)}}{\mu(t)h(t)} \right] \Psi, \quad \theta(t) = \frac{h(t)}{\mu(t)} = a\mu(t). \quad (692)$$

The linear substitution used in the QCPM proof treats θ as constant, although it varies with t . The clock that normalizes the kinetic term is instead

$$\tau(t) = \frac{1}{\eta} \int_0^t \theta(u) du = \frac{a}{\eta} \int_0^t \mu(u) du. \quad (693)$$

Indeed, $dt/d\tau = \eta/\theta(t)$, so (692) becomes

$$i\partial_\tau \tilde{\Psi} = \left[-\frac{1}{2}\nabla^2 + \frac{f_{\mu(t(\tau))}}{h(t(\tau))^2} \right] \tilde{\Psi}. \quad (694)$$

Combining (689)–(693) yields the exact corrected identity

$$\Lambda := \|V\|_{\infty,1} = \frac{1}{a\eta} \int_0^1 \frac{\sup_{z \in \Omega_D} f_{\mu(t)}(z)}{\mu(t)^3} dt. \quad (695)$$

This identity does not use a lower-bound argument. The clock correction by itself works in the authors' favor: it lowers the norm weight from the constant- θ argument's μ^{-4} to μ^{-3} . The ε^{-3} blow-up below is driven entirely by the false global supremum bound.

The schedule in Algorithm 1 is

$$g(t) = c_e^{-1} \int_0^t \exp\left[-\frac{1}{\tau(1-\tau)}\right] d\tau, \quad \mu(t) = \varepsilon + (1-\varepsilon)(1-g(t)), \quad (696)$$

where c_e normalizes $g(1) = 1$. The bump integrand is extended by zero at both endpoints. It is continuous, nonnegative, and positive on $(0, 1)$, so $c_e > 0$.

Lemma 15.2 (Width of the flat endpoint). *For $0 < \varepsilon \leq \min\{1/4, c_e\}$, put $u_\varepsilon = 1/(2\log(1/\varepsilon))$. Then $\mu(t) \leq 2\varepsilon$ for every $t \in [1 - u_\varepsilon, 1]$.*

Proof. For $0 < u < 1$, substitute $v = 1 - \tau$ in the tail of (696). Since $(1-v)v \leq v$,

$$1 - g(1 - u) \leq \frac{1}{c_e} \int_0^u e^{-1/v} dv \leq \frac{u}{c_e} e^{-1/u}. \quad (697)$$

At $u = u_\varepsilon$, the exponential equals ε^2 . The last quantity in (697) is at most ε : $\varepsilon \leq 1/4$ implies $0 < u_\varepsilon \leq 1/2$, and $u_\varepsilon \varepsilon \leq c_e$ follows from $\varepsilon \leq c_e$. For $t \geq 1 - u_\varepsilon$, monotonicity then gives $\mu(t) \leq \varepsilon + (1 - \varepsilon)\varepsilon \leq 2\varepsilon$. $\square$

Theorem 15.3 (Corrected QCPM simulator-norm scale). *For the choices in Algorithm 1 and $0 < \varepsilon \leq \min\{1/4, c_e\}$,*

$$\Lambda \geq \frac{d}{128a\eta\varepsilon^3 \log(1/\varepsilon)}. \quad (698)$$

If $\Omega_D = [0, D]^d$, a valid global upper certificate is

$$\Lambda \leq \frac{F_D}{a\eta\varepsilon^3}, \quad F_D = \frac{d}{2} [D(D\|M\|_\infty + \|q\|_\infty) + 1]^2. \quad (699)$$

Here $\|M\|_\infty$ is the maximum absolute row sum.

Proof. On the interval supplied by lemma 15.2, take $\varepsilon \leq 1/4$. Proposition 15.1 gives $\sup_{\Omega_D} f_{\mu(t)} \geq d/8$, while $\mu(t)^{-3} \geq (2\varepsilon)^{-3}$. The interval has length $1/(2\log(1/\varepsilon))$. Substitution in (695) proves (698).

For the upper bound, every $z \in [0, D]^d$ satisfies

$$|s_i(z)| \leq D\|M\|_\infty + \|q\|_\infty.$$

Hence $|F_i(z) - \mu|$ is at most $D(D\|M\|_\infty + \|q\|_\infty) + 1$ for $0 < \mu \leq 1$, and $f_\mu(z) \leq F_D$. Since $\mu(t) \geq \varepsilon$, the exact identity (695) gives (699). $\square$

Since $a^{-1} = 2R_1C_{d,\delta}/\gamma^2$, [equation \(698\)](#) gives the simulator-norm lower bound

$$\Omega\left(\frac{R_1dC_{d,\delta}}{\gamma^2\eta\varepsilon^3\log(1/\varepsilon)}\right). \quad (700)$$

By contrast, the bound claimed in v2 is

$$\|V\|_{\infty,1} \leq \frac{K\gamma^2}{\theta\eta}, \quad O\left(\frac{K\gamma^2}{a\eta\varepsilon}\right) = O\left(\frac{R_1C_{d,\delta}}{\eta\varepsilon}\right) \quad (K = O(1)).$$

Thus γ^2 cancels against $1/a$. Conditional on all other simulation hypotheses, [\(699\)](#) supplies the norm-based upper certificate

$$O\left(\frac{F_D R_1 C_{d,\delta}}{\gamma^2 \eta \varepsilon^3}\right)$$

before polylogarithmic factors and potential-evaluation cost.

The calculation has a schedule-independent form.

Corollary 15.4 (Norm–speed tradeoff). *Let $\mu : [0, 1] \rightarrow [\varepsilon, 1]$ be L -Lipschitz, with $\mu(0) = 1$, $\mu(1) = \varepsilon$, and $0 < L < \infty$. In particular, a differentiable schedule with $|\dot{\mu}| \leq L$ satisfies this hypothesis by the mean value theorem. Retain $h = a\mu^2$. For $\varepsilon \leq 1/4$,*

$$\Lambda \geq \frac{d}{64a\eta L \varepsilon^2}. \quad (701)$$

Proof. The endpoint values imply $L \geq 1 - \varepsilon$, hence $\varepsilon/L \leq 1$. On the terminal interval $[1 - \varepsilon/L, 1]$, the Lipschitz bound gives $\mu(t) \leq \varepsilon + L(1 - t) \leq 2\varepsilon$. Throughout that interval, [proposition 15.1](#) gives $\sup_{\Omega_D} f_\mu \geq d/8$, and $\mu^{-3} \geq (2\varepsilon)^{-3}$. Equation [\(695\)](#) proves the result. $\square$

No monotonicity assumption is needed for this bound.

Taking L large weakens [\(701\)](#), but moves L into the derivative, smoothness, and adiabatic estimates. The corollary alone does not optimize that tradeoff, and it does not exclude a theorem that rigorously exploits localization.

As a numerical check, adaptive quadrature gives $c_e = 0.007029858406609657$. For

$$J(\varepsilon) = \varepsilon^3 \log(1/\varepsilon) \int_0^1 (1 - \mu(t))^2 \mu(t)^{-3} dt,$$

the computed values are

ε	$J(\varepsilon)$
10^{-2}	0.77120646
10^{-3}	0.89055595
10^{-4}	0.95036732

The sampled values are consistent with a limit of one; no limit is deduced from this computation. Retaining only the witness $f_\mu(e) = d(1 - \mu)^2/2$ gives

$$\Lambda \geq \Lambda_e := \frac{d}{2a\eta} \int_0^1 (1 - \mu(t))^2 \mu(t)^{-3} dt \geq \frac{d}{128a\eta\varepsilon^3 \log(1/\varepsilon)} \quad (0 < \varepsilon \leq \min\{1/4, c_e\}).$$

The proof certifies $J(\varepsilon) \geq 1/64$ in this range and makes no claim that this constant is sharp. The quadrature is a sanity check on the proved scale, not an estimate of the full supremum or part of the proof.

Formal verification of the norm correction. The companion Lean development in `formal/QipmFormal/QCPM/` checks the finite-dimensional potential and initialization identity, the compact-domain supremum bound, robustness under scalar shifts, the corrected clock and its inverse, and the integral change of variables. It also checks the actual normalized bump schedule, the explicit endpoint threshold, both lower-bound constants, and the fixed-box upper certificate. The headline bounds use the spatial supremum of the stated potential; their continuity and integrability are proved rather than assumed as extra properties of that supremum. The initialization row equations $Me + q = e$ and inclusion of the required points in the fixed domain are explicit hypotheses. The source correspondence is to Algorithm 1 of v2, whose normalization differs from its equation (23). The verification does not formalize Schrödinger evolution, the external simulation or adiabatic theorems, or an unconditional query lower bound.

The norm and clock errors invalidate the stated complexity derivation, not the QCPM idea itself, and (700) is not an unconditional query lower bound for every implementation of the evolution. Chakrabarti et al. analyze missing error controls in the pseudo-spectral real-space simulation [40]. Gribling et al. cite that analysis and separately identify the absence of a suitable directly applicable infinite-dimensional adiabatic theorem [70]. We do not claim those broader criticisms as new. Even after correcting the norm, the following obligations remain:

- The schedule in (696) is nonconstant and C^∞ -flat, but it is not analytic through either endpoint. An analytic function with its zero endpoint Taylor series would be locally constant. The quoted exponential adiabatic theorem assumes analytic continuation through the closed interval.
- The adiabatic theorem must be applied to the generator in (692); the derivative discussion analyzes a differently scaled Hamiltonian and does not establish uniform derivative constants for time-dependent h .
- The exact potential (688) is generally quartic. The displayed Gaussian state and oscillator gap concern its quadratic approximation. A dimension-, μ -, and input-uniform transfer to the exact ground state and spectral gap is not established.
- The proposition and algorithm use different choices of h ; the proposition's choice has no γ or δ dependence and therefore cannot by itself provide arbitrary neighborhood opening and failure probability.
- The initial exact ground state is assumed rather than prepared. Knowing its minimizer e does not prepare the ground state of a quartic Hamiltonian.
- The real-space theorem assumes a smooth periodic potential and periodic boundary conditions. Restriction of (688) to a box neither makes it periodic nor controls the boundary and truncation errors.
- The box size must retain its solution-radius dependence unless an input-size promise bounds the central path.

A localized or clipped-potential repair would need leakage, spurious-state, uniform-gap, boundary, and comparison estimates; concentration alone does not supply them.

15.2 Sparse metric, dense inverse metric

The proposal of Gribling, Apers, Nieuwboer, and Walter replaces the Euclidean Laplacian by the Laplace–Beltrami operator of the Hessian metric [70]. If $G(x) = \nabla^2 \phi(x)$, its principal part is

$$L\psi = \frac{1}{\sqrt{\det G}} \sum_{i,j} \partial_i \left((G^{-1})_{ij} \sqrt{\det G} \partial_j \psi \right). \quad (702)$$

Sparse constraints do not make these standard-coordinate coefficients entrywise sparse.

Theorem 15.5 (Path-simplex inverse metric). *Let*

$$\mathcal{D}_n = \{x \in \mathbb{R}^n : 0 < x_1 < \cdots < x_n < 1\}$$

and use the logarithmic barrier

$$\phi(x) = -\log x_1 - \sum_{i=1}^{n-1} \log(x_{i+1} - x_i) - \log(1 - x_n). \quad (703)$$

Every inequality contains at most two variables, and the constraint-intersection graph is a path. At the analytic center $\bar{x}_i = i/(n+1)$, the Hessian $G = \nabla^2 \phi(\bar{x})$ is tridiagonal, but

$$(G^{-1})_{ij} = \frac{\min(i, j)(n+1 - \max(i, j))}{(n+1)^3} > 0 \quad (704)$$

for every i, j . For $i, j \in [(n+1)/4], \lfloor 3(n+1)/4 \rfloor$, these entries are $\Theta(1/n)$, so $\Theta(n^2)$ coefficients are of comparable order.

Proof. Write the $n+1$ slacks as $s_0 = x_1$, $s_i = x_{i+1} - x_i$ for $1 \leq i < n$, and $s_n = 1 - x_n$. At $\bar{x}$, every slack is $1/(n+1)$. If B is the signed path-incidence Jacobian of this affine slack map, then

$$G = B^T \text{diag}(s^{-2}) B = (n+1)^2 T_n,$$

where T_n has diagonal entries two and adjacent off-diagonal entries minus one. The discrete Green identity is

$$(T_n^{-1})_{ij} = \frac{\min(i, j)(n+1 - \max(i, j))}{n+1}.$$

Scaling gives (704). In the middle half, both factors in its numerator are bounded above by n and below by a fixed positive multiple of n , proving the final assertion. $\square$

Thus a direct standard-coordinate, nondivergence-form finite-difference stencil for the principal symbol in (702) has $\Omega(n^2)$ mixed-derivative directions at the analytic-center grid point, although the input has only $O(n)$ nonzeros. This is a densification warning, not a simulation or query lower bound. Indeed, at an arbitrary interior point put $r_k = s_k^2$, $R = \sum_{k=0}^n r_k$, and let $i \leq j$. Weighted path inversion gives

$$(G(x)^{-1})_{ij} = \frac{(\sum_{k=0}^{i-1} r_k)(\sum_{k=j}^n r_k)}{R}. \quad (705)$$

Every strict off-diagonal block is rank one, and prefix sums give an $O(n)$ matrix–vector apply. The witness therefore rules out only the inference from sparse Hessian support to an entrywise sparse inverse-metric tensor. It does not rule out implicit, transform-based, or coordinate-adapted simulation.

Abe and Nagai parameterize their Riemannian Hamiltonian simulation by the kinetic-matrix sparsity and separately assume sparse-access oracles for that matrix; they note that constructing those black-box oracles is generally nontrivial [2]. Theorem 15.5 shows only that input-level $O(n)$ sparsity is lost: the direct stencil has $\Theta(n^2)$ terms. On a tensor grid with q points per coordinate it still has at most $2n^2 + 1$ neighbors per row, sparse relative to the q^n -dimensional grid. Thus the theorem does not rule out sparse-Hamiltonian access or refute the separately assumed Abe–Nagai oracles. Gribling et al. likewise leave the simulation and qubit-realization questions open.

15.3 A hidden-sign start-oracle separation

We reuse, without reconstructing it, the path-sparse hidden-sign LP (683) from section 14.5. Recall that $p_0 = 1$, $p_i = \prod_{j=1}^i \varsigma_j$, and $q_i = u_i + v_i$.

Theorem 15.6 (Value, output, and central-start separation). *For the family (683):*

1. *It is strictly primal–dual feasible and has the unique optimum*

$$\text{OPT} = N + 1 + \alpha p_N. \quad (706)$$

Estimating its shifted value $\text{OPT} - (N + 1)$ to additive error less than α requires $\Omega(N)$ bounded-error raw sparse queries.

2. *Every classical nonnegative output $\hat{x}$ satisfying $\|A_\varsigma \hat{x} - b\|_1 < 1/2$ determines p_N and therefore also requires $\Omega(N)$ queries.*
3. *The primal central path has the public support-two orthonormal null basis stated in section 14.5, and its reduced log-barrier Hessian has condition number one for every $\mu > 0$.*
4. *At $\mu = 3/4$, one lookup at the final cell of a reusable random-access value oracle for the exact primal central point reveals p_N . If setup and that lookup use respectively Q_{setup} and Q_{lookup} raw queries, then $Q_{\text{setup}} + Q_{\text{lookup}} = \Omega(N)$. In particular, query-free invocations after preprocessing force $Q_{\text{setup}} = \Omega(N)$.*

Proof. The feasibility argument and central-path calculations are given throughout section 14.5, with the displayed formulas in (684)–(686). In particular, feasibility fixes $d_i = p_i$ and leaves $q_i \geq 1$, so the objective is $\sum_i q_i + \alpha p_N$ and has the unique minimum (706). The shifted value reveals parity.

For an approximate output define $r_0 = \hat{d}_0 - 1$ and $r_i = \hat{d}_i - \varsigma_i \hat{d}_{i-1}$. Telescoping gives

$$\hat{d}_N - p_N = \sum_{j=0}^N \left(\prod_{k=j+1}^N \varsigma_k \right) r_j, \quad (707)$$

so $|\hat{d}_N - p_N| < 1/2$ under the asserted residual bound. The sign of $\hat{d}_N = \hat{u}_N - \hat{v}_N$ is therefore p_N .

Finally, (685) gives $q = 2$ at $\mu = 3/4$. The final pair is $(3/2, 1/2)$ or its swap according to p_N , so one coordinate-oracle lookup determines parity even with coordinate error below $1/2$. Since quantum parity has query complexity $\Theta(N)$, each claimed raw-query lower bound, including the combined setup–lookup bound, follows from the local sign-oracle reduction used throughout section 9. $\square$

This theorem charges an original-form feasible start. It does not apply to infeasible-start or homogeneous self-dual methods, which can supply a different universal start; nor does it say that the scalar reduced Newton solve itself is hard.

15.4 Dead ends and false shortcuts

First inverse versus filtering. A small value of $\|H\| \|H^{-1}f\| / \|f\|$ does not imply a fast DLS filtering solve. Filtering depends on the second inverse through $\|H\| \|H^{-2}f\| / \|H^{-1}f\|$, and active–inactive coupling can restore the full condition-number exponent. The exact identities and sparse sharpness pair are in sections 5.1 and 5.2.

Sparse support versus inverse access. Sparse Hessian or KKT support does not imply a sparse inverse, a cheap block encoding after elimination, or low circuit depth on bounded-range hardware. The factor-normalization and inverse-compilation frontiers in [section 12.1](#) prove the access cost; the bounded-range depth claim requires the bounded-occupancy geometric input placement of [section 14.5](#). The dense metric in [section 15.2](#) gives a separate coordinate-level warning. None of these statements says that every structured inverse is expensive.

Condition one versus the surrounding maps. A condition-one reduced Newton matrix does not make the affine feasible offset, coordinate gauge, transformed right-hand side, recovery map, or requested output free. The LP separations in [sections 9.4](#) and [9.5.1](#) place parity in the affine offset while leaving the reduced geometry public. The full-KKT and preconditioner dichotomies in [sections 9.6](#) and [12.3](#) show how input dependence can relocate among these objects.

Exact Schur preconditioning. An exact Schur transformation can make the abstract product constantly conditioned while relocating input dependence into construction, transformed RHS loading, or recovery. The complete ledger is proved in [sections 9.6](#) and [12.1](#). Supplying the preconditioned product itself as a unit-cost oracle is a strictly stronger input model, not a consequence of sparse access to the original system.

State output versus coordinate access. A normalized solution state is not a reusable coordinate oracle. Its norm and global phase are absent, while coherent diagonal construction also needs controlled and adjoint access and pays a normalization factor. The tight state-to-diagonal product appears in [section 12.4](#); the implicit-coordinate escape route in [section 13.3](#) states the stronger interface it actually constructs.

One-step bounds versus a fixed trajectory. Reapplying a one-step lower bound at every point of one fixed LP trajectory is invalid without freshness, a memory restriction, an adaptivity/causality restriction, or mandatory explicit output. The fresh-oracle direct sum is [theorem 12.14](#); the surrounding discussion shows how caching a static input can erase later raw-query cost.

Residual bounds versus projective variation. Ordinary short-step neighborhood and relative OSS residual bounds do not imply finite projective variation. [Section 13.1](#) gives a sparse cycling counterexample and the additional hypotheses needed for reuse. The refined audit in [section 13.2](#) also corrects an earlier version’s proposed sharpness family: its convergent leakage and transverse-forcing ledgers still allow linearly growing realized variation, and a separate oscillating-displacement family shows that adding the $O(\mu_t)$ tail bound on face leakage does not help either; finite realized variation is therefore stated as an independent hypothesis rather than derived from the ledgers.

Expanderization. Replacing a path by an expander does not evade the bounded-degree local-linear mixture or cut/resistance obstruction. Those no-go results are [sections 10.1](#) and [10.3](#). In the favorable well-spread regime, [section 13.4](#) gives the matching classical sampling shortcut; making rare columns influential instead moves cost into normalization or one-time search.

Scalar gap versus density visibility. A constant scalar objective gap can coexist with a trace-normalized central density that is only $O(1/G)$ from a public state. Thus an output lower bound must state its accuracy and normalization contract. The zero-query boundary and the mass-visibility inequalities are proved in [sections 6.5](#) and [6.6](#); the dilution example is [remark 7.1](#).

Pairwise distance versus a common decoder. Pairwise distinguishability of states indexed by an unknown winner set does not provide one public measurement that decodes every set. The failed collision shortcut loses a factor $1/k$. Section 8.1 repairs this only by measuring a proposed label and verifying that label with fresh input queries; the verifier, not pairwise trace distance alone, supplies the common decision procedure.

15.5 Disposition of remaining algorithmic candidates

To state the factorized rectangular observation precisely, let B_μ have full row rank, let $f = B_\mu h$ lie in $\text{range}(B_\mu)$, and let $\alpha_{B_\mu} \geq \|B_\mu\|$ be its block-encoding normalization. Define

$$\rho_{B_\mu}(f) = \alpha_{B_\mu} \frac{\|(B_\mu B_\mu^T)^{-1} f\|_2}{\sqrt{f^T (B_\mu B_\mu^T)^{-1} f}}. \quad (708)$$

This pseudoinverse parameter is distinct from the nonsingular filtering parameter in (86). In the mixed strictly-complementary exact-centering asymptotic, assume $B_\mu B_\mu^T = \mu^{-1}C + \mu D + o(\mu)$, $f = b \in U := \text{range } C$, and the generic active-inactive coupling of section 5.1, so that

$$(B_\mu B_\mu^T)^{-1} b = \mu(p, -g) + o(\mu), \quad b^T (B_\mu B_\mu^T)^{-1} b = \Theta(\mu).$$

With the essential normalization $\alpha_{B_\mu} = \Theta(\mu^{-1/2})$, substitution in (708) gives the proved algebraic observation $\rho_{B_\mu}(b) = \Theta(1)$.

Here $D_\mu = (XS^{-1})^{1/2}$, $B_\mu = AD_\mu$, and $h = (XS)^{-1/2}r_c$ is the scaled complementarity right-hand side; thus $f = B_\mu h$ is the row-space right-hand side denoted “ Bh ”. The two scaled direction components are

$$u = \Pi_{\ker B_\mu} h, \quad v = B_\mu^+ f = B_\mu^+ (B_\mu h),$$

with $u + v = h$. They recover the physical directions through $\Delta x = D_\mu u$ and $\Delta s = D_\mu^{-1} v$. The constant value of (708) is not an end-to-end QIPM speedup. Six obligations remain: a DLS-quality rank-deficient rectangular filter theorem with its cutoff and error law; preparation of $f = B_\mu h$; coherent construction, norm control, and cancellation-safe physical recovery of u and v ; inexact-neighborhood propagation; maintenance of the changing diagonal oracles; and a complete output ledger with a strong classical comparison. Li’s instance-dependent rectangular minimum-norm algorithm already uses closely related second-inverse geometry [94], so the primitive also substantially collides with prior work.

A further candidate, the augmented spectral Newton sketch, instead samples the augmented matrix $[D_N^{1/2} A^T, v_N]$, where D_N and v_N denote the sketch’s Newton scaling and regression target. Under its stated row-access assumptions, a $(1 \pm \varepsilon)$ embedding of the augmented Gram matrix gives a valid least-squares embedding lemma at $\tilde{O}(\sqrt{nm}/\varepsilon)$ row-query scale; the infeasible-step analysis, implicit update mechanism, and end-to-end comparison remain open, and the direction overlaps the factorized rectangular candidate above.

State-to-diagonal lifting is a useful but known primitive, not a new result: Rattew and Rebertost give the amplitude-to-diagonal block encoding, with the phase-aligned coherent access assumptions recorded in section 12.4 [121]. Likewise, the projective-plus-implicit expander candidate is closed by the dichotomy in section 13.4: bounded contributions admit a comparable classical sketch, while rare influential columns incur normalization or one-time search cost.

16 Discussion and Open Problems

The motivating question was whether QIPMs beat classical IPMs on sparse problems. The results support a more useful answer than a single yes or no. They separate three layers that are often

combined in informal complexity claims: the geometry of the central path, the access model for the resulting linear-algebra objects, and the interface by which one iteration produces the next. Each layer has its own obstruction and its own escape routes.

16.1 What the program establishes

Geometry. For barrier-Newton methods, part of the condition growth is intrinsic to the optimization problem. [Theorem 3.3](#) lower-bounds the reduced barrier Hessian at a fixed objective gap for every self-concordant barrier, and [theorem 3.5](#) shows that log and log-det barriers attain the resulting order up to instance constants. [Theorem 3.4](#) refines the two extreme eigenvalues into a geometric description of the full spectrum. This settles the barrier-redesign question in the stated minimax sense: changing the barrier cannot generally eliminate a sublevel-chord obstruction.

That conclusion is not the folklore claim that every central path has $\kappa = \Theta(\mu^{-2})$. The rate depends on sublevel geometry. A positive-dimensional optimal face gives inverse-square gap growth, a curved unique optimum can give inverse-linear growth, a unique LP vertex can remain bounded, and attained singularity-degree behavior can give fractional laws [corollaries 3.6](#) and [3.8](#). Nor does a large condition number alone determine solver cost. [Propositions 4.1](#) and [4.2](#) show that classical Krylov convergence can be polylogarithmic in an arbitrarily large inter-cluster ratio. The path’s own right-hand side may avoid the weak modes as well [proposition 4.5](#) and [corollary 4.6](#).

Access models. Quantum cost depends on how the matrix is supplied, not only on its spectrum. The clustered comparison in [theorem 4.3](#) separates plain normalized block-encoding access from access to a factor $H = G^\dagger G$. Plain access retains the known worst-case $\tilde{\Theta}(\kappa)$ query cost even on two-point spectra in sufficiently large dimension. Norm-tight factor access has a square-root polynomial route, but only an overlap-qualified end-to-end solve; a generic square-root factor-access QLS would contradict the same plain-access hard instance because its square root factor is easy to query. The exact coupling and spectral-dimension laws in [theorems 5.1](#) and [5.4](#) further show that right-hand-side mass determines whether truncation or filtering can improve on κ .

Preconditioning obeys the same discipline. A whitening factor may exchange condition for subnormalization; a congruence may exchange transformed condition for physical recovery sensitivity; and an input-dependent perfect preconditioner may move the hard information into setup or RHS preparation [theorems 12.2](#), [12.4](#), [12.7](#) and [12.8](#). These are product and ledger statements. They do not say that preconditioning is useless. They say that a favorable transformed condition is not, by itself, a complexity bound.

Interfaces. The strongest recurring obstruction is often loading or recovery rather than inversion. The prefix LP has a condition-one reduced Hessian along its path, yet preparing a robust original-primal direction requires linear raw queries [theorem 9.7](#). The SDP families distinguish an intrinsic value gap [theorem 11.1](#), parity-hard primal and complete-KKT solution states [theorem 11.2](#), and a public Schur solve with an original-coordinate loading frontier [theorems 11.3](#) and [11.4](#). The canonical KKT block encoding itself is exact at one raw query per call; the linear first-synthesis bound instead applies to the tangent projector [section 11.1.2](#). A normalized state is also not a reusable coordinate oracle: producing the changing diagonal for the next Newton system obeys the state-to-diagonal and state-to-next-solve products in [theorems 12.9](#) and [12.13](#).

The condensation results make the interface visible rather than implicit. Winner mass controls both the reduced-Hessian condition and the trace distance from a public loser state [theorems 6.5](#) and [6.6](#). The bounded-incidence holonomy realization then has matching query bounds for value and compact winner output [theorems 6.7](#) and [7.3](#), while central-state preparation changes across the condensation phases [section 7.7](#) and [equation \(243\)](#). The verified-sample decoder turns trace-close mixed output into a public, one-sided decision procedure, and parity collapse gives

the intervening accuracy curve [theorems 8.1 and 8.7](#) and [corollary 8.9](#). The lower bound is tied to a specified output accuracy; a scalar objective gap alone would not supply it.

Together these results give a three-layer verdict. Conditioning is governed by sublevel geometry but may be algorithmically benign when its spectrum or right-hand side is structured. Quantum performance is governed by the operator and state access actually supplied. End-to-end performance is governed by the loading, recovery, update, certification, and output interfaces that connect iterations. Sparsity can help at each layer, but no single notion of sparsity controls all three.

16.2 What the program does not establish

The paper proves neither an unconditional sparse-QIPM advantage nor an unconditional sparse-QIPM lower bound. The tall-row result of Apers–Gribling remains the clearest positive quantum–classical row-query separation in the literature considered here [\[10\]](#); our results do not weaken it. Conversely, none of the parity or holonomy families rules out a method with a different oracle, a weaker but still useful output, a condition-number-free optimization primitive, or a structure-aware preconditioner outside the stated class.

The barrier theorem is about the reduced Hessian in an orthonormal tangent basis. It is not automatically a theorem about an augmented KKT system, primal–dual scaling, or a Hamiltonian central-path method. The clustered QLS theorem is access specific and includes dimension and overlap qualifications. The LP and SDP loading theorems charge raw coefficient queries; a supplied parity, prefix, tangent-projector, or preconditioned-product oracle is a stronger input model. The online direct sum requires fresh independent input, whereas a fixed instance can be cached [theorem 12.14](#) and [section 12.5](#). The physical light-cone bound concerns locally placed data and output with bounded site occupancy, not all-to-all query access [theorem 14.13](#).

Likewise, the positive modules are components rather than a hidden complete algorithm. Projective reuse, as analyzed here, needs finite realized variation and a fixed residual contract; active-range equilibration and face repair need a certified tail and spectral gaps; compressed dual output needs a genuinely smaller retained dimension and a charged coordinate compiler; and the block-angular module needs a low-parameter projected barrier and persistent local representation [section 13.7](#). These qualifications are not technical decorations: the counterexamples in [sections 13.1, 13.2 and 13.4](#) fail precisely when one of them is removed.

The five referee rules introduced in [section 13](#) are therefore the standard we propose for future positive claims:

1. Expand loading, normalization, RHS preparation, recovery, and every output compiler into the declared access model; a small spectral condition alone makes none of them free.
2. State the data-dependent tail threshold, separation margin, and active singular gaps used by any strict-complementarity argument.
3. Claim a dimensional benefit from compressed dual output only when the retained dimension is asymptotically smaller and the classical comparator is charged under the same oracle and output contract.
4. Treat a coherent coordinate oracle as substantive output, especially when its values permit constant-overhead conversion to a solution state.
5. Before claiming that a dual-only certificate contradicts a primal lower bound, give an explicit low-query reduction from that certificate to the hard primal output contract.

To these five rules one should add the comparison discipline used throughout the paper: match residual and precision contracts, charge setup and reuse on both sides, and compare with the best classical method for the same structure. A query advantage is not automatically a gate or time advantage, and a comparison of two upper bounds is not a separation.

16.3 Collected open problems

The following list collects every explicit open-problem environment in [sections 3, 4, 8, 9](#) and [11](#) and the additional frontiers left in the technical discussion. They are grouped by the layer at which new information is needed.

16.3.1 Geometry and conditioning

- G1. **Degenerate-vertex constants.** Determine the limiting spectrum, not only boundedness, of the canonical reduced log Hessian at a unique degenerate LP vertex directly from active-set geometry [open problem 3.12](#).
- G2. **Facial characterization of chord growth.** Characterize the exact objective-sublevel chord growth $D(g)$ of an SDP from facial data and decide when the singularity-degree exponent is attained by objective sublevel diameters [open problem 3.13](#).
- G3. Relate the blockwise relative stationarity and trace-residual neighborhood used by robust condensation to the algorithm-specific inexact Newton residual of [definition 2.8](#), the unresolved condensation η -channel [section 6.11](#).

16.3.2 Quantum access and state conversion

- A1. **Exact subnormalization shift.** Decide whether plain access to H permits a block encoding of $I - H$ with unit subnormalization without paying the apparent uniform-amplification cost at the shifted spectral edge [open problem 4.4](#). This is the promise-sensitive form of the normalization question of Orsucci and Dunjko [118].
- A2. **General-inner composition.** Prove or refute the $\sqrt{\delta} Q(f) \sqrt{G/k}$ small-success composition lower bound for a general inner predicate f , beyond the parity-collapse proof [open problem 8.11](#).
- A3. Determine the minimax output radius and query curve when the public loser internal states differ by η_{int} , including the below-condensation regime [section 8.4](#).
- A4. Resolve the state-conversion sliver $\delta = o(k/G)$, where a public uniform-label output may already satisfy part of the contract and winner mass alone does not determine the boundary [section 8.4](#).
- A5. **Intrinsically nonabelian word hardness.** Prove a local-symbol query lower bound for a genuinely noncommuting word promise and transfer it to conjugacy-class output in the group-holonomy SDP, rather than restricting to a cyclic subgroup [open problem 11.8](#).

16.3.3 Hard families and gadget frontiers

- F1. **Bounded-scale full-KKT parity amplifier.** Construct or rule out a linear-size standard-form LP with bounded row and column degree, bounded coefficients with one-bit-local input dependence, public simply preparable Newton right-hand sides, primal and slack starts bounded above and below by positive constants, a coordinatewise bounded complete Newton direction, and a parity-hard primal component with constant mass in that complete direction state [open problem 9.14](#).
- F2. Determine whether an arbitrary sparse bounded-range equality LP can support a nonlinear normalized-state decoder with a fixed global residual tube without introducing nonlocal coefficient access [section 10.6](#).

- F3. Determine whether a nonorthogonal mixed-mode gadget can solve the same bounded-scale complete-direction problem while charging every input-dependent source and recovery map [sections 9.9](#) and [10.6](#).

16.3.4 Positive algorithmic frontier

- P1. Exhibit one explicit family and access model satisfying all six requirements simultaneously—stable direct normalization and inverse, stable forcing and RHS preparation, finite projective variation or a low-parameter barrier, cheap static construction, useful compressed output, and a classical lower bound surviving randomized preprocessing—as required in [section 13.7](#).
- P2. Find a structured access model that supplies a constant-normalized quantum operator without at the same time supplying a bounded-leverage classical sampling distribution, and realize it on an explicit sparse LP family [section 13.4](#).
- P3. Construct a projected block-angular barrier with parameter $\tilde{O}(k + \text{poly}(p))$ whose local center and derivatives are recomputable from one scenario block and the explicit border [section 13.5](#).

The list deliberately does not include directions already closed by a frontier theorem. For example, expanderizing a bounded-degree local-linear parity gadget does not evade the mixture and resistance bounds, and ordinary short-step residual control does not imply finite projective variation [sections 10.6](#) and [13.1](#). A new proposal must leave a proved premise, not merely relabel the same construction.

16.4 Outlook

The most promising next result would be small and complete rather than broad and schematic: one structured family, one explicit access model, one useful output contract, a full path ledger, and a matched classical lower bound or algorithmic comparison. The six-desiderata problem above is deliberately hard because each individual module already has a classical or quantum counterweight. Stable spectral geometry is not enough if RHS preparation is hard; cheap coherent access is not enough if classical sampling sees the same average; compressed output is not enough if the next iterate requires full materialization.

There are also concrete nearer-term targets. The exact-subnormalization question would clarify whether the plain/factor separation in clustered QLS is an artifact of the current construction or an access-model invariant. A general-inner small-success theorem would turn the state-conversion curve from a parity-specific result into a reusable composition principle. A genuinely nonabelian word bound would separate matrix-valued holonomy from its abelian restriction. A projected block-angular barrier would connect the border module—whose $\sqrt{N}$ block-count exponent is optimal in the block-oracle model—to an end-to-end QIPM candidate.

The central methodological lesson is that a future sparse-QIPM advantage should not be advertised by one favorable parameter. It should survive the whole diagram: sublevel geometry constrains the orthonormal-tangent reduced barrier Hessian; the access model determines what transformations are available; and the output interface determines whether the computation can continue. The present paper supplies laws and counterexamples at every arrow in that diagram, together with conditional modules where the arrows can be made cheap. Closing even one of the open problems above would sharpen the boundary; closing the six-desiderata problem would finally give the end-to-end verdict that this paper does not claim.

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